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Panorama Getting Started

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Panorama Overview

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

The Panorama® management server provides centralized monitoring and administration for multiple next-generation firewalls, WildFire® appliances, and appliance clusters. Managing a distributed network of firewalls individually can be complex, time-consuming, and lacks a unified view of network activity. Panorama solves these challenges by offering a single location to oversee all applications, users, and content traversing your network, significantly increasing operational efficiency and enabling you to enforce consistent application enablement policies.To meet diverse network needs, it is available in multiple models, including the virtual appliance and various M-Series hardware appliances. To simplify large-scale deployments, the server offers powerful features for centralized configuration and deployment. You can use templates and template stacks to configure common base network and device settings, while device groups allow for the centralized administration of globally shared and local policy rules.Panorama also streamlines routine maintenance by allowing you to centrally deploy licenses, PAN-OS® software, and dynamic content updates across all managed devices from one location. To safely delegate responsibilities, Role-Based Access Control (RBAC) enables distributed administration by defining granular privileges to restrict or grant access to specific firewalls and configuration areas.For comprehensive visibility, it aggregates logs from all managed firewalls, providing an essential audit trail of network activity and configuration changes. Using integrated tools like the Application Command Center (ACC), log viewers, and robust reporting capabilities, you can centrally analyze traffic patterns, investigate security incidents, and identify potential threats across the entire network. Additionally, the management server optimizes large networks through data redistribution, which seamlessly shares essential User-ID™ data and authentication timestamps with managed firewalls to ensure consistent policy enforcement and maximize device capacity.

About Panorama

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

Panorama enables you to effectively configure, manage, and monitor your Palo Alto Networks firewalls with central oversight. The three main areas in which Panorama adds value are:

- **Centralized configuration and deployment**

  To simplify central management and rapid deployment of the firewalls and WildFire appliances on your network, use Panorama to pre-stage the firewalls and WildFire appliances for deployment. You can then assemble the firewalls into groups, and create templates to apply a base network and device configuration and use device groups to administer globally shared and local policy rules. See Centrally Manage Firewall Configuration and Updates with Panorama.
- **Aggregated logging with central oversight for analysis and reporting**

  Collect information on activity across all the managed firewalls on the network and centrally analyze, investigate and report on the data. This comprehensive view of network traffic, user activity, and the associated risks empowers you to respond to potential threats using the rich set of policies to securely enable applications on your network. See Centralized Logging and Reporting.
- **Distributed administration**

  Enables you to delegate or restrict access to global and local firewall configurations and policies. See Role-Based Access Control for delegating appropriate levels of access for distributed administration.

Six Panorama Models are available: the Panorama virtual appliance, M-600 appliance, M-500 appliance, and M-200 appliance are supported in PAN-OS 10.0 and later releases. The M-300 appliance and M-700 appliance are supported in PAN-OS 10.2 and later releases.Panorama Centralized Management illustrates how you can deploy Panorama in a high availability (HA) configuration to manage firewalls.

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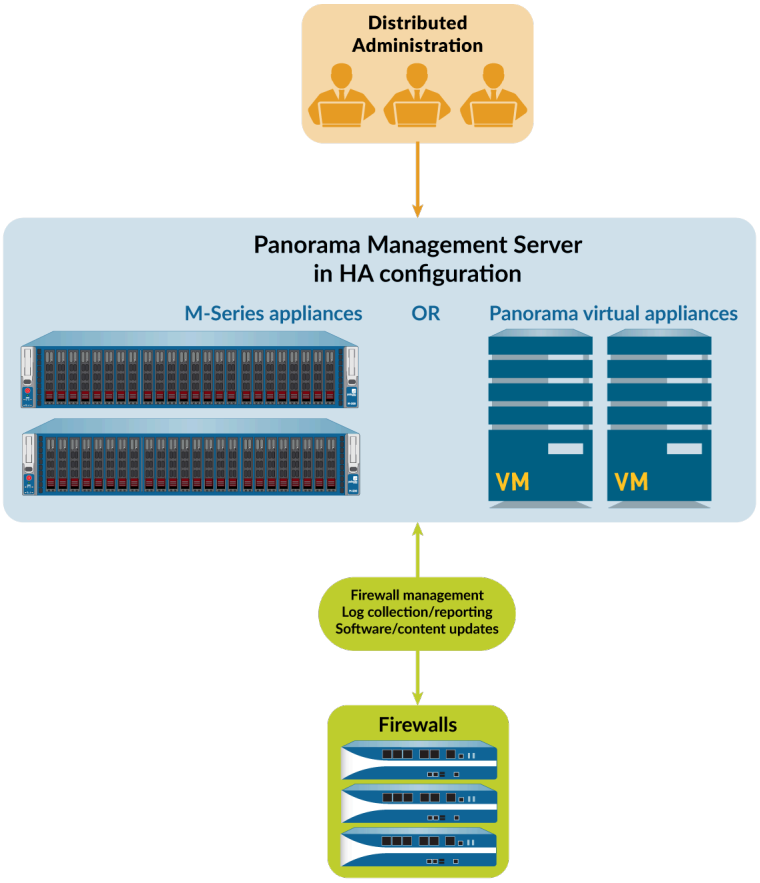

Figure 1: Panorama Centralized Management

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Panorama Models

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

Panorama is available as one of the following virtual or physical appliances, each of which supports licenses for managing up to 25, 100, or 1,000 firewalls. Additionally, M-600 and M-700 appliances support licenses for managing up to 5,000 firewalls and similarly resourced Panorama virtual appliances support licenses for managing up to 2,500 firewalls:

- **Panorama virtual appliance**—This model provides simple installation and facilitates server consolidation for sites that need a virtual management appliance. You can install Panorama on Alibaba Cloud, Amazon Web Services (AWS), AWS GovCloud, Microsoft Azure, Google Cloud Platform (GCP), KVM, Hyper-V, Oracle Cloud Infrastructure (OCI), a VMware ESXi server, or on VMware vCloud Air. The virtual appliance can collect firewall logs locally at rates of up to 20,000 logs per second and can manage Dedicated Log Collectors for higher logging rates. The virtual appliance can function as a dedicated management server, a Panorama management server with local log collection capabilities, or as a Dedicated Log Collector. For the supported interfaces, log storage capacity, and maximum log collection rates, see the Setup Prerequisites for the Panorama Virtual Appliance. You can deploy the virtual appliance in the following modes:
  - **Panorama mode**

    In this mode, the Panorama virtual appliance supports a local Log Collector with 1 to 12 virtual logging disks (see Deploy Panorama Virtual Appliances with Local Log Collectors). Each logging disk has 2TB of storage capacity for a total maximum of 24TB on a single virtual appliance and 48TB on a high availability (HA) pair. Only Panorama mode enables you to add multiple virtual logging disks without losing logs on existing disks. Panorama mode also provides the benefit of faster report generation. In Panorama mode, the virtual appliance does not support NFS storage.
  - **Management Only mode**

    In this mode, the Panorama virtual appliance is a dedicated management appliance for your managed devices and Dedicated Log Collectors. Additionally, an appropriately resourced Panorama virtual appliance can manage up to 2,500 firewalls in this mode. The Panorama virtual appliance has no log collection capabilities except for config and system logs and requires a Dedicated Log Collector to these store logs. By default, the virtual appliance in Management Only mode has only one disk partition for all data so all logs forwarded to a Panorama virtual appliance in Management Only mode are dropped. Therefore, to store the log data from your managed appliances, you must configure log forwarding in order to store the log data from your managed devices. For more information, see Increased Device Management Capacity Requirements.
  - **Log Collector mode**

    The Panorama virtual appliance functions as a Dedicated Log Collector. If multiple firewalls forward large volumes of log data, a Panorama virtual appliance in Log Collector mode provides increased scale and performance. In this mode, the appliance does not have a web

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interface for administrative access; it has only a command line interface (CLI). However, you can manage the appliance using the web interface of the Panorama management server. CLI access to a Panorama virtual appliance in Log Collector mode is necessary only for initial setup and debugging. For configuration details, see Deploy Panorama with Dedicated Log Collectors.

- *As of September 1, 2025, the legacy mode is no longer supported on Panorama. For additional information, see the official end-of-life notice.*

- **M-Series appliance**—The M-200, M-300, M-500, M-600, and M-700 appliances are dedicated hardware appliances intended for large-scale deployments. In environments with high logging rates (over 10,000 logs per second) and log retention requirements, these appliances enable scaling of your log collection infrastructure. For the supported interfaces, log storage capacity, and maximum log collection rates, see M-Series Appliance Interfaces. All M-Series models share the following attributes:
  - RAID drives to store firewall logs and RAID 1 mirroring to protect against disk failures
  - SSD to store the logs that Panorama and Log Collectors generate
  - MGT, Eth1, Eth2, and Eth3 interfaces that support 1Gbps throughput
  - Redundant, hot-swappable power supplies
  - front-to-back airflow

  The M-500 and M-600 appliances have the following additional attributes, which make them more suitable for data centers:

  - Eth4 and Eth5 interfaces that support 10Gbps throughput

  Additionally, the following attribute makes the M-600 and M-700 appliances more suitable for large-scale firewall deployments:

  - The M-600 and M-700 appliances in Management Only mode can manage up to 5,000 firewalls.

  You can deploy the M-Series appliances in the following modes:

  - **Panorama mode**

    The appliance functions as a Panorama management server to manage firewalls and Dedicated Log Collectors. The appliance also supports a local Log Collector to aggregate firewall logs. Panorama mode is the default mode. For configuration details, see Deploy Panorama M-Series Appliances with Local Log Collectors.

  - **Management Only mode**

    The Panorama appliance is a dedicated management appliance for your managed devices and Dedicated Log Collectors. The Panorama appliance has no log collection capabilities except for config and system logs and your deployment requires a Dedicated Log Collector to store these logs. By default, the Panorama appliance in Management Only mode has only one disk partition for all data so all logs forwarded to a Panorama virtual appliance in Management Only mode are dropped. Therefore, to store the log data from your managed

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appliances, you must configure log forwarding in order to store the log data from your managed devices.

- **Log Collector mode**

  The appliance functions as a Dedicated Log Collector. If multiple firewalls forward large volumes of log data, an M-Series appliance in Log Collector mode provides increased scale and performance. In this mode, the appliance does not have a web interface for administrative access; it has only a command line interface (CLI). However, you can manage the appliance using the web interface of the Panorama management server. CLI access to an M-Series appliance in Log Collector mode is necessary only for initial setup and debugging. For configuration details, see Deploy Panorama with Dedicated Log Collectors.

For more details and specifications for the M-Series appliances, see the M-Series Appliance Hardware Reference Guides.

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Centrally Manage Firewall Configuration and Updates with Panorama

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

Panorama™ uses *device groups* and *templates* to group firewalls into logical sets that require similar configuration. You use device groups and templates to centrally manage all configuration elements, policies, and objects on the managed firewalls. Panorama also enables you to centrally manage licenses, software (PAN-OS® software, SSL-VPN client software, GlobalProtect™ agent/app software), and content updates (Applications, Threats, WildFire®, and Antivirus). All device group, template, and template stack configuration objects are required to have a unique name.

In the event an unforeseen restart of your managed firewall or Panorama occurs, all uncommitted configuration changes in your device groups and templates are preserved locally until you successfully commit the changes. A restart can be the restart of the firewall or Panorama or of a PAN-OS management process related to configuration management. For firewalls or Panorama in a high availability (HA) configuration, the uncommitted configuration changes do not automatically sync across the HA peers in the event of an unforeseen restart.

Switch Context Between Panorama and Firewall Views

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

The Panorama™ web interface enables you to toggle between a Panorama-centric view and a firewall-centric view using the **Context** drop-down at the top-left of every tab. Set the **Context** to **Panorama** to manage firewalls centrally or switch context to the web interface of a specific firewall to configure it locally. The similarity of the Panorama and firewall web interfaces enables you to seamlessly move between them to monitor and manage firewalls.

The **Context** drop-down lists only the firewalls that are connected to Panorama. For a Device Group and Template administrator, the drop-down lists only the connected firewalls that are within the Access Domains assigned to that administrator. To search a long list, use the Filters within the drop-down.

For firewalls in a high availability (HA) configuration, the icons have colored backgrounds to indicate the HA state (as follows). Knowing the HA state is useful when selecting a firewall context. For example, you generally make firewall-specific configuration changes on an active firewall.

- **Green**

  Active

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- **Yellow**

  Passive or the firewall is initiating (the initiating state lasts for up to 60 seconds after boot up).
- **Red**

  The firewall is non-functional (error state), suspended (an administrator disabled the firewall), or tentative (for a link or path monitoring event in an active/active HA configuration).

When you configure an admin role profile for a Device Group and Template admin, you must assign a **Device Admin Role** that is pushed to your managed firewalls to context switch between the Panorama and firewall web interface.

During the context switch, Panorama validates if the admin has access to a specific vsys or for all vsys. If the admin has access to all vsys, then Panorama uses the device admin role context switch. If the admin has access to one or some of the vsys, then Panorama uses the vsys admin role to context switch.

Total Configuration Size for Panorama

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

The total configuration file size of Panorama™ M-Series and virtual appliances is an important piece of the performance metric when determining which M-Series appliance or the minimum amount of virtual resources you need to allocate on your Panorama virtual appliance to ensure that you meet your Security requirements. Exceeding the supported total configuration file size of the Panorama management server results in reduced performance when performing configuration changes, commits, and pushes to managed firewalls.

Starting with PAN-OS 11.1, both Panorama and managed firewalls proactively monitor configuration size and alerts administrators when it nears or exceeds the recommended threshold. The lower-end platforms with smaller configuration limits, might hit the warning during the config push from Panorama.

The Panorama management server in Panorama mode supports a total configuration file size of 80MB for all template, device group, and Panorama-specific configurations. The maximum configuration file size supported by Panorama in Management Only mode depends on the Panorama model or resources you allocate to the Panorama virtual appliance. Refer to the table below for the recommended maximum configuration file size based on the Panorama M-Series appliance model or on the resources you allocate to the Panorama virtual appliance.

Panorama Model	Virtual Resources Required	Maximum Configuration File Size in Management Only Mode	Maximum Configuration File Size in Panorama Mode
M-200	N/A	120 MB	80 MB
M-300		150 MB	

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Panorama Model	Virtual Resources Required	Maximum Configuration File Size in Management Only Mode	Maximum Configuration File Size in Panorama Mode
M-500		120 MB	
M-600		150 MB	
M-700		180 MB	
Panorama Virtual Appliance Refer to the Setup Prerequisites for the Panorama Virtual Appliance for additional setup information.	• 16 vCPU • 128GB memory	120 MB	
	• 56 vCPU • 256GB memory	150 MB	

Monitor Configuration Size

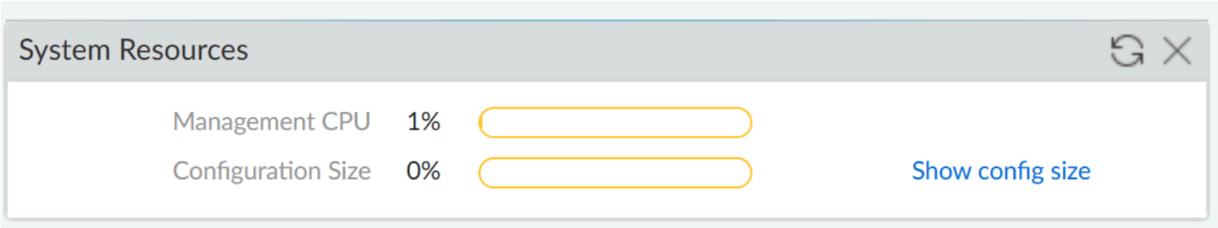
Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

(PAN-OS 11.1 and later releases) You can track the total configuration size as part of your regular system health checks.

STEP 1 | View Configuration Size on the Dashboard.

1. Log in to the Panorama web interface.
2. In the **Dashboard**, locate the **System Resources** section.

   The percentage comparison of the current configuration size against the maximum supported limit appears.



STEP 2 | Drill Down into Configuration Components.

1. Select **Dashboard** > **System Resources**, and click the **Configuration Size** link.

   The **Configuration Size Distribution** window provides a detailed breakdown of your configuration by listing all components and their respective sizes. This helps you

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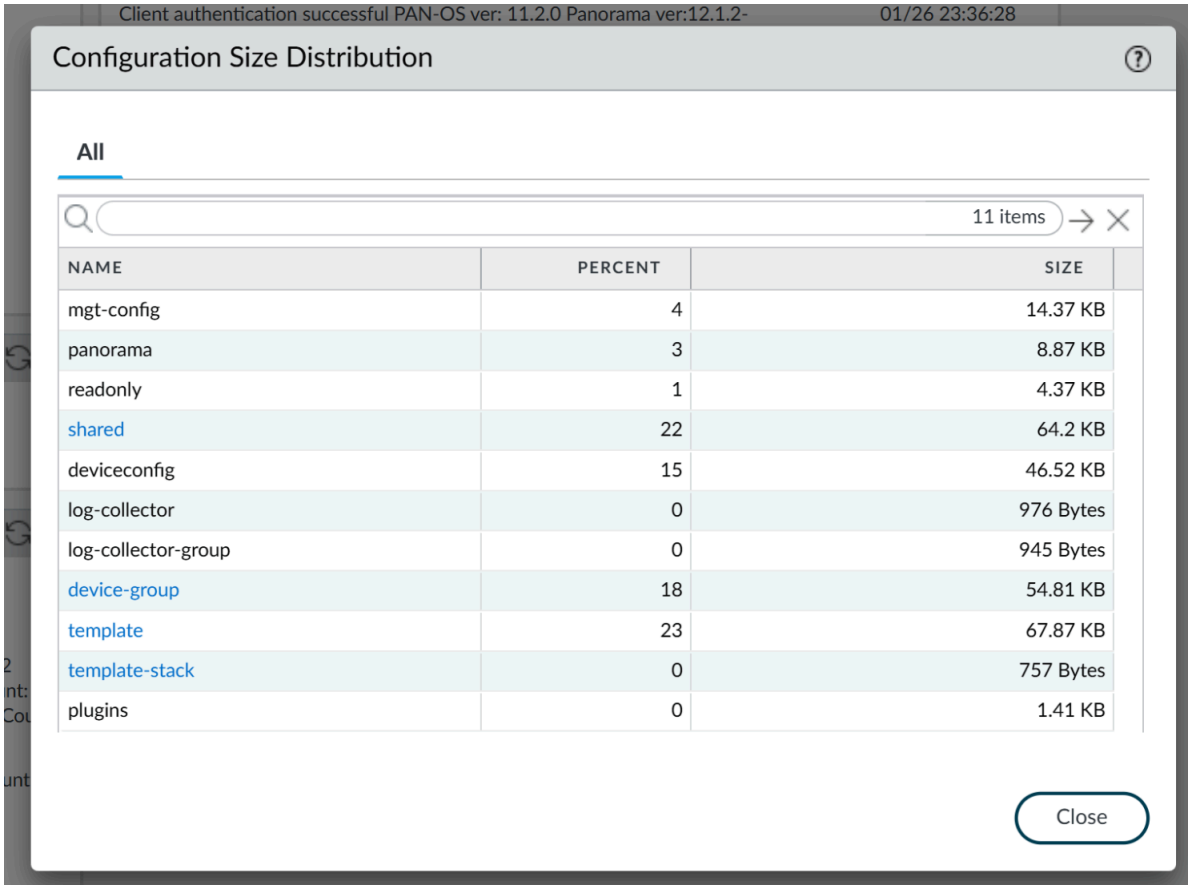
determine exactly which device groups or templates constitute the majority of the configuration.



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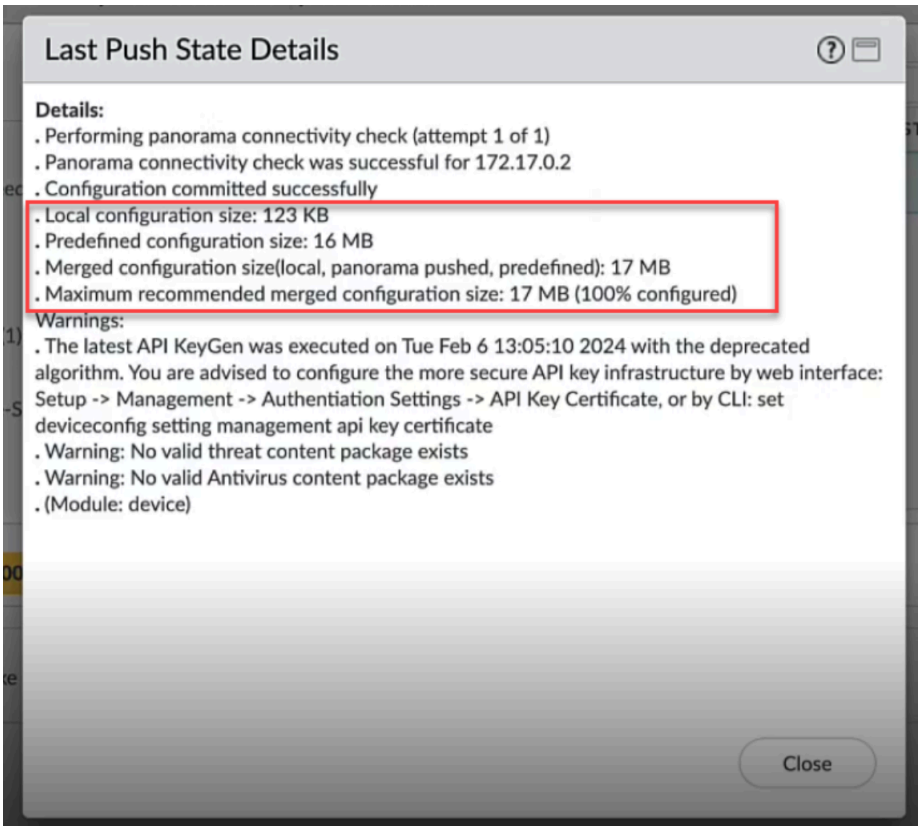
STEP 3 | Review post-commit impact.

1. Perform a configuration change and initiate a **Commit**.
2. Wait for the commit to complete successfully.
3. Review the latest push details.

   You can view the impact of the specific change on the configuration size.



*The following screenshot is specific to the **Commit-All** or **Push** operations; ensure you select the correct scope before finalizing.*

4.

Operational Commands

Use the following CLI commands to retrieve detailed size metrics:

- - **`show management-server candidate config-size`**: Displays the size of the current candidate configuration.
  - **`show management-server last-committed config-size`**: Displays the size of the running configuration.
  - **`show config size distribution`**: Provides a percentage and megabyte breakdown of configuration nodes for remediation.

The low-end firewalls are equipped with configuration file size capacity of 25–30 MB. The pre-defined configurations occupy approximately 75% of this available space. Upon installing standard content updates, the device reaches near-maximum capacity.

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Templates and Template Stacks

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

You use templates and template stacks to configure the settings that enable firewalls to operate on the network. Templates are the basic building blocks you use to configure the **Network** and **Device** tabs on Panorama™. You can use templates to define interface and zone configurations, to manage the server profiles for logging and syslog access, or to define VPN configurations. Template stacks give you the ability to layer multiple templates and create a combined configuration. Template stacks simplify management because they allow you to define a common base configuration for all devices attached to the template stack and they give you the ability to layer templates to create a combined configuration. This enables you to define templates with location- or function-specific settings and then stack the templates in descending order of priority so that firewalls inherit the settings based on the order of the templates in the stack.

Both templates and template stacks support variables. Variables allow you to create placeholder objects with their value specified in the template or template stack based on your configuration needs. Create a template or template stack variable to replace IP addresses, Group IDs, and interfaces in your configurations. Template variables are inherited by the template stack and you can override them to create a template stack variable. However, templates do not inherit variables defined in the template stack. When a variable is defined in the template or template stack and pushed to the firewall, the value defined for the variable is displayed on the firewall.

Use templates to accommodate firewalls that have unique settings. Alternatively, you can push a broader, common base configuration and then override certain pushed settings with firewall-specific values on individual firewalls. When you override a setting on the firewall, the firewall saves that setting to its local configuration and Panorama no longer manages the setting. To restore template values after you override them, use Panorama to force the template or template stack configuration onto the firewall. For example, after you define a common NTP server in a template and override the NTP server configuration on a firewall to accommodate a local time zone, you can later revert to the NTP server defined in the template.

When defining a template stack, consider assigning firewalls that are the same hardware model and require access to similar network resources, such as gateways and syslog servers. This enables you to avoid the redundancy of adding every setting to every template stack. The following figure illustrates an example configuration in which you assign data center firewalls in the Asia-Pacific (APAC) region to a stack with global settings, one template with APAC-specific settings, and one template with data center-specific settings. To manage firewalls in an APAC branch office, you can then re-use the global and APAC-specific templates by adding them to another stack that includes a template with branch-specific settings. Templates in a stack have a configurable priority order that ensures Panorama pushes only one value for any duplicate setting. Panorama evaluates the templates listed in a stack configuration from top to bottom with higher templates having priority. The following figure illustrates a data center stack in which the data center template has a higher priority than the global template: Panorama pushes the idle timeout value from the data center template and ignores the value from the global template.

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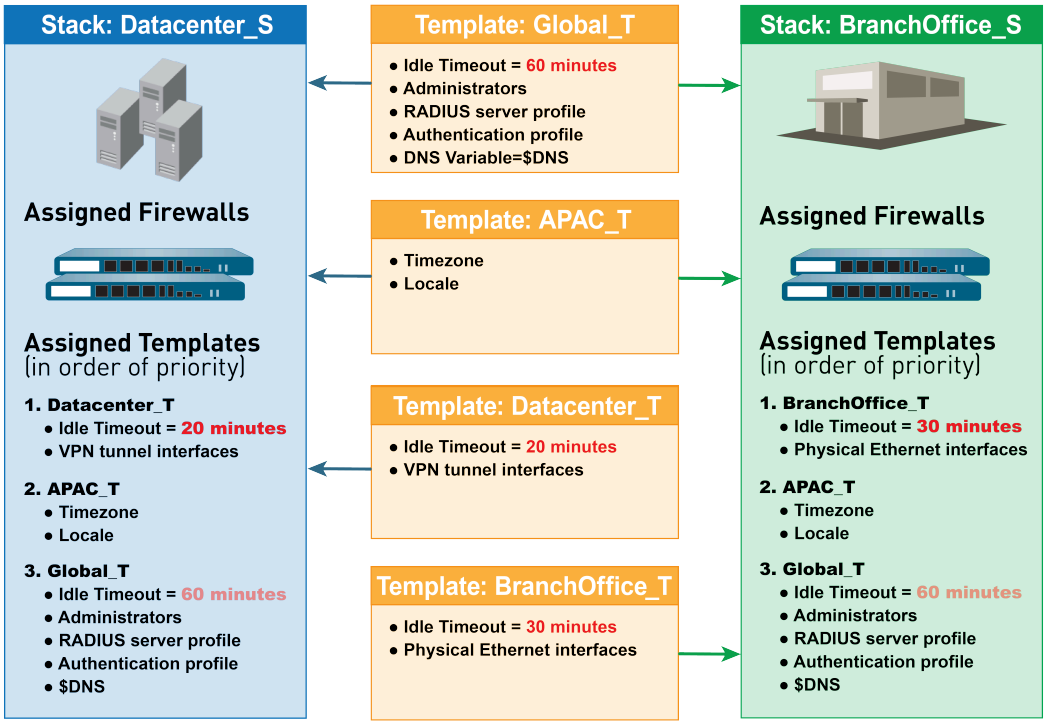

Figure 2: Template Stacks

You cannot use templates or template stacks to set firewall modes: virtual private network (VPN) mode, multiple virtual systems (multi-vsys) mode, or operational modes (normal or FIPS-CC mode). For details, see Template Capabilities and Exceptions. However, you can assign firewalls that have non-matching modes to the same template or stack. In such cases, Panorama pushes mode-specific settings only to firewalls that support those modes. As an exception, you can configure Panorama to push the settings of the default vsys in a template to firewalls that don't support virtual systems or that don't have any virtual systems configured.

For the relevant procedures, see Manage Templates and Template Stacks.

Device Groups

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

To use Panorama effectively, you have to group the firewalls in your network into logical units called *device groups*. A device group enables grouping based on network segmentation, geographic location, organizational function, or any other common aspect of firewalls that require similar policy configurations. Using device groups, you can configure policy rules and the objects they reference. You can organize device group hierarchically, with shared rules and objects at the top, and device group-specific rules and objects at subsequent levels. This enables you to create a hierarchy of rules that enforce how firewalls handle traffic. For example, you can define a set of shared rules as a corporate acceptable use policy. Then, to allow only regional offices to access peer-to-peer traffic such as BitTorrent, you can define a device group rule that Panorama pushes only to the regional offices (or define a shared security rule and target it to the regional offices).

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For the relevant procedures, see Manage Device Groups. The following topics describe device group concepts and components in more detail:

Device Group Hierarchy

You can create a device group hierarchy to nest device groups in a tree hierarchy of up to four levels, with lower-level groups inheriting the settings (policy rules and objects) of higher-level groups. At the bottom level, a device group can have parent, grandparent, and great-grandparent device groups (*ancestors*). At the top level, a device group can have child, grandchild, and great-grandchild device groups (*descendants*). All device groups inheriting settings from the *Shared* location—a container at the top of the hierarchy for configurations that are common to all device groups.

Creating a device group hierarchy enables you to organize firewalls based on common policy requirements without redundant configuration. For example, you could configure shared settings that are global to all firewalls, configure device groups with function-specific settings at the first level, and configure device groups with location-specific settings at lower levels. Without a hierarchy, you would have to configure both function- and location-specific settings for every device group in a single level under Shared.

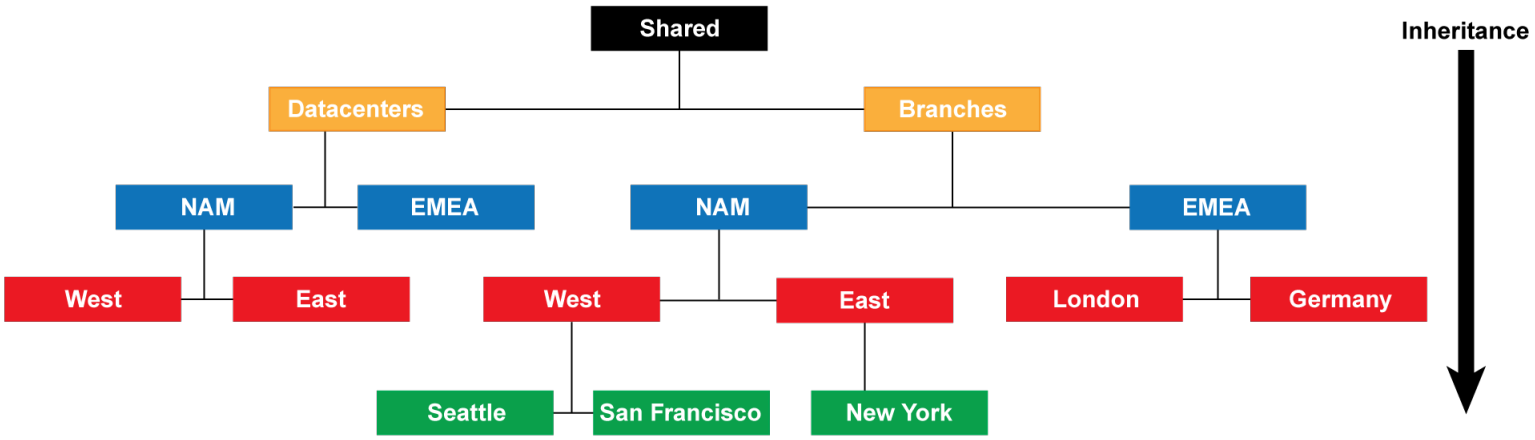

Figure 3: Device Group Hierarchy

For details on the order in which firewalls evaluate policy rules in a device group hierarchy, see Device Group Policies. For details on overriding the values of objects that device groups inherit from ancestor device groups, see Device Group Objects.

In a multiple Panorama plugin deployment to perform, a device group containing firewalls deployed in a particular hypervisor cannot be the child or parent of a device group containing firewalls deployed in a different hypervisor. For example, if Panorama receives IP address updates from VMware NSX-V and AWS, you cannot create a device group of NSX-V VM-Series firewalls that is a child of an AWS VM-Series firewall device group.

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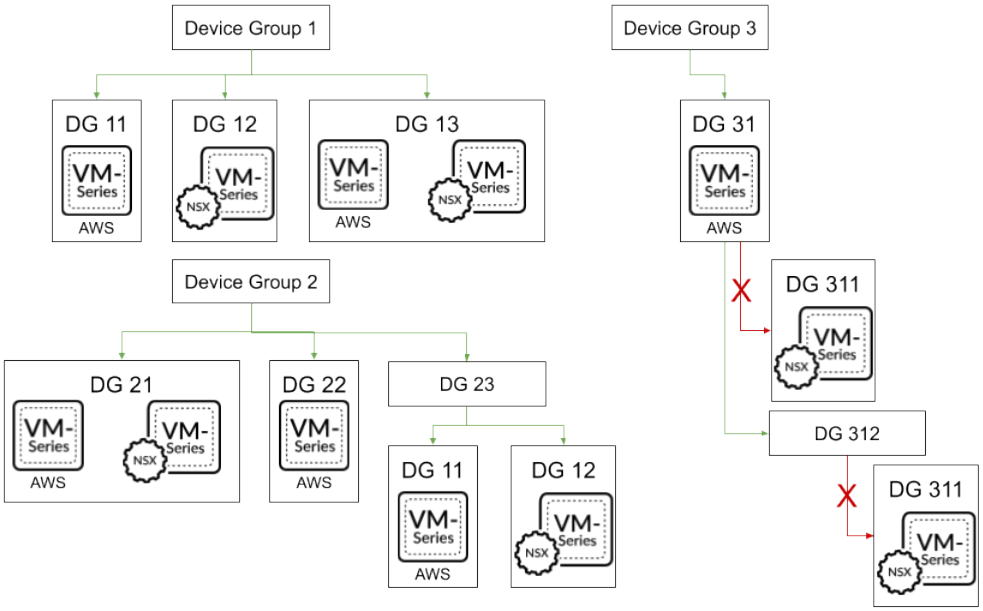
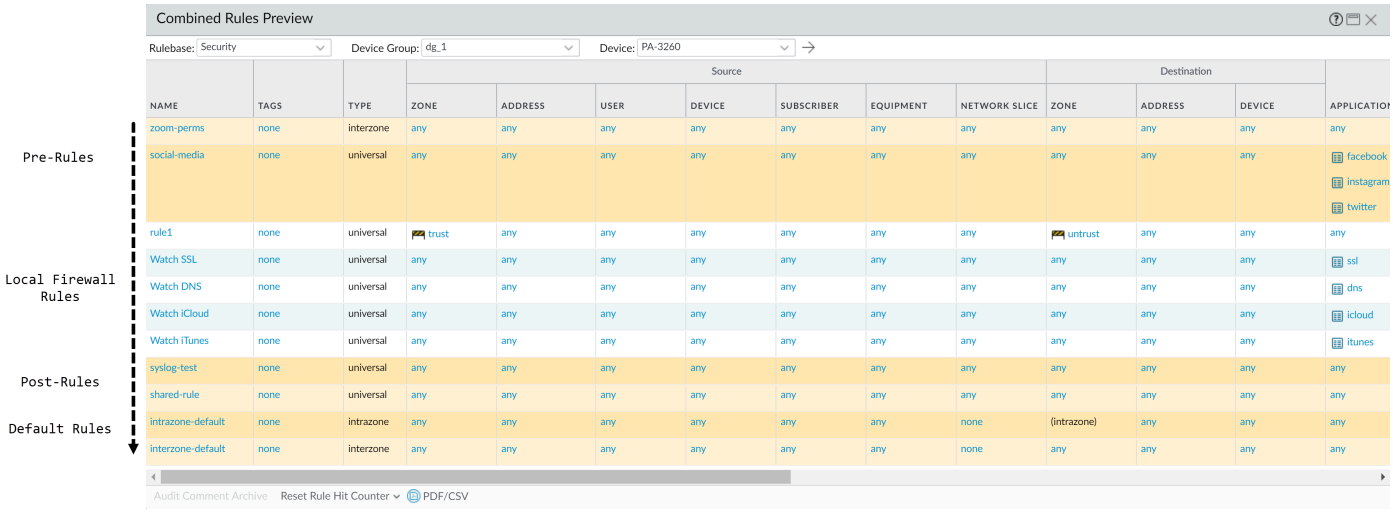


Device Group Policies

Device groups provide a way to implement a layered approach for managing policies across a network of managed firewalls. A firewall evaluates policy rules by layer (shared, device group, and local) and by type (pre-rules, post-rules, and default rules) in the following order from top to bottom. When the firewall receives traffic, it performs the action defined in the first evaluated rule that matches the traffic and disregards all subsequent rules. To change the evaluation order for rules within a particular layer, type, and rulebase (for example, shared Security pre-rules), see Manage the Rule Hierarchy.

Whether you view rules on a firewall or in Panorama, the web interface displays them in evaluation order. All the shared, device group, and default rules that the firewall inherits from Panorama are shaded orange. Local firewall rules display between the pre-rules and post-rules.



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Evaluation Order	Rule Scope and Description	Administration Device
Shared pre-rules Device group pre-rules	Panorama pushes shared pre-rules to all the firewalls in all device groups. Panorama pushes device group-specific pre-rules to all the firewalls in a particular device group and its descendant device groups. If a firewall inherits rules from device groups at multiple levels in the device group hierarchy, it evaluates pre-rules in the order of highest to lowest level. This means the firewall first evaluates shared rules and last evaluates the rules of device groups with no descendants. You can use pre-rules to enforce the acceptable use policy of an organization. For example, a pre-rule might block access to specific URL categories or allow Domain Name System (DNS) traffic for all users.	These rules are visible on firewalls but you can only manage them in Panorama.
Local firewall rules	Local rules are specific to a single firewall or virtual system (vsys).	A local firewall administrator, or a Panorama administrator who switches to a local firewall context, can edit local firewall rules.
Device group post-rules Shared post-rules	Panorama pushes shared post-rules to all the firewalls in all device groups. Panorama pushes device group-specific post-rules to all the firewalls in a particular device group and its descendant device groups. If a firewall inherits rules from device groups at multiple levels in the device group hierarchy, it evaluates post-rules in the order of lowest to highest level. This means the firewall first evaluates the rules of device groups with no descendants and last evaluates shared rules. Post-rules typically include rules to deny access to traffic based on the App-ID™ signatures, User-ID™	These rules are visible on firewalls but you can only manage them in Panorama.

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Evaluation Order	Rule Scope and Description	Administration Device
	information (users or user groups), or service.	
intrazone-default interzone-default	The default rules apply only to the Security rulebase, and are predefined on Panorama (at the Shared level) and the firewall (in each vsys). These rules specify how PAN-OS handles traffic that doesn't match any other rule. The intrazone-default rule allows all traffic within a zone. The interzone-default rule denies all traffic between zones. If you override default rules, their order of precedence runs from the lowest context to the highest: overridden settings at the firewall level take precedence over settings at the device group level, which take precedence over settings at the Shared level.	Default rules are initially read-only, either because they are part of the predefined configuration or because Panorama pushed them to firewalls. However, you can override the rule settings for tags, action, logging, and security profiles. The context determines the level at which you can override the rules: • Panorama—At the Shared or device group level, you can override default rules that are part of the predefined configuration. • Firewall—You can override default rules that are part of the predefined configuration on the firewall or vsys, or that Panorama pushed from the Shared location or a device group.

Device Group Objects

Objects are configuration elements that policy rules reference, for example: IP addresses, URL categories, security profiles, users, services, and applications. Rules of any type (pre-rules, post-rules, default rules, and rules locally defined on a firewall) and any rulebase (Security, NAT, QoS, Policy Based Forwarding, Decryption, Application Override, Captive Portal, and DoS Protection) can reference objects. You can reuse an object in any number of rules that have the same scope as that object in the Device Group Hierarchy. For example, if you add an object to the Shared location, all rules in the hierarchy can reference that *shared object* because all device groups inherit objects from Shared. If you add an object to a particular device group, only the rules in that device group and its descendant device groups can reference that *device group object*. If object values in a device group must differ from those inherited from an ancestor device group, you can Override inherited object values (see Step Override inherited object values). You can also Revert to Inherited Object Values at any time. When you Create Objects for Use in Shared or Device Group Policy once and use them many times, you reduce administrative overhead and ensure consistency across firewall policies.

You can configure how Panorama handles objects system-wide:

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- **Pushing unused objects**

  By default, Panorama pushes all objects to firewalls regardless of whether any shared or device group policy rules reference the objects. Optionally, you can configure Panorama to push only referenced objects. For details, see Manage Unused Shared Objects.

- **Precedence of ancestor and descendant objects**

  By default, when device groups at multiple levels in the hierarchy have an object with the same name but different values (because of overrides, as an example), policy rules in a descendant device group use the object values in that descendant instead of object values inherited from ancestor device groups or Shared. Optionally, you can reverse this order of precedence to push values from Shared or the highest ancestor containing the object to all descendant device groups. For details, see Manage Precedence of Inherited Objects.

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Centralized Logging and Reporting using Panorama

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

Panorama aggregates logs from all managed firewalls and provides visibility across all the traffic on the network. It also provides an audit trail for all policy modifications and configuration changes made to the managed firewalls. In addition to aggregating logs, Panorama can forward them as SNMP traps, email notifications, syslog messages, and HTTP payloads to an external server.

For centralized logging and reporting, you also have the option to use the cloud-based Strata Logging Service that is architected to work seamlessly with Panorama. The Strata Logging Service allows your managed firewalls to forward logs to the Strata Logging Service infrastructure instead of to Panorama or to the managed Log Collectors, so you can augment your existing distributed log collection setup or to scale your current logging infrastructure without having to invest time and effort yourself.

The Application Command Center (ACC) on Panorama provides a single pane for unified reporting across all the firewalls. It enables you to centrally Monitor Network Activity, to analyze, investigate, and report on traffic and security incidents. On Panorama, you can view logs and generate reports from logs forwarded to the Strata Logging Service, Panorama or to the managed Log Collectors, if configured, or you can query the managed firewalls directly. For example, you can generate reports about traffic, threat, and/or user activity in the managed network based on logs stored on Panorama (and the managed collectors) or by accessing the logs stored locally on the managed firewalls, or in the Strata Logging Service.

If you don't Configure Log Forwarding to Panorama or the Strata Logging Service, you can schedule reports to run on each managed firewall and forward the results to Panorama for a combined view of user activity and network traffic. Although reports don't provide a granular drill-down on specific information and activities, they still provide a unified monitoring approach.

Managed Collectors and Collector Groups

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

Panorama uses Log Collectors to aggregate logs from managed firewalls. When generating reports, Panorama queries the Log Collectors for log information, providing you visibility into all the network activity that your firewalls monitor. Because you use Panorama to configure and manage Log Collectors, they are also known as *managed collectors*. Panorama can manage two types of Log Collectors:

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- **Local Log Collector**

  This type of Log Collector runs locally on the Panorama management server. Only an M-700, M-600, M-500, M-300, or M-100 appliance, or Panorama virtual appliance in Panorama mode supports a local Log Collector.

- **Dedicated Log Collector**

  This is an M-700, M-600, M-500, M-300, M-200, or M-100 appliance or Panorama virtual appliance in Log Collector mode. You can use an M-Series appliance in Panorama mode or a Panorama virtual appliance in Panorama or Legacy (ESXi and vCloud Air) mode to manage Dedicated Log Collectors. To use the Panorama web interface for managing Dedicated Log Collectors, you must add them as managed collectors. Otherwise, administrative access to a Dedicated Log Collector is only available through its CLI using the predefined administrative user (*admin*) account. Dedicated Log Collectors don't support additional administrative user accounts.

You can use either or both types of Log Collectors to achieve the best logging solution for your environment (see Local and Distributed Log Collection).

A Collector Group is 1 to 16 managed collectors that operate as a single logical log collection unit. If the Collector Group contains Dedicated Log Collectors, Panorama uniformly distributes the logs across all the disks in each Log Collector and across all Log Collectors in the group. This distribution optimizes the available storage space. To enable a Log Collector to receive logs, you must add it to a Collector Group. You can enable log redundancy by assigning multiple Log Collectors to a Collector Group (see Caveats for a Collector Group with Multiple Log Collectors). The Collector Group configuration specifies which managed firewalls can send logs to the Log Collectors in the group.

To configure Log Collectors and Collector Groups, see Manage Log Collection.

Local and Distributed Log Collection

Before you Configure Log Forwarding to Panorama, you must decide whether to use local Log Collectors, Dedicated Log Collectors, or both.

A local Log Collector is easy to deploy because it requires no additional hardware or virtual machine instance. In a high availability (HA) configuration, you can send logs to the local Log Collector on both Panorama peers; the passive Panorama doesn't wait for failover to start collecting logs.

Dedicated Log Collectors are M-700, M-600, M-300, M-200, or Panorama virtual appliance in Log Collector mode. Because they perform only log collection, not firewall management, Dedicated Log Collectors allow for a more robust environment than local Log Collectors. Dedicated Log Collectors provide the following benefits:

- Enable the Panorama management server to use more resources for management functions instead of logging.
- Provide high-volume log storage on a dedicated hardware appliance.
- Enable higher logging rates.
- Provide horizontal scalability and redundancy with RAID 1 storage.
- Optimize bandwidth resources in networks where more bandwidth is available for firewalls to send logs to nearby Log Collectors than to a remote Panorama management server.

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- Enable you to meet regional regulatory requirements (for example, regulations might not allow logs to leave a particular region).

Distributed Log Collection illustrates a topology in which the Panorama peers in an HA configuration manage the deployment and configuration of firewalls and Dedicated Log Collectors.

You can deploy the Panorama management server in an HA configuration but not the Dedicated Log Collectors.

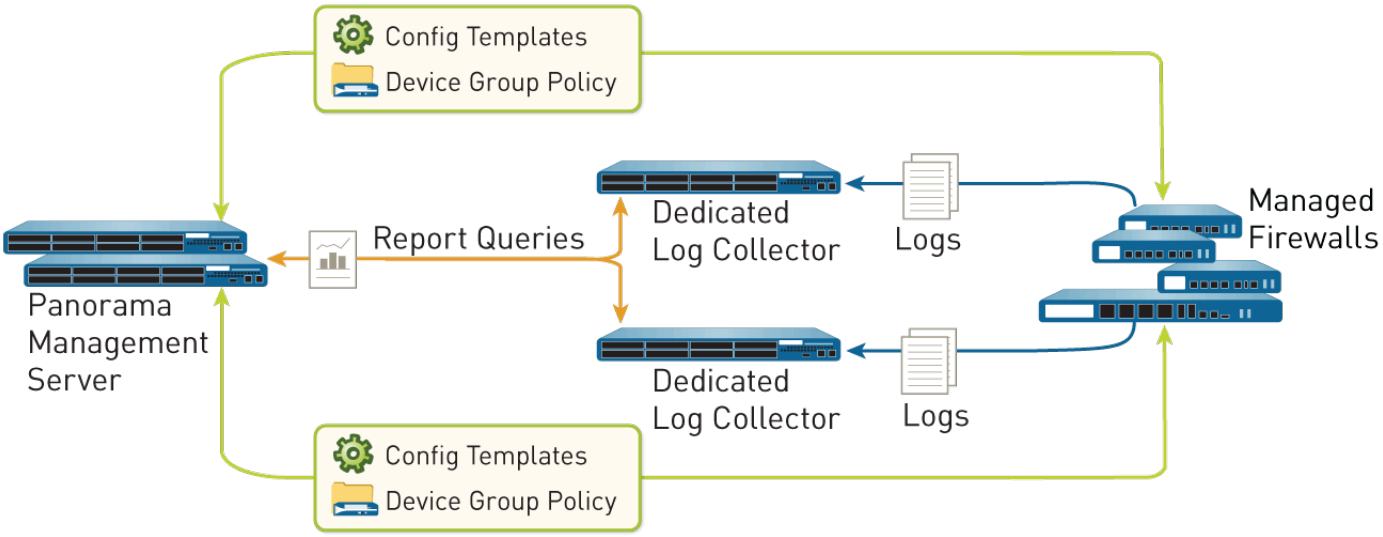

Figure 4: Distributed Log Collection

Caveats for a Collector Group with Multiple Log Collectors

You can Configure a Collector Group with multiple Log Collectors (up to 16) to ensure log redundancy, increase the log retention period, and accommodate logging rates that exceed the capacity of a single Log Collector (see Panorama Models for capacity information). In any single Collector Group, all the Log Collectors must run on the same Panorama model: all M-700 appliances, all M-600 appliances, all M-500 appliances, all M-300 appliances, all M-200 appliances, or all Panorama virtual appliances. For example, if a single managed firewall generates 48 TB of logs, the Collector Group that receives those logs will require at least three Log Collectors that are M-300 appliances or one Log Collector that is an M-700 appliance or similarly resourced Panorama virtual appliance.

A Collector Group with multiple Log Collectors uses the available storage space as one logical unit and uniformly distributes the logs across all its Log Collectors. The log distribution is based on the disk capacity of the Log Collectors (see Panorama Models) and a hash algorithm that dynamically decides which Log Collector owns the logs and writes to disk. Although Panorama uses a preference list to prioritize the list of Log Collectors to which a managed firewall can forward logs, Panorama does not necessarily write the logs to the first Log Collector specified in the preference list. For example, consider the following preference list:

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Managed Firewall	Log Forwarding Preference List Defined in a Collector Group
FW1	L1,L2,L3
FW2	L4,L5,L6

Using this list, FW1 will forward logs to L1 so long as that primary Log Collector is available. However, based on the hash algorithm, Panorama might choose L2 as the owner who writes the logs to its disks. If L2 becomes inaccessible or has a chassis failure, FW1 won't know because it can still connect to L1.

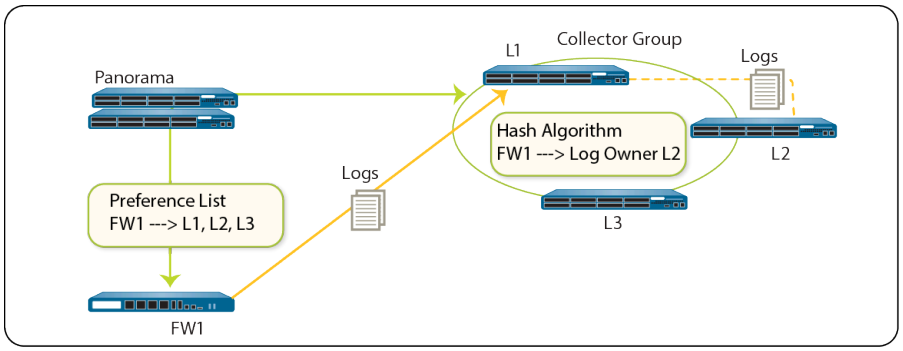

Figure 5: Example - Typical Log Collector Group Setup

In the case where a Collector Group has only one Log Collector and the Log Collector fails, the firewall stores the logs to its HDD/SSD (the available storage space varies by firewall model). As soon as connectivity is restored to the Log Collector, the firewall resumes forwarding logs where it left off before the failure occurred.

In the case of a Collector Group with multiple Log Collectors, the firewall does not buffer logs to its local storage if only one Log Collector is down. In the example scenario where L2 is down, FW1 continues sending logs to L1, and L1 stores the log data that would be sent to L2. Once L2 is back up, L1 no longer stores log data intended for L2 and distribution resumes as expected. If one of the Log Collectors in a Collector Group goes down, the logs that would be written to the down Log Collector are redistributed to the next Log Collector in the preference list.

Palo Alto Networks recommends adding at least three Log Collectors to a Collector Group to avoid split brain and log ingestion issues should one Log Collector go down. See the changes to default Collector Group behavior for more information.

Two Log Collectors in a Collector Group are supported, but the Collector Group becomes non-operational if one Log Collector goes down.

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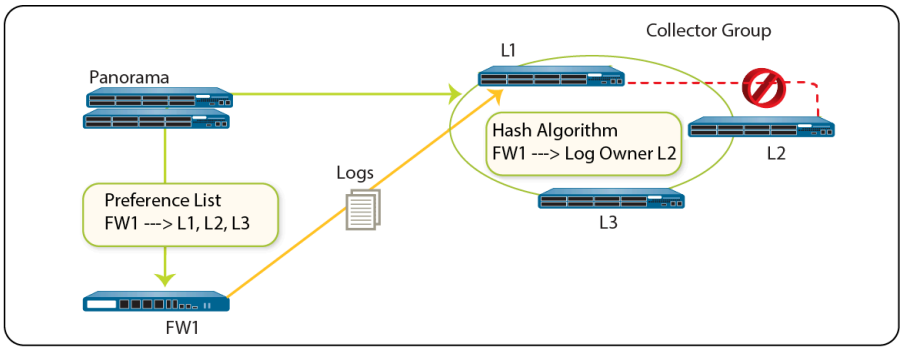

Figure 6: Example - When a Log Collector Fails

Palo Alto Networks recommends the following mitigations if using multiple Log Collectors in a Collector Group:

- Enable log redundancy when you Configure a Collector Group. This ensures that no logs are lost if any one Log Collector in the Collector Group becomes unavailable. Each log will have two copies and each copy will reside on a different Log Collector. Log redundancy is available only if each Log Collector has the same number of logging disks.

  Because enabling redundancy creates more logs, this configuration requires more storage capacity. When a Collector Group runs out of space, it deletes older logs.

  Enabling redundancy doubles the log processing traffic in a Collector Group, which reduces its maximum logging rate by half, as each Log Collector must distribute a copy of each log it receives.

- Obtain an On-Site-Spare (OSS) to enable prompt replacement if a Log Collector failure occurs.
- In addition to forwarding logs to Panorama, configure forwarding to an external service as standby storage. The external service can be a syslog server, email server, SNMP trap server, or HTTP server.

  If you enable log redundancy in a Collector Group and also configure syslog forwarding to an external service, be aware that syslog forwarding sends both copies of each redundant log. This results in duplicate logs being forwarded to your syslog server. Each log will be sent twice—once from each Log Collector that stores a copy—effectively doubling the volume of logs forwarded via syslog.

  To avoid receiving duplicate logs on your syslog server, consider one of the following approaches:

  - *Disable log redundancy if your primary goal is syslog forwarding and you can accept the risk of potential log loss during Log Collector failures.*
  - *Configure your syslog server to deduplicate incoming logs based on log attributes such as serial number, session ID, and timestamp.*
  - *Configure syslog forwarding on only one Log Collector in the Collector Group, though this does not provide redundancy for syslog forwarding itself.*

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Log Forwarding Options

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

By default, each firewall stores its log files locally. To use Panorama for centralized log monitoring and report generation, you must Configure Log Forwarding to Panorama. Logs are forwarded over the management interface by default unless you configure a dedicated service route to forward logs. Panorama supports forwarding logs to either a Log Collector, the Strata Logging Service, or both in parallel. You can also use external services for archiving, notification, or analysis by forwarding logs to the services directly from the firewalls or from Panorama. External services include the syslog servers, email servers, SNMP trap servers, or HTTP-based services. In addition to forwarding firewall logs, you can forward the logs that the Panorama management server and Log Collectors generate. The Panorama management server, Log Collector, or firewall that forwards the logs converts them to a format that is appropriate for the destination (syslog message, email notification, SNMP trap, or HTTP payload).

Palo Alto Networks firewalls and Panorama support the following log forwarding options. Before choosing an option, consider the logging capacities of your Panorama Models and Determine Panorama Log Storage Requirements.

- Forward logs from firewalls to Panorama and from Panorama to external services—This configuration is best for deployments in which the connections between firewalls and external services have insufficient bandwidth to sustain the logging rate, which is often the case when the connections are remote. This configuration improves firewall performance by offloading some processing to Panorama.

  You can configure each Collector Group to forward logs to different destinations.

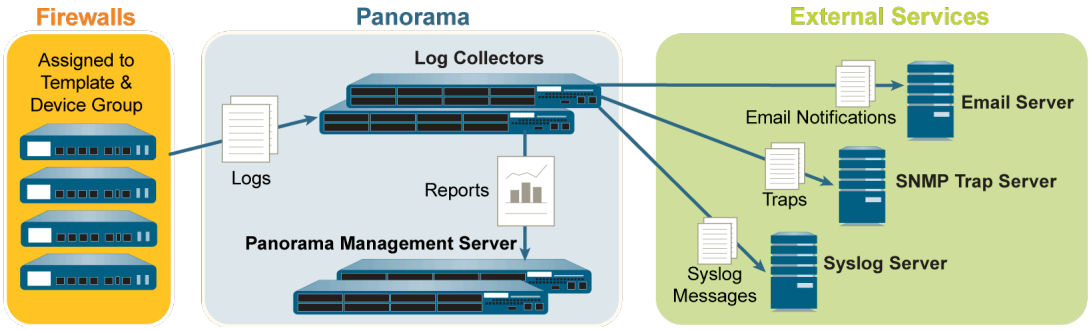

Figure 7: Log Forwarding to Panorama and then to External Services

- Forward logs from firewalls to Panorama and to external services in parallel—In this configuration, both Panorama and the external services are endpoints of separate log forwarding flows; the firewalls don't rely on Panorama to forward logs to external services. This configuration is best for deployments in which the connections between firewalls and external services have sufficient bandwidth to sustain the logging rate, which is often the case when the connections are local.

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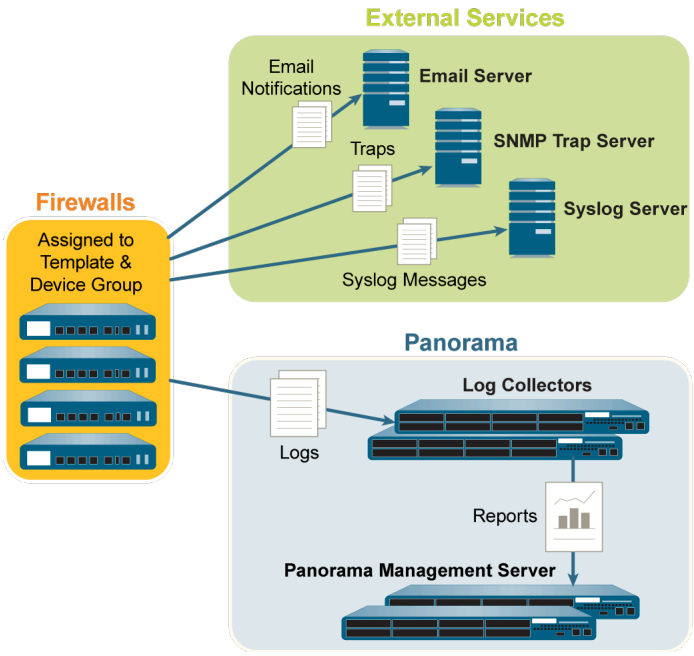

Figure 8: Log Forwarding to External Services and Panorama in Parallel

Centralized Reporting

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

Panorama aggregates logs from all managed firewalls and enables reporting on the aggregated data for a global view of application use, user activity, and traffic patterns across the entire network. As soon as the firewalls are added to Panorama, the ACC can display all traffic traversing your network. With logging enabled, clicking into a log entry in the ACC provides direct access to granular details about the application.

For generating reports, Panorama uses two sources: the local Panorama database and the remote firewalls that it manages. The Panorama database refers to the local storage on Panorama that is allocated for storing both summarized logs and some detailed logs. If you have a distributed Log Collection deployment, the Panorama database includes the local storage on Panorama and all the managed Log Collectors. Panorama summarizes the information—traffic, application, threat — collected from all managed firewalls at 15-minute intervals. Using the local Panorama database allows for faster response times, however, if you prefer to not forward logs to Panorama, Panorama can directly access the remote firewall and run reports on data that is stored locally on the managed firewalls.

Panorama offers more than 40 predefined reports that can be used as is, or they can be customized by combining elements of other reports to generate custom reports and report groups that can be saved. Reports can be generated on demand, on a recurring schedule, and can be scheduled for email delivery. These reports provide information on the user and the context so that you correlate events and identify patterns, trends, and potential areas of interest. With the

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integrated approach to logging and reporting, the ACC enables correlation of entries from multiple logs relating to the same event.

For more information, see Monitor Network Activity.

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Data Redistribution Using Panorama

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

With data redistribution, you only have to configure each source once, then you can redistribute multiple data types to as many clients as needed. This helps you to scale your network so that you can easily add or remove source and clients as your network needs change.

Data redistribution also provides granularity by redistributing only the types of information to only the firewalls or Panorama management systems that you specify. You can use subnets, ranges, and regions to further reduce network traffic and maximize device capacity.

One of the key benefits of the Palo Alto Networks firewall is that it can enforce policies and generate reports based on usernames and tags instead of IP addresses. The challenge for large-scale networks is ensuring every firewall that enforces policies and generates reports has the mappings and tags that apply for all of your policy rules. Additionally, every firewall that enforces Authentication Policy requires a complete, identical set of authentication timestamps for your user base. Whenever users authenticate to access services and applications, individual firewalls record the associated timestamps but don't automatically share them with other firewalls to ensure consistency. Data redistribution solves these challenges for large-scale networks by enabling you to redistribute the necessary data. However, instead of setting up extra connections to redistribute the data between firewalls, you can leverage your Panorama infrastructure to Redistribute Data to Managed Firewalls.

The infrastructure has existing connections that enable you to redistribute data in layers, from firewalls to Panorama. Panorama can then redistribute the information to the firewalls that enforce policies and generate reports.

Each firewall or Panorama management server can receive data from up to 100 redistribution points. The redistribution points can be other firewalls or Panorama management servers. However, you can also use Windows-based User-ID agents to perform the mapping and redistribute the information to firewalls. Only the firewalls record authentication timestamps when user traffic matches Authentication policy rules.

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Role-Based Access Control

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

Role-based access control (RBAC) enables you to define the privileges and responsibilities of administrative users (administrators). Every administrator must have a user account that specifies a role and authentication method. Administrative Roles define access to specific configuration settings, logs, and reports within Panorama and firewall contexts. For Device Group and Template administrators, you can map roles to Access Domains, which define access to specific device groups, templates, and firewalls (through context switching). By combining each access domain with a role, you can enforce the separation of information among the functional or regional areas of your organization. For example, you can limit an administrator to monitoring activities for data center firewalls but allow that administrator to set policies for test lab firewalls. By default, every Panorama appliance (virtual appliance or M-Series appliance) has a predefined administrative account (admin) that provides full read-write access (superuser access) to all functional areas and to all device groups, templates, and firewalls. For each administrator, you can define an authentication profile that determines how Panorama verifies user access credentials.

Instead of using the default account for all administrators, it is a best practice to create a separate administrative account for each person who needs access to the administrative or reporting functions on Panorama. This provides better protection against unauthorized configuration changes and enables Panorama to log and identify the actions of each administrator.

- Administrative Roles
- Authentication Profiles and Sequences
- Access Domains
- Administrative Authentication

Administrative Roles

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

You configure administrator accounts based on the security requirements of your organization, any existing authentication services that your network uses, and the required administrative roles. A *role* defines the type of system access that is available to an administrator. You can define and restrict access as broadly or granularly as required, depending on the security requirements of your organization. For example, you might decide that a data center administrator can have access to all device and networking configurations, but a security administrator can control only security policy definitions, while other key individuals can have limited CLI or XML API access. The role types are:

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- **Dynamic Roles**

  These are built-in roles that provide access to Panorama and managed firewalls. When new features are added, Panorama automatically updates the definitions of dynamic roles; you never need to manually update them. The following table lists the access privileges associated with dynamic roles.

Dynamic Role	Privileges
Superuser	Full read-write access to Panorama
Superuser (read-only)	Read-only access to Panorama
Panorama administrator	Full access to Panorama except for the following actions: • Create, modify, or delete Panorama or firewall administrators and roles. • Export, validate, revert, save, load, or import a configuration in the **Device** > **Setup** > **Operations** page. • Configure **Scheduled Config Export** functionality in the **Panorama** tab. • **Generate Tech Support File**, **Generate Stats Dump File**, and **Download Core Files** (**Panorama** > **Support**)

- **Admin Role Profiles**

  To provide more granular access control over the functional areas of the web interface, CLI, and XML API, you can create custom roles. When new features are added to the product, you must update the roles with corresponding access privileges: Panorama does not automatically add new features to custom role definitions. You select one of the following profile types when you configure an admin role profile.

Admin Role Profile	Description
Panorama	For these roles, you can assign read-write access, read-only access, or no access to all the Panorama features that are available to the superuser dynamic role except the management of Panorama administrators and Panorama roles. For the latter two features, you can assign read-only access or no access, but you cannot assign read-write access. An example use of a Panorama role would be for security administrators who require access to security policy definitions, logs, and reports on Panorama. Custom Panorama admin roles have the following limitations: • No access to **Reboot Panorama** (**Panorama** > **Setup** > **Operations**)

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Admin Role Profile	Description
	• No access to **Generate Tech Support File**, **Generate Stats Dump File**, and **Download Core Files** (**Panorama** > **Support**)

Device Group and Template	For these roles, you can assign read-write access, read-only access, or no access to specific functional areas within device groups, templates, and firewall contexts. By combining these roles with Access Domains, you can enforce the separation of information among the functional or regional areas of your organization. Device Group and Template roles have the following limitations:
	• No access to the CLI or XML API
	• No access to configuration or system logs
	• No access to VM information sources
	• No access to **Reboot Panorama** (**Panorama** > **Setup** > **Operations**)
	• No access to **Generate Tech Support File**, **Generate Stats Dump File**, and **Download Core Files** (**Panorama** > **Support**)
	• In the **Panorama** tab, access is limited to:
	  • Device deployment features (read-write, read-only, or no access)
	  • The device groups specified in the administrator account (read-write, read-only, or no access)
	  • The templates and managed firewalls specified in the administrator account (read-only or no access)
	An example use of this role would be for administrators in your operations staff who require access to the device and network configuration areas of the web interface for specific device groups and/or templates.

Authentication Profiles and Sequences

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

An authentication profile defines the authentication service that validates the login credentials of administrators when they access Panorama. The service can be local authentication or an external authentication service. Some services (SAML, TACACS+, and RADIUS) provide the option to manage both authentication and authorization for administrative accounts on the external server instead of on Panorama. In addition to the authentication service, the authentication profile defines options such as Kerberos single sign-on (SSO) and SAML single logout (SSO).

Some networks have multiple databases (such as TACACS+ and LDAP) for different users and user groups. To authenticate administrators in such cases, configure an authentication sequence—a ranked order of authentication profiles that Panorama matches an administrator against during

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login. Panorama checks against each profile in sequence until one successfully authenticates the administrator. An administrator is denied access only if authentication fails for all the profiles in the sequence.

Access Domains

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

Access domains control administrative access to specific Device Groups and templates, and also control the ability to sswitch context to the web interface of managed firewalls. Access domains apply only to administrators with Device Group and Template roles. Mapping Administrative Roles to access domains enables very granular control over the information that administrators access on Panorama. For example, consider a scenario where you configure an access domain that includes all the device groups for firewalls in your data centers and you assign that access domain to an administrator who is allowed to monitor data center traffic but who is not allowed to configure the firewalls. In this case, you would map the access domain to a role that enables all monitoring privileges but disables access to device group settings. Additionally, Device Group and Template admins can perform administrative tasks for managed firewalls in their access domain such as viewing the configuration and system logs, perform configuration audits, review pending tasks, and directly access firewall operations such as reboot, generating a tech support file, executing a stats dump, and exporting a core file.

You configure access domains in the local Panorama configuration and then assign them to administrative accounts and roles. You can perform the assignment locally or use an external SAML, TACACS+, or RADIUS server. Using an external server enables you to quickly reassign access domains through your directory service instead of reconfiguring settings on Panorama. To use an external server, you must define a server profile that enables Panorama to access the server. You must also define Vendor-Specific Attributes (VSAs) on the RADIUS or TACACS+ server, or SAML attributes on the SAML IdP server.

For example, if you use a RADIUS server, you would define a VSA number and value for each administrator. The value defined has to match the access domain configured on Panorama. When an administrator tries to log in to Panorama, Panorama queries the RADIUS server for the administrator access domain and attribute number. Based on the response from the RADIUS server, the administrator is authorized for access and is restricted to the firewalls, virtual systems, device groups, and templates that are assigned to the access domain.

For the relevant procedures, see:

- Configure an Access Domain
- Configure RADIUS Authentication for Panorama Administrators
- Configure TACACS+ Authentication for Panorama Administrators
- Configure SAML Authentication for Panorama Administrator

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Administrative Authentication

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

You can configure the following types of authentication and authorization (Administrative Roles and Access Domains) for Panorama administrators:

Authentication Method	Authorization Method	Description
Local	Local	The administrative account credentials and authentication mechanisms are local to Panorama. You use Panorama to assign administrative roles and access domains to the accounts. To further secure the accounts, you can create a password profile that defines a validity period for passwords and set Panorama-wide password complexity settings. For details, see Configure Local or External Authentication for Panorama Administrators.
SSH Keys	Local	The administrative accounts are local to Panorama, but authentication to the CLI is based on SSH keys. You use Panorama to assign administrative roles and access domains to the accounts. For details, see Configure an Administrator with SSH Key-Based Authentication for the CLI.
Certificates	Local	The administrative accounts are local to Panorama, but authentication to the web interface is based on client certificates. You use Panorama to assign administrative roles and access domains to the accounts. For details, see Configure a Panorama Administrator with Certificate-Based Authentication for the Web Interface.
External service	Local	The administrative accounts you define locally on Panorama serve as references to the accounts defined on an external Multi-Factor Authentication, SAML, Kerberos, TACACS+, RADIUS, or LDAP server. The external server performs authentication. You use Panorama to assign administrative roles and access domains to the accounts. For details, see Configure Local or External Authentication for Panorama Administrators.
External	External service	The administrative accounts are defined only on an external SAML, TACACS+, or RADIUS server. The server performs both authentication and authorization. For authorization, you define Vendor-Specific Attributes (VSAs) on the TACACS+ or RADIUS server, or SAML attributes on the SAML server. Panorama maps the attributes to administrator roles and access domains that you define on Panorama. For details, see:

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Authentication Method	Authorization Method	Description
		• Configure SAML Authentication for Panorama Administrators • Configure TACACS+ Authentication for Panorama Administrators • Configure RADIUS Authentication for Panorama Administrators

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Panorama Commit, Validation, and Preview Operations

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

When you are ready to activate changes that you made to the candidate configuration on Panorama or to push changes to the devices that Panorama manages (firewalls, Log Collectors, and WildFire appliances and appliance clusters), you can Preview, Validate, or Commit Configuration Changes. For example, if you add a Log Collector to the Panorama configuration, firewalls cannot send logs to that Log Collector until you commit the change to Panorama and then push the change to the Collector Group that contains the Log Collector.

You can filter changes by administrator or *location* and then commit, push, validate, or preview only those changes. The location can be specific device groups, templates, Collector Groups, Log Collectors, shared settings, or the Panorama management server.

When you commit changes, they become part of the running configuration. Changes that you haven't committed are part of the candidate configuration. Panorama queues commit requests so that you can initiate a new commit while a previous commit is in progress. Panorama performs the commits in the order they are initiated but prioritizes auto-commits that are initiated by Panorama (such as FQDN refreshes). However, if the queue already has the maximum number of administrator-initiated commits (10), you must wait for Panorama to finish processing a pending commit before initiating a new one. You can Use the Panorama Task Manager (Tasks) to cancel pending commits or to see details about commits that are pending, in progress, completed, or failed. To check which changes a commit will activate, you can run a commit preview.

When you initiate a commit, Panorama checks the validity of the changes before activating them. The validation output displays conditions that block the commit (errors) or that are important to know (warnings). For example, validation could indicate an invalid route destination that you need to fix for the commit to succeed. The validation process enables you to find and fix errors before you commit (it makes no changes to the running configuration). This is useful if you have a fixed commit window and want to be sure the commit will succeed without errors.

When you preview your configuration commit, any configuration object added between existing any other existing object is displayed as a modified configuration object rather than an added configuration object. For example, `Address1` and `Address2` are existing Address objects. A Panorama admin later creates `Address3` and adds the Address object between `Address1` and `Address2`. When the Panorama admin goes to preview the configuration changes, `Address3` is displayed as a modified configuration object.

Automated commit recovery is enabled by default, allowing the managed firewalls to locally test the configuration pushed from Panorama to verify that the new changes do not break the connection between Panorama and the managed firewall. If the committed configuration breaks the connection between Panorama and a managed firewall then the firewall automatically fails the commit and the configuration is reverted to the previous running configuration and the Shared Policy or Template Status (**Panorama** > **Managed Devices** > **Summary**) gets out of sync depending on which configuration objects were pushed. Additionally, the managed firewalls test their connection to Panorama every 60 minutes and if a managed firewall detects that it can no

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longer successfully connect to Panorama then it reverts its configuration to the previous running configuration.

For details on candidate and running configurations, see Manage Panorama and Firewall Configuration Backups.

To prevent multiple administrators from making configuration changes during concurrent sessions, see Manage Locks for Restricting Configuration Changes.

When pushing configurations to managed firewalls, Panorama pushes the running configuration. Because of this, Panorama does not let you push changes to managed firewalls until you first commit the changes to Panorama.

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Plan Your Panorama Deployment

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

- Determine the management approach. Do you plan to use Panorama to centrally configure and manage the policies, to centrally administer software, content and license updates, and/or centralize logging and reporting across the managed firewalls in the network?

  If you already deployed and configured Palo Alto Networks firewalls on your network, determine whether to transition the firewalls to centralized management. This process requires a migration of all configuration and policies from your firewalls to Panorama. For details, see Transition a Firewall to Panorama Management.
- Verify the Panorama and firewall software versions. Panorama can manage firewalls running PAN-OS versions that match the Panorama version or are earlier than the Panorama version. See Panorama Management Compatibility for more information.
- (Multi-vsys firewalls) If you already deployed and configured multi-vsys Palo Alto Networks firewalls on your network, Palo Alto Networks recommends you transition and manage all vsys configurations of the multi-vsys firewall from Panorama. This is required to avoid commit issues on the multi-vsys firewall and allows you to take advantage of the optimized shared object pushes from Panorama.
- (Multi-vsys firewalls) Delete or rename any locally configured firewall **Shared** object that has an identical name to an object in the Panorama **Shared** configuration. Otherwise, configuration pushes from Panorama fail after the upgrade and display the error `<object-name> is already in use`.
- Determine your authentication method between Panorama and its managed devices and high availability peer. By default, Panorama uses predefined certificates to authenticate the SSL connections used for management and inter-device communication. However, you can configure custom certificate-based authentication to enhance the security of the SSL connections between Panorama, firewalls, and log collectors. By using custom certificates, you can establish a unique chain of trust to ensure mutual authentication between Panorama and the devices it manages. You can import the certificates from your enterprise public key infrastructure (PKI) or generate it on Panorama.
- Plan to use Panorama in a high availability configuration; set it up as an active/passive high availability pair. See Panorama High Availability.
- Plan how to accommodate network segmentation and security requirements in a large-scale deployment. By default, Panorama running on an M-Series appliance uses the management (MGT) interface for administrative access to Panorama and for managing devices (firewalls, Log Collectors, and WildFire appliances and appliance clusters), collecting logs, communicating with Collector Groups, and deploying software and content updates to devices. However, to improve security and enable network segmentation, you can reserve the MGT interface for administrative access and use dedicated M-Series Appliance Interfaces (Eth1, Eth2, Eth3, Eth4, and Eth5) for the other services.

- ☐ For meaningful reports on network activity, plan a logging solution:
  - Verify the resource allocation for your Panorama virtual appliance deployed in Log Collector mode on AWS or Azure. The Panorama virtual appliance does not retain Log Collector mode if resized. This results in log data loss.
  - Estimate the log storage capacity your network needs to meet security and compliance requirements. Consider such factors as the logging capacities of your Panorama Models, network topology, number of firewalls sending logs, type of log traffic (for example, URL Filtering and Threat logs versus Traffic logs), the rate at which firewalls generate logs, and the number of days for which you want to store logs on Panorama. For details, see Determine Panorama Log Storage Requirements.
  - Do you need to forward logs to external services (such as a syslog server) in addition to Panorama? See Log Forwarding Options.
  - Do you want to own or manage your own log storage on premises, or do you want to leverage the Strata Logging Service provided by Palo Alto Networks?
  - If you need a long-term storage solution, do you have a Security Information and Event Management (SIEM) solution, such as Splunk or ArcSight, to which you can forward logs?
  - Do you need redundancy in logging?

    If you configure a Collector Group with multiple Log Collectors, you can enable redundancy to ensure that no logs are lost if any one Log Collector becomes unavailable (see Caveats for a Collector Group with Multiple Log Collectors).
- ☐ Determine which role-based access privileges administrators require to access managed firewalls and Panorama. See Configure Administrative Access to Panorama.
- ☐ Plan the required Device Groups. Consider whether to group firewalls based on function, security policy, geographic location, or network segmentation. An example of a function-based device group is one that contains all the firewalls that a Research and Development team uses. Consider whether to create smaller device groups based on commonality, larger device groups to scale more easily, or a Device Group Hierarchy to simplify complex layers of administration.
- ☐ Plan a layering strategy for administering policies. Consider how firewalls inherit and evaluate policy rules within the Device Group Hierarchy, and how to best implement shared rules, device-group rules, and firewall-specific rules to meet your network needs. For visibility and centralized policy management, consider using Panorama for administering rules even if you need firewall-specific exceptions for shared or device group rules. If necessary, you can Push a Policy Rule to a Subset of Firewalls within a device group.
- ☐ Plan the organization of your firewalls based on how they inherit network configuration settings from Templates and Template Stacks. For example, consider assigning firewalls to templates based on hardware models, geographic proximity, and similar network needs for time zones, a DNS server, and interface settings.

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Determine Panorama Log Storage Requirements

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

When you Plan Your Panorama Deployment, estimate how much log storage capacity Panorama requires to determine which Panorama Models to deploy, whether to expand the storage on those appliances beyond their default capacities, whether to deploy Dedicated Log Collectors, and whether to Configure Log Forwarding from Panorama to External Destinations. When log storage reaches the maximum capacity, Panorama automatically deletes older logs to create space for new ones.

Perform the following steps to determine the approximate log storage that Panorama requires. For details and use cases, refer to Panorama Sizing and Design Guide.

STEP 1 | Determine the log retention requirements of your organization.

Factors that affect log retention requirements include:

- IT policy of your organization
- Log redundancy—If you enable log redundancy when you Configure a Collector Group, each log will have two copies, which doubles your required log storage capacity.
- Regulatory requirements, such as those specified by the Payment Card Industry Data Security Standard (PCI DSS), Sarbanes-Oxley Act, and Health Insurance Portability and Accountability Act (HIPAA).

If your organization requires the removal of logs after a certain period, you can set the expiration period for each log type. You can also set a storage quota for each log type as a percentage of the total space if you need to prioritize log retention by type. For details, see Manage Storage Quotas and Expiration Periods for Logs and Reports.

STEP 2 | Determine the average daily logging rates.

Do this multiple times each day at peak and non-peak times to estimate the average. The more often you sample the rates, the more accurate your estimate.

1. Display the current log generation rate in logs per second:
   - If Panorama is not yet collecting logs, access the CLI of each firewall, run the following command, and calculate the total rates for all the firewalls. This command displays the number of logs received in the last second.

     ```
     > debug log-receiver statistics
     ```

   - If Panorama is already collecting logs, run the following command at the CLI of each appliance that receives logs (Panorama management server or Dedicated Log

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Collector) and calculate the total rates. This command gives the average logging rate for the last five minutes.

```
> debug log-collector log-collection-stats show incoming-logs
```

You can also use an SNMP manager to determine the logging rates of Log Collectors (see the panLogCollector MIB, OID 1.3.6.1.4.1.25461.1.1.6) and firewalls (see the panDeviceLogging, OID 1.3.6.1.4.1.25461.2.1.2.7).

2. Calculate the average of the sampled rates.
3. Calculate the daily logging rate by multiplying the average logs-per-second by 86,400.

STEP 3 | Estimate the required storage capacity.

This formula provides only an estimate; the exact amount of required storage will differ from the formula result.

Use the formula:

[(<logs_per_second> x 86400) x <days_of retention>] x <average_log_size> ÷ (1024 x 1024 x 1024)

The average log size varies considerably by log type. However, you can use 489 bytes as an approximate average log size.

For example, if Panorama must store logs for 30 days with a log rate of 1,500 LPS, then the required log storage capacity is: **[(1500 x 86400) x 30] x 489 ÷ (1024 x 1024 x 1024) = 1770GB**.

The results above are calculations for the detailed logs only with the default quota settings reserve 60% of the available storage for detail logs. This means that the calculated number represents 60% of the storage used by the Log Collector. To calculate the total storage required, divide this number by .60: **1770 ÷ .6 = 2951GB**.

One third, roughly 33%, of the avilable disk space is allocated to logd formatted logs to support upgrades, downgrades, and to support fixing database corruption. To calculate the total storage, divide the storage required by .66: **2951 ÷ .66 = 4471 total storage**.

STEP 4 | Next steps...

If you determine that Panorama requires more log storage capacity:

- Expand Log Storage Capacity on the Panorama Virtual Appliance.
- Increase Storage on the M-Series Appliance.

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Deploy Panorama

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

The following task list summarizes the steps to get started with Panorama. For an example of how to use Panorama for central management, see Use Case: Configure Firewalls Using Panorama.

STEP 1 | (M-Series appliance only) Rack mount the appliance.

STEP 2 | Perform initial configuration to enable network access to Panorama. See Set Up the Panorama Virtual Appliance or Set Up the M-Series Appliance.

STEP 3 | Register Panorama and Install Licenses.

STEP 4 | Install Content and Software Updates for Panorama.

STEP 5 | (Recommended) Set up Panorama in a high availability configuration. See Panorama High Availability.

STEP 6 | Add a Firewall as a Managed Device.

STEP 7 | Add a Device Group or Create a Device Group Hierarchy, Add a Template, and (if applicable) Configure a Template Stack.

STEP 8 | (Optional) Configure log forwarding to Panorama and/or to external services. See Manage Log Collection.

STEP 9 | Monitor Network Activity using the visibility and reporting tools on Panorama.

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Cheat Sheet: Manage Prisma Access by Using Panorama

Where Can I Use This?	What Do I Need?
• Panorama	❑ Prisma Access License

You can manage your Prisma Access deployment using either Panorama or Strata Cloud Manager.

The set of supported features is not identical across Prisma Access managed by Panorama and Prisma Access managed by Strata Cloud Manager. Refer to thecompatibility metrics to determine which features are supported with each option. This helps you decide on the management interface.

Important: You cannot switch management interfaces after you activate your Prisma Access license. Therefore, determine your interface choice before you start.

For a comprehensive overview of Prisma Access features and management options, refer to Prisma Access Overview.

Here's what you need to know to use Panorama to manage Prisma Access. The following tables summarize the topics and steps.

Activation & Initial Onboarding

Deploy and activate the service.

Reference Topic	Description
Activate Your Prisma Access License	Complete the initial steps required to deploy and activate the Prisma Access service.
Set Up Prisma Access	Configure the foundational settings required for Prisma Access deployment.

Activation and Onboarding

Reference Topic	Description
Activate Your Prisma Access License	Complete the initial steps required to deploy and activate the Prisma Access service.
Set Up Prisma Access	Configure the foundational settings required for Prisma Access deployment.
How to Manage Prisma Access	Understand the high-level steps for Prisma Access management

Core Configuration and Security

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Reference Topic	Description
Configure a Service Connection in Prisma Access (Panorama)	Configure a connection between Prisma Access and your data center/headquarters.
Enable Private App Security (Panorama)	Enable security for privately hosted applications.
Configure App Acceleration in Prisma Access (Panorama)	Configure settings to enhance application performance.
Remote Networks—High Performance	Configurations for high-performance remote network connectivity.

Operations and Maintenance (Panorama Focused)

Reference Topic	Description
Prisma Access Visibility and Monitoring	Monitor the status, health, and traffic flowing through Prisma Access.
Prisma Access and Associated Services Processing Locations	Access information regarding the geographic locations where Prisma Access and associated services are processed.
Prisma Access Infrastructure Management	Manage the underlying Prisma Access infrastructure.
Prisma Access APIs	Accessing and utilizing the management API.
Migrate Prisma Access from Panorama to Strata Cloud Manager	Moving the management plane from Panorama to Strata Cloud Manager.
Prisma Access Releases and Upgrades	The process for managing and applying Prisma Access software updates.

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Set Up Panorama

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

For centralized reporting and cohesive policy management across all the firewalls on your network, you can deploy the Panorama™ management server as a virtual appliance or as a dedicated M-Series hardware appliance. These appliances can operate in distinct modes like Panorama mode or Management Only mode.

To begin, you must evaluate essential planning phases, determine log storage requirements based on daily logging rates, and optimally structure large-scale firewall deployments by configuring an active and passive High Availability (HA) pair for fault tolerance.

Next, you can follow comprehensive instructions to install and initially configure Panorama across various cloud and hypervisor environments, including AWS, Azure, and ESXi, or as dedicated hardware.

After the core system is initialized with a management IP address, you must register the deployment and install the necessary management and support licenses. These licenses enable you to oversee managed firewalls and retrieve PAN-OS software and dynamic content updates.

Finally, to ensure security and communication integrity, you can install device certificates on both the Panorama server and any dedicated log collectors.

You can also deploy custom certificates to establish mutual authentication between HA peers and configure Role-Based Access Control (RBAC) to securely govern administrative access to the web interface, CLI, and XML API.

Manage Large-Scale Firewall Deployments with Panorama

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	No prerequisites needed

Panorama™ provides multiple options to manage a large-scale firewall deployment. For consolidation of all management functions, Panorama supports management of up to 5,000 firewalls using an M-600, M-700 appliance, or Panorama virtual appliance on ESXi in Management Only mode or up to 2,500 firewalls with a Panorama virtual appliance in Management Only mode. To simplify the deployment and operational management of a large-scale firewall deployment greater than 5,000 firewalls, the Panorama Interconnect plugin allows you to manage multiple Panorama management server Nodes from a single Panorama Controller.

Determine the Optimal Large-Scale Firewall Deployment Solution

To ease the operational burden of managing the configuration of your large-scale firewall deployment, Palo Alto Networks provides different firewall management options to best suit your deployment scenario.

If your large-scale firewall deployment is composed of one or very few Panorama management servers, you can deploy an M-600, M-700 appliance, or Panorama virtual appliance on ESXi to manage up to 5,000 firewalls, or Panorama virtual appliance to manage up to 2,500 firewalls, to leverage all Panorama capabilities from a single Panorama management server. The Increased Device Management Capacity for M-Series and Panorama Virtual Appliance is ideal for vertically scaled deployments where you manage a large number of firewalls from a single Panorama management server rather than deploying multiple Panorama management servers to manage fewer firewalls.

As of September 1, 2025 the Panorama Interconnect Plugin is no longer supported. For additional information regarding this change and alternatives, see the official end-of-life notice.

If your large-scale firewall deployment is composed of multiple Panorama management servers with similar configurations, the Panorama Interconnect plugin allows you to manage multiple Panorama Nodes from a single Panorama Controller. This plugin simplifies the deployment and operational management of large scale firewall deployments because you can centrally manage policy and configuration from a Panorama Controller. From the Panorama Controller, the device group and template stack configuration is synchronized to the Panorama Nodes and pushed to managed devices. The Panorama Interconnect plugin is ideal for horizontally-scaled firewall deployments with multiple distributed Panorama management servers.

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Increased Device Management Capacity for M-Series and Panorama Virtual Appliance

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Device management licence (per-device) • Support license • Outbound Internet Access • Customer Support Portal (CSP) Account • Panorama Superuser Role

You can manage up to 5,000 firewalls using a single M-600, M-700 appliance, or Panorama™ virtual appliance installed on VMware ESXi, or up to 2,5000 firewalls with all other supported Panorama virtual appliances in order to reduce the management footprint of your large-scale firewall deployment.

Increased Device Management Capacity Requirements

You can manage up to 5,000 firewalls using a single M-600, M-700 appliance, or Panorama™ virtual appliance installed on VMware ESXi, or up to 2,500 firewalls with all other supported Panorama virtual appliances. For managing such large deployments from a single Panorama management server alleviates the operational complexity of configuration management and reduces the security and compliance risk of managing multiple Panorama management servers. For details about the maximum number of template stacks that Panorama supports, see Template Stacks.

For log collection, a single Panorama management server is ideal because it provides a centralized location to view and analyze log data from managed devices rather than requiring you to access each individual Panorama management server. To provide redundancy in the event of system or network failure, Palo Alto Networks recommends deploying two Panorama management servers in a high availability (HA) configuration. For Panorama system and config logs, an additional disk with a minimum 92GB capacity is required. This additional disk is automatically detected by the Panorama virtual appliance when Panorama is rebooted and mounted as a partition for system and config log storage.

For generating pre-defined reports, you must enable Panorama to use Panorama data for pre-defined reports. This generates pre-defined reports using log data already collected by Panorama or the Dedicated Log Collector, which reduces the resource utilization when generating reports. Enabling this setting is required, otherwise Panorama performance may be impacted, and Panorama may become unresponsive.

To manage up to 5,000 firewalls, the Panorama management server must meet the following minimum requirements:

Requirement	5,000 Firewalls	2,500 Firewalls
Model	M-600	All supported Panorama hypervisors. For more

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Requirement	5,000 Firewalls	2,500 Firewalls
	M-700 VMware ESXi	information, see Panorama Models.
Panorama Mode	Management Only	Management Only
System Disk	Used to store the operating system files, system logs, software updates, and content updates. • **M-Series Appliances**—240GB SSD • **ESXi**—224GB You must manually increase the system disk to 224GB.	• 81GB—Used to store the operating system files and system logs. • Additional disk with a minimum 92GB capacity used for storing Panorama system and config logs.
CPUs	56	32
Memory	256GB	256GB
Log Collection	Local log collection is not supported. See Deploy Panorama with Dedicated Log Collectors to set up log collection.	
Logging and Reporting	Enable the **Use Panorama Data for Pre-Defined Reports** setting (**Panorama** > **Setup** > **Management** > **Logging and Reporting Settings** > **Log Export and Reporting**)	

Install Panorama for Increased Device Management Capacity

Activate the device management license to manage more than 1,000 firewalls from a single M-600 Panorama™ management server or a single Panorama virtual appliance.

STEP 1 | Contact your Palo Alto Networks sales representative to obtain the Panorama device management license that enables you to manage up to 5,000 firewalls.

- If you are deploying an M-600 appliance, obtain the `PAN-M-600-P-1K` device management license.
- If you are deploying an M-700 appliance, obtain the `PAN-M-700-P-1K` device management license.
- If you are deploying a Panorama virtual appliance, obtain the `PAN-PRA-1000` device management license.

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STEP 2 | Set up the Panorama management server.

- (M-600 and M-700 appliances only) Set Up the M-Series Appliance.

or

- Set Up the Panorama Virtual Appliance.

STEP 3 | Increase the System Disk for Panorama on an ESXi Server to 224GB.

A 224GB system disk is required for a Panorama virtual appliance installed on VMware ESXi to manage up to 5,000 firewalls. Review the Increased Device Management Capacity Requirements for more information.

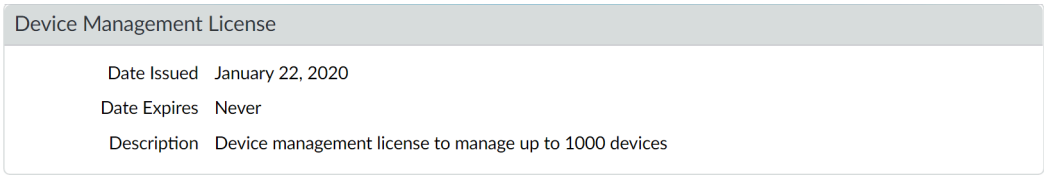
STEP 4 | Change the Panorama management server to Management Only mode if Panorama is not already in this mode.

- Begin at Step 5 to Set Up an M-Series Appliance in Management Only Mode.
- Set up a Panorama Virtual Appliance in Management Only Mode.

STEP 5 | Register your Panorama management server and install licenses.

1. Register Panorama.
2. Activate a Panorama Support License.
3. Activate the device management license on the Panorama management server.
   - Activate/Retrieve a Firewall Management License on the M-Series Appliance.
   - Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is not Internet-connected.
   - Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is Internet-connected.

STEP 6 | Select **Panorama** > **Licenses** and verify that the device management license is successfully activated.



If you are activating a new device management license on a Panorama, you can manage up to 5,000 firewalls with an M-600, M-700 appliance, or Panorama virtual appliance on ESXi, or up to 2,500 firewalls with a Panorama virtual appliance, but the Description still displays `Device management license to manage up to 1000 devices or more`.

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Set Up the Panorama Virtual Appliance

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Device management licence • Support license • Outbound Internet Access • Customer Support Portal (CSP) Account

You can set up the Panorama virtual appliance across a variety of platforms. Before beginning, review the necessary setup prerequisites to ensure your infrastructure is ready. Next, install the virtual appliance on a wide range of supported hypervisors and public cloud environments, including VMware (ESXi and vCloud Air), Alibaba Cloud, AWS, Azure, Google Cloud Platform (GCP), KVM, Hyper-V, and Oracle Cloud Infrastructure (OCI).

After the installation is complete, you can perform the initial configuration and select the specific operating mode that fits your network deployment. You can set up the virtual appliance in standard Panorama mode, Management Only mode, or as a dedicated Log Collector. For more details about the Panorama virtual appliance and its modes, see Panorama Models.

You also have the flexibility to configure the appliance with a local log collector, if needed. As your network management and logging demands grow, you can scale the resources of your virtual appliance.

You can complete the virtual appliance setup and follow instructions to convert your appliance's licensing or deployment type if your requirements change. This includes transitioning an evaluation Panorama to a production deployment, converting to VM-Flex licensing, or moving a production Panorama to an Enterprise License Agreement (ELA).

These topics assume you are familiar with the public and private hypervisor products required to create the virtual appliance, and don't cover any related concepts or terminology.

- Setup Prerequisites for the Panorama Virtual Appliance
- Install the Panorama Virtual Appliance
- Perform Initial Configuration of the Panorama Virtual Appliance
- Set Up The Panorama Virtual Appliance as a Log Collector
- Set Up the Panorama Virtual Appliance with Local Log Collector
- Set up a Panorama Virtual Appliance in Panorama Mode
- Set up a Panorama Virtual Appliance in Management Only Mode
- Expand Log Storage Capacity on the Panorama Virtual Appliance
- Increase CPUs and Memory on the Panorama Virtual Appliance
- Increase the System Disk on the Panorama Virtual Appliance
- Complete the Panorama Virtual Appliance Setup

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- Convert Your Panorama Virtual Appliance

Setup Prerequisites for the Panorama Virtual Appliance

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Device management licence • Support license • Outbound Internet Access • Customer Support Portal (CSP) Account

Complete the following tasks before you Install the Panorama Virtual Appliance:

- ☐ Use a dedicated hypervisor host for Panorama and do not share resources with other virtual machines.

  Do not take or use snapshots of the virtual machine.

- ☐ Use your browser to access the Palo Alto Networks Customer Support web site and Register Panorama You will need the Panorama serial number that you received in the order fulfillment email. After registering Panorama, you can access the Panorama software downloads page.
- ☐ PAN-OS 12.1.2 and later releases require a minimum system disk size of 224 GB. Ensure you upgrade the system disk before installation. For more information, see Increase the System Disk on the Panorama Virtual Appliance.
- ☐ Review the supported Panorama hypervisors to verify the hypervisor meets the minimum version requirements to deploy Panorama.
- ☐ If you will install Panorama on a VMware ESXi server, verify that the server meets the minimum requirements as listed in the System Requirements for the Panorama Virtual Appliance. These requirements apply to Panorama 5.1 and later releases. The requirements vary based on whether you will run the virtual appliance in Panorama mode or Management Only mode. For details on the modes, see Panorama Models.

  If you install Panorama on VMware vCloud Air, you set the system settings during installation.

Review the minimum resource requirements for deploying the Panorama virtual appliance on Alibaba Cloud, Amazon Web Services (AWS), AWS GovCloud, Microsoft Azure, Google Cloud Platform (GCP), Hyper-V, KVM, Oracle Cloud Infrastructure (OCI), and VMware ESXi to ensure that the virtual machine meets the minimum required resources for the desired mode (Panorama, Management Only, or Log Collector). The minimum resource requirements for the Panorama virtual appliance are designed to help you achieve the maximum number of logs per second (LPS) for log collection in Panorama and Log Collector mode. If you add or remove virtual logging disks that results in a configuration that does not meet or exceed the number of virtual logging disks recommended (below), your LPS will be reduced.

If the minimum resource requirements are not met for Panorama mode when you Install the Panorama Virtual Appliance, Panorama defaults to Management Only mode for all supported public (Alibaba Cloud, AWS, AWS GovCloud, Azure, GCP, and OCI) and private (Hyper-V,

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KVM, and VMware ESXi) hypervisors. If the minimum resource requirements are not met for Management Only mode, Panorama defaults to Maintenance mode for all supported public hypervisors, Hyper-V, and KVM.

Table 1: System Requirements for the Panorama Virtual Appliance

Requirement	Panorama Virtual Appliance in Management Only Mode	Panorama Virtual Appliance in Panorama Mode	Panorama Virtual Appliance in Log Collector Mode
Virtual hardware version	• **VMware ESXi and vCloud Air**—64-bit kernel-based VMware ESXi 6.0, 6.5, 6.7, 7.0, or 8.0. The supported version of the virtual hardware family type (also known as the VMware virtual hardware version) on the ESXi server is vmx-10. *The Panorama virtual appliance for ESXi **does not support** the following:* • *Creation of quiesced snapshots.* *Disable **Quiesce guest file system** in the vSphere client or set the **`quiesce`** flag to 0 or false in the vSphere CLI before creating a snapshot of your virtual Panorama appliance.* • *VMware vMotion to migrate a Panorama virtual appliance from one ESXi server to another.* • **Hyper-V**—Windows Server 2016 with Hyper-V role or Hyper-V 2016, Windows Server 2019 with Hyper-V role or Hyper-V 2019, and Windows Server 2022 with Hyper-V role or Hyper-V 2022. • **KVM**—Ubuntu version 16.04 or CentOS7 In Panorama mode, the virtual appliance running on any ESXi version supports up to 12 virtual logging disks with 2TB of log storage each, for a total maximum capacity of 24TB.		
(ESXi and vCloud Air only) Client computer	To install the Panorama virtual appliance and manage its resources, you must install a VMware vSphere Client or VMware Infrastructure Client that is compatible with your ESXi server.		
System disk	• (PAN-12.1.2 and later releases) — The Panorama virtual appliance requires a minimum system disk size of **224 GB** to upgrade to or run the 12.1.2 and later versions. • **Default**—81GB		

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<table>
<tr><th>Requirement</th><th>Panorama Virtual Appliance in Management Only Mode</th><th>Panorama Virtual Appliance in Panorama Mode</th><th>Panorama Virtual Appliance in Log Collector Mode</th></tr>
<tr><td></td><td colspan="3">• Upgraded—224GB
An upgraded system disk is required for SD-WAN.
(Panorama and Log Collector mode) An upgraded system disk is required if you added more than 8 logging disks.
For log storage, Panorama uses virtual logging disks instead of the system disk or an NFS datastore.
Panorama must be initially installed with the Default system disk size, with the option to increase the system disk size after initial installation.</td></tr>
<tr><td rowspan="2">CPUs, memory, and logging disks</td><td rowspan="2">• Manage up to 500 managed devices
  • 16 CPUs
  • 64GB memory
  • Local log storage not supported
• Manage up to 1,000 managed devices
  • 32 CPUs
  • 128GB memory
  • Local log storage not supported
• To manage more than 1,000 firewalls, see Increased Device Management Capacity Requirements.</td><td>The minimum resources below are required to achieve the specified logging rate.
• Up to 10,000 logs/sec (LPS):
  • 16 CPUs
  • 64GB memory
  • 4x2TB logging disks
  • Manage up to 500 managed devices
• Up to 20,000 log/sec (LPS)
  • 32 CPUs
  • 128GB memory
  • 8x2TB logging disks
  • Manage up to 1,000 managed devices</td><td>The minimum resources below are required to achieve the specified logging rate.
• Up to 15,000 log/sec (LPS)
  • 16 CPUs
  • 64GB memory
  • 4x2TB logging disks
• Up to 25,000 logs/sec (LPS)
  • 32 CPUs
  • 128GB memory
  • 8x2TB logging disks</td></tr>
<tr><td colspan="2">The first logging disk on the Panorama virtual appliance must be 2TB in order to add additional logging disks. If the first logging disk is smaller than 2TB, you are unable to add additional logging disks.</td></tr>
<tr><td>Minimum CPUs and memory</td><td>• 16 CPUs
• 64GB memory</td><td colspan="2">The minimum resources below do not take LPS into consideration and are only required for the Panorama virtual appliance to function based on the number of logging disks added. Palo Alto Networks recommends you refer to the recommended resources above.</td></tr>
</table>

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Requirement	Panorama Virtual Appliance in Management Only Mode	Panorama Virtual Appliance in Panorama Mode	Panorama Virtual Appliance in Log Collector Mode
		For larger Panorama deployments, be aware that you may be under-provisioning your Panorama. This may lead to impacted performance and may cause Panorama to become unresponsive depending on the number of firewalls managed, the configuration size, the number of administrators logged in to Panorama, and the volume of logs ingested. • **2TB to 8TB**—16 CPUs, 64GB memory • **10TB to 24TB**— 16 CPUs, 128GB memory	
Log storage capacity	Panorama in Management Only mode requires log forwarding to a Dedicated Log Collector.	2TB to 24TB	2TB to 24TB

Supported Interfaces

Interfaces can be used for device management, log collection, Collector Group communication, licensing and software updates. The Panorama virtual appliance supports up to six interfaces (MGT and Eth1 - Eth5).

Table 2: Supported interfaces for public hypervisors

Function	Alibaba Cloud	Amazon Web Services (AWS) and AWS GovCloud		Microsoft Azure	Google Cloud Platform (GCP)	OCI
Device Management	Any interface supported	Any interface supported	Any interface supported	Any interface supported	Any interface supported	Any interface supported
Device Log Collection	Any interface supported	Any interface supported	Any interface supported	Any interface supported	Any interface supported	Any interface supported
Collector Group Communication	Any interface supported	Any interface supported	Any interface supported	Any interface supported	Any interface supported	Any interface supported

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Function	Alibaba Cloud	Amazon Web Services (AWS) and AWS GovCloud		Microsoft Azure	Google Cloud Platform (GCP)	OCI
Licensing and Software Updates	MGT interface only	MGT interface only	MGT interface only	MGT interface only	MGT interface only	MGT interface only

Table 3: Supported Interfaces for Private Hypervisors

Function	KVM	Hyper-V	VMware (ESXi, vCloud Air)
Device Management	Any interface supported	Any interface supported	Any interface supported
Device Log Collection	Any interface supported	Any interface supported	Any interface supported
Collector Group Communication	Any interface supported	Any interface supported	Any interface supported
Licensing and Software Updates	Any interface supported	Any interface supported	Any interface supported

Install the Panorama Virtual Appliance

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Device management licence • Support license • Panorama Virtual Appliance license (serial number) • Customer Support Portal (CSP) Account

You can deploy the Panorama virtual appliance across a range of on-premises hypervisors and public cloud environments.

Before installation, decide whether to run the virtual appliance in Panorama mode, Management Only mode or Log Collector mode. Each mode has different resource requirements, as described in Setup Prerequisites for the Panorama Virtual Appliance. You must complete the prerequisites before starting the installation.

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As of September 1, 2025, the legacy mode is no longer supported on Panorama. For additional information, see the official end-of-life notice.

Installation procedures vary by platform, but the underlying requirements and configuration goals remain consistent across environments. Each platform-specific topic in this section provides the steps needed to deploy the virtual appliance image, configure initial networking, and prepare the instance for use. Follow the topic that corresponds to your target environment and refer to the prerequisite guidance if you are unsure which mode or resource profile applies to your deployment.

As new hypervisor and cloud platforms are validated and supported, additional installation topics will be added to this section.

Install Panorama on VMware

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Device management licence • Support license • Outbound Internet Access • Customer Support Portal (CSP) Account

You can install the Panorama virtual appliance on the ESXi and vCloud Air VMware platforms.

- Install Panorama on an ESXi Server
- Install Panorama on vCloud Air

Support for VMware Tools on the Panorama Virtual Appliance

VMware Tools is bundled with the software image (ovf) for the Panorama virtual appliance. The support for VMware Tools allows you to use the vSphere environment—vCloud Director and vCenter server—for the following:

- View the IP address assigned to the Panorama management interface.
- View resource utilization metrics on hard disk, memory, and CPU. You can use these metrics to enable alarms or actions on the vCenter server or vCloud Director.
- Graceful shutdown and restart of Panorama using the power off function on the vCenter server or vCloud Director.
- Enables a heartbeat mechanism between the vCenter server and Panorama for verifying that Panorama is functioning, or if the firewall/Panorama is rebooting. If the firewall goes into maintenance mode, heartbeats are disabled so that the vCenter server does not shut down the firewall. Disabling heartbeats allows the firewall to stay operational in maintenance mode when it cannot not send heartbeats to the vCenter server.

Install Panorama on an ESXi Server

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Device management licence

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Where Can I Use This?	What Do I Need?
	• Support license • Panorama Virtual Appliance license (serial number) • Customer Support Portal (CSP) Account

Use these instructions to install a new Panorama virtual appliance on a VMware ESXi server. For upgrades to an existing Panorama virtual appliance, skip to Install Content and Software Updates for Panorama.

STEP 1 | Download the Panorama 11.1 base image Open Virtual Appliance (OVA) file.

1. Log in to the Palo Alto Networks Support Portal.
2. Select **Updates** > **Software Updates** and filter by **Panorama Base Images** to download the OVA file (`Panorama-ESX-11.1.0.ova`).

STEP 2 | Install Panorama.

1. Launch the VMware vSphere Client and connect to the VMware server.
2. Select **File** > **Deploy OVF Template**.
3. **Browse** to select the Panorama OVA file and click **Next**.
4. Confirm that the product name and description match the downloaded version, and click **Next**.
5. Enter a descriptive name for the Panorama virtual appliance, and click **Next**.
6. Select a datastore location (system disk) on which to install the Panorama image. See the Setup Prerequisites for the Panorama Virtual Appliance for the supported **Default** system disk sizes. After selecting the datastore, click **Next**.

   *The Panorama virtual appliance must be initially installed with the **Default** system disk size. Installing the Panorama virtual appliance with a system disk larger than the **Default** system disk size is not supported and may result in limited utilization. You have the option to increase the system disk size after initial installation*
7. Select **Thick Provision Lazy Zeroed** as the disk format, and click **Next**.
8. Specify which networks in the inventory to use for the Panorama virtual appliance, and click **Next**.
9. Confirm the selected options, click **Finish** to start the installation process, and click **Close** when it finishes. Do not power on the Panorama virtual appliance yet.

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STEP 3 | Configure resources on the Panorama virtual appliance.

1. Right-click the Panorama virtual appliance and **Edit Settings**.
2. In the **Hardware** settings, allocate the CPUs and memory as necessary.

   *The virtual appliance boots up in Panorama mode if you allocate sufficient **CPUs** and **Memory** and add a virtual logging disk (later in this procedure). Otherwise, the appliance boots up in Management Only mode. For details on the modes, see Panorama Models.*
3. Set the **SCSI Controller** to **LSI Logic Parallel**.
4. (Optional) Add a virtual logging disk.

   This step is required in the following scenarios:

   - *In Panorama mode to store logs on a dedicated logging disk.*
   - *Manage your SD-WAN deployment in Management Only mode.*

   1. **Add** a disk, select **Hard Disk** as the hardware type, and click **Next**.
   2. **Create a new virtual disk** and click **Next**.
   3. Set the **Disk Size** to exactly 2TB.

      In Panorama mode, you can later add additional logging disks (for a total of 12) with 2TB of storage each. Expanding the size of a logging disk that is already added to Panorama is not supported.
   4. Select your preferred **Disk Provisioning** disk format.

      Consider your business needs when selecting the disk provisioning format. For more information regarding the disk provisioning performance considerations, refer to the VMware Thick vs Thin Disks and All Flash Arrays document, or additional VMware documentation.

      *When adding multiple logging disks, it is a best practice to select the same **Disk Provisioning** format for all disks to avoid any unexpected performance issues that may arise.*
   5. Select **Specify a datastore or datastore structure** as the location, **Browse** to a datastore that has sufficient storage, click **OK**, and click **Next**.
   6. Select a SCSI **Virtual Device Node** (you can use the default selection) and click **Next**.

      Panorama will fail to boot if you select a format other than SCSI.
   7. Verify that the settings are correct and click **Finish**.
5. Click **OK** to save your changes.

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STEP 4 | Power on the Panorama virtual appliance.

1. In the vSphere Client, right-click the Panorama virtual appliance and select **Power** > **Power On**. Wait for Panorama to boot up before continuing.
2. Log in to the Panorama virtual appliance CLI from the ESXi console:
   1. Right-click the Panorama virtual appliance and select **Open Console**.
   2. Enter your username and password to log in (default is **admin** for both).

STEP 5 | Configure a new administrative password for the Panorama virtual appliance.

You must configure a unique administrative password before you can access the web interface or CLI of the Panorama virtual appliance. The new password must be a minimum of eight characters and include a minimum of one lowercase character, one uppercase character, and one number or special character.

When you first log in to the Panorama CLI, you are prompted to enter the **Old Password** and the **New Password** for the `admin` user before you can continue.

STEP 6 | Verify the Panorama is running the correct system mode.

```
admin> show system info
```

In the output, the `system-mode` indicates either `panorama` or `management-only` mode.

STEP 7 | Register the Panorama virtual appliance and activate the device management license and support licenses.

1. (VM Flex Licensing Only) Provisioning the Panorama Virtual Appliance Serial Number.

   When leveraging VM Flex licensing, this step is required to generate the Panorama virtual appliance serial number needed to register the Panorama virtual appliance with the Palo Alto Networks Customer Support Portal (CSP).
2. Register Panorama.

   You must register the Panorama virtual appliance using the serial number provided by Palo Alto Networks in the order fulfillment email.

   This step is not required when leveraging VM Flex licensing as the serial number is automatically registered with the CSP when generated.
3. Activate the firewall management license.
   - Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is Internet-connected.
   - Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is not Internet-connected.
4. Activate a Panorama Support License.

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STEP 8 | Increase the System Disk for Panorama on an ESXi Server if you intend to use the Panorama virtual appliance for the following:

- Manage your SD-WAN deployment in Panorama mode.
- Requires additional storage space for dynamic updates when managing large-scale firewall deployments.

STEP 9 | Complete configuring the Panorama virtual appliance for your deployment needs.

- For Panorama in Log Collector Mode.
  1. Add a Virtual Disk to Panorama on an ESXi Server as needed.

     Adding at least one virtual logging disk is required before you can change the Panorama virtual appliance to Log Collector mode.
  2. Begin at Step 6 to switch to Log Collector mode.

     *Enter the Public IP address of the Dedicated Log Collector when you add the Log Collector as a managed collector to the Panorama management server. You cannot specify the **IP Address**, **Netmask**, or **Gateway**.*
- For Panorama in Panorama mode.
  1. Add a Virtual Disk to Panorama on an ESXi Server.

     Adding at least one virtual logging disk is required before you can change the Panorama virtual appliance to Panorama mode.
  2. Set up a Panorama Virtual Appliance in Panorama Mode.
  3. Configure a Managed Collector.
- For Panorama in Management Only mode.
  1. Set up a Panorama Virtual Appliance in Management Only Mode.
  2. Configure a Managed Collector. to add a Dedicated Log Collector to the Panorama virtual appliance.

     Management Only mode does not support local log collection, and requires a Dedicated Log Collector to store managed device logs.
- For SD-WAN deployments.
  1. Increase the System Disk for Panorama on an ESXi Server

     To leverage SD-WAN on Panorama deployed on ESXi, you must increase the system disk to 224GB.

     You cannot migrate back to a 81GB system disk after successfully increasing the system disk to 224GB.
  2. Set up a Panorama Virtual Appliance in Management Only Mode.
  3. Add a Virtual Disk to Panorama on an ESXi Server.

     To leverage SD-WAN, you must add a single 2TB logging disk to Panorama in Management Only mode.

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Install Panorama on vCloud Air

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Device management licence • Support license • Panorama Virtual Appliance license (serial number) • Customer Support Portal (CSP) Account

Use these instructions to install a new Panorama virtual appliance in VMware vCloud Air. If you are upgrading a Panorama virtual appliance deployed in vCloud Air, skip to Install Content and Software Updates for Panorama.

STEP 1 | Download the Panorama 11.1 base image Open Virtual Appliance (OVA) file.

1. Go to the Palo Alto Networks software downloads site. (If you can't log in, go to the Palo Alto Networks Customer Support web site for assistance.)
2. In the Download column in the Panorama Base Images section, download the Panorama 11.1 release OVA file (`Panorama-ESX-10.0.0.ova`).

STEP 2 | Import the Panorama image to the vCloud Air catalog.

For details on these steps, refer to the OVF Tool User's Guide.

1. Install the OVF Tool on your client system.
2. Access the client system CLI.
3. Navigate to the OVF Tool directory (for example, C:\Program Files\VMware\VMware OVF Tool).
4. Convert the OVA file to an OVF package:

```
ovftool.exe <OVA-file-pathname> <OVF-file-pathname>
```

5. Use a browser to access the vCloud Air web console, select your **Virtual Private Cloud OnDemand** location, and record the browser URL. You will use the URL information to complete the next step. The URL format is: **`https://<virtual-cloud-location>.vchs.vmware.com/compute/cloud/org/<vCloud-account-number>/#/catalogVAppTemplateList?catalog=<catalog-ID>`**.
6. Import the OVF package, using the information from the vCloud Air URL to complete the <virtual#cloud#location>, <vCloud#account#number>, and <catalog#ID> variables. The other variables are your vCloud Air username and domain <user>@<domain>, a virtual data center <datacenter>, and a vCloud Air template <template>.

```
ovftool.exe -st="OVF" "<OVF-file-pathname>"
 "vcloud://<user>@<domain>:password@<virtual-cloud-
location>.vchs.vmware.com?vdc=<datacenter>&org=<vCloud-
```

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```
account-number>&vappTemplate=<template>.ovf&catalog=default-
catalog"
```

STEP 3 | Install Panorama.

1. Access the vCloud Air web console and select your **Virtual Private Cloud OnDemand** region.
2. Create a Panorama virtual machine. For the steps, refer to Add a Virtual Machine from a Template in the vCloud Air Documentation Center. Configure the **CPU**, **Memory** and **Storage** as follows:
   - Set the **CPU** and **Memory** based on whether the virtual appliance mode: see Setup Prerequisites for the Panorama Virtual Appliance.
   - Set the **Storage** to configure the Panorama virtual appliance system disk. See Setup Prerequisites for the Panorama Virtual Appliance for the supported disk sizes based on the Panorama virtual appliance mode. For better logging and reporting performance, select the **SSD-Accelerated** option.

     To increase the log storage capacity, you must Add a Virtual Disk to Panorama on vCloud Air. In Panorama mode, the virtual appliance does not use the system disk for log storage; you must add a virtual logging disk.

STEP 4 | Create vCloud Air NAT rules on the gateway to allow inbound and outbound traffic for the Panorama virtual appliance.

Refer to Add a NAT Rule in the vCloud Air Documentation Center for the detailed instructions:

1. Add a NAT rule that allows Panorama to receive traffic from the firewalls and allows administrators to access Panorama.
2. Add a NAT rule that allows Panorama to retrieve updates from the Palo Alto Networks update server and to access the firewalls.

STEP 5 | Create a vCloud Air firewall rule to allow inbound traffic on the Panorama virtual appliance.

Outbound traffic is allowed by default.

Refer to Add a Firewall Rule in the vCloud Air Documentation Center for the detailed instructions.

STEP 6 | Power on the Panorama virtual appliance if it isn't already on.

In the vCloud Air web console, select the **Virtual Machines** tab, select the Panorama virtual machine, and click **Power On**.

You are now ready to Perform Initial Configuration of the Panorama Virtual Appliance.

Set Up Panorama on Alibaba Cloud

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Device management licence • Support license

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Where Can I Use This?	What Do I Need?
	• Panorama Virtual Appliance license (serial number) • Customer Support Portal (CSP) Account

Set up a Panorama™ virtual appliance on Alibaba Cloud to centrally managed the configuration of physical and VM-Series firewalls.

Upload the Panorama Virtual Appliance Image to Alibaba Cloud

Complete the following procedure to upload a Panorama™ management server qcow2 file for KVM and create a custom image that you need to launch the Panorama virtual appliance. Uploading and creating the image is required only once. You can use the same image for all subsequent deployments of the Panorama virtual appliance.

STEP 1 | Download the Panorama qcow2 file for KVM from the Palo Alto Networks Customer Support Portal (CSP).

1. Log in to the Palo Alto Networks CSP.
2. Select **Updates** > **Software Updates** and select **Panorama Base Images** from the software updates filter drop-down.
3. Download the latest version of the `Panorama-KVM` qcow2 file.

STEP 2 | Log in to the Alibaba Cloud Console.

STEP 3 | Create an Object Storage Service (OSS) bucket for the Panorama virtual appliance image.

1. From the Alibaba Cloud menu, select **Object Storage Service** > **Buckets** and **Create Bucket**.
2. Enter a descriptive **Bucket Name**.
3. Select the bucket **Region**.

   This region must be in the same region you plan on deploying your Panorama virtual appliance and in the same region as the firewalls you plan to manage with Panorama.
4. Configure the remaining OSS bucket settings as needed.
5. Click **OK**.

   You are automatically taken to the OSS bucket Overview page after successful creation.

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STEP 4 | Upload the qcow2 file to the OSS bucket.

1. In the OSS bucket Overview, select **Files** and **Upload** the qcow2 file you downloaded in the previous step.
2. For **Upload To** target, select **Current**.
3. For the **File ACL**, select **Inherited from Bucket**.
4. Click **Select Files** and select the qcow2 file.

   Alternatively, you can drag and drop the qcow2 file into the **Files to Upload** section.
5. **Upload** the qcow2 file.

   A Task List window appears displaying the upload Status. Continue to the next step after the qcow2 file upload **Status** displays **Uploaded**.

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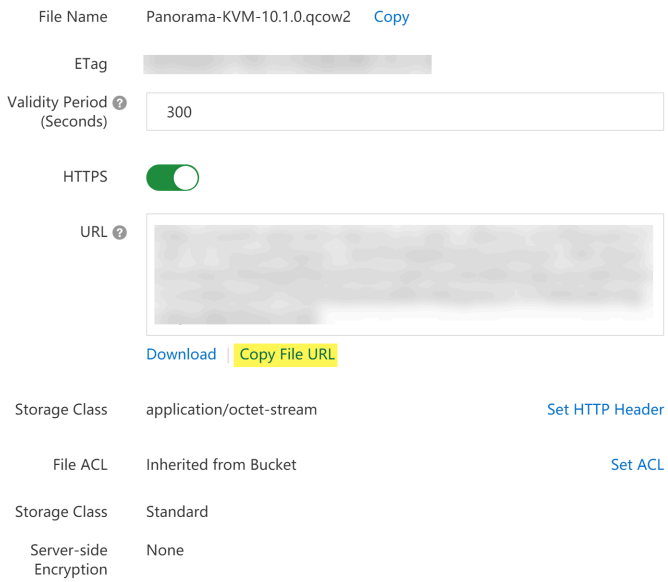
STEP 5 | Make the qcow2 file a bootable image.

1. In the OSS bucket Overview, select **Files** and click the qcow2 file you uploaded to view the file Details.
2. Click **Copy File URL** and exit the file Details.



3. From the Alibaba Cloud menu, select **Elastic Compute Service** > **Instances & Images** > **Images** and **Import Image**.
4. Paste the **OSS Object Address** for the qcow2 file.

   This the file URL you copied in the previous step.
5. Enter an **Image Name**.
6. For the **Operating System/Platform**, select **Linux CentOS**.
7. For the **System Disk (GiB)**, enter **81**.
8. For the **System Architecture**, select **x86_64**.
9. For the **Image Format**, select **QCOW2**.
10. Click **OK**.

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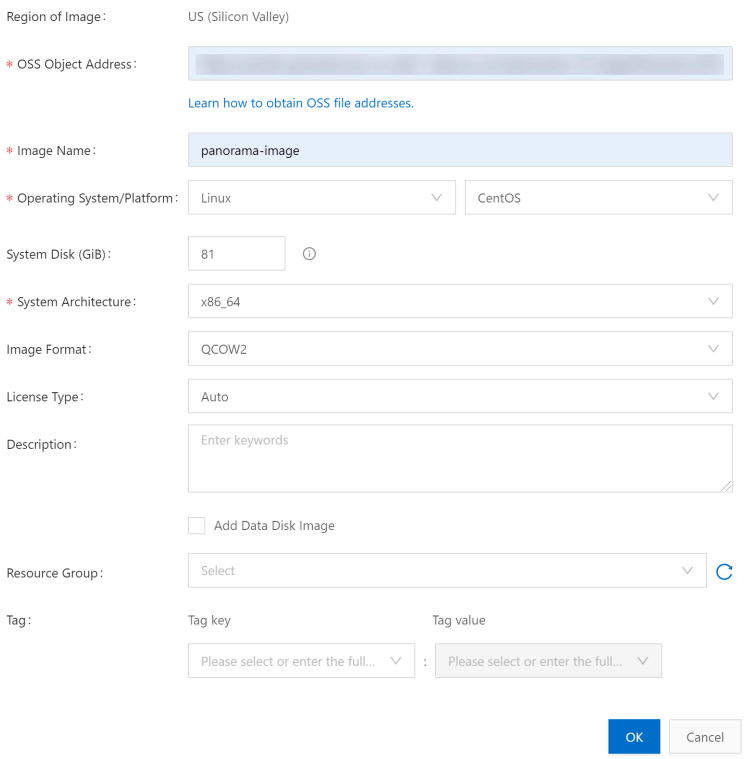

Install Panorama on Alibaba Cloud

Use the Elastic Compute Service (ECS) to create a Panorama™ virtual appliance instance on Alibaba Cloud. An ECS instance supports a single NIC by default and automatically attached an Elastic Network Interface (ENI) to it. You must manually upload a Panorama virtual appliance qcow2 image downloaded from the Palo Alto Networks Customer Supported Portal (CSP) to Alibaba Cloud to successfully install the Panorama virtual appliance on Alibaba Cloud.

A Panorama virtual appliance deployed on Alibaba Cloud is Bring Your Own License (BYOL), supports all deployment modes (Panorama, Log Collector, and Management Only), and shares the same processes and functionality as the M-Series hardware appliances. For more information on Panorama modes, see Panorama Models.

Review the Setup Prerequisites for the Panorama Virtual Appliance to determine the correct Elastic Computer Service (ECS) instance type for your needs. The virtual resources requirement for the Panorama virtual appliance is based on the total number of firewalls managed by the Panorama virtual appliance and the required Logs Per Second (LPS) for forwarding logs from your managed firewalls to your Log Collector.

Palo Alto Networks supports the following instance types.

- `ecs.g5.xlarge`, `ecs.g5.2xlarge`, `ecs.g5.4xlarge`
- `ecs.sn2ne.xlarge`, `ecs.sn2ne.2xlarge`, `ecs.sn2ne.4xlarge`

Under-provisioning the Panorama virtual appliance will impact management performance. This includes the Panorama virtual appliance becoming slow or unresponsive depending on how under-provisioning the Panorama virtual appliance is.

STEP 1 | Log in to the Alibaba Cloud Console.

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STEP 2 | Upload the Panorama Virtual Appliance Image to Alibaba Cloud.

STEP 3 | Set up the virtual private cloud (VPC) for your network needs.

Whether you launch the Panorama virtual appliance in an existing VPC or you create a new VPC, the Panorama virtual appliance must be able to receive traffic from other instances in the VPC and perform inbound and outbound communication between the VPC and the internet as needed.

Refer to the Alibaba Cloud VPC documentation for more information.

1. Create a VPC and Configure Networks or use an existing VPC.
2. Verify that the network and security components are appropriately defined.
   - Create an internet gateway to enable internet access to the subnet of your Panorama virtual appliance. Internet access is required to install software and content updates, activate licenses, and leverage Palo Alto Networks cloud services. Otherwise, you must manually install updates and activate licenses.
   - Create subnets. Subnets are segments of the IP address range assigned to the VPC in which you can launch Alibaba Cloud instances. It is recommended that the Panorama virtual appliance belong to the management subnet so that you can configure it to access the internet if needed.
   - Add routes to the route table for a private subnet to ensure traffic can be routed across subnets in the VPC and from the internet if applicable.

     Ensure you create routes between subnets to allow communication between:

     - Panorama, managed firewalls, and Log Collectors.
     - (Optional) Panorama and the internet.
   - Ensure that the following ingress security rules are allowed for the VPC to manage VPC traffic. The ingress traffic source for each rule is unique to your deployment topology.

     See Ports Used for Panorama for more information.

     - Allow SSH (port **22**) traffic to enable access to the Panorama CLI.
     - Allow HTTPS (port **443** and **27280**) traffic to enable access to the Panorama web interface.
     - Allow traffic on port **3978** to enable communication between Panorama, manage firewalls, and managed Log Collectors. This port is also used by Log Collectors to forward logs to Panorama.
     - Allow traffic on port **28443** to enable managed firewalls to get software and content updates from Panorama.

STEP 4 | Select **Elastic Compute Service** > **Instances & Images** > **Instances** and click **Create Instance** in the upper right corner.

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STEP 5 | Create the Panorama virtual appliance instance.

1. Select **Custom Launch**.
2. Configure the Panorama virtual appliance instance.
   - **Billing Method**—Select the desired subscription method for the instance.
   - **Region**—Select a region of your choice. The region you select must provide on of the supported instance types.
   - **Instance Type**—Select one of the supported instance types. You can select Type-based Selection to search for the instance type.
   - **Image**— Select **Custom Image** and select the Panorama virtual appliance image you uploaded.
   - **Storage**—Choose a disk type and enter **81**GiB as the system disk capacity.
   - (Optional) **Add Disk**—Add additional logging disks.

     If you intend to use the Panorama virtual appliance in Panorama mode or as a Dedicated Log Collector, add the virtual logging disks during the initial deployment. By default, the Panorama virtual appliance is in Panorama mode for the initial deployment when you meet the Panorama mode resource requirements and have added at least one virtual logging disk. Otherwise, the Panorama virtual appliance defaults to Management Only mode. Change the Panorama virtual appliance to Management Only mode if you just want to manage devices and Dedicated Log Collectors, and to not collect logs locally.

     The Panorama virtual appliance on Alibaba Cloud only supports 2TB logging disks, and in total supports up to 24TB of log storage. You are unable to add a logging disk smaller than 2TB, or a logging disk with a size not divisible by the 2TB logging disk requirement. The Panorama virtual appliance partitions logging disks larger than 2TB into 2TB partitions.
   - (Optional) **Snapshot**—Specify how often a snapshot is automatically taken of the Panorama virtual appliance instance to prevent risks and accidental data deletion.
   - **Duration**—Specify the duration for the Panorama virtual appliance instance.

STEP 6 | Configure the Panorama virtual appliance network settings.

1. Select **Next: Networking**.
2. Configure the network settings for the Panorama virtual appliance instance.
   - **Network Type**—Select the VPC and management VSwitch you created.
   - **Public IP Address**—If you do not have a public IP address, enable (check) **Assign Public IPv4 Address** and a public IPv4 address is automatically assigned to the Panorama virtual appliance instance.

     If you must use a specific IP address, or an address in a specific range, you can request a custom IP address. Refer to the Elastic IP Address User Guide.
   - **Security Group**—Select the management security group you created and enable **Port 443 (HTTPS)**, **Port 22**, and **Port 3389**.
   - **Elastic Network Interface**—No configuration needed. The Management interface is already attached to eth0.

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STEP 7 | Configure the Panorama virtual appliance instance system settings.

1. Select **Next: System Configurations**.
2. Configure system settings for the Panorama virtual appliance instance.
   - **Logon Credentials**—Select **Key Pair** and select the key pair. If a key pair has not already been created, select **Create Key Pair** to create an new key pair on Alibaba Cloud or import an existing key pair.

     Password authentication is not supported.
   - **Instance Name**—Enter a descriptive name for the Panorama virtual appliance. This the name displayed for the instance throughout the Alibaba Cloud Console.
   - **Host**—Enter a hostname for the Panorama virtual appliance instance.

STEP 8 | (Optional) Select **Next: Grouping** to configuring grouping for all Alibaba Cloud resources associated with the Panorama virtual appliance instance.

STEP 9 | Select **Preview** to view the configuration before ordering.

STEP 10 | View and check the **ECS Terms of Service** and **Product Terms of Service**.

STEP 11 | **Create Instance** to create the Panorama virtual appliance instance.

When prompted, click **Console** to view the instance creation status.

STEP 12 | Allocate Elastic IP (EIP) addresses.

The EIP is a public IP address used to connect to the Panorama virtual appliance.

This step is required only if you want to enable internet access for the Panorama virtual appliance.

1. Select **Elastic Compute Service** > **Network & Security** > **VPC** > **Elastic IP Addresses** > **Elastic IP Addresses**.

   Select **Create EIP** if you do not have any existing EIPs.
2. In the **Actions** column, select **Bind Resource** to bind an EIP to any interface exposed to the Internet.

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STEP 13 | Log in to the Panorama CLI using the SSH to configure the Panorama virtual appliance network settings.

You must configure the `admin` password, system IP address , netmask, and default gateway. Additionally, you must add the Alibaba Cloud DNS servers to successfully connect to the Palo Alto Networks update server.

*You can also access the Panorama CLI from the Alibaba console. To access the Panorama CLI from the Alibaba console, select **Elastic Compute Service** > **Instances & Images** > **Instances** and select the Panorama virtual appliance instance. In the Instance Details, select **Connect**.*

You are prompted to create a VCN password for the Panorama virtual appliance instance on first connection from the Alibaba VCN. Be sure to save this password as it cannot be recovered and is required to connect using the VCN or update the password in the future.

STEP 14 | Configure a new administrative password for the Panorama virtual appliance.

You must configure a unique administrative password before you can access the web interface or CLI of the Panorama virtual appliance. To access the CLI, the private key used to launch the Panorama virtual appliance is required.

The new password must be a minimum of eight characters and include a minimum of one lowercase character, one uppercase character, and one number or special character.

Configure a new password using the following commands and follow the on screen prompts:

```
admin> configure
admin# set mgt-config users admin password
```

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STEP 15 | Configure the initial network settings for the Panorama virtual appliance.

```
admin> configure
```

```
admin#  set deviceconfig system type static
```

```
admin#  set deviceconfig system ip-address <instance-private-IP
 address> netmask <netmask> default-gateway <default-gateway-IP>
```

The default gateway on Alibaba Cloud ends in ***.253****. For example, if the private IP address for your Panorama virtual appliance instance is 192.168.100.20, the default gateway is 192.168.100.253.*

```
admin#  set deviceconfig system dns-setting servers primary
 100.100.2.136
```

```
admin#  set deviceconfig system dns-setting servers secondary
 100.100.2.138
```

```
admin# commit
```

STEP 16 | Register the Panorama virtual appliance and activate the device management license and support licenses.

1. (VM Flex Licensing Only) Provisioning the Panorama Virtual Appliance Serial Number.

   When leveraging VM Flex licensing, this step is required to generate the Panorama virtual appliance serial number needed to register the Panorama virtual appliance with the Palo Alto Networks Customer Support Portal (CSP).
2. Register Panorama.

   You must register the Panorama virtual appliance using the serial number provided by Palo Alto Networks in the order fulfillment email.

   This step is not required when leveraging VM Flex licensing as the serial number is automatically registered with the CSP when generated.
3. Activate the firewall management license.
   - Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is Internet-connected.
   - Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is not Internet-connected.
4. Activate a Panorama Support License.

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STEP 17 | Complete configuring the Panorama virtual appliance for your deployment needs.

- (Management Only mode) Set up a Panorama Virtual Appliance in Management Only Mode.
- (Log Collector mode) Begin at Step 6 to Switch from Panorama mode to Log Collector mode.

  *Enter the Public IP address of the Dedicated Log Collector when you Add the Log Collector as a managed collector to the Panorama management server. You cannot specify the **IP Address**, **Netmask**, or **Gateway**.*
- (Panorama and Management Only mode) Configure a Managed Collector to add a Dedicated Log Collector to the Panorama virtual appliance. Management Only mode does not support local log collection, and requires a Dedicated Log Collector to store managed device logs.

STEP 18 | Complete configuring the Panorama virtual appliance for your deployment needs.

- For Panorama in Log Collector Mode.
  1. Add a Virtual Disk to Panorama on Alibaba Cloud as needed.

     Adding at least one virtual logging disk is required before you can change the Panorama virtual appliance to Log Collector mode.
  2. Begin at Step 6 to switch to Log Collector mode.

     *Enter the Public IP address of the Dedicated Log Collector when you add the Log Collector as a managed collector to the Panorama management server. You cannot specify the **IP Address**, **Netmask**, or **Gateway**.*
- For Panorama in Panorama mode.
  1. Add a Virtual Disk to Panorama on Alibaba Cloud as needed.

     Adding at least one virtual logging disk is required before you can change the Panorama virtual appliance to Panorama mode.
  2. Set up a Panorama Virtual Appliance in Panorama Mode.
  3. Configure a Managed Collector.
- For Panorama in Management Only mode.
  1. Set up a Panorama Virtual Appliance in Management Only Mode.
  2. Configure a Managed Collector to add a Dedicated Log Collector to the Panorama virtual appliance.

     Management Only mode does not support local log collection, and requires a Dedicated Log Collector to store managed device logs.

Install Panorama on AWS

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Device management licence • Support license

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Where Can I Use This?	What Do I Need?
	• Panorama Virtual Appliance license (serial number) • Customer Support Portal (CSP) Account

You can now deploy Panorama™ and a Dedicated Log Collector on Amazon Web Services (AWS). Panorama deployed on AWS is Bring Your Own License (BYOL), supports all deployment modes (Panorama, Log Collector, and Management Only), and shares the same processes and functionality as the M-Series hardware appliances. For more information on Panorama modes, see Panorama Models.

STEP 1 | Log in to AWS Web Service console and select the EC2 Dashboard.

- Amazon Web Service Console
- AWS GovCloud Web Service Console

STEP 2 | Set up the virtual private cloud (VPC) for your network needs.

Whether you launch the Panorama virtual appliance in an existing VPC or you create a new VPC, the Panorama virtual appliance must be able to receive traffic from other instances in the VPC and perform inbound and outbound communication between the VPC and the internet as needed.

Refer to the AWS VPC documentation for instructions on creating a VPC and setting it up for access.

1. Create a new VPC or use an existing VPC. Refer to the AWS Getting Started documentation
2. Verify that the network and security components are appropriately defined.
   - Create an internet gateway to enable internet access to the subnet of your Panorama virtual appliance. Internet access is required to install software and content updates, activate licenses, and leverage Palo Alto Networks cloud services. Otherwise, you must manually install updates and activate licenses.
   - Create subnets. Subnets are segments of the IP address range assigned to the VPC in which you can launch AWS instances. It is recommended that the Panorama virtual

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appliance belong to the management subnet so that you can configure it to access the internet if needed.

- Add routes to the route table for a private subnet to ensure traffic can be routed across subnets in the VPC and from the internet if applicable.

  Ensure you create routes between subnets to allow communication between:

  - Panorama, managed firewalls, and Log Collectors.
  - (Optional) Panorama and the internet.
- Ensure that the following inbound security rules are allowed for the VPC to manage VPC traffic. The ingress traffic source for each rule is unique to your deployment topology.

  See Ports Used for Panorama for more information.

  - Allow SSH (port **22**) traffic to enable access to the Panorama CLI.
  - Allow HTTPS (port **443**) traffic to enable access to the Panorama web interface.
  - Allow traffic on port **3978** to enable communication between Panorama, manage firewalls, and managed Log Collectors. This port is also used by Log Collectors to forward logs to Panorama.
  - Allow traffic on port **28443** to enable managed firewalls to get software and content updates from Panorama.

STEP 3 | Deploy Panorama on Amazon Web Services.

1. Select **Services** > **EC2** > **Instances** and **Launch Instance**.
2. Select **AWS Marketplace**, search for **`Palo Alto Networks Panorama`**, and **Select** the Panorama AMI and **Continue**.
3. Choose the **EC2 instance type** for allocating the resources required for the Panorama virtual appliance, and click **Next: Configure Instance Details**. Review the Setup Prerequisites for the Panorama Virtual Appliance for resource requirements.

   If you plan to use the Panorama virtual appliance as a Dedicated Log Collector, ensure that you configure the appliance with the required resources during initial deployment. The Panorama virtual appliance does not remain in Log Collector mode if you resize the virtual machine after you deploy it, and this results in a loss of log data.
4. Configure the instance details.

   1. Select **Next: Configure Instance Details**.
   2. For the **Network**, select the VPC.
   3. Select the **Subnet**.
   4. To **Auto-assign Public IP** select **Enable.**

      This IP must be accessible by the firewalls you plan to manage using Panorama. This allows you to obtain a publicly accessible IP address for the management interface of the Panorama virtual appliance. You can later attach an Elastic IP address to the management interface. Unlike the public IP address that is disassociated from the virtual appliance when the instance is terminated, the Elastic IP address provides

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persistence and you can the IP address to a new (or replacement) instance of the Panorama virtual appliance without the need to reconfigure the IP address whenever the Panorama virtual appliance instance is powered off.

5. Configure any additional instance details as needed.

5. (Optional) Configure the Panorama virtual appliance storage.

   1. Select **Next: Add Storage**.
   2. **Add New Volume** to add additional log storage.

      (SD-WAN only) If you plan on managing your SD-WAN deployment in Management Only mode, you must add a 2TB logging disk.

      If you intend to use the Panorama virtual appliance in Panorama mode or as a Dedicated Log Collector, add the virtual logging disks during the initial deployment. By default, the Panorama virtual appliance is in Panorama mode for the initial deployment when you meet the Panorama mode resource requirements and have added at least one virtual logging disk. Otherwise, the Panorama virtual appliance defaults to Management Only mode. Change the Panorama virtual appliance to Management Only mode if you just want to manage devices and Dedicated Log Collectors, and to not collect logs locally.

      The Panorama virtual appliance on AWS only supports 2TB logging disks, and in total supports up to 24TB of log storage. You are unable to add a logging disk smaller than 2TB, or a logging disk with a size not divisible by the 2TB logging disk requirement. The Panorama virtual appliance partitions logging disks larger than 2TB into 2TB partitions.

6. (Optional) Select **Next: Add Tags** and add one or more tags as metadata to help you identify and group the Panorama virtual appliance. For example, add a **Name** tag with a **Value** that helps you identify which firewalls the Panorama virtual appliance manages.
7. Configure the instance security group.

   1. Select **Next: Configure Security Group**.
   2. **Select an existing security group** to assign a security group for the Panorama virtual appliance instance.
   3. Select the security group you previously created.

      You can select the `default` security group to allow all inbound and outbound traffic types.

8. **Review and Launch** the Panorama virtual appliance instance to verify that your selections are accurate before you **Launch.**
9. Select an existing key pair or create a new one and acknowledge the disclaimer.

   If you created a new key from AWS, download and save the key to a safe location. The file extension is `.pem`. You must load the public key into PuTTYgen *and save it in `.ppk` format. You cannot regenerate this key if lost.*

   It takes about 30 minutes to finish deploying the Panorama virtual appliance after you launch it on AWS. Deploying the Panorama virtual appliance may take longer depending

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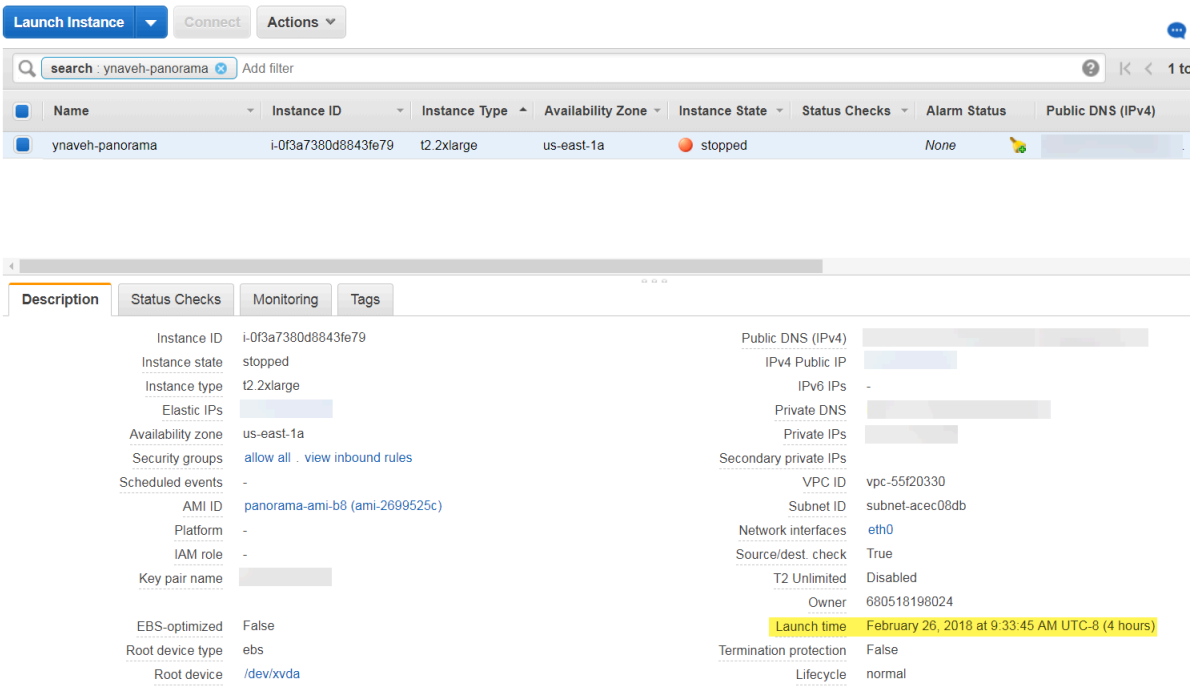
on the number and size of the disks attached to the instance. View the Launch Time by selecting the Panorama virtual appliance instance (**Instances**).



If you plan to use the Panorama virtual appliance as a Dedicated Log Collector, ensure that you provision the appliance with the required resources. The Panorama virtual appliance does not remain in Log Collector mode if you resize the virtual machine after you deploy it and this results in a loss of log data.

STEP 4 | Shut down the Panorama virtual appliance.

1. On the EC2 Dashboard, select **Instances**.
2. Select the Panorama virtual appliance and click **Instance State** > **Stop Instance**.

STEP 5 | Create or assign an Elastic IP (EIP) address to the management interface.

1. Select **Services** > **EC2** > **Elastic IPs** and **Allocate Elastic IP address**.
2. Select a **Network Border Group** to specify the logical group of zones from where the public IP4v address is advertised.
3. For the Public IPv4 address pool, select **Amazon's pool of IPv4 addresses**.
4. **Allocate** the EIP.
5. Click the IPv4 address in the Allocated IPv4 address column and **Associate Elastic IP address**.
6. Select the Panorama virtual appliance **Instance.**
7. Select the Panorama virtual appliance **Private IP address** to with which to associate the EIP.

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STEP 6 | Power on the Panorama virtual appliance.

1. On the EC2 Dashboard, select **Instance**.
2. From the list, select the Panorama virtual appliance and click **Actions** > **Instance State** > **Start**.

STEP 7 | Configure a new administrative password for the Panorama virtual appliance.

You must configure a unique administrative password before you can access the web interface of the Panorama virtual appliance. To access the CLI, the private key used to launch the Panorama virtual appliance is required.

The new password must be a minimum of eight characters and include a minimum of one lowercase character, one uppercase character, and one number or special character.

- If you have an SSH service installed on your computer:

1. Enter the following command to log into the Panorama virtual appliance:

```
ssh -i <private_key.ppk> admin@<public-ip_address>
```

2. Configure a new password using the following commands and follow the on screen prompts:

```
admin> configure
```

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```
admin# set mgt-config users admin password
```

3. If you need to activate a BYOL, set the DNS server IP address so that the Panorama virtual appliance can access the Palo Alto Networks licensing server. Enter the following command to set the DNS server IP address:

```
admin# set deviceconfig system dns-setting servers
 primary <ip_address>
```

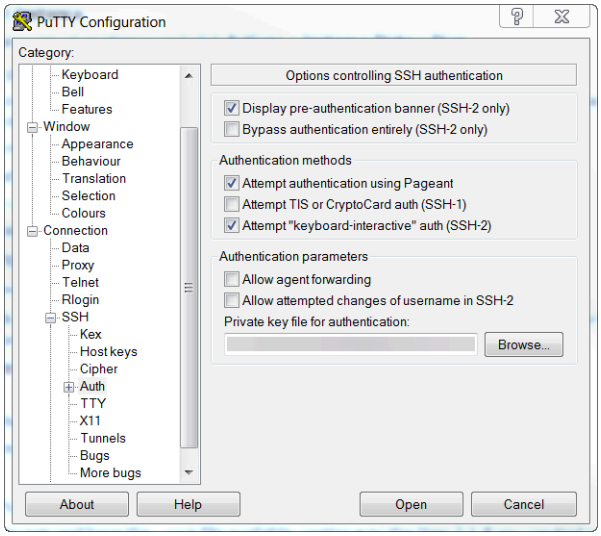
4. Commit your changes with the command:

```
admin# commit
```

5. Terminate the SSH session.

- If you are using PuTTY to SSH into the Panorama virtual appliance:

1. If you are using an existing key pair and have the `.ppk` file available, continue to the Step 7.3. If you created a new key pair or have only the `.pem` file of the existing key pair, open PuTTYgen and **Load** the `.pem` file.
2. **Save the private key** to a local accessible destination.
3. Open PuTTY and select **SSH** > **Auth** and then **Browse** to the `.ppk` file you saved in the previous step.



4. Select **Sessions** and enter the public IP address of the Panorama virtual appliance. Click **Open** and click **Yes** when the security prompt appears.
5. Log in as admin when prompted.
6. Configure a new password using the following commands and follow the onscreen prompts:

```
admin> configure
```

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```
admin# set mgt-config users admin password
```

7. Set the DNS server IP address so that the Panorama virtual appliance can access the Palo Alto Networks licensing server. Enter the following command to set the DNS server IP address:

```
admin# set deviceconfig system dns-setting servers
 primary <ip_address>
```

8. Commit your changes with the command:

```
admin# commit
```

9. Terminate the SSH session.

STEP 8 | Register the Panorama virtual appliance and activate the device management license and support licenses.

1. (VM Flex Licensing Only) Provisioning the Panorama Virtual Appliance Serial Number.

   When leveraging VM Flex licensing, this step is required to generate the Panorama virtual appliance serial number needed to register the Panorama virtual appliance with the Palo Alto Networks Customer Support Portal (CSP).

2. Register Panorama.

   You must register the Panorama virtual appliance using the serial number provided by Palo Alto Networks in the order fulfillment email.

   This step is not required when leveraging VM Flex licensing as the serial number is automatically registered with the CSP when generated.

3. Activate the firewall management license.
   - Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is Internet-connected.
   - Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is not Internet-connected.
4. Activate a Panorama Support License.

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STEP 9 | Complete configuring the Panorama virtual appliance for your deployment needs.

- For Panorama in Log Collector Mode.
  1. Add a Virtual Disk to Panorama on AWS as needed.

     Adding at least one virtual logging disk is required before you can change the Panorama virtual appliance to Log Collector mode.
  2. Begin at Step 6 to switch to Log Collector mode.

     *Enter the Public IP address of the Dedicated Log Collector when you add the Log Collector as a managed collector to the Panorama management server. You cannot specify the **IP Address**, **Netmask**, or **Gateway**.*
- For Panorama in Panorama mode.
  1. Add a Virtual Disk to Panorama on AWS.

     Adding at least one virtual logging disk is required before you can change the Panorama virtual appliance to Panorama mode.
  2. Set up a Panorama Virtual Appliance in Panorama Mode.
  3. Configure a Managed Collector.
- For Panorama in Management Only mode.
  1. Set up a Panorama Virtual Appliance in Management Only Mode.
  2. Configure a Managed Collector to add a Dedicated Log Collector to the Panorama virtual appliance.

     Management Only mode does not support local log collection, and requires a Dedicated Log Collector to store managed device logs.

Install Panorama on AWS GovCloud

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Device management licence • Support license • Panorama Virtual Appliance license (serial number) • Customer Support Portal (CSP) Account

You can now deploy Panorama™ and a Dedicated Log Collector on Amazon Web Services (AWS) GovCloud. AWS GovCloud is an isolated AWS region that meets the regulatory and compliance requirements of the US government agencies and customers. Panorama deployed on AWS GovCloud is Bring Your Own License (BYOL), supports all deployment modes (Panorama, Log Collector, and Management Only). For more information on Panorama modes, see Panorama Models.

To secure your workloads that contain all categories of Controlled Unclassified Information (CUI) data and government-oriented, publicly available data in the AWS GovCloud (US) region, the Panorama virtual appliance provides the same security features offered in the standard AWS

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public cloud on AWS GovCloud. The Panorama virtual appliance on AWS GovCloud and the standard AWS public cloud support the same features and capabilities.

Review the Setup Prerequisites for the Panorama Virtual Appliance to review the supported EC2 instance types. Once you are ready, refer to Install Panorama on AWS to install the Panorama virtual appliance on AWS GovCloud.

See the following procedures to add additional logging storage to your Panorama virtual appliance, or to increase the allocated CPU cores and memory:

- Add a Virtual Disk to Panorama on AWS
- Increase CPUs and Memory for Panorama on AWS

Install Panorama on Azure

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Device management licence • Support license • Panorama Virtual Appliance license (serial number) • Customer Support Portal (CSP) Account

You can now deploy Panorama™ and a Dedicated Log Collector on Microsoft Azure. Panorama deployed on Azure is Bring Your Own License (BYOL), supports all deployment modes (Panorama, Log Collector, and Management Only), and shares the same processes and functionality as the M-Series hardware appliances. For more information on Panorama modes, see Panorama Models.

STEP 1 | Log into to the Microsoft Azure portal.

STEP 2 | Set up the virtual network for your network needs.

Whether you launch the Panorama virtual appliance in an existing virtual network or you create a new virtual network, the Panorama virtual appliance must be able to receive traffic from other instances in the virtual network and perform inbound and outbound communication between the virtual network and the internet as needed.

Refer to the Microsoft Azure Virtual Network documentation for more information.

1. Create a Virtual Network or use an existing virtual network.
2. Verify that the network and security components are appropriately defined.
   - Create a NAT gateway if you want to enable only outbound internet access for the subnet to which the Panorama virtual appliance belongs.
   - Create subnets. Subnets are segments of the IP address range assigned to the VNet in which you can launch Microsoft Azure instances. It is recommended that

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the Panorama virtual appliance belong to the management subnet so that you can configure it to access the internet if needed.

- Add routes to the route table for a private subnet to ensure traffic can be routed across subnets in the VNet and from the internet if applicable.

  Ensure you create routes between subnets to allow communication between:

  - Panorama, managed firewalls, and Log Collectors.
  - (Optional) Panorama and the internet.
- Ensure that the following ingress security rules are allowed for the VNet to manage VNet traffic. The ingress traffic source for each rule is unique to your deployment topology.

  See Ports Used for Panorama for more information.

  - Allow SSH (port **22**) traffic to enable access to the Panorama CLI.
  - Allow HTTPS (port **443**) traffic to enable access to the Panorama web interface.
  - Allow traffic on port **3978** to enable communication between Panorama, manage firewalls, and managed Log Collectors. This port is also used by Log Collectors to forward logs to Panorama.
  - Allow traffic on port **28443** to enable managed firewalls to get software and content updates from Panorama.

STEP 3 | Deploy the Panorama virtual appliance.

1. In the Azure Dashboard, select **Virtual machines** and **Add** a new virtual machine.
2. Search for Palo Alto Networks and select the latest Panorama virtual appliance image.
3. **Create** the Panorama virtual appliance.

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STEP 4 | Configure the Panorama virtual appliance.

1. Select your Azure **Subscription**.
2. Select the Azure **Resource Group** to contain all your Azure instance resources.
3. Enter a **Virtual machine name** for the Panorama virtual appliance.
4. Select the **Region** for the Panorama virtual appliance to be deployed in.
5. (Optional) Select the **Availability options**. See How to use availability sets for more information.
6. Select the **Image** used to deploy the Panorama management server. **Browse all public and private images** to deploy the Panorama management server from the Panorama image on the Azure marketplace.
7. Configure the Panorama virtual appliance size. Review the Setup Prerequisites for the Panorama Virtual Appliance for sizing requirements.

   If you plan to use the Panorama virtual appliance as a Dedicated Log Collector, ensure that you configure the appliance with the required resources during initial deployment. The Panorama virtual appliance does not remain in Log Collector mode if you resize the virtual machine after you deploy it, and this results in a loss of log data.
8. Configure the unique Panorama virtual appliance administrator credentials.

   You must configure a unique administrative password before you can access the web interface and CLI of the Panorama virtual appliance.

   1. Enter a **Username** for the Panorama virtual appliance administrator. To ensure that your username is secure, admin is not a valid entry.
   2. Enter a **Password** or copy and paste an **SSH public key** for securing administrative access to the Panorama virtual appliance.

      You must enable SSH key authentication if you plan to use this instance of the Panorama virtual appliance in FIPS-CC operational mode. Although you can deploy the Panorama virtual appliance using a username and password, you will be unable to authenticate using the username and password after changing the operational mode to FIPS-CC. After resetting to FIPS-CC mode, you must use the SSH key to log in and can then configure a username and password that you can use for subsequently logging in to the Panorama web interface. For details on creating the SSH key, refer to the Azure documentation.
9. Configure the Panorama virtual appliance instance **Networking**

   1. Select an existing **Virtual network** or create a new virtual network.
   2. Configure the **Subnet**. The subnet is dependent on the virtual network you selected or created in the previous step. If you selected an existing virtual network, you can choose one of the subnets for the selected virtual network.
   3. Select an existing **Public IP address** or create a new one. This creates the management interface used to access your Panorama virtual appliance.
   4. Select an existing **NIC network security group** or create a new security group. Network security groups control traffic to the virtual machine. Make sure that HTTPS and SSH are allowed for the Inbound rules.

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10. Configure the instance **Management** settings.
    1. Select whether to enable **Auto-shutdown**. Auto-shutdown allows you to configure a daily time to automatically shut down the virtual machine that you disable auto-shutdown to avoid the possibility that a new public IP address gets assigned to the virtual machine, that logs are dropped, that logs are not or that you are unable to manage your firewalls while the Panorama virtual appliance is shut down.
    2. Select whether to enable boot **Monitoring**. Select the Diagnostic storage account if enabled. Monitoring automatically sends boot-up diagnostic logs to your Diagnostics storage account. For more information, see Overview of Monitoring in Microsoft Azure.
    3. Configure any other settings as needed.
11. Review the summary, accept the terms of use and privacy policy, and **Create** the Panorama virtual appliance.

STEP 5 | Verify that you the Panorama virtual appliance has been successfully deployed.

1. Select **Dashboard** > **Resource Groups** and select the resource group containing the Panorama virtual appliance.
2. Under Settings, select **Deployments** for the virtual machine deployment status.

It takes about 30 minutes to deploy the Panorama virtual appliance. Launching the Panorama virtual appliance may take longer depending on the resources configured for the virtual machine. Microsoft Azure does not permit the ICMP protocol to test whether it deployed successfully.

If you plan to use the Panorama virtual appliance as a Dedicated Log Collector, ensure that you correctly configured the appliance the required resources. The Panorama virtual appliance does not remain in Log Collector mode if you resize the virtual machine after you deploy it and this results in a loss of log data.

STEP 6 | Configure a static Public IP address.

1. On the Azure portal, select **Virtual machines** and select the Panorama virtual machine.
2. Select **Overview** and click the **Public IP address**.
3. Under Assignment, select **Static** and **Save** the new IP address configuration.

STEP 7 | Log in to the web interface of the Panorama virtual appliance.

1. On the Azure portal, in **All Resources**, select the Panorama virtual appliance and view the public IP address located in the Overview section.
2. Use a secure (https) connection from your web browser to log in to the Panorama virtual appliance using the public IP address.
3. Enter the username and password of the Panorama virtual appliance. You are prompted with a certificate warning. Accept the certificate warning and continue to the web page.

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STEP 8 | Register the Panorama virtual appliance and activate the device management license and support licenses.

1. (VM Flex Licensing Only) Provisioning the Panorama Virtual Appliance Serial Number.

   When leveraging VM Flex licensing, this step is required to generate the Panorama virtual appliance serial number needed to register the Panorama virtual appliance with the Palo Alto Networks Customer Support Portal (CSP).
2. Register Panorama.

   You must register the Panorama virtual appliance using the serial number provided by Palo Alto Networks in the order fulfillment email.

   This step is not required when leveraging VM Flex licensing as the serial number is automatically registered with the CSP when generated.
3. Activate the firewall management license.
   - Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is Internet-connected.
   - Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is not Internet-connected.
4. Activate a Panorama Support License.

STEP 9 | Complete configuring the Panorama virtual appliance for your deployment needs.

- For Panorama in Log Collector Mode.
  1. Add a Virtual Disk to Panorama on Azure as needed.

     Adding at least one virtual logging disk is required before you can change the Panorama virtual appliance to Log Collector mode.
  2. Begin at Step 6 to switch to Log Collector mode.

     *Enter the Public IP address of the Dedicated Log Collector when you add the Log Collector as a managed collector to the Panorama management server. You cannot specify the **IP Address**, **Netmask**, or **Gateway**.*
- For Panorama in Panorama mode.
  1. Add a Virtual Disk to Panorama on Azure.

     Adding at least one virtual logging disk is required before you can change the Panorama virtual appliance to Panorama mode.
  2. Set up a Panorama Virtual Appliance in Panorama Mode.
  3. Configure a Managed Collector.
- For Panorama in Management Only mode.
  1. Set up a Panorama Virtual Appliance in Management Only Mode.
  2. Configure a Managed Collector to add a Dedicated Log Collector to the Panorama virtual appliance.

     Management Only mode does not support local log collection, and requires a Dedicated Log Collector to store managed device logs.

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Install Panorama on Google Cloud Platform

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Device management licence • Support license • Panorama Virtual Appliance license (serial number) • Customer Support Portal (CSP) Account

You can now deploy Panorama™ and a Dedicated Log Collector on Google Cloud Platform (GCP). Panorama deployed on GCP is Bring Your Own License (BYOL), supports all deployment modes (Panorama, Log Collector, and Management Only), and shares the same processes and functionality as the M-Series hardware appliances. For more information on Panorama modes, see Panorama Models.

To deploy the Panorama virtual appliance on GCP, you need to build a custom image. To begin this process, you must download the Panorama `tar.gz` from the Palo Alto Networks Customer Support portal and upload it to a GCP storage bucket. You can then create the custom image and use the image to deploy the Panorama virtual appliance on GCP.

STEP 1 | Download the Panorama virtual appliance image.

1. Log in to the Palo Alto Networks Support Portal.
2. Select **Updates** > **Software Updates** and filter by **Panorama Base Images**.
3. Download the latest version of the Panorama on GCP `tar.gz` image.

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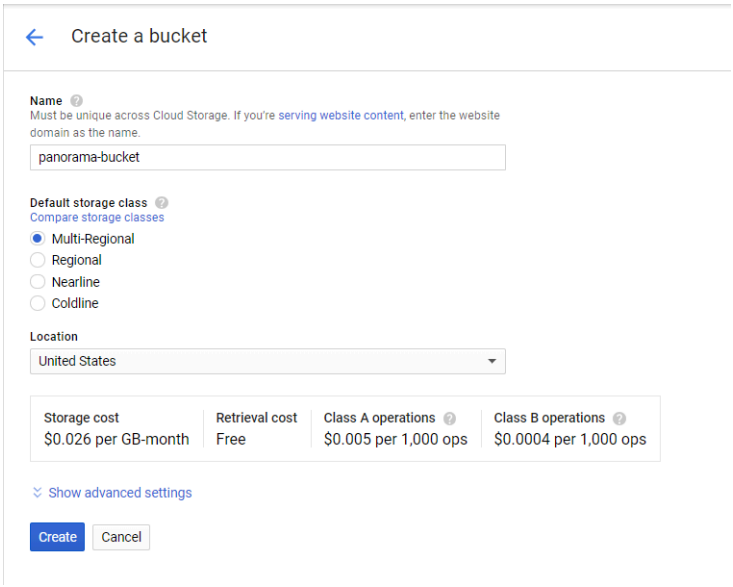
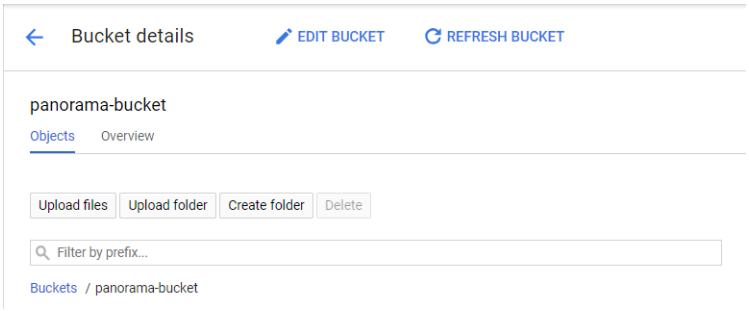
STEP 2 | Upload the Panorama virtual appliance image to the Google Cloud Platform.

1. Log in to the Google Cloud Console.
2. From the **Products and Services** menu, select **Storage**.
3. Click **Create Bucket**, configure the new storage bucket and click **Create.**



4. Select the storage bucket you created in the previous step, click **Upload files**, and select the Panorama virtual appliance image you downloaded.



5. From the **Products and Services** menu, select **Compute Engine** > **Images**.
6. Click **Create Image** and create the Panorama virtual appliance image:
   1. **Name** the Panorama virtual appliance image.
   2. In the **Source** field, select **Cloud Storage file** from the drop-down menu.
   3. Click **Browse** and navigate to the storage bucket where you uploaded the Panorama virtual appliance image, and **Select** the uploaded image.
   4. **Create** the Panorama virtual appliance image.

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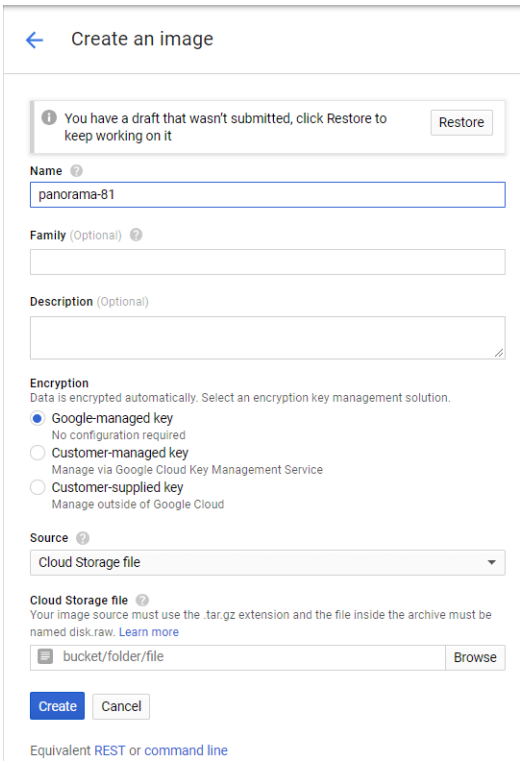

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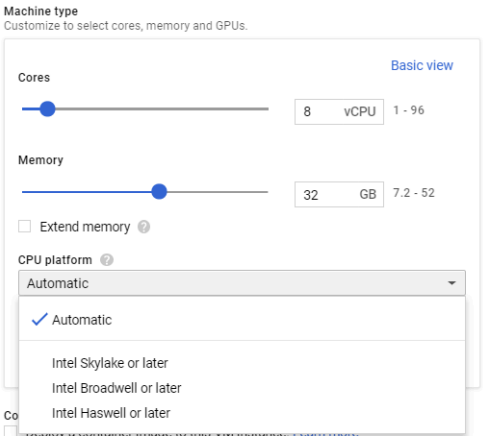
STEP 3 | Configure the Panorama virtual appliance.

1. From the **Products and Services** menu and select the **Compute Engine**.
2. Click **Create Instance** to begin deploying the Panorama virtual appliance.
3. Add a descriptive **Name** to easily identify the Panorama virtual appliance.
4. Select the **Region** and **Zone** where you want to deploy the Panorama virtual appliance.
5. Allocate the **Machine Type** and **Customize** the CPU cores, memory and CPU platform. Review the Setup Prerequisites for the Panorama Virtual Appliance for minimum resource requirements.

   If you plan to use the Panorama virtual appliance as a Dedicated Log Collector, ensure that you configure the appliance with the required resources during initial deployment. The Panorama virtual appliance does not remain in Log Collector mode if you resize the virtual machine after you deploy it, and this results in a loss of log data.

   The GCP zone selection determines the CPU platforms available to you. For more information, refer to Regions and Zones for details.



6. Configure the Panorama boot disk.
   1. For the **Boot Disk**, click **Change** > **Custom image** and select the Panorama image file you uploaded in Step 2
   2. Review the boot disk **Size** and verify the system disk is **81**GB.

      *The Panorama virtual appliance must be initially installed with the **Default** system disk size. Installing the Panorama virtual appliance with a system disk larger than the **Default** system disk size is not supported and may result in limited utilization. You have the option to increase the system disk size after initial installation*

   3. Click **Select** to save your configuration.
7. Under **Identity and API access**, select **Allow full access to all Cloud APIs**.

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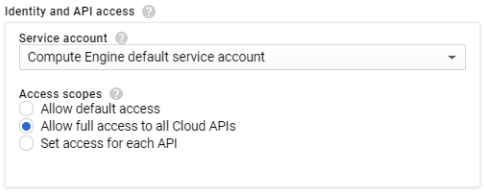
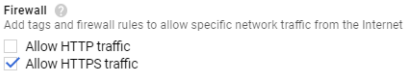


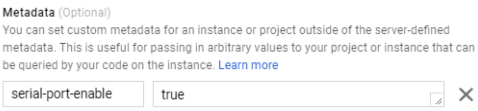
8. Under **Firewall**, select **Allow HTTPS traffic**.



STEP 4 | Expand **Management, security, disks, networking, sole tenancy** Management, security, disks, networking, sole tenancy .

STEP 5 | Enable access to the serial port so you can manage the Panorama virtual appliance.

1. Select **Management**.
2. Enter the following name-value pair as Metadata:
   `serial-port-enable true`



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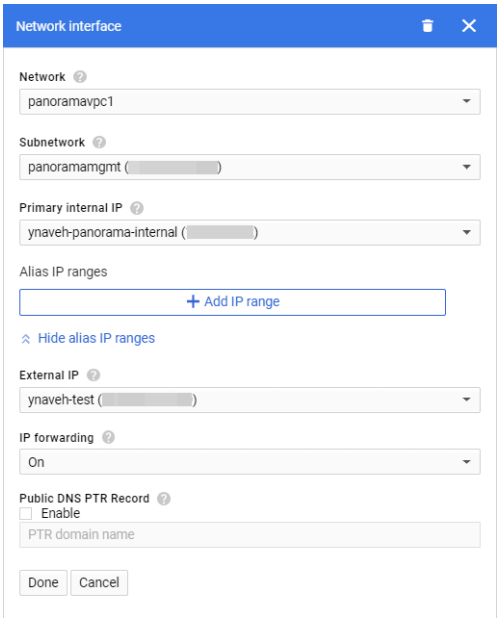
STEP 6 | Reserve a static IP address for the management interface.

Reserve static internal and external IP addresses for the management interface in the event that if the Panorama virtual appliance is rebooted, your managed devices do not lose connection to the Panorama virtual appliance when the IP addresses are reassigned.

For more information on how to reserve IP addresses, refer to Reserving a Static Internal IP Address and Reserving a Static External IP Address.

1. Select **Networking**.
2. **Edit** the network interface.

3. Select the Panorama virtual appliance **Network**.
4. Select the Panorama virtual appliance **Subnetwork**. Instances in the same subnetwork will communicate with each other using their internal IP addresses.
5. Set the **Primary internal IP** address.
   - **Ephemeral (Automatic)**— Automatically assign a primary internal IP address.
   - **Ephemeral (Custom)**—Configure a custom IP range that GCP uses to assign a primary internal IP address.
   - **Reserve a static internal IP address**—Manually configure a static primary internal IP address.
6. Set the **External IP** address.
   - **Ephemeral**—Automatically assign an external IP address from a shared IP pool.
   - Select an existing reserved external IP address.
   - **Create IP address**—Reserve an external IP address.
7. Set **IP forwarding** to **On** to allow the Panorama virtual appliance to receive packets from non-matching destinations or source IP addresses.



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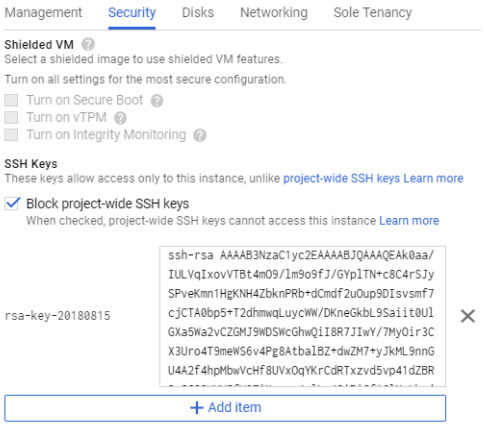
STEP 7 | Configure the SSH key. You need an SSH key to access the Panorama virtual appliance CLI to configure the administrative user password after the initial deployment.

- **PuTTY Users**

  1. Select **Security**.
  2. Select the **Block project-wide SSH keys** box. Only instance keys are currently supported for logging in to the Panorama virtual appliance after initial deployment.
  3. Paste the SSH key in the comment box. For information on the correct SSH key format and how to generate SSH keys for GCP, refer to Managing SSH keys in Metadata.

     When generating the SSH key, save the private key in .ppk format. The private key is required to log in to the Panorama virtual appliance after the initial deployment before you can configure the administrative password.



- **Linux and macOS Users**

  1. Generate the SSH key from the CLI of your Linux device.

     ```
     ssh-keygen -C admin@panorama -f <panorama_key_name>
     ```

     Where **admin@panorama** is a comment GCP requires and **<panorama_key_name>** is the name of the key file being generated.
  2. Create an output file for the SSH key.

     ```
     cat <panorama_key_name>.pub
     ```

     After the output file for the SSH key is created, manually copy the SSH key contents.
  3. Paste the public key into the SSH Keys section of the GCP instance creation.

STEP 8 | (Optional) Add additional storage for log collection. Repeat this step as needed to add additional virtual logging disks.

If you intend to use the Panorama virtual appliance in Panorama mode or as a Dedicated Log Collector, add the virtual logging disks during initial deployment. By default, the Panorama virtual appliance is in Panorama mode for the initial deployment when you meet the Panorama mode resource requirements and have added at least one virtual logging disk. Otherwise, the

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Panorama virtual appliance defaults to Management Only mode in which you can manage devices and Dedicated Log Collectors, and cannot collect logs locally.

The Panorama virtual appliance on GCP only supports 2TB logging disks, and in total supports up to 24TB of log storage. You are unable to add a logging disk smaller than 2TB, or a logging

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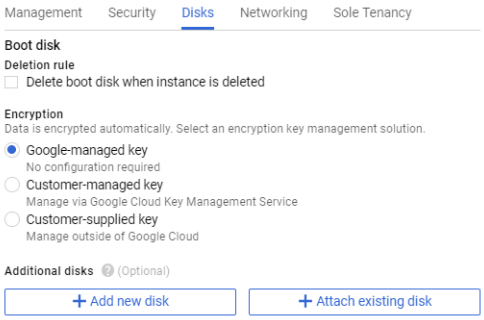
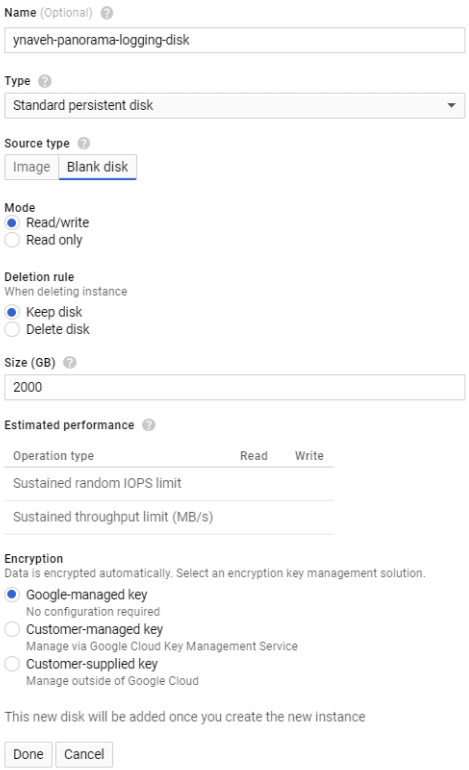
disk with a size not divisible by the 2TB logging disk requirement. The Panorama virtual appliance partitions logging disks larger than 2TB into 2TB partitions.

1. Select **Disks** > **Add new disk**.



2. Enter the **Name**.
3. Expand the **Type** drop-down menu and select the desired type.
4. For the **Source type**, select **Blank disk**.
5. For the **Mode**, select **Read/write**.
6. Select the **Deletion rule** to configure whether to delete the virtual logging disk if the Panorama virtual appliance instance is deleted. To
7. Set the **Size (GB)** of the virtual logging disk.
8. Set your preferred **Encryption** solution for the data on the virtual logging disk.
9. Click **Done**.



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STEP 9 | **Create** the Panorama virtual appliance. The Panorama virtual appliances takes roughly 10 minutes to boot up after initial deployment.

STEP 10 | Configure a new administrative password for the Panorama virtual appliance.

You must configure a unique administrative password before you can access the web interface of the Panorama virtual appliance. To access the CLI, use the private key to launch the Panorama virtual appliance.

The new password must be a minimum of eight characters and include a minimum of one lowercase character, one uppercase character, and one number or special character.

- If you have an SSH service installed on your computer:
  1. Enter the following command to log into the Panorama virtual appliance:
     - Windows Devices

       ```
       ssh -i <private_key.ppk> <username>@<public-ip_address>
       ```

     - Linux Devices

       ```
       ssh -i <prive_key.ppk> -oHostKeyAlgorithms=+ssh-rsa
        <username>@<public-ip_address>
       ```

       Including *-oHostKeyAlgorithms=+ssh-rsa* is required to specify the host key type. An error is displayed if this is not included in the SSH login command.
  2. Configure a new password using the following commands and follow the onscreen prompts:

     ```
     admin> configure
     ```

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```
admin# set mgt-config users admin password
```

3. If you have a BYOL that you need to, set the DNS server IP address so that the Panorama virtual appliance can access the Palo Alto Networks licensing server. Enter the following command to set the DNS server IP address:

```
admin# set deviceconfig system dns-setting servers
 primary <ip_address>
```

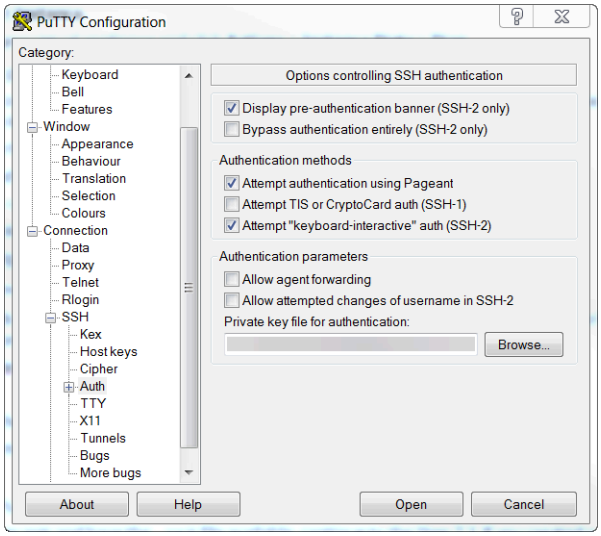
4. Commit your changes:

```
admin# commit
```

5. Terminate the SSH session.

- If you are using PuTTY to SSH into the Panorama virtual appliance:

1. If you are using an existing key pair and have the `.ppk` file available, continue to Step 11.3. If you created a new key pair or only have the `.pem` file of the existing key pair, open PuTTYgen and **Load** the `.pem` file.
2. **Save the private key** to a local accessible destination.
3. Open PuTTY and select **SSH** > **Auth** and **Browse** for the `.ppk` file saved in the previous step.



4. Select **Sessions** and enter the public IP address of the Panorama virtual appliance. Then **Open** and click **Yes** when the security prompt appears.
5. Login as admin when prompted.
6. Configure a new password using the following commands and follow the on screen prompts:

```
admin> configure
```

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```
admin# set mgt-config users admin password
```

7. Set the DNS server IP address so that the Panorama virtual appliance can access the Palo Alto Networks licensing server. Enter the following command to set the DNS server IP address:

```
admin# set deviceconfig system dns-setting servers
 primary <ip_address>
```

8. Commit your changes with the command:

```
admin# commit
```

9. Terminate the SSH session.

STEP 11 | Register the Panorama virtual appliance and activate the device management license and support licenses.

1. (VM Flex Licensing Only) Provisioning the Panorama Virtual Appliance Serial Number.

   When leveraging VM Flex licensing, this step is required to generate the Panorama virtual appliance serial number needed to register the Panorama virtual appliance with the Palo Alto Networks Customer Support Portal (CSP).

2. Register Panorama.

   You must register the Panorama virtual appliance using the serial number provided by Palo Alto Networks in the order fulfillment email.

   This step is not required when leveraging VM Flex licensing as the serial number is automatically registered with the CSP when generated.

3. Activate the firewall management license.
   - Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is Internet-connected.
   - Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is not Internet-connected.
4. Activate a Panorama Support License.

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STEP 12 | Complete configuring the Panorama virtual appliance for your deployment needs.

- For Panorama in Log Collector Mode.
  1. Add a Virtual Disk to Panorama on Google Cloud Platform as needed.

     Adding at least one virtual logging disk is required before you can change the Panorama virtual appliance to Log Collector mode.
  2. Begin at Step 6 to switch to Log Collector mode.

     *Enter the Public IP address of the Dedicated Log Collector when you add the Log Collector as a managed collector to the Panorama management server. You cannot specify the **IP Address**, **Netmask**, or **Gateway**.*
- For Panorama in Panorama mode.
  1. Add a Virtual Disk to Panorama on Google Cloud Platform.

     Adding at least one virtual logging disk is required before you can change the Panorama virtual appliance to Panorama mode.
  2. Set up a Panorama Virtual Appliance in Panorama Mode.
  3. Configure a Managed Collector.
- For Panorama in Management Only mode.
  1. Set up a Panorama Virtual Appliance in Management Only Mode.
  2. Configure a Managed Collector to add a Dedicated Log Collector to the Panorama virtual appliance.

     Management Only mode does not support local log collection, and requires a Dedicated Log Collector to store managed device logs.
- For SD-WAN deployments.
  1. Increase the System Disk for Panorama on Google Cloud Platform

     To leverage SD-WAN on Panorama deployed on GCP, you must increase the the system disk to 224GB.

     You cannot migrate back to a 81GB system disk after successfully increasing the system disk to 224GB.
  2. Set up a Panorama Virtual Appliance in Management Only Mode.
  3. Add a Virtual Disk to Panorama on Google Cloud Platform.

     To leverage SD-WAN, you must add a single 2TB logging disk to Panorama in Management Only mode.

Install Panorama on KVM

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Device management licence • Support license

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Where Can I Use This?	What Do I Need?
	• Panorama Virtual Appliance license (serial number) • Customer Support Portal (CSP) Account

You can now deploy Panorama™ and a Dedicated Log Collector on KVM. Panorama deployed on KVM is Bring Your Own License (BYOL), supports all deployment modes (Panorama, Log Collector, and Management Only), and shares the same processes and functionality as the M-Series hardware appliances. For more information on Panorama modes, see Panorama Models.

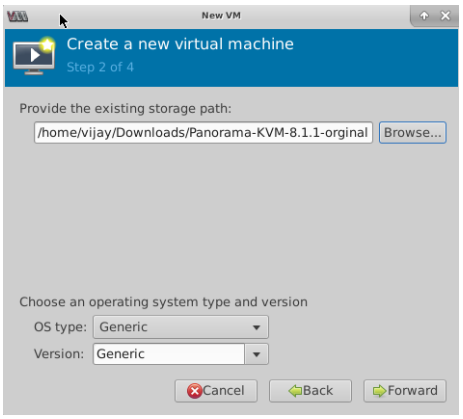
STEP 1 | Download the Panorama 11.1 base image QCOW2 file.

1. Log in to the Palo Alto Networks Support Portal.
2. Select **Updates** > **Software Updates** and filter by **Panorama Base Images** to download the QCOW2 file (`Panorama-KVM-11.1.0.qcow2`).

STEP 2 | Create a new virtual machine image and add the Panorama virtual appliance image for KVM to the Virtual Machine Manager.

1. On the Virtual Machine Manager, select **Create a new virtual machine**.
2. Select **Import Existing disk image** and click **Forward**.

3. **Browse** and select the Panorama virtual appliance image volume and **Choose volume**.
4. Click **Forward**.



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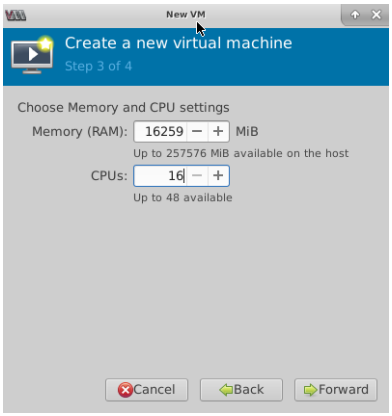
STEP 3 | Configure the memory and CPU settings.

Review the Setup Prerequisites for the Panorama Virtual Appliance for minimum resource requirements.

If you plan to use the Panorama virtual appliance as a Dedicated Log Collector, ensure that you configure the appliance with the required resources during initial deployment. The Panorama virtual appliance does not remain in Log Collector mode if you resize the virtual machine after you deploy it, and this results in a loss of log data.

1. Configure the **Memory** based on the requirements for the desired operational mode.

   The Virtual Machine Manager may use MiB (mebibyte) to allocate memory depending on the version you are running. If MiB is used, be sure to correctly convert your required memory allocation to avoid under provisioning the Panorama virtual appliance.
2. Configure the **CPU** based on the requirements for the desired operational mode.
3. Click **Forward**.



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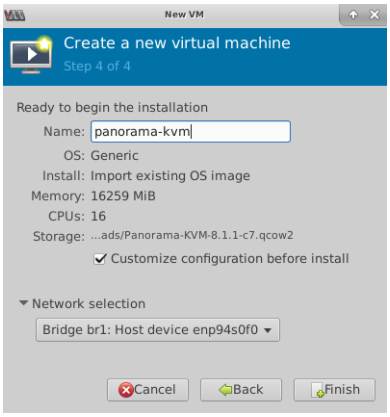
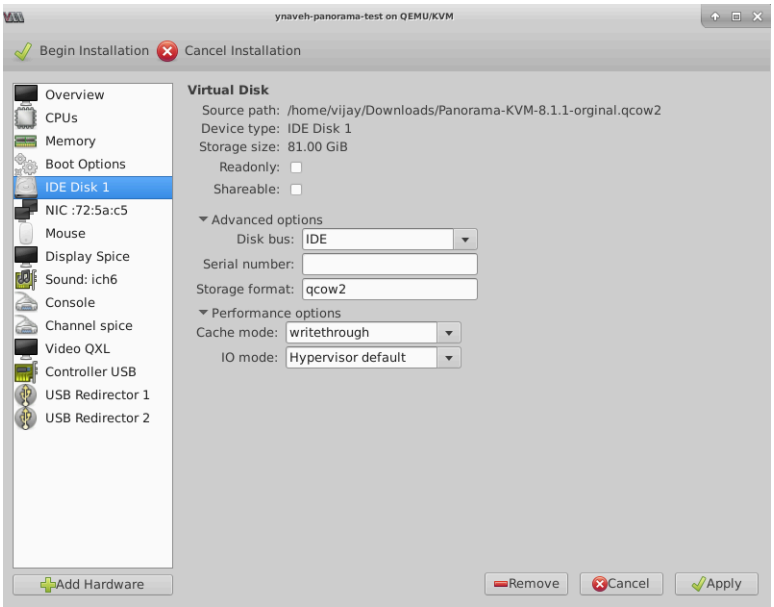
STEP 4 | Name the Panorama virtual appliance, enable configuration customization, and select the management interface bridge.

1. Enter a descriptive **Name** for the Panorama virtual appliance.
2. **Customize configuration before install**.
3. Make a **Network selection**—select the bridge for the management interface and accept the default settings.
4. Click **Finish**.



STEP 5 | Configure the virtual system disk settings.

1. Select **IDE Disk 1**, go to **Advanced options**, and select the following:
   - **Disk Bus**—**VirtIO** or **IDE**, depending on your configuration.
   - **Storage format**—**qcow2**
2. Go to **Performance options** and set **Cache mode** to **writethrough**. This setting improves installation time and execution speed on the Panorama virtual appliance.
3. Click **Apply**.



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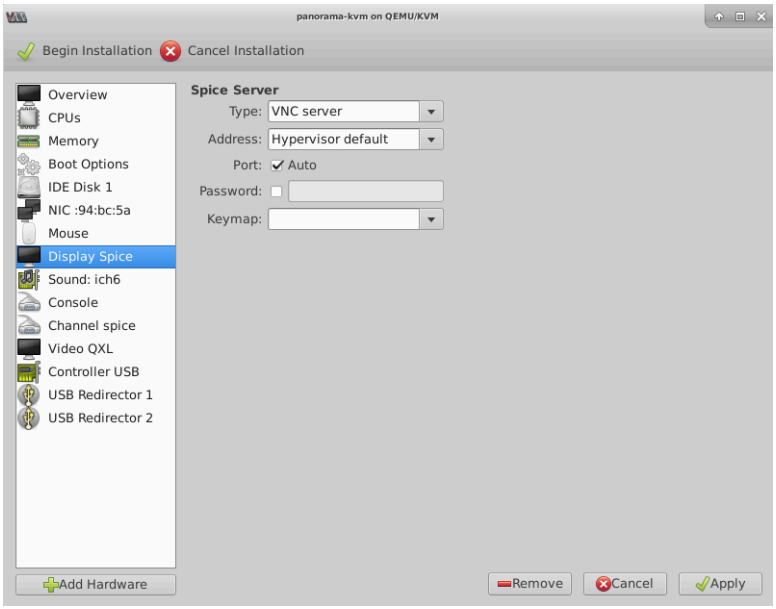
STEP 6 | Configure the virtual machine console display to use the VNC server to interact with the virtual machine.

1. Select **Display Spice**.

   *Continue to the next step if **Display VNC** is listed in the Hardware list because the virtual machine is already configured to use the VNC server for the display.*

2. In the **Type** drop-down, select **VNC server**.
3. Click **Apply**.



STEP 7 | (Optional) Add additional storage for log collection. Repeat this step as needed to add additional virtual logging disks.

If you intend to use the Panorama virtual appliance in Panorama mode or as a Dedicated Log Collector, add the virtual logging disks during the initial deployment. By default, the Panorama virtual appliance is in Panorama mode for the initial deployment when you meet the Panorama mode resource requirements and have added at least one virtual logging disk. Otherwise, the Panorama virtual appliance defaults to Management Only mode. Change the Panorama virtual appliance to Management Only mode if you just want to manage devices and Dedicated Log Collectors, and to not collect logs locally.

The Panorama virtual appliance on KVM only supports 2TB logging disks, and in total supports up to 24TB of log storage. You are unable to add a logging disk smaller than 2TB, or a logging

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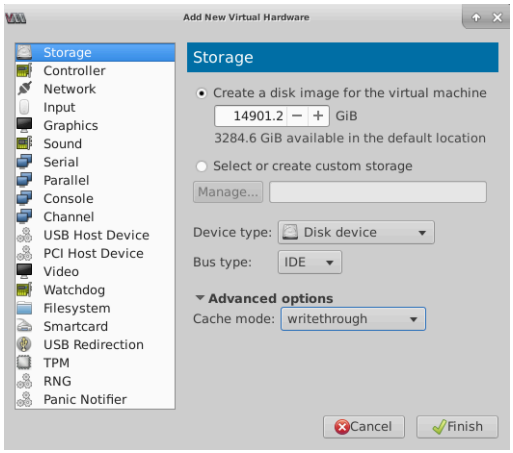
disk with a size not divisible by the 2TB logging disk requirement. The Panorama virtual appliance partitions logging disks larger than 2TB into 2TB partitions.

1. **Add Hardware**.
2. Configure the new **Storage** disk:
   1. **Create a disk image for a virtual machine** and configure the virtual disk storage capacity to 14901.2 GiB (this is equivalent to 2TB).

      The Virtual Machine Manager may use GiB (gibibyte) to allocate memory depending on the version you are running. If GiB is used, be sure to correctly convert the required storage capacity to avoid under provisioning the virtual logging disk and sending the Panorama virtual appliance into maintenance mode.
   2. Set the **Device type** to **Disk** device.
   3. Set the **Bus type** to **VirtIO** or **IDE**, depending on your configuration.
   4. Go to **Advanced options** and set **Cache mode** to **writethrough**.
3. Click **Finish**.



STEP 8 | **Begin Installation** (Begin Installation). The Panorama virtual appliances takes approximately 10 minutes to boot.

STEP 9 | Open a connection to the console of the Panorama virtual appliance.

You are prompted to log in to the firewall using the default username and password: **`admin`/`admin`**.

STEP 10 | Configure a new administrative password for the Panorama virtual appliance.

You must configure a unique administrative password before you can access the web interface or CLI of the Panorama virtual appliance. The new password must be a minimum of eight characters and include a minimum of one lowercase character, one uppercase character, and one number or special character.

When you first log in to the Panorama CLI, you are prompted to enter the **Old Password** and the **New Password** for the `admin` user before you can continue.

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STEP 11 | Configure the network access settings for the management interface.

1. Enter configuration mode using the following command:

   ```
   admin> configure
   ```

2. Use the following commands to configure and enable access to the management interface:

   ```
   admin# set deviceconfig system type static
   admin# set deviceconfig system ip-address <Panorama-IP>
    netmask <netmask> default-gateway <gateway-IP> dns-setting
    servers primary <DNS-IP>
   ```

   where <*Panorama-IP*> is the IP address you want to assign to the management interface, <*netmask*> is the subnet mask, <*gateway-IP*> is the IP address of the network gateway, and <*DNS-IP*> is the IP address of the DNS server.

   ```
   admin# commit
   ```

STEP 12 | Register the Panorama virtual appliance and activate the device management license and support licenses.

1. (VM Flex Licensing Only) Provisioning the Panorama Virtual Appliance Serial Number.

   When leveraging VM Flex licensing, this step is required to generate the Panorama virtual appliance serial number needed to register the Panorama virtual appliance with the Palo Alto Networks Customer Support Portal (CSP).

2. Register Panorama.

   You must register the Panorama virtual appliance using the serial number provided by Palo Alto Networks in the order fulfillment email.

   This step is not required when leveraging VM Flex licensing as the serial number is automatically registered with the CSP when generated.

3. Activate the firewall management license.

   - Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is Internet-connected.
   - Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is not Internet-connected.

4. Activate a Panorama Support License.

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STEP 13 | Complete configuring the Panorama virtual appliance for your deployment needs.

- For Panorama in Log Collector Mode.
  1. Add a Virtual Disk to Panorama on KVM as needed.

     Adding at least one virtual logging disk is required before you can change the Panorama virtual appliance to Log Collector mode.
  2. Begin at Step 6 to switch to Log Collector mode.

     *Enter the Public IP address of the Dedicated Log Collector when you add the Log Collector as a managed collector to the Panorama management server. You cannot specify the **IP Address**, **Netmask**, or **Gateway**.*
- For Panorama in Panorama mode.
  1. Add a Virtual Disk to Panorama on KVM.

     Adding at least one virtual logging disk is required before you can change the Panorama virtual appliance to Panorama mode.
  2. Set up a Panorama Virtual Appliance in Panorama Mode.
  3. Configure a Managed Collector.
- For Panorama in Management Only mode.
  1. Set up a Panorama Virtual Appliance in Management Only Mode.
  2. Configure a Managed Collector to add a Dedicated Log Collector to the Panorama virtual appliance.

     Management Only mode does not support local log collection, and requires a Dedicated Log Collector to store managed device logs.

Install Panorama on Hyper-V

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Device management licence • Support license • Panorama Virtual Appliance license (serial number) • Customer Support Portal (CSP) Account

You can now deploy Panorama™ and a Dedicated Log Collector on Hyper-V. Panorama deployed on Hyper-V is Bring Your Own License (BYOL), supports all deployment modes (Panorama, Log Collector, and Management Only), and shares the same processes and functionality as the M-Series hardware appliances. For more information on Panorama modes, see Panorama Models. Panorama virtual appliance and virtual Dedicated Log Collector on Hyper-V is available only on PAN-OS 8.1.3 and later releases.

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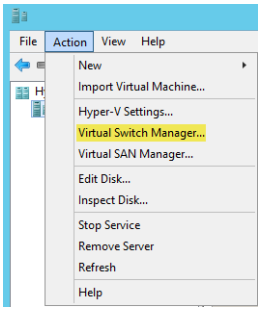
STEP 1 | Download the Panorama 11.1 base image VHDX file.

1. Log in to the Palo Alto Networks Support Portal.
2. Select **Updates** > **Software Updates** and filter by **Panorama Base Images** to download the VHDX file (`Panorama-HPV-11.1.0.vhdx`).

STEP 2 | Set up any vSwitch(es) that you will need. For more information, review the Virtual Switch Types for more information.

1. From Hyper-V Manager, select the host and select **Action** > **Virtual Switch Manager** to open the Virtual Switch Manager window.



2. Under **Create virtual switch**, select the type of vSwitch to create and click **Create Virtual Switch**.

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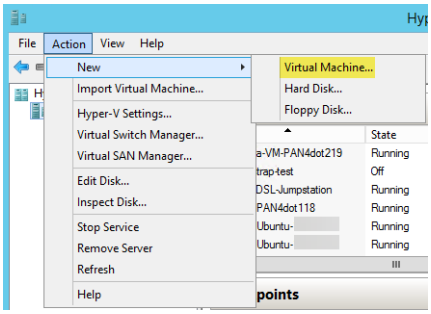
STEP 3 | Install the Panorama virtual appliance.

1. On the Hyper-V Manager, select the host and select **Action** > **New** > **Virtual Machine.** Configure the following settings in the New Virtual Machine Wizard:



   1. Choose a **Name** and **Location** for the Panorama virtual appliance. The Panorama virtual appliance stores the VHDX file at the specified location.
   2. Choose **Generation 1**. This is the default option and the only version supported.
   3. For **Startup Memory**, assign the memory based on the intended system mode. See the Setup Prerequisites for the Panorama Virtual Appliance for the memory requirements for each mode.

      Do not enable dynamic memory; the Panorama virtual appliance requires static memory allocation.
   4. Configure **Networking**. Select an external vSwitch to connect the management interface on the firewall.
   5. To connect the **Virtual Hard Disk**, select **Use an existing virtual hard disk** and browse to the VHDX file you downloaded earlier.
   6. Review the summary and click **Finish**.

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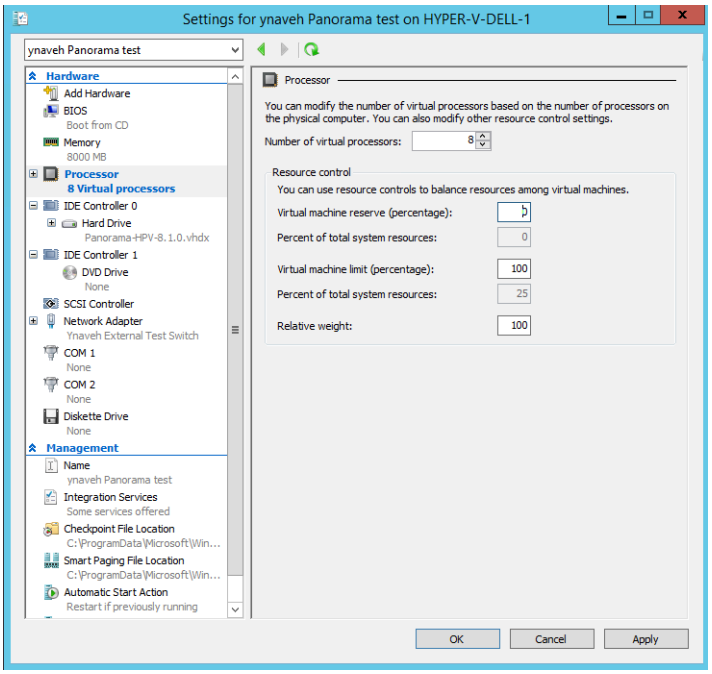
STEP 4 | Allocate the Panorama virtual appliance CPU cores.

Review the Setup Prerequisites for the Panorama Virtual Appliance for minimum resource requirements.

If you plan to use the Panorama virtual appliance as a Dedicated Log Collector, ensure that you configure the appliance with the required resources during initial deployment. The Panorama virtual appliance does not remain in Log Collector mode if you resize the virtual machine after you deploy it, and this results in a loss of log data.

1. In the **Hardware** list, select **Processor**.
2. Edit the currently allocated **Number of virtual processors**.



STEP 5 | Connect at least one network adapter for the dataplane interface on the firewall. Repeat this to create additional network interfaces on the Panorama virtual appliance.

1. Select **Settings** > **Hardware** > **Add Hardware** and select the **Hardware type** for your network adapter.

   Legacy Network Adapter and SR-IOV are not supported. If selected, the VM-Series firewall will boot into maintenance mode.

2. Click **OK**.

STEP 6 | (Optional) Add additional storage for log collection. Repeat this step as needed to add additional virtual logging disks. If you want to deploy the Panorama virtual appliance in Management Only mode, continue to Step 6.

If you intend to use the Panorama virtual appliance in Panorama mode or as a Dedicated Log Collector, add the virtual logging disks during the initial deployment. By default, the Panorama virtual appliance is in Panorama mode for the initial deployment when you meet the Panorama mode resource requirements and have added at least one virtual logging disk. Otherwise, the Panorama virtual appliance defaults to Management Only mode. Change the Panorama virtual

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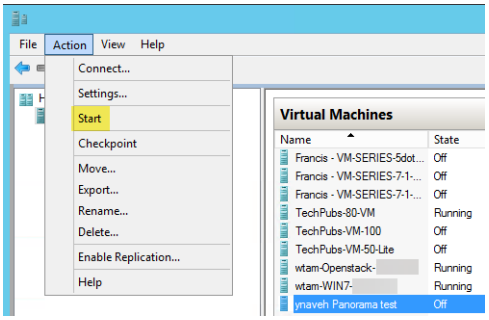
appliance to Management Only mode if you just want to manage devices and Dedicated Log Collectors, and to not collect logs locally.

The Panorama virtual appliance on Hyper-V only supports 2TB logging disks, and in total supports up to 24TB of log storage. You are unable to add a logging disk smaller than 2TB, or a logging disk with a size not divisible by the 2TB logging disk requirement. The Panorama virtual appliance partitions logging disks larger than 2TB into 2TB partitions.

1. On the Hyper-V Manager, select the host and select **Action** > **New** > **Hard Disk.**
2. If you see the Before You Begin prompt, click **Next** to begin adding the virtual logging disk.
3. For the Disk Format, select **VHDX**. Click **Next**. to continue
4. For the Disk Type, select **Fixed Size** or **Dynamically Expanding** based on your needs. Click **Next** to continue.
5. Specify the **Name** and **Location** for the virtual logging disk file. Click **Next** to continue.
6. To configure the disk, select **Create a new virtual hard disk** and enter the disk size. Click **Next** to continue.
7. Review the Summary and **Finish** adding the virtual hard logging disk.

STEP 7 | Power on the Panorama virtual appliance.

1. Select the Panorama virtual appliance instance from the list of **Virtual Machines**.
2. Select **Action** > **Start** to power on the Panorama virtual appliance.



STEP 8 | Connect to the Panorama virtual appliance console from the Hyper-V Manager.

1. In the **Virtual Machines** list, select the Panorama virtual appliance.
2. Select **Actions** > **Connect** and enter the username and password to log in (default is admin for both).

STEP 9 | Configure a new administrative password for the Panorama virtual appliance.

You must configure a unique administrative password before you can access the web interface or CLI of the Panorama virtual appliance. The new password must be a minimum of eight characters and include a minimum of one lowercase character, one uppercase character, and one number or special character.

When you first log in to the Panorama CLI, you are prompted to enter the **Old Password** and the **New Password** for the `admin` user before you can continue.

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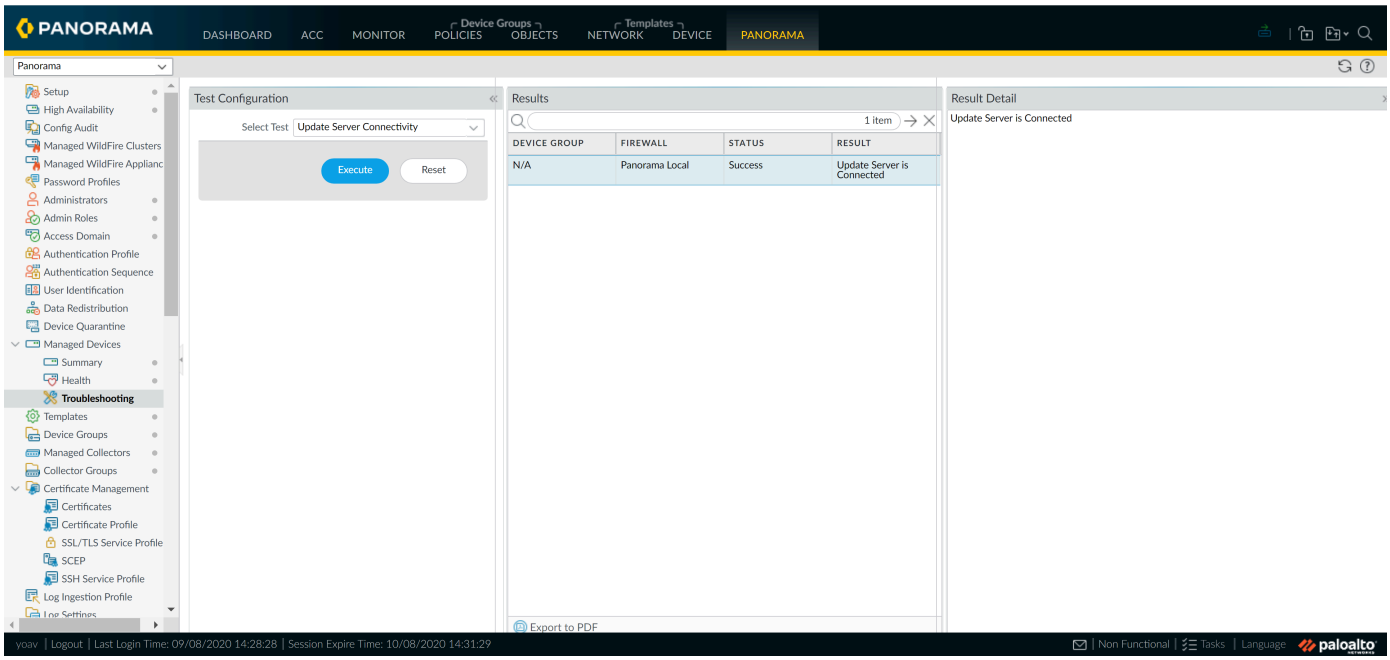
STEP 10 | Configure the IP address of the management interface.

1. Enter the following commands, where *<Panorama-IP>* is the IP address you want to assign to the Panorama management interface, *<netmask>* is the subnet mask, *<gateway-IP>* is the IP address of the network gateway, and *<DNS-IP>* is the IP address of the DNS server:

```
admin> configure
admin# set deviceconfig system ip-address <Panorama-IP>
 netmask <netmask> default-gateway <gateway-IP> dns-setting
 servers primary <DNS-IP>
admin# commit
admin# exit
```

2. Troubleshoot Connectivity to Network Resources to verify network access to external services required for firewall management, such as the default gateway, DNS server, and the Palo Alto Networks Update Server, as shown in the following example:



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STEP 11 | Register the Panorama virtual appliance and activate the device management license and support licenses.

1. (VM Flex Licensing Only) Provisioning the Panorama Virtual Appliance Serial Number.

   When leveraging VM Flex licensing, this step is required to generate the Panorama virtual appliance serial number needed to register the Panorama virtual appliance with the Palo Alto Networks Customer Support Portal (CSP).
2. Register Panorama.

   You must register the Panorama virtual appliance using the serial number provided by Palo Alto Networks in the order fulfillment email.

   This step is not required when leveraging VM Flex licensing as the serial number is automatically registered with the CSP when generated.
3. Activate the firewall management license.
   - Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is Internet-connected.
   - Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is not Internet-connected.
4. Activate a Panorama Support License.

STEP 12 | Complete configuring the Panorama virtual appliance for your deployment needs.

- For Panorama in Log Collector Mode.
  1. Add a Virtual Disk to Panorama on Hyper-V as needed.

     Adding at least one virtual logging disk is required before you can change the Panorama virtual appliance to Log Collector mode.
  2. Begin at Step 6 to switch to Log Collector mode.

     *Enter the Public IP address of the Dedicated Log Collector when you add the Log Collector as a managed collector to the Panorama management server. You cannot specify the **IP Address**, **Netmask**, or **Gateway**.*
- For Panorama in Panorama mode.
  1. Add a Virtual Disk to Panorama on Hyper-V.

     Adding at least one virtual logging disk is required before you can change the Panorama virtual appliance to Panorama mode.
  2. Set up a Panorama Virtual Appliance in Panorama Mode.
  3. Configure a Managed Collector.
- For Panorama in Management Only mode.
  1. Set up a Panorama Virtual Appliance in Management Only Mode.
  2. Configure a Managed Collector to add a Dedicated Log Collector to the Panorama virtual appliance.

     Management Only mode does not support local log collection, and requires a Dedicated Log Collector to store managed device logs.

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Set Up Panorama on Oracle Cloud Infrastructure (OCI)

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Device management licence • Support license • Panorama Virtual Appliance license (serial number) • Customer Support Portal (CSP) Account

Set up a Panorama™ virtual appliance on Oracle Cloud Infrastructure (OCI) to centrally managed the configuration of physical and VM-Series firewalls.

Upload the Panorama Virtual Appliance Image to OCI

Complete the following procedure to upload a Panorama qcow2 file for KVM and create a custom image that you need to launch the Panorama virtual appliance. Uploading and creating the image is required only once. You can use the same image for all subsequent deployments of the Panorama virtual appliance.

STEP 1 | Download the Panorama qcow2 file for KVM from the Palo Alto Networks Customer Support Portal (CSP).

1. Log in to the Palo Alto Networks CSP.
2. Select **Updates** > **Software Updates** and select **Panorama Base Images** from the software updates filter drop-down.
3. Download the latest version of the `Panorama-KVM` qcow2 image.

STEP 2 | Log in to the Oracle Cloud Infrastructure console.

STEP 3 | Create a storage bucket for the qcow2 file.

1. Select **Object Storage** > **Object Storage** and **Create Bucket**.
2. Enter a descriptive **Bucket Name**.
3. For the Storage Tier, select **Standard**.
4. **Create Bucket**.

STEP 4 | Upload the qcow2 image to the OCI storage bucket.

1. Click the storage bucket you created in the previous step to view the bucket details.
2. Click **Upload** and select the qcow2 image you downloaded from the Palo Alto Networks CSP.
3. **Upload** the image.

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STEP 5 | Create a pre-authenticated request for the qcow2 file.

This is required to create the object URL used in the creation of the custom image for the Panorama virtual appliance.

1. Select **Object Storage** > **Object Storage** and click the storage bucket you created in the previous step.
2. Select **Pre-Authenticated Requests** > **Create Pre-Authenticated Request**.
3. Enter a descriptive **Name** for your Pre-Authenticated Request.
4. Select **Object** and enter the qcow2 image name for the **Object Name**.
5. **Create Pre-Authenticated Request**.
6. For the Access Type, select **Permit object reads and writes**.
7. Enter an **Expiration** date and time.
8. **Create Pre-Authenticated Request**.
9. In the Pre-Authenticated Request Details, copy the Pre-Authenticated Request URL.

   The Pre-Authenticated Request URL is required to create the custom image and must be copied when displayed to you.

   The Pre-Authenticated Request URL is only displayed after the request is created and is not shown again.
10. **Close** the Pre-Authenticated Request Details after you copy the URL.

STEP 6 | Import the qcow2 file and create a custom Panorama virtual appliance image.

1. Select **Compute** > **Custom Images** and **Import Image**.
2. Enter a descriptive **Name** for your image.
3. Select **Import from an Object Storage URL** and paste the object storage URL.
4. For the Image type, select **QCOW2**.
5. For the Launch Mode, select **Paravirtualized Mode**.
6. **Import Image**.

Install Panorama on Oracle Cloud Infrastructure (OCI)

Create a Panorama™ virtual appliance instance on Oracle Cloud Infrastructure (OCI). An OCI instance supports a single NIC by default. You must manually upload a Panorama virtual appliance qcow2 image downloaded from the Palo Alto Networks Customer Supported Portal (CSP) to OCI to successfully install the Panorama virtual appliance on OCI.

A Panorama virtual appliance deployed on OCI is Bring Your Own License (BYOL), supports all deployment modes (Panorama, Log Collector, and Management Only), and shares the same processes and functionality as the M-Series hardware appliances. For more information on Panorama modes, see Panorama Models.

A machine running a Linux operating system is required successfully install the Panorama on OCI. To successfully install Panorama on OCI, you must generate a `.pub` key using OpenSSH. Additionally, you can only use a Linux machine to log into the Panorama CLI for the initial network configuration.

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Review the Setup Prerequisites for the Panorama Virtual Appliance to determine the virtual resources required for your needs. The virtual resources requirement for the Panorama virtual appliance is based on the total number of firewalls managed by the Panorama virtual appliance and the required Logs Per Second (LPS) for forwarding logs from your managed firewalls to your Log Collector.

Under-provisioning the Panorama virtual appliance will impact management performance. This includes the Panorama virtual appliance becoming slow or unresponsive depending on how under-provisioning the Panorama virtual appliance is.

STEP 1 | Log in to the Oracle Cloud Infrastructure console.

STEP 2 | Set up the Virtual Cloud Network (VCN) for your network needs.

Whether you launch the Panorama virtual appliance in an existing VCN or you create a new VCN, the Panorama virtual appliance must be able to receive traffic from other instances in the VCN and perform inbound and outbound communication between the VCN and the internet as needed.

Refer to the OCI VCN documentation for more information.

1. Configure a VCN or use an existing VCN.
2. Verify that the network and security components are appropriately defined.
   - Create an internet gateway to enable internet access to the subnet of your Panorama virtual appliance. Internet access is required to install software and content updates, activate licenses, and leverage Palo Alto Networks cloud services. Otherwise, you must manually install updates and activate licenses.

     If the Panorama virtual appliance instance is part of a private subnet, you can configure a NAT gateway to enable only outbound internet access for the subnet.
   - Create subnets. Subnets are segments of the IP address range assigned to the VCN in which you can launch OCI instances. It is recommended that the Panorama virtual

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appliance belong to the management subnet so that you can configure it to access the internet if needed.

- Add routes to the route table for a private subnet to ensure traffic can be routed across subnets in the VCN and from the internet if applicable.

  Ensure you create routes between subnets to allow communication between:

  - Panorama, managed firewalls, and Log Collectors.
  - (Optional) Panorama and the internet.
- Ensure that the following ingress security rules are allowed for the VCN to manage VCN traffic. The ingress traffic source for each rule is unique to your deployment topology.

  See Ports Used for Panorama for more information.

  - Allow SSH (port **22**) traffic to enable access to the Panorama CLI.
  - Allow HTTPS (port **443** and **28270**) traffic to enable access to the Panorama web interface.
  - Allow traffic on port **3978** to enable communication between Panorama, manage firewalls, and managed Log Collectors. This port is also used by Log Collectors to forward logs to Panorama.
  - Allow traffic on port **28443** to enable managed firewalls to get software and content updates from Panorama.

STEP 3 | Select **Compute** > **Instances** and **Create Instance**.

STEP 4 | Enter a descriptive **Name** for the Panorama virtual appliance image.

STEP 5 | Select the **Availability domain**.

STEP 6 | Select the Palo Alto Networks Panorama image.

*See Upload the Panorama Virtual Appliance Image to OCI to upload and maintain your own Panorama virtual appliance **Custom Image** on OCI.*

1. Under Image and shape, select **Change Image**.
2. For the Image Source, select **Partner Image**.

   If you maintain your own Panorama virtual appliance image, select **Custom Image** instead and select the Panorama virtual appliance image you uploaded to OCI.
3. Search for **`Palo Alto Networks Panorama`** and select (check) the image.

   *Skip this step if you selected **Custom Image** in the previous step.*

   PAN-OS 10.2.0 is the default PAN-OS version.
4. **Select Image**.

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STEP 7 | Configure the instance resources.

Refer to the Setup Prerequisites for the Panorama Virtual Appliance for more information for the minimum resources required based on your Panorama usage needs.

1. Under Image and shape, select **Change Shape**.
2. Select the shape with number of CPUs, amount of RAM, and number of interfaces you require.
3. **Select Shape**.

STEP 8 | Configure the instance Networking settings.

1. For the Network, **Select existing virtual cloud network** and select the VCN.
2. For the Subnet, **Select existing subnet** and select the subnet.

   It is recommended to deploy the Panorama virtual appliance instance in a management subnet to safely allow internet access if needed.
3. (Optional) For the Public IP Address, select **Assign a public IPv4 address** if you want to make the Panorama virtual appliance accessible from outside the VCN.

STEP 9 | Configure the Panorama virtual appliance instance boot volume.

1. For the Boot volume, **specify a custom boot volume size**.
2. For the Boot volume size, enter **81**.

STEP 10 | **Create** the Panorama virtual appliance image.

STEP 11 | Log in to the Panorama virtual appliance CLI from the OCI console.

1. Generate a SSH Key for Panorama on OCI.
2. In the OCI console, select **Instances** and select the Panorama virtual appliance instance.
3. Select **Console Connection** and **Create Console Connection**.
4. Select **Upload public key files (.pub)** and upload the public SSH key you generated to **Create Console Connection**.
5. In the Instance Details screen, expand the Console Connection options and **Copy Serial Connection for Linux/Mac**.
6. On your Linux machine, open a terminal and paste the serial connection.

STEP 12 | Configure a new administrative password for the Panorama virtual appliance.

You must configure a unique administrative password before you can access the web interface or CLI of the Panorama virtual appliance. The new password must be a minimum of eight characters and include a minimum of one lowercase character, one uppercase character, and one number or special character.

When you first log in to the Panorama CLI, you are prompted to enter the **Old Password** and the **New Password** for the `admin` user before you can continue.

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STEP 13 | Configure the system IP address settings for the Panorama virtual appliance.

1. Configure the initial network settings for the Panorama virtual appliance.

```
admin> configure
```

```
admin#  set deviceconfig system type static
```

```
admin#  set deviceconfig system ip-address <instance-private-
IP address> netmask <netmask> default-gateway <default-
gateway-IP>
```

```
admin#  set deviceconfig system dns-setting servers primary
 <primary-dns-IP>
```

```
admin#  set deviceconfig system dns-setting servers secondary
 <secondary-dns-IP>
```

```
admin# commit
```

2. Verify you can log in to the Panorama web interface.

   If you cannot log in to the Panorama web interface, review your route table and VCN security rules to ensure the correct routes and security rules are created.

STEP 14 | Register the Panorama virtual appliance and activate the device management license and support licenses.

1. (VM Flex Licensing Only) Provisioning the Panorama Virtual Appliance Serial Number.

   When leveraging VM Flex licensing, this step is required to generate the Panorama virtual appliance serial number needed to register the Panorama virtual appliance with the Palo Alto Networks Customer Support Portal (CSP).

2. Register Panorama.

   You must register the Panorama virtual appliance using the serial number provided by Palo Alto Networks in the order fulfillment email.

   This step is not required when leveraging VM Flex licensing as the serial number is automatically registered with the CSP when generated.

3. Activate the firewall management license.

   - Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is Internet-connected.
   - Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is not Internet-connected.

4. Activate a Panorama Support License.

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STEP 15 | Complete configuring the Panorama virtual appliance for your deployment needs.

- For Panorama in Log Collector Mode.
  1. Add a Virtual Disk to Panorama on Oracle Cloud Infrastructure (OCI) as needed.

     Adding at least one virtual logging disk is required before you can change the Panorama virtual appliance to Log Collector mode.
  2. Begin at Step 6 to switch to Log Collector mode.

     *Enter the Public IP address of the Dedicated Log Collector when you add the Log Collector as a managed collector to the Panorama management server. You cannot specify the **IP Address**, **Netmask**, or **Gateway**.*
- For Panorama in Panorama mode.
  1. Add a Virtual Disk to Panorama on Oracle Cloud Infrastructure (OCI).

     Adding at least one virtual logging disk is required before you can change the Panorama virtual appliance to Panorama mode.
  2. Set up a Panorama Virtual Appliance in Panorama Mode.
  3. Configure a Managed Collector.
- For Panorama in Management Only mode.
  1. Set up a Panorama Virtual Appliance in Management Only Mode.
  2. Configure a Managed Collector to add a Dedicated Log Collector to the Panorama virtual appliance.

     Management Only mode does not support local log collection, and requires a Dedicated Log Collector to store managed device logs.

Generate a SSH Key for Panorama on OCI

To connect to the Panorama™ virtual appliance installed on Oracle Cloud Infrastructure (OCI), you must generate a public and private SSH key on a Linux machine. You use the generated SSH key to log in to the Panorama CLI to set up a new administrative password and configure the Panorama network settings.

A Linux machine is required to generate the SSH key and access the Panorama CLI for the initial configuration. Generating a SSH from OCI or third-party applications such as PuTTygen is not supported.

STEP 1 | Open the terminal on your Linux machine.

STEP 2 | Navigate to the hidden `.ssh` directory.

```
admin:~$ cd ~/.ssh
```

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STEP 3 | Generate an SSH key in the `.ssh` directory.

```
admin:~/.ssh$ ssh-keygen
```

When prompted, save the key in the default `.ssh` directory. A password for the key is optional.

The default name for the private key is `id_rsa` and the default name for the public key is `id_rsa.pub`.

STEP 4 | Copy the public key from the `.ssh` directory to your home directory.

This step is required to upload the public key to OCI.

```
admin: ~/.ssh$ cp id_rsa.pub ~
```

Perform Initial Configuration of the Panorama Virtual Appliance

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Device management licence • Support license • Outbound Internet Access • Customer Support Portal (CSP) Account

Based on your Panorama model, use the Alibaba Cloud Console, AWS, Azure, GCP, or OCI web interface, KVM Virtual Machine Manager, Hyper-V Manager, VMware vSphere Client, or vCloud Air web console to set up network access to the Panorama virtual appliance. By default, the Panorama virtual appliance is deployed in Panorama mode. For unified reporting, consider using Greenwich Mean Time (GMT) or Coordinated Universal Time (UTC) as the uniform time zone across Panorama and all the managed firewalls and Log Collectors.

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STEP 1 | Gather the required information from your network administrator.

Collect the following information for the management (MGT) interface:

- IP address for the management (MGT) interface

  The default management interface IP address is 192.168.1.1. if you do not configure the management interface as described when you install the Panorama virtual appliance.

- Netmask
- Default gateway
- DNS server IP address

  To complete the configuration of the MGT interface, you must specify the IP address, netmask (for IPv4) or prefix length (for IPv6), and default gateway. If you omit settings (such as the default gateway), you can access Panorama only through the console port for future configuration changes. As a best practice, always commit a complete MGT interface configuration.

STEP 2 | Access the console of the Panorama virtual appliance.

Panorama uses the MGT interface for management traffic, high availability synchronization, log collection, and communication within Collector Groups.

Starting with PAN-OS 9.0.4, the default `admin` credentials are no longer supported. When you first install the Panorama virtual appliance, you are required to log in to the Panorama CLI to configure a unique `admin` password.

*If this is the first time you are logging in to the Panorama CLI, you are prompted to enter the **Old Password** and the **New Password** for the `admin` user before you can continue with the initial configuration of the Panorama virtual appliance.*

1. Access the console.

   On an ESXi server:

   1. Launch the VMware vSphere Client.
   2. Select the **Console** tab for the Panorama virtual appliance and press enter to access the login screen.

   On vCloud Air:

   1. Access the vCloud Air web console and select your **Virtual Private Cloud OnDemand** region.
   2. Select the **Virtual Machines** tab, right-click the Panorama virtual machine, and select **Open In Console**.

2. Enter your username and password to log in (default is admin for both).

   On Alibaba Cloud, AWS, Azure, GCP, KVM, Hyper-V, and OCI:

   - Log in to the Panorama CLI.

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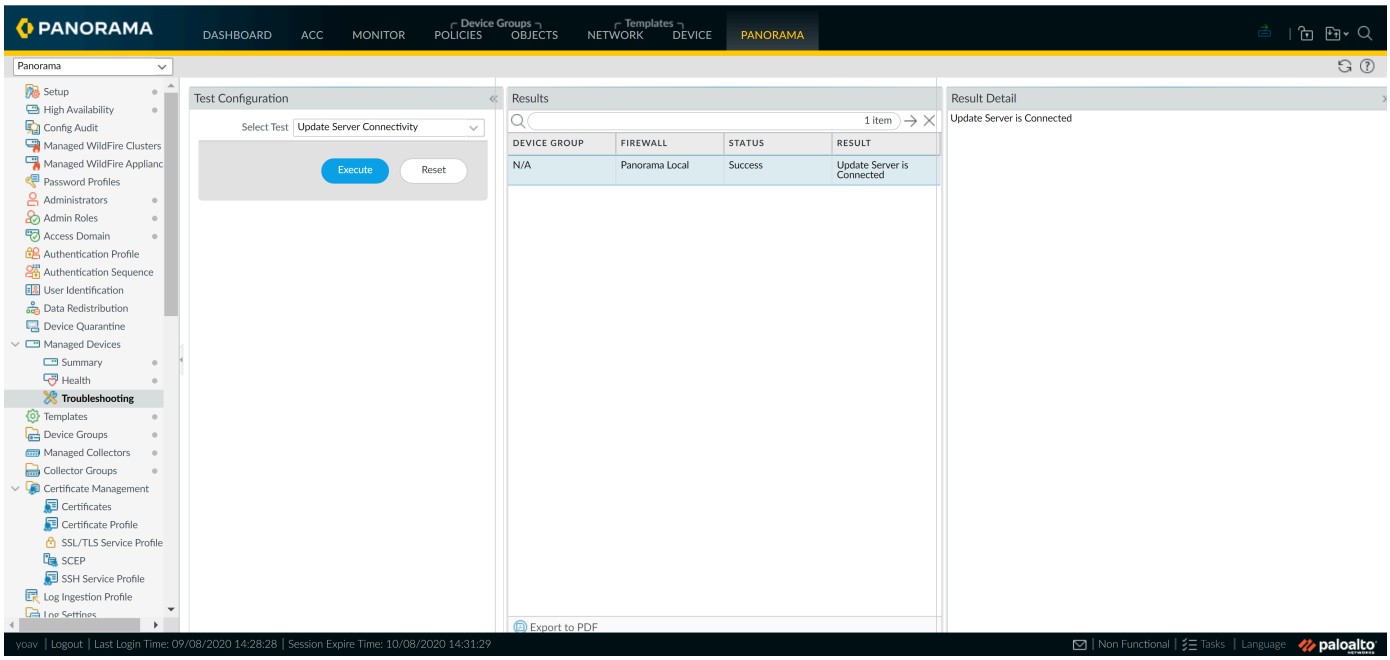
STEP 3 | Configure the network access settings for the MGT interface.

Panorama uses the MGT interface for management traffic, high availability synchronization, log collection, and communication within Collector Groups.

1. Enter the following commands, where *<Panorama-IP>* is the IP address you want to assign to the Panorama management interface, *<netmask>* is the subnet mask, *<gateway-IP>* is the IP address of the network gateway, and *<DNS-IP>* is the IP address of the DNS server:

```
> configure
# set deviceconfig system ip-address <Panorama-IP>
 netmask <netmask> default-gateway <gateway-IP> dns-setting
 servers primary <DNS-IP>
# commit
# exit
```

2. Troubleshoot Connectivity to Network Resources to verify network access to external services required for firewall management, such as the default gateway, DNS server, and the Palo Alto Networks Update Server, as shown in the following example:



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STEP 4 | Configure the general settings.

1. Using a secure connection (HTTPS) from a web browser, log in to the Panorama web interface using the IP address and password you assigned to the management interface (https://<IP address>).
2. Select **Panorama** > **Setup** > **Management** and edit the General Settings.
3. Enter a **Hostname** for the server and enter the network **Domain** name. The domain name is just a label; Panorama doesn't use it to join the domain.
4. Align the clock on Panorama and the managed firewalls to use the same **Time Zone**, for example GMT or UTC. If you plan to use the Strata Logging Service, you must configure NTP so that Panorama can stay in sync with the Strata Logging Service.

   Timestamps are recorded when Panorama receives the logs and the managed firewalls generate the logs. Aligning the time zones on Panorama and the firewalls ensures that the timestamps are synchronized and the process of querying logs and generating reports on Panorama is harmonious.
5. Enter the **Latitude** and **Longitude** to enable accurate placement of the Panorama management server on the world map.
6. Enter the **Serial Number** you received in the order fulfillment email.
7. Click **OK** to save your changes.

STEP 5 | (Optional) Modify the management interface settings.

To configure connectivity to Panorama using an IPv6 IP address, you must configure both an IPv4 and IPv6 to successfully configure Panorama using an IPv6 IP address. Panorama does not support configuring the management interface with only an IPv6 IP address.

1. Select **Panorama** > **Setup** > **Interfaces** and click **Management**.
2. If your firewalls connect to the Panorama management server using a public IP address that is translated to a private IP address (NAT), enter the public IP in the **Public IP Address** field, and the private IP in the **IP Address** field to push both addresses to your firewalls.
3. Select which Network Connectivity Services to allow on the interface (such as **SSH** access).

   *Don't select **Telnet** or **HTTP**. These services use plaintext and are less secure than the other services.*
4. Click **OK** to save your changes to the interface.

STEP 6 | Commit your configuration changes.

Select **Commit** > **Commit to Panorama** and **Commit** your changes.

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STEP 7 | Next steps...

1. If necessary, Expand Log Storage Capacity on the Panorama Virtual Appliance.
2. (Best Practice) Replace the default certificate that Panorama uses to secure HTTPS traffic over the management (MGT) interface.
3. Activate a Panorama Support License.
4. Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is Internet-connected.
5. Install Content and Software Updates for Panorama.
6. Configure Administrative Access to Panorama

Set Up The Panorama Virtual Appliance as a Log Collector

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Device management licence • Support license • Outbound Internet Access • Customer Support Portal (CSP) Account

If you want a dedicated virtual appliance for log collection, configure a Panorama virtual appliance on ESXi, Alibaba Cloud, AWS, AWS GovCloud, Azure, Google Cloud Platform, KVM, Hyper-V, or Oracle Cloud Infrastructure (OCI) in Log Collector mode. To do this, you first perform the initial configuration of the virtual appliance in Panorama mode, which includes licensing, installing software and content updates, and configuring the management (MGT) interface. You then switch the Panorama virtual appliance to Log Collector mode and complete the Log Collector configuration. Additionally, if you want to use dedicated M-Series Appliance Interfaces (recommended) instead of the MGT interface for log collection and Collector Group communication, you must first configure the interfaces for the Panorama management server, then configure them for the Log Collector, and then perform a Panorama commit followed by a Collector Group commit.

Perform the following steps to set up a new virtual appliance as a Log Collector or to convert an existing virtual appliance that was previously deployed as a Panorama management server.

Switching the virtual appliance from Panorama mode to Log Collector mode reboots the appliance, deletes the local Log Collector, deletes any existing log data, and deletes all configurations except the management access settings. Switching the mode does not delete licenses, software updates, or content updates.

STEP 1 | Set up the Panorama virtual appliance management server that will manage the Log Collector if you have not already done so.

Perform one of the following tasks:

- Set Up the Panorama Virtual Appliance
- Set Up the M-Series Appliance

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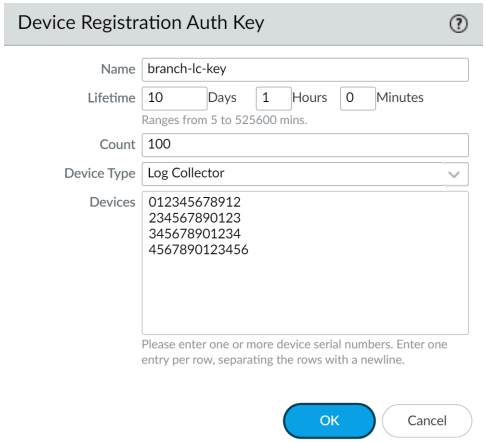
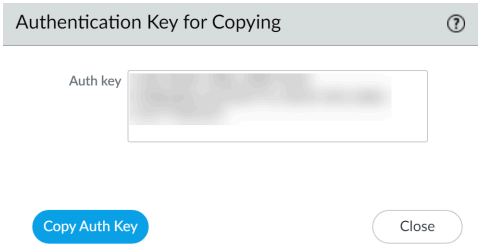
STEP 2 | On the Panorama management server, create a device registration authentication key to securely add the Dedicated Log Collector to Panorama management.

1. Log in to the Panorama Web Interface.
2. Select **Panorama** > **Device Registration Auth Key** and **Add** a new authentication key.
3. Configure the authentication key.
   - **Name**—Add a descriptive name for the authentication key.
   - **Lifetime**—Specify the key lifetime for how long you can use the authentication key to onboard new Log Collectors.
   - **Count**—Specify how many times you can use the authentication key to onboard new Log Collectors.
   - **Device Type**—Specify that this authentication key is used to authenticate only a **Log Collector**.

     *You can select **Any** to use the device registration authentication key to onboard firewalls, Log Collectors, and WildFire appliances.*
   - (Optional) **Devices**—Enter one or more device serial numbers to specify for which Log Collectors the authentication key is valid.
4. Click **OK**.



5. **Copy Auth Key** and **Close**.



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STEP 3 | Record the management IP addresses of the Panorama management server.

If you deployed Panorama in a high availability (HA) configuration, you need the IP address of each HA peer.

1. Log in to the web interface of the Panorama management server.
2. Record the **IP Address** of the solitary (non-HA) or active (HA) Panorama by selecting **Panorama** > **Setup** > **Management** and checking the Management Interface Settings.
3. For an HA deployment, record the **Peer HA IP Address** of the passive Panorama by selecting **Panorama** > **High Availability** and checking the Setup section.

STEP 4 | Set up the Panorama virtual appliance that will serve as a Dedicated Log Collector.

If you previously deployed this appliance as a Panorama management server, you can skip this step because the MGT interface is already configured and the licenses and updates are already installed.

The Panorama virtual appliance in Log Collector mode does not have a web interface for configuration tasks, only a CLI. Therefore, before changing the mode on the Panorama virtual appliance, use the web interface in Panorama mode to:

1. Set up the Panorama virtual appliance in one of the following supported hypervisors:
   - Install Panorama on an ESXi Server
   - Install Panorama on Alibaba Cloud
   - Install Panorama on AWS
   - Install Panorama on AWS GovCloud
   - Install Panorama on Azure
   - Install Panorama on Google Cloud Platform
   - Install Panorama on Hyper-V
   - Set Up Panorama on Oracle Cloud Infrastructure (OCI)
2. Perform Initial Configuration of the Panorama Virtual Appliance.
3. Register Panorama and Install Licenses.
4. Install Content and Software Updates for Panorama.

STEP 5 | (Panorama on Azure only) Modify the admin password.

The Dedicated Log Collector supports only the admin Administrator user in order to change in to Log Collector mode. Modify the admin password to allow you to log in using the admin Administrator user.

1. Log in to the Panorama Web Interface.
2. Select **Panorama** > **Administrators** and select **admin**.
3. Enter the **Password**, **Confirm Password** and click **OK**.
4. Select **Commit** > **Commit to Panorama** and **Commit** your changes.

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STEP 6 | (Panorama on AWS and Azure only) Delete all users, except for the admin user.

1. Log in to the Panorama Web Interface as admin.
2. Select **Panorama** > **Administrators**.
3. Select the existing Administrators, except admin, and **Delete**.
4. Select **Commit** > **Commit to Panorama** and **Commit** your changes.

STEP 7 | Log in to the Panorama CLI.

STEP 8 | Switch from Panorama mode to Log Collector mode.

1. Switch to Log Collector mode by entering the following command:

```
> request system system-mode logger
```

2. Enter **Y** to confirm the mode change. The virtual appliance reboots. If the reboot process terminates your terminal emulation software session, reconnect to the virtual appliance to see the Panorama login prompt.

   *If you see a **CMS Login** prompt, this means the Log Collector has not finished rebooting. Press Enter at the prompt without typing a username or password.*

3. Log back in to the CLI.
4. Verify that the switch to Log Collector mode succeeded:

```
> show system info | match system-mode
```

   If the mode change succeeded, the output displays:

```
system-mode: logger
```

STEP 9 | Enable connectivity between the Log Collector and Panorama management server.

Enter the following commands at the Log Collector CLI, where <*IPaddress1*> is for the MGT interface of the solitary (non-HA) or active (HA) Panorama and <*IPaddress2*> is for the MGT interface of the passive (HA) Panorama, if applicable.

```
> configure
# set deviceconfig system panorama-server <IPaddress1> panorama-
server-2 <IPaddress2>
# commit
# exit
```

STEP 10 | Add the device registration authentication key to the Dedicated Log Collector.

```
admin> request authkey set <auth-key>
```

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STEP 11 | Record the serial number of the Log Collector.

You need the serial number to add the Log Collector as a managed collector on the Panorama management server.

1. At the Log Collector CLI, enter the following command to display its serial number.

   ```
   > show system info | match serial
   ```

2. Record the serial number.

STEP 12 | Add the Log Collector as a managed collector to the Panorama management server.

1. Select **Panorama** > **Managed Collectors** and **Add** a managed collector.
2. In the **General** settings, enter the serial number (**Collector S/N**) you recorded for the Log Collector.
3. In the **Panorama Server IP** field, enter the IP address or FQDN of the solitary (non-HA) or active (HA) Panorama. For HA deployments, enter the IP address or FQDN of the passive Panorama peer in the **Panorama Server IP 2** field.

   These IP addresses must specify a Panorama interface that has **Device Management and Device Log Collection** services enabled. By default, these services are enabled only on the MGT interface. However, you might have enabled the services on other interfaces when you Set Up the M-Series Appliance that is a Panorama management server.
4. Select **Interfaces**, click **Management**, and enter the **Public IP Address**of the Dedicated Log Collector.
5. Click **OK** twice to save your changes to the Log Collector.
6. Select **Commit** > **Commit to Panorama** and **Commit** your changes to the Panorama configuration.
7. Verify that **Panorama** > **Managed Collectors** lists the Log Collector you added. The Connected column displays a check mark to indicate that the Log Collector is connected to Panorama. You might have to wait a few minutes before the page displays the updated connection status.

   At this point, the Configuration Status column displays Out of Sync and the Run Time Status column displays disconnected. The status will change to In Sync and connected after you configure a Collector Group.

STEP 13 | Enable the logging disks.

1. Select **Panorama** > **Managed Collectors** and edit the Log Collector.
2. Select **Disks** and **Add** each disk.
3. Click **OK** to save your changes.
4. Select **Commit** > **Commit to Panorama** and **Commit** your changes to the Panorama configuration.

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STEP 14 | (Recommended) Configure the **Ethernet1**, **Ethernet2**, **Ethernet3**, **Ethernet4**, and **Ethernet5** interfaces if the Panorama management server and Log Collector will use them for **Device Log Collection** (receiving logs from firewalls) and **Collector Group Communication**.

If you previously deployed the Log Collector as a Panorama management server and configured these interfaces, you must reconfigure them because switching to Log Collector mode would have deleted all configurations except the management access settings.

1. Configure each interface on the Panorama management server (other than the MGT interface) if you haven't already:
   1. Select **Panorama** > **Setup** > **Interfaces** and click the Interface Name.
   2. Select <*interface-name*> to enable the interface.
   3. Complete one or both of the following field sets based on the IP protocols of your network:
      - For ESXi
        - IPv4—**Public IP Address**, **IP Address**, **Netmask**, and **Default Gateway**

          IPv6—**IPv6 Address/Prefix Length** and **Default IPv6 Gateway**
      - For Alibaba Cloud, AWS, Azure, GCP, and OCI
        - **Public IP address**
   4. Select the Device Management Services that the interface supports:

      Device Management and Device Log Collection—You can assign one or more interfaces.

      Collector Group Communication—You can assign only one interface.

      Device Deployment (software and content updates)—You can assign only one interface.
   5. Click **OK** to save your changes.
2. Configure each interface on the Log Collector (other than the MGT interface):
   1. Select **Panorama** > **Managed Collectors** and edit the Log Collector.
   2. Select **Interfaces** and click the name of the interface.
   3. Select <*interface-name*> to enable the interface.
   4. Complete one or both of the following field sets based on the IP protocols of your network:
      - For ESXi
        - IPv4—**Public IP Address**, **IP Address**, **Netmask**, and **Default Gateway**

          IPv6—**IPv6 Address/Prefix Length** and **Default IPv6 Gateway**
      - For Alibaba Cloud, AWS, Azure, GCP, and OCI
        - **Public IP address**
   5. Select the Device Management Services that the interface supports:

      Device Log Collection—You can assign one or more interfaces.

      Collector Group Communication—You can assign only one interface.

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6. Click **OK** to save your changes to the interface.

3. Click **OK** to save your changes to the Log Collector.
4. Select **Commit** > **Commit to Panorama** and **Commit** your changes to the Panorama configuration.

STEP 15 | (Optional) If your deployment is using custom certificates for authentication between Panorama and managed devices, deploy the custom client device certificate. For more information, see Configure Authentication Using Custom Certificates.

1. Select **Panorama** > **Certificate Management** > **Certificate Profile** and choose the certificate profile from the drop-down or click **New Certificate Profile** to create one.
2. Select **Panorama** > **Managed Collectors** > **Add** > **Communication** for a Log Collector.
3. Select the **Secure Client Communication** check box.
4. Select the type of device certificate the Type drop-down.
   - If you are using a local device certificate, select the **Certificate** and **Certificate Profile** from the respective drop-downs.
   - If you are using SCEP as the device certificate, select the **SCEP Profile** and **Certificate Profile** from the respective drop-downs.
5. Click **OK**.

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STEP 16 | (Optional) Configure Secure Server Communication Sn a Log Collector. For more information, see Configure Authentication Using Custom Certificates.

1. Select **Panorama** > **Managed Collectors** > **Add** > **Communication**.
2. Verify that the **Custom Certificate Only** check box is not selected. This allows you to continue managing all devices while migrating to custom certificates.

   When the Custom Certificate Only check box is selected, the Log Collector does not authenticate and cannot receive logs from devices using predefined certificates.
3. Select the SSL/TLS service profile from the **SSL/TLS Service Profile** drop-down. This SSL/TLS service profile applies to all SSL connections between the Log Collector and devices sending it logs.
4. Select the certificate profile from the **Certificate Profile** drop-down.
5. Select **Authorize Client Based on Serial Number** to have the server check clients against the serial numbers of managed devices. The client certificate must have the special keyword $UDID set as the CN to authorize based on serial numbers.
6. In **Disconnect Wait Time (min)**, enter the number of minutes Panorama should wait before breaking and reestablishing the connection with its managed devices. This field is blank by default and the range is 0 to 44,640 minutes.

   The disconnect wait time does not begin counting down until you commit the new configuration.
7. (Optional) Configure an authorization list.
   1. Click **Add** under Authorization List.
   2. Select the **Subject** or **Subject Alt Name** as the Identifier type.
   3. Enter an identifier of the selected type.
   4. Click **OK**.
   5. Select **Check Authorization List** to enforce the authorization list.
8. Click **OK**.
9. Select **Commit** > **Commit to Panorama**.

STEP 17 | Assign the Log Collector to a Collector Group.

1. Configure a Collector Group. You must perform a Panorama commit and then a Collector Group commit to synchronize the Log Collector configuration with Panorama and to put

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the Eth1, Eth2, Eth3, Eth4, and Eth5 interfaces (if you configured them) in an operational state on the Log Collector.

In any single Collector Group, all the Log Collectors must run on the same Panorama model: all M-700 appliances, all M-600 appliances, all M-500 appliances, all M-300 appliances, all M-200 appliances, or all Panorama virtual appliances.

*As a best practice, **Enable log redundancy across collectors** if you add multiple Log Collectors to a single Collector group. This option requires each Log Collector to have the same number of logging disks.*

2. Select **Panorama** > **Managed Collectors** to verify that the Log Collector configuration is synchronized with Panorama.

   The Configuration Status column should display In Sync and the Run Time Status column should display connected.

3. Access the Log Collector CLI and enter the following command to verify that its interfaces are operational:

   ```
   > show interface all
   ```

   The output displays the `state` as `up` for each interface that is operational.

4. If the Collector Group has multiple Log Collectors, troubleshoot connectivity to network resources to verify they can communicate with each other by performing a Ping connectivity test for each interface that the Log Collectors use. For the `source` IP address, specify the interface of one of the Log Collectors. For the `host` IP address, specify the matching interface of another Log Collector in the same Collector Group.

STEP 18 | Next steps...

To enable the Log Collector to receive firewall logs:

1. Configure Log Forwarding to Panorama.
2. Verify Log Forwarding to Panorama.

Set Up the Panorama Virtual Appliance with Local Log Collector

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Device management licence • Support license • Outbound Internet Access • Customer Support Portal (CSP) Account

If the Panorama virtual appliance is in Legacy mode after you upgrade from a Panorama 8.0 or earlier release to a Panorama 8.1 (or later) release, switch to Panorama mode in order to create a

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local Log Collector, add multiple logging disks without losing existing logs. increase log storage up to 24TB, and enable faster report generation.

Once you change from Legacy mode to Panorama mode, Legacy mode will no longer be available.

After upgrading to Panorama 8.1, the first step is to increase the system resources on the virtual appliance to the minimum required for Panorama mode. Panorama reboots when you increase resources, so perform this procedure during a maintenance window. You must install a larger system disk (81GB), increase CPUs and memory based on the log storage capacity, and add a virtual logging disk. The new logging disk must have at least as much capacity as the appliance currently uses in Legacy mode and cannot be less than 2TB. Adding a virtual disk enables you to migrate existing logs to the Log Collector and enables the Log Collector to store new logs.

If Panorama is deployed in an HA configuration, perform the following steps on the secondary peer first and then on the primary peer.

STEP 1 | Determine which system resources you need to increase before the virtual appliance can operate in Panorama mode.

You must run the command specified in this step even if you have determined that Panorama already has adequate resources.

1. Access the Panorama CLI:
   1. Use terminal emulation software such as PuTTY to open an SSH session to the IP address that you specified for the Panorama MGT interface.
   2. Log in to the CLI when prompted.
2. Check the resources you must increase by running the following command:

```
> request system system-mode panorama
```

Enter **y** when prompted to continue. The output specifies the resources you must increase. For example:

```
Panorama mode not supported on current system disk of size
 52.0 GB.
Please attach a disk of size 81.0 GB, then use 'request system
 clone-system-disk' to migrate the current system disk
Please add a new virtual logging disk with more than 50.00 GB
 of storage capacity.
Not enough CPU cores: Found 4 cores, need 8 cores
```

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STEP 2 | Increase the CPUs and memory, and replace the system disk with a larger disk.

1. Access the VMware ESXi vSphere Client, select **Virtual Machines**, right-click the Panorama virtual appliance, and select **Power** > **Power Off**.
2. Right-click the Panorama virtual appliance and **Edit Settings**.
3. Select **Memory** and enter the new **Memory Size**.
4. Select **CPUs** and specify the number of CPUs (the **Number of virtual sockets** multiplied by the **Number of cores per socket**).
5. Add a virtual disk.

   You will use this disk to replace the existing system disk.

   1. In the **Hardware** settings, **Add** a disk, select **Hard Disk** as the hardware type, and click **Next**.
   2. **Create a new virtual disk** and click **Next**.
   3. Set the **Disk Size** to exactly 81GB and select the **Thick Provision Lazy Zeroed** disk format.
   4. Select **Specify a datastore or datastore structure** as the location, **Browse** to a datastore of at least 81GB, click **OK**, and click **Next**.
   5. Select a SCSI **Virtual Device Node** (you can use the default selection) and click **Next**.

      Panorama will fail to boot if you select a format other than SCSI.

   6. Verify that the settings are correct and then click **Finish** and **OK**.
6. Right-click the Panorama virtual appliance and select **Power** > **Power On**. Wait for Panorama to reboot before continuing.
7. Return to the Panorama CLI and copy the data from the original system disk to the new system disk:

   ```
   > request system clone-system-disk target sdb
   ```

   Enter **y** when prompted to continue.

   The copying process takes around 20 to 25 minutes, during which Panorama reboots. When the process finishes, the output tells you to shut down Panorama.
8. Return to the vSphere Client console, right-click the Panorama virtual appliance, and select **Power** > **Power Off**.
9. Right-click the Panorama virtual appliance and **Edit Settings**.
10. Select the original system disk, click **Remove**, select **Remove from virtual machine**, and click **OK**.
11. Right-click the Panorama virtual appliance and **Edit Settings**.
12. Select the new system disk, set the **Virtual Device Node** to **SCSI (0:0)**, and click **OK**.
13. Right-click the Panorama virtual appliance and select **Power** > **Power On**. Before proceeding, wait for Panorama to reboot on the new system disk (around 15 minutes).

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STEP 3 | Add a virtual logging disk.

This is the disk to which you will migrate existing logs.

1. In the VMware ESXi vSphere Client, right-click the Panorama virtual appliance and select **Power** > **Power Off**.
2. Right-click the Panorama virtual appliance and **Edit Settings**.
3. Repeat the steps to Add a virtual disk. Set the **Disk Size** to a multiple of 2TB based on the amount of log storage you need. The capacity must be at least as large as the existing virtual disk or NFS storage that Panorama currently uses for logs. The disk capacity must be a multiple of 2TB and can be up to 24TB. For example, if the existing disk has 5TB of log storage, you must add a new disk of at least 6TB.

   After you switch to Panorama mode, Panorama will automatically divide the new disk into 2TB partitions, each of which will function as a separate virtual disk.
4. Right-click the Panorama virtual appliance and select **Power** > **Power On**. Wait for Panorama to reboot before continuing.

STEP 4 | Switch from Legacy mode to Panorama mode.

After switching the mode, the appliance reboots again and then automatically creates a local Log Collector and Collector Group. The existing logs won't be available for querying or reporting until you migrate them later in this procedure.

1. Return to the Panorama CLI and run the following command.

   ```
   > request system system-mode panorama
   ```

   Enter **y** when prompted to continue. After rebooting, Panorama automatically creates a local Log Collector (named Panorama) and creates a Collector Group (named default) to contain it. Panorama also configures the virtual logging disk you added and divides it into separate 2TB disks. Wait for the process to finish and for Panorama to reboot (around five minutes) before continuing.
2. Log in to the Panorama web interface.
3. In the **Dashboard**, **General Information** settings, verify that the **Mode** is now **panorama**.

   In an HA deployment, the secondary peer is in a suspended state at this point because its mode (Panorama) does not match the mode on the primary peer (Legacy). You will un-

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suspend the secondary peer after switching the primary peer to Panorama mode later in this procedure.

4. Select **Panorama** > **Collector Groups**to verify that the **default** collector group has been created, and that the local Log Collector is part of the default collector group.
5. Push the configuration to the managed devices.
   - If there are no pending changes:
     1. Select **Commit** > **Push to Devices** and **Edit Selections**.
     2. Select **Collector Group** and make sure the **default** collector group is selected.
     3. Click **OK** and **Push**.
   - If you have pending changes:
     1. Select **Commit** > **Commit and Push** and **Edit Selections**.
     2. Verify that your **Device Group** devices and **Templates** are included.
     3. Select **Collector Group** and make sure the **default** collector group is selected.
     4. Click **OK** and **Commit and Push**.
6. Select **Panorama** > **Managed Collectors** and verify that the columns display the following information for the local Log Collector:
   - Collector Name—This defaults to the Panorama hostname. It should be listed under the **default** Collector Group.
   - Connected—Check mark
   - Configuration Status—In sync
   - Run Time Status—connected

STEP 5 | (HA only) Switch the primary Panorama from Legacy mode to Panorama mode.

This step triggers failover.

1. Repeat Step 1 through Step 4 on the primary Panorama.

   Wait for the primary Panorama to reboot and return to an active HA state. If preemption is not enabled, you must manually fail back: select **Panorama** > **High Availability** and, in the Operational Commands section, **Make local Panorama functional**.
2. On the primary Panorama, select **Dashboard** and, in the High Availability section, **Sync to peer**, click **Yes**, and wait for the **Running Config** to display **Synchronized** status.
3. On the secondary Panorama, select **Panorama** > **High Availability** and, in the Operational Commands section, **Make local Panorama functional**.

   This step is necessary to bring the secondary Panorama out of its suspended HA state.

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STEP 6 | Migrate existing logs to the new virtual logging disks.

If you deployed Panorama in an HA configuration, perform this only on the primary peer.

Palo Alto Networks recommends migrating existing logs to the new virtual logging disks during your maintenance window. The log migration requires a large number of the Panorama virtual appliance CPU cores to execute and impacts Panorama operational performance.

1. Return to the Panorama CLI.
2. Start the log migration:

   ```
   > request logdb migrate vm start
   ```

   The process duration varies by the volume of log data you are migrating. To check the status of the migration, run the following command:

   ```
   > request logdb migrate vm status
   ```

   When the migration finishes, the output displays: `migrationhas been done`.
3. Verify that the existing logs are available.
   1. Log in to the Panorama web interface.
   2. Select **Panorama** > **Monitor**, select a log type that you know matches some existing logs (for example, **Panorama** > **Monitor** > **System**), and verify that the logs display.

STEP 7 | Next steps...

Configure log forwarding to Panorama so that the Log Collector receives new logs from firewalls.

Set up a Panorama Virtual Appliance in Panorama Mode

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Device management licence • Support license • Outbound Internet Access • Customer Support Portal (CSP) Account

Panorama mode allows the Panorama™ virtual appliance to operate as a Panorama management server with local log collection capabilities. By default, the Panorama virtual appliance is deployed in Panorama mode when at least one virtual logging disk is attached to a Panorama virtual appliance.

STEP 1 | Log in to the Panorama CLI.

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STEP 2 | Switch to Panorama mode.

1. Change to Panorama mode:

```
> request system system-mode panorama
```

2. Enter **Y** to confirm the mode change. The Panorama virtual appliance reboots. If the reboot process terminates your terminal emulation software session, reconnect to the Panorama virtual appliance to see the Panorama login prompt.

   If you see a `CMS Login` prompt, this means the Panorama virtual appliance has not finished rebooting. Press Enter at the prompt without typing a username or password.

STEP 3 | Verify that the switch to Panorama mode succeeded.

1. Log back in to the CLI.
2. Verify that the switch to Panorama mode succeeded:

```
> show system info | match system-mode
```

   If the mode change succeeded, the output displays:

```
> system mode:panorama
```

Set up a Panorama Virtual Appliance in Management Only Mode

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Device management licence • Support license • Outbound Internet Access • Customer Support Portal (CSP) Account

Management Only mode allows the Panorama virtual appliance to operate strictly as a Panorama management server without local log collection capabilities. By default, the Panorama virtual appliance is in Panorama mode for the initial deployment. It is recommended to change the Panorama virtual appliance to Management Only immediately after the initial deployment because changing to Management Only mode requires that there are no logs being forwarded to the Panorama management server because the Panorama virtual appliance in Management Only mode does not support log collection. After you change to Management Only mode, any existing log data stored on the Panorama virtual appliance becomes inaccessible, and the ACC and reporting features cannot query the logs stored on the Panorama virtual appliance.

*If you configured a local Log Collector, the local Log Collector still exists on Panorama when you change to Management Only mode despite having no log collection capabilities. Deleting the local Log Collector (**Panorama** > **Managed Collectors**) deletes the Eth1/1 interface configuration the local Log Collector uses by default. If you decide to delete the local Log Collector, you must* reconfigure the Eth1/1 interface.

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STEP 1 | Log in to the Panorama CLI.

STEP 2 | Switch to Management Only mode.

1. Change to Management Only mode:

```
> request system system-mode management-only
```

2. Enter **Y** to confirm the mode change. The Panorama virtual appliance reboots. If the reboot process terminates your terminal emulation software session, reconnect to the Panorama virtual appliance to see the Panorama login prompt.

   If you see a `CMS Login` prompt, this means the Panorama virtual appliance has not finished rebooting. Press Enter at the prompt without typing a username or password.

STEP 3 | Verify that the switch to Management Only mode succeeded.

1. Log back in to the CLI.
2. Verify that the switch to Management Only mode succeeded:

```
> show system info | match system-mode
```

   If the mode change succeeded, the output displays:

```
> system mode:management-only
```

Expand Log Storage Capacity on the Panorama Virtual Appliance

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Device management licence • Support license • Customer Support Portal (CSP) Account

As of September 1, 2025, the legacy mode is no longer supported on Panorama. For additional information, see the official end-of-life notice.

After you Perform Initial Configuration of the Panorama Virtual Appliance, the available log storage capacity and the options for expanding it depend on the virtual platform (VMware ESXi, vCloud Air, Alibaba Cloud, AWS, AWS GovCloud, Azure, Google Cloud Platform, KVM, Hyper-V, or OCI) and mode (Panorama or Log Collector mode): see Panorama Models for details.

To expand the log storage capacity on the Panorama virtual appliance, you must add additional logging disks. Expanding the log storage capacity of an existing logging disk is not supported, and Panorama does not recognize the additional storage capacity. For example; if you added a 2TB logging disk, and then expanded that existing logging disk to 4TB, Panorama continues to recognize the logging disk as having 2TB of storage capacity and ignores the additional 2TB of storage capacity.

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For additional log storage, you can also forward firewall logs to Dedicated Log Collectors (see Configure a Managed Collector) or Configure Log Forwarding from Panorama to External Destinations.

Before expanding log storage capacity on Panorama, Determine Panorama Log Storage Requirements.

- Add a Virtual Disk to Panorama on an ESXi Server
- Add a Virtual Disk to Panorama on vCloud Air
- Add a Virtual Disk to Panorama on Alibaba Cloud
- Add a Virtual Disk to Panorama on AWS
- Add a Virtual Disk to Panorama on Azure
- Add a Virtual Disk to Panorama on Google Cloud Platform
- Add a Virtual Disk to Panorama on KVM
- Add a Virtual Disk to Panorama on Hyper-V
- Add a Virtual Disk to Panorama on Oracle Cloud Infrastructure (OCI)

Add a Virtual Disk to Panorama on an ESXi Server

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Panorama virtual appliance operating in Panorama mode or Log Collector mode on VMware ESXi • Initial logging disk of exactly 2TB

To expand log storage capacity on the Panorama virtual appliance, you can add virtual logging disks. If the appliance is in Panorama mode, you can add 1 to 12 virtual logging disks of 2TB each or 1 24TB logging disk, for a maximum total of 24TB. Additionally, it is recommended to add logging disks with the same disk provisioning format to avoid any unexpected performance that may arise from having multiple disk with different provisioning formats.

If Panorama loses connectivity to the new virtual disk, Panorama might lose logs during the failure interval.

To allow for redundancy, use the virtual disk in a RAID configuration. RAID10 provides the best write performance for applications with high logging characteristics.

If necessary, you can Replace the Virtual Disk on an ESXi Server.

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STEP 1 | Add additional disks to Panorama

In all modes, the first logging disk on the Panorama VM must be at least 2TB in order to add additional disks. If the first logging disk is smaller than 2TB, you will be unable to add additional disk space.

1. Access the VMware vSphere Client and select **Virtual Machines**.
2. Right-click the Panorama virtual appliance and select **Power** > **Power off**.
3. Right-click the Panorama virtual appliance and select **Edit Settings**.
4. Click **Add** in the **Hardware** tab to launch the Add Hardware wizard.
5. Select **Hard Disk** as the hardware type and click **Next**.
6. **Create a new virtual disk** and click **Next**.
7. Set the **Disk Size**. If the Panorama virtual appliance is in Panorama mode, set the size to at least 2TB.

   In Panorama mode, you can add disk sizes larger than 2TB and Panorama will automatically create as many 2TB partitions as possible. For example, if disk sdc was 24TB, it will create 12 2TB partitions. These disks will be named sdc1-12.
8. Select the **Disk Provisioning** format and click **Next**.
9. **Specify a datastore or datastore structure**, **Browse** to a datastore with enough space for the specified **Disk Size**, click **OK**, and click **Next**.
10. Select a SCSI **Virtual Device Node** (you can use the default selection) and click **Next**.

    The selected node must be in SCSI format; Panorama will fail to boot if you select another format.
11. Verify that the settings are correct and then click **Finish** and **OK**.

    The new disk appears in the list of devices for the virtual appliance.
12. Repeat Step 4 through Step 11 to add additional disks to the Panorama virtual appliance if necessary.
13. Right click the Panorama virtual appliance and select **Power** > **Power On**. The virtual disk initializes for first-time use. The size of the new disk determines how long initialization takes.

STEP 2 | Configure each disk.

The following example uses the sdc virtual disk.

1. Log in to the Panorama CLI.
2. Enter the following command to view the disks on the Panorama virtual appliance:

   `show system disk details`

   The user will see the following response:

```
 Name
: sdb
 State : Present
 Size : 2048000 MB
```

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```
Status : Available
Reason : Admin enabled
Name : sdc
State : Present
Size : 2048000 MB
Status : Available
Reason : Admin disabled
```

3. Enter the following command and confirm the request when prompted for all disks with the `Reason : Admin disabled` response:

   `request system disk add sdc`

   *The **`request system disk add`** command is not available on a Panorama management server in Management Only mode because logging is not supported in this mode. If you do not see the command, Set up a Panorama Virtual Appliance in Panorama Mode to enable the logging disks. Once in Panorama mode, Log in to the Panorama CLI and continue to Step 4 to verify the disk addition.*

4. Enter the **`show system disk details`** command to verify the status of the disk addition. Continue to Step 3 when all newly added disk responses display `Reason : Admin enabled`.

STEP 3 | Make disks available for logging.

1. Log in to the Panorama web interface.
2. Select **Panorama** > **Managed Collectors** and edit the Log Collector.
3. Select **Disks** and Add each newly added disk.
4. Click **OK**.
5. Select **Commit** > **Commit to Panorama**.

   For Panorama in an Active/Passive high availability (HA) configuration, wait for HA sync to complete before continuing.

6. Select **Commit** > **Push to Devices** and push the changes to the Collector Group the Log Collector belongs to.

STEP 4 | Configure Panorama to receive logs.

This step is intended for new Panorama deployments in Panorama mode. If you are adding logging disks to an existing Panorama virtual appliance, continue to Step 5.

1. Configure a Managed Collector.
2. Configure a Collector Group.
3. Configure Log Forwarding to Panorama.

STEP 5 | Verify that the Panorama Log Storage capacity has been increased.

1. Log in to the Panorama web interface.
2. Select **Panorama** > **Collector Groups** and select the Collector Group that the Panorama virtual appliance belongs to.
3. Verify that the **Log Storage** capacity accurately displays the disk capacity.

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Add a Virtual Disk to Panorama on vCloud Air

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Panorama virtual appliance operating in Panorama mode or Log Collector mode on vCloud Air • Initial logging disk of exactly 2TB

You can add virtual logging disks to expand log storage capacity on the Panorama™ virtual appliance. If the appliance is in Panorama mode, you can add 1 to 12 virtual logging disks of 2TB each or 1 24TB logging disk, for a maximum total of 24TB.

If Panorama loses connectivity to the new virtual disk, Panorama might lose logs for the duration of the failure.

If necessary, you can Replace the Virtual Disk on an ESXi Server.

STEP 1 | Add additional disks to Panorama.

In all modes, the first logging disk on the Panorama VM must be at least 2TB to add additional disks. If the first logging disk is less than 2TB, you will be unable to add additional disk space.

1. Access the vCloud Air web console and select your **Virtual Private Cloud On Demand** region.
2. Select the Panorama virtual appliance in the **Virtual Machines** tab.
3. **Add another disk** (**Actions** > **Edit Resources**).
4. Set the **Storage** size. If the Panorama virtual appliance is in Panorama mode, set the size to at least 2TB.

   In Panorama mode, you can add disk sizes larger than 2TB and Panorama will automatically create as many 2TB partitions as possible. For example, if disk sdc was 24TB, Panorama will create 12 2TB partitions. These disks will be named sdc1 through sdc12.
5. Set the storage tier to **Standard** or **SSD-Accelerated**.
6. Repeat the previous steps to add additional disks to the Panorama virtual appliance as needed.
7. **Save** your changes.

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STEP 2 | Configure each disk.

The following example uses the sdc virtual disk.

1. Log in to the Panorama CLI.
2. Enter the following command to view the disks on the Panorama virtual appliance:
   `show system disk details`

   The user will see the following response:

```
 Name
: sdb
 State : Present
 Size : 2048000 MB
 Status : Available
 Reason : Admin enabled
 Name : sdc
 State : Present
 Size : 2048000 MB
 Status : Available
 Reason : Admin disabled
```

3. Enter the following command and confirm the request when prompted for all disks with the `Reason : Admin disabled` response:
   `request system disk add sdc`

   *The **`request system disk add`** command is not available on a Panorama management server in Management Only mode because logging is not supported in this mode. If you do not see the command, Set up a Panorama Virtual Appliance in Panorama Mode to enable the logging disks. Once in Panorama mode, Log in to the Panorama CLI and continue to Step 4 to verify the disk addition.*
4. Enter the **`show system disk details`** command to verify the status of the disk addition. Continue to the next step when all newly added disk responses display `Reason : Admin enabled`.

STEP 3 | Make disks available for logging.

1. Log in to the Panorama web interface.
2. Select **Panorama** > **Managed Collectors** and edit the Log Collector.
3. Select **Disks** and **Add** each new disk.
4. Click **OK**.
5. Select **Commit** > **Commit to Panorama**.

   For Panorama in an Active/Passive high availability (HA) configuration, wait for HA sync to complete before continuing.
6. Select **Commit** > **Push to Devices** and push the changes to the Collector Group the Log Collector belongs to.

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STEP 4 | Configure Panorama to receive logs.

This step is intended for new Panorama deployments in Panorama mode. If you are adding logging disks to an existing virtual Panorama appliance, continue to the next step.

1. Configure a Managed Collector.
2. Configure a Collector Group.
3. Configure Log Forwarding to Panorama.

STEP 5 | Verify that the Panorama Log Storage capacity has been increased.

1. Log in to the Panorama web interface.
2. Select **Panorama** > **Collector Groups** and select the Collector Group to which the virtual Panorama appliance belongs.
3. Verify that the **Log Storage** capacity accurately displays your new disk capacity.

Add a Virtual Disk to Panorama on Alibaba Cloud

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Panorama virtual appliance operating in Panorama mode or Log Collector mode on Alibaba • Initial logging disk of exactly 2TB

After you Install Panorama on Alibaba Cloud, add additional virtual logging disks to expand log storage capacity on the Panorama™ virtual appliance for logs generated by managed firewalls. You can add virtual disks to a local Log Collector for a Panorama virtual appliance in Panorama mode or for a Dedicated Log Collector. To add virtual disks, you must have access to the Alibaba Cloud Console, the Panorama command-line interface (CLI), and the Panorama web interface.

The Panorama virtual appliance on Alibaba Cloud supports only 2TB logging disks and, in total, supports up to 24TB of log storage. You cannot add a logging disk smaller than 2TB or a logging disk of a size that is not evenly divisible by 2TB because the Panorama virtual appliance partitions logging disks in to 2TB partitions. For example, if you attach a 4TB logging disk, Panorama will create two 2TB partitions. However, you cannot add a 5TB logging disk because the leftover 1TB is not supported as a partition.

STEP 1 | Log in to the Alibaba Cloud Console.

STEP 2 | Select **Elastic Compute Service** > **Instances & Images** > **Instances** and navigate to the Panorama virtual appliance instance.

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STEP 3 | Add a virtual logging disk to Panorama.

In all modes, the first logging disk on the Panorama VM must be at least 2TB in order to add additional disks. If the first logging disk is smaller than 2TB, you will be unable to add additional disk space.

1. In the Actions column, select **Manage**.
2. Select **Cloud Disk** and **Create Disk**.
3. Configure the virtual logging disk.
   - **Attach**—Select **Attach to ECS Instance**.
   - **ECS Instance**—Select the region and the Panorama virtual appliance instance.
   - **Storage**—Select type of virtual disk and enter the disk capacity.
   - (Optional) **Quantity**—Specify how many virtual disks to create. By default, 1 virtual disk is created. When creating multiple logging disks, be sure that the sum of all virtual disks does not exceed 24TB.
   - **Terms of Service**—Review the Alibaba Cloud Terms of Service and check after you have reviewed.
4. **Preview** the virtual disk creation.
5. **Create** the new virtual disk.

   A status window displays after you create the new virtual disk. After the virtual disk is successfully created, **Go to the Disk List** to confirm the disk is successfully created.

STEP 4 | Configure each disk.

The following example uses the sdc virtual disk.

1. Log in to the Panorama CLI.
2. Enter the following command to view the disks on the Panorama virtual appliance:
   `show system disk details`

   The user will see the following response:

```
 Name : sdb
State : Present
Size : 2048000 MB
Status : Available
```

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```
Reason : Admin disabled
```

3. Enter the following command and confirm the request when prompted for all disks with the `Reason : Admin disabled` response:
   `request system disk add sdc`

   > *The **`request system disk add`** command is not available on a Panorama management server in Management Only mode because logging is not supported in this mode. If you do not see the command, Set up a Panorama Virtual Appliance in Panorama Mode to enable the logging disks. Once in Panorama mode, log in to the Panorama CLI and continue to the next step to verify the disk addition.*

4. Enter the **`show system disk details`** command to verify the status of the disk addition. Continue to the next step when all newly added disk responses display `Reason : Admin enabled`.

STEP 5 | Make disks available for logging.

1. Log in to the Panorama web interface.
2. Edit a Log Collector (**Panorama** > **Managed Collectors**).
3. Select **Disks** and **Add** each newly added disk.
4. Click **OK**.
5. Select **Commit** > **Commit to Panorama**.

   > *For Panorama in an Active/Passive high availability (HA) configuration, wait for HA sync to complete before continuing.*

6. Select **Commit** > **Push to Devices** and push the changes to the Collector Group the Log Collector belongs to.

STEP 6 | (New Panorama deployments in Panorama mode only) Configure Panorama to receive logs.

If you are adding logging disks to an existing Panorama virtual appliance, skip to step 6.

1. Configure a Collector Group.
2. Configure Log Forwarding to Panorama

STEP 7 | Verify that the Panorama Log Storage capacity is increased.

1. Log in to the Panorama web interface.
2. Select the Collector Group to which the Panorama virtual appliance belongs (**Panorama** > **Collector Groups**).
3. Verify that the **Log Storage** capacity accurately displays the disk capacity.

Add a Virtual Disk to Panorama on AWS

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Panorama virtual appliance operating in Panorama mode or Log Collector mode on AWS

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Where Can I Use This?	What Do I Need?
	• Initial logging disk of exactly 2TB

After you Install Panorama on AWS or Install Panorama on AWS GovCloud, add virtual logging disks to the Panorama™ virtual appliance instance to provide storage for logs generated by managed firewalls. You can add virtual disks to a local log Collector for a Panorama virtual appliance in Panorama mode or for a Dedicated Log Collector. To add virtual disks, you must have access to the Amazon Web Service Console, the Panorama command-line interface (CLI), and the Panorama web interface.

The Panorama virtual appliance on AWS supports only 2TB logging disks and, in total, supports up to 24TB of log storage. You cannot add a logging disk smaller than 2TB or a logging disk of a size that is not evenly divisible by 2TB because the Panorama virtual appliance partitions logging disks in to 2TB partitions. For example, if you attach a 4TB logging disk, Panorama will create two 2TB partitions. However, you cannot add a 5TB logging disk because the leftover 1TB is not supported as a partition.

STEP 1 | Log in to AWS Web Service console and select the EC2 Dashboard.

- Amazon Web Service Console
- AWS GovCloud Web Service Console

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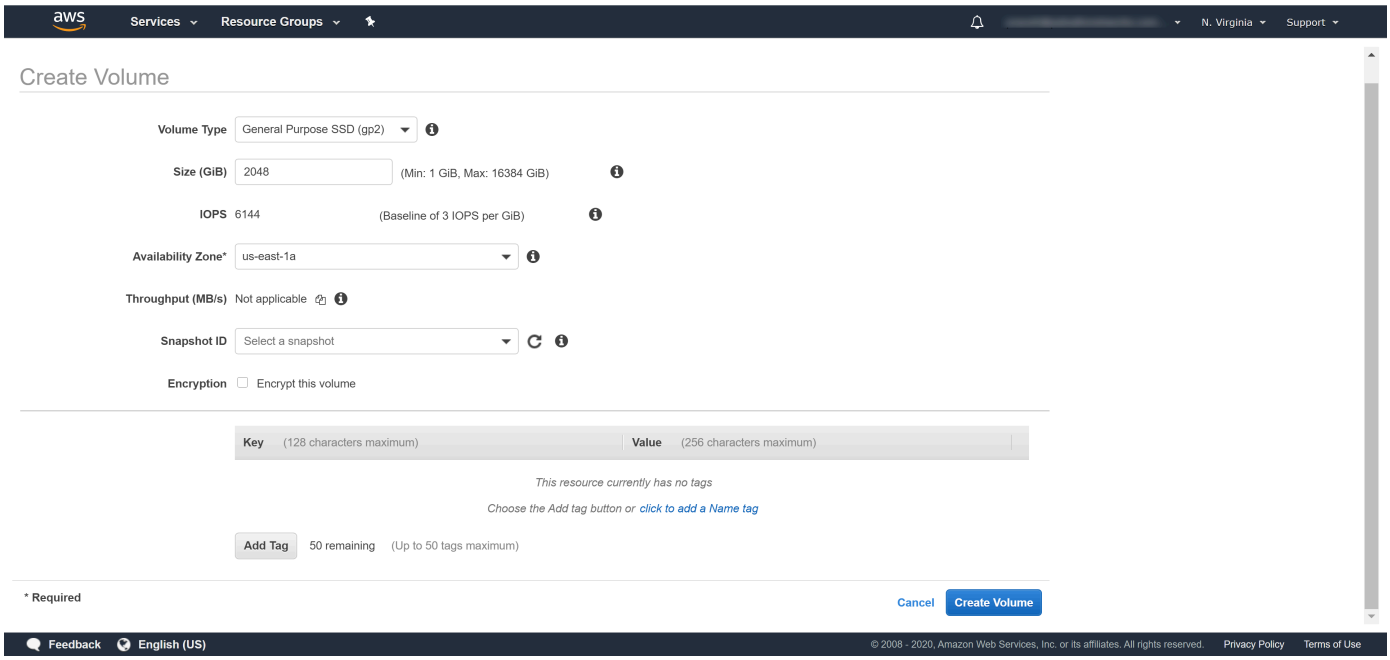
STEP 2 | Add a virtual logging disk to Panorama.

In all modes, the first logging disk on the Panorama VM must be at least 2TB in order to add additional disks. If the first logging disk is smaller than 2TB, you will be unable to add additional disk space.

1. On the EC2 Dashboard, select **Volumes** and **Create Volume**:
   - Select your preferred Volume Type. For general purpose use, select **General Purpose SSD (GP2)**.
   - Configure the **Size** of the volume as 2048 GiB.
   - Select the same Availability Zone that your Panorama virtual appliance instance is located in.
   - (Optional) Encrypt the volume.
   - (Optional) Add tags to your volume.
2. Click **Create Volume**.



3. In the Volumes page, select the volume you, and select **Actions** > **Attach Volume**.
4. Attach the Panorama virtual appliance Instance.
   1. Select your Panorama **Instance**.
   2. Specify the **Device name** for the logging disk volume you created.

STEP 3 | Configure each disk.

The following example uses the sdc virtual disk.

1. Log in to the Panorama CLI.
2. Enter the following command to view the disks on the Panorama virtual appliance:
   `show system disk details`

   The user will see the following response:

```
Name : nvme1n1
State : Present
Size : 2048000 MB
Status : Available
Reason : Admin enabled
Name : nvme2n1
State : Present
Size : 2048000 MB
Status : Available
Reason : Admin disabled
```

3. Enter the following command and confirm the request when prompted for all disks with the `Reason : Admin disabled` response:
   `request system disk add nvme2n1`

   *The **`request system disk add`** command is not available on a Panorama management server in Management Only mode because logging is not supported in this mode. If you do not see the command, Set up a Panorama Virtual Appliance in Panorama Mode to enable the logging disks. Once in Panorama mode, Log in to the Panorama CLI and continue to Step 4 to verify the disk addition.*
4. Enter the **`show system disk details`** command to verify the status of the disk addition. Continue to the next step when all newly added disk responses display `Reason : Admin enabled`.

STEP 4 | Make disks available for logging.

1. Log in to the Panorama web interface.
2. Add the Log Collector to Panorama (**Panorama** > **Managed Collectors**).
3. Select **Commit** > **Commit to Panorama**.
4. Edit the Log Collector (**Panorama** > **Managed Collectors**) and select **Disks**. Select the disks that do not match the local disk names.
5. **Add** each newly added disk and click **OK**.
6. Select **Commit** > **Commit to Panorama**.

   For Panorama in an Active/Passive high availability (HA) configuration, wait for HA sync to complete before continuing.
7. Select **Commit** > **Push to Devices** and push the changes to the Collector Group the Log Collector belongs to.

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STEP 5 | (New Panorama deployments in Panorama mode only) Configure Panorama to receive logs.

If you are adding logging disks to an existing Panorama virtual appliance, skip to step 6.

1. Configure a Collector Group.
2. Configure Log Forwarding to Panorama.

STEP 6 | Verify that the Panorama Log Storage capacity is increased.

1. Log in to the Panorama web interface.
2. Select the Collector Group to which the Panorama virtual appliance belongs (**Panorama** > **Collector Groups**).
3. Verify that the **Log Storage** capacity accurately displays the disk capacity.

Add a Virtual Disk to Panorama on Azure

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Panorama virtual appliance operating in Panorama mode or Log Collector mode on Azure • Initial logging disk of exactly 2TB

After you Install Panorama on Azure, add virtual logging disks to the Panorama™ virtual appliance instance to provide storage for logs generated by managed firewalls. You can add virtual disks to a local log Collector for a Panorama virtual appliance in Panorama mode or for a Dedicated Log Collector. To add virtual disks, you must have access to the Microsoft Azure portal, the Panorama command-line interface (CLI), and the Panorama web interface.

The Panorama virtual appliance on Azure supports only 2TB logging disks and, in total, supports up to 24TB of log storage. You cannot add a logging disk smaller than 2TB or a logging disk of a size that is not evenly divisible by 2TB because the Panorama virtual appliance partition logging disks in to 2TB partitions. For example, if you attach a 4TB logging disk, Panorama will create two 2TB partitions. However, you cannot add a 5TB logging disk because the leftover 1TB is not supported as a partition.

STEP 1 | Log in to the Microsoft Azure portal.

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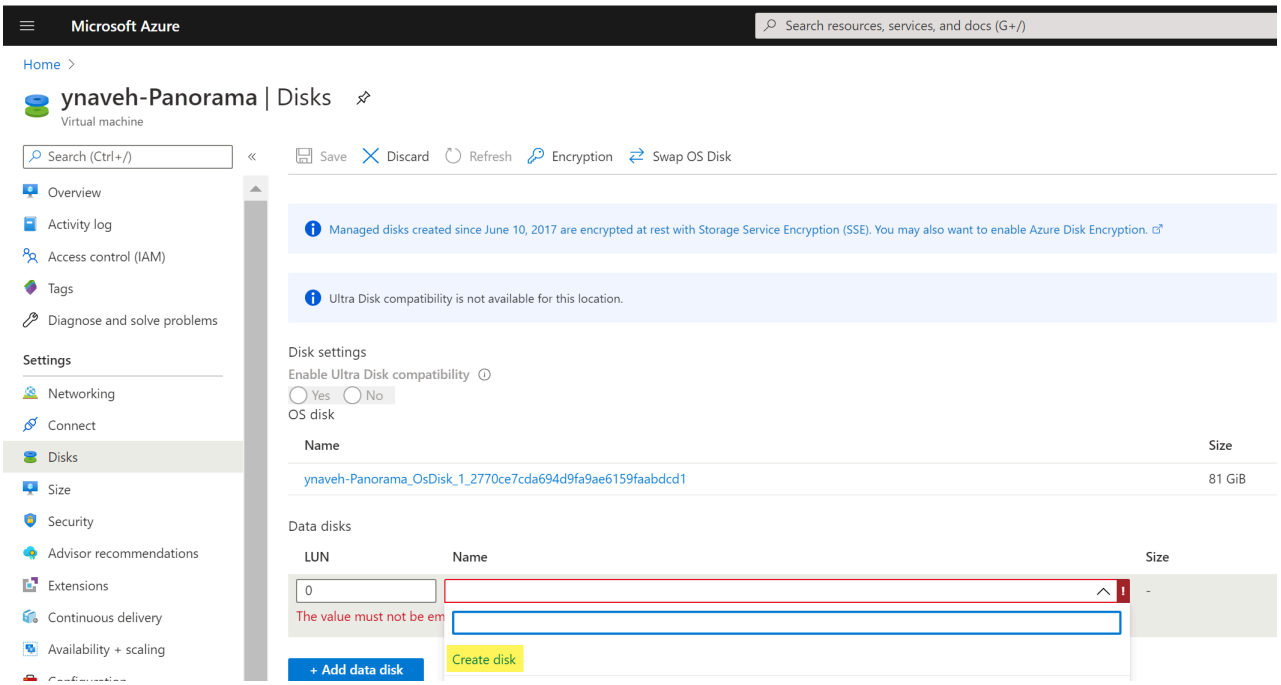
STEP 2 | Add a virtual logging disk to Panorama.

In all modes, the first logging disk on the Panorama VM must be at least 2TB in order to add additional disks. If the first logging disk is smaller than 2TB, you will be unable to add additional disk space.

1. In the Azure Dashboard, select the Panorama **Virtual Machines** to which you want to add a logging disk.
2. Select **Disks**.
3. **+Add data disk**.
4. In the drop-down for the new disk, **Create disk**.



5. Configure the logging disk.
   1. Enter the disk **Name**.
   2. Select the Resource group. If you **Create new** resource groups, enter the group name.
   3. Verify the **Account type** (this field is automatically populated).
   4. In the **Source type** drop-down, select **None**.
   5. Select **Change Size** and select a 2048 GiB logging disk.
   6. **Create** the new logging disk.

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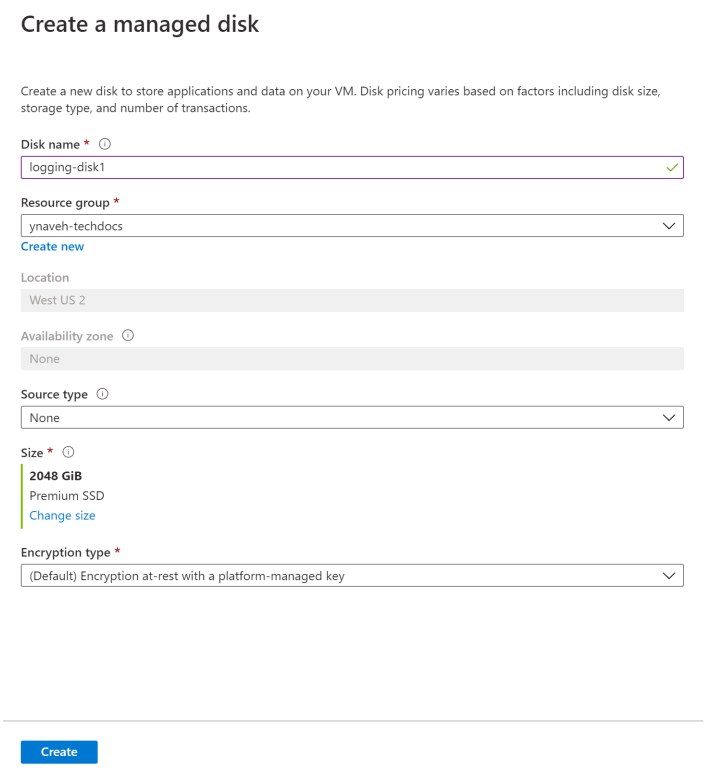
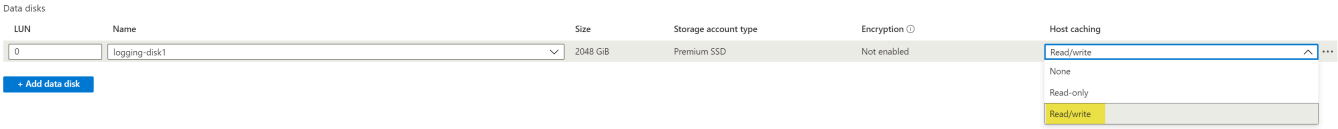


7. For the **Host caching**, select **Read/write**.



STEP 3 | Enable each disk.

The following example uses the sdc virtual disk.

1. Log in to the Panorama CLI.
2. Enter the following command to view the disks on the Panorama virtual appliance:

   `show system disk details`

   The user will see the following response:

```
 Name
: sdb
 State : Present
 Size : 2048000 MB
 Status : Available
 Reason : Admin enabled
 Name : sdc
 State : Present
 Size : 2048000 MB
 Status : Available
```

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```
Reason : Admin disabled
```

3. Enter the following command and confirm the request when prompted for all disks with the `Reason : Admin disabled` response:
   `request system disk add sdc`

   > *The **`request system disk add`** command is not available on a Panorama management server in Management Only mode because logging is not supported in this mode. If you do not see the command, Set up a Panorama Virtual Appliance in Panorama Mode to enable the logging disks. Once in Panorama mode, Log in to the Panorama CLI and continue to Step 4 to verify the disk addition.*

4. Enter the **`show system disk details`** command to verify the status of the disk addition. Continue to the next step when all newly added disk responses display `Reason : Admin enabled`.

STEP 4 | Make disks available for logging.

1. Log in to the Panorama web interface.
2. Edit a Log Collector (**Panorama** > **Managed Collectors**)
3. Select **Disks** and **Add** each newly added disk.
4. Click **OK**.
5. Select **Commit** > **Commit to Panorama**.

   > *For Panorama in an Active/Passive high availability (HA) configuration, wait for HA sync to complete before continuing.*

6. Select **Commit** > **Push to Devices** and push the changes to the Collector Group the Log Collector belongs to.

STEP 5 | (New Panorama deployments in Panorama mode only) Configure Panorama to receive logs.

If you are adding logging disks to an existing Panorama virtual appliance, skip to step 6.

1. Configure a Collector Group.
2. Configure Log Forwarding to Panorama.

STEP 6 | Verify that the Panorama Log Storage capacity is increased.

1. Log in to the Panorama web interface.
2. Select the Collector Group to which the Panorama virtual appliance belongs (**Panorama** > **Collector Groups**).
3. Verify that the **Log Storage** capacity accurately displays the disk capacity.

Add a Virtual Disk to Panorama on Google Cloud Platform

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Panorama virtual appliance operating in Panorama mode or Log Collector mode on Google Cloud Platform

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Where Can I Use This?	What Do I Need?
	• Initial logging disk of exactly 2TB

After you Install Panorama on Google Cloud Platform, add virtual logging disks to the Panorama™ virtual appliance instance to provide storage for logs generated by managed firewalls. You can add virtual disks to a local log Collector for a Panorama virtual appliance in Panorama mode or for a Dedicated Log Collector. The Panorama virtual appliance on Google Cloud Platform supports only 2TB logging disks and, in total, supports up to 24TB of log storage. You cannot add a logging disk smaller than 2TB or a logging disk of a size that is not evenly divisible by 2TB because the Panorama virtual appliance partitions logging disks in to 2TB partitions. For example, if you attach a 4TB logging disk, Panorama will create two 2TB partitions. However, you cannot add a 5TB logging disk because the leftover 1TB is not supported as a partition.

STEP 1 | Log in to the Google Cloud Console.

STEP 2 | Add the virtual logging disk.

In all modes, the first logging disk on the Panorama VM must be at least 2TB in order to add additional disks. If the first logging disk is smaller than 2TB, you will be unable to add additional disk space.

1. In the Products & Services menu, select and then **Edit** the Panorama virtual appliance instance (**Compute Engine** > **VM Instances**).
2. In the Additional Disks section, **Add Item**.
3. **Create disk** (**Name** drop-down).

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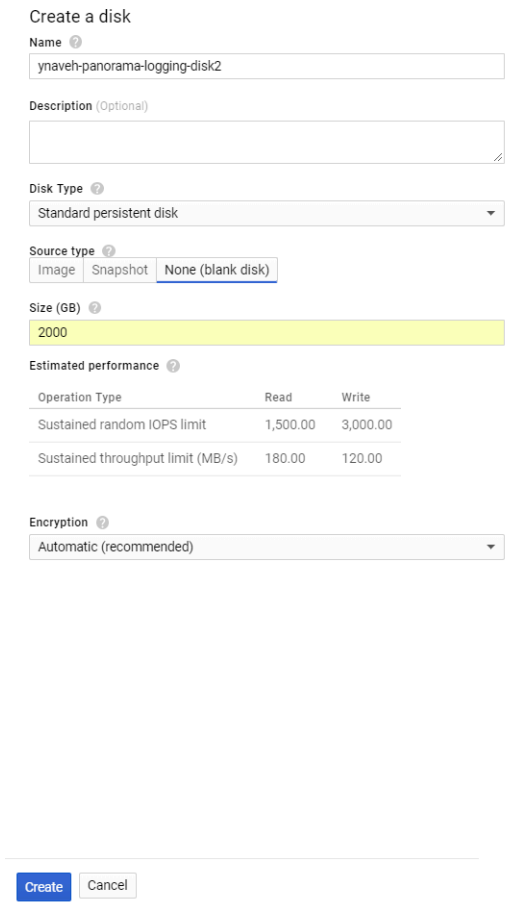
STEP 3 | Configure the virtual logging disks.

1. Enter the **Name**.
2. Expand the **Disk Type** drop-down menu and select the desired type.
3. For the **Source type**, select **None (blank disk)**.
4. Set the **Size (GB)** of the virtual logging disk.
5. Click **Create**.



6. **Save** the changes to update the Panorama virtual appliance instance.

STEP 4 | Configure each disk.

The following example uses the sdc virtual disk.

1. Log in to the Panorama CLI.
2. Enter the following command to view the disks on the Panorama virtual appliance:
   `show system disk details`

   The user will see the following response:

```
 Name
: sdb
 State : Present
 Size : 2048000 MB
```

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```
Status : Available
Reason : Admin enabled
Name : sdc
State : Present
Size : 2048000 MB
Status : Available
Reason : Admin disabled
```

3. Enter the following command and confirm the request when prompted for all disks with the `Reason : Admin disabled` response:

   `request system disk add sdc`

   *The **`request system disk add`** command is not available on a Panorama management server in Management Only mode because logging is not supported in this mode. If you do not see the command, Set up a Panorama Virtual Appliance in Panorama Mode to enable the logging disks. Once in Panorama mode, Log in to the Panorama CLI and continue to Step 4 to verify the disk addition.*

4. Enter the **`show system disk details`** command to verify the status of the disk addition. Continue to the next step when all newly added disk responses display `Reason : Admin enabled`.

STEP 5 | Make disks available for logging.

1. Log in to the Panorama web interface.
2. Edit a Log Collector (**Panorama** > **Managed Collectors**).
3. Select **Disks** and **Add** each newly added disk.
4. Click **OK**.
5. Select **Commit** > **Commit to Panorama**.

   For Panorama in an Active/Passive high availability (HA) configuration, wait for HA sync to complete before continuing.

6. Select **Commit** > **Push to Devices** and push the changes to the Collector Group the Log Collector belongs to.

STEP 6 | (New Panorama deployments in Panorama mode only) Configure Panorama to receive logs.

If you are adding logging disks to an existing Panorama virtual appliance, skip to step 7.

1. Configure a Collector Group.
2. Configure Log Forwarding to Panorama.

STEP 7 | Verify that the Panorama Log Storage capacity is increased.

1. Log in to the Panorama web interface.
2. Select the Collector Group to which the Panorama virtual appliance belongs (**Panorama** > **Collector Groups**).
3. Verify that the **Log Storage** capacity accurately displays the disk capacity.

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Add a Virtual Disk to Panorama on KVM

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Panorama virtual appliance operating in Panorama mode or Log Collector mode on KVM • Initial logging disk of exactly 2TB

After you Install Panorama on KVM, add virtual logging disks to the Panorama™ virtual appliance instance to provide storage for logs generated by managed firewalls. You can add virtual disks to a local log Collector for a Panorama virtual appliance in Panorama mode or for a Dedicated Log Collector. The Panorama virtual appliance on KVM supports only 2TB logging disks and, in total, supports up to 24TB of log storage. You cannot add a logging disk smaller than 2TB or a logging disk of a size that is not evenly divisible by 2TB because the Panorama virtual appliance partitions logging disks in to 2TB partitions. For example, if you attach a 4TB logging disk, Panorama will create two 2TB partitions. However, you cannot add a 5TB logging disk because the leftover 1TB is not supported as a partition.

STEP 1 | **Shutdown** the Panorama virtual appliance instance on the Virtual Machine Manager.

STEP 2 | Double-click the Panorama virtual appliance instance in the Virtual Machine Manager and **Show virtual hardware details** .

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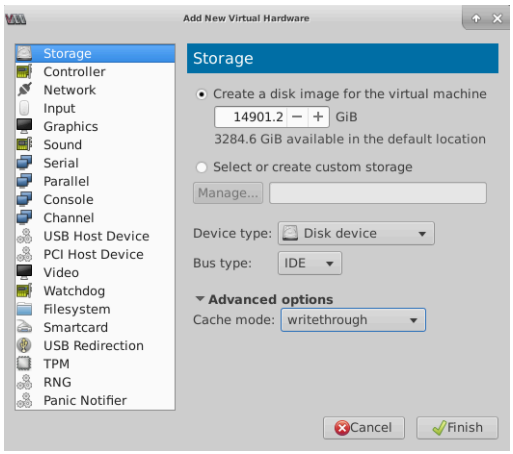
STEP 3 | Add the virtual logging disk. Repeat this step as many times as needed.

In all modes, the first logging disk on the Panorama VM must be at least 2TB in order to add additional disks. If the first logging disk is smaller than 2TB, you will be unable to add additional disk space.

1. **Create a disk image for a virtual image** (**Add Hardware** > **Storage**) and configure the virtual disk storage capacity to the appropriate 2TB value:2000GB or 14901.2GiB depending on your Virtual Machine Manager.

   Depending on the version, some Virtual Machine Managers use GiB (gibibyte) to allocate memory. Be sure you correctly convert the required storage capacity to avoid under provisioning the virtual logging disk and sending the Panorama virtual appliance into maintenance mode.
2. In the **Device type** drop-down, select **Disk device**.
3. In the **Bus type** drop-down, select **VirtIO** or **IDE** based on your configuration.
4. Expand **Advanced options** and, in the **Cache mode** drop-down, select **writethrough**.
5. Click **Finish**.



STEP 4 | **Power on** the Panorama virtual appliance instance.

STEP 5 | Configure each disk.

The following example uses the sdc virtual disk.

1. Log in to the Panorama CLI.
2. Enter the following command to view the disks on the Panorama virtual appliance:
   `show system disk details`

   The user will see the following response:

```
 Name
: sdb
 State : Present
 Size : 2048000 MB
 Status : Available
 Reason : Admin enabled
```

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```
Name : sdc
State : Present
Size : 2048 MB
Status : Available
Reason : Admin disabled
```

3. Enter the following command and confirm the request when prompted for all disks with the `Reason : Admin disabled` response:

   `request system disk add sdc`

   > *The **`request system disk add`** command is not available on a Panorama management server in Management Only mode because logging is not supported in this mode. If you do not see the command, Set up a Panorama Virtual Appliance in Panorama Mode to enable the logging disks. Once in Panorama mode, Log in to the Panorama CLI and continue to Step 4 to verify the disk addition.*

4. Enter the **`show system disk details`** command to verify the status of the disk addition. Continue to the next step when all newly added disk responses display `Reason : Admin enabled`.

STEP 6 | Make disks available for logging.

1. Log in to the Panorama web interface.
2. Edit a Log Collector (**Panorama** > **Managed Collectors**).
3. Select **Disks** and **Add** each newly added disk.
4. Click **OK**.
5. Select **Commit** > **Commit to Panorama**.

   > *For Panorama in an Active/Passive high availability (HA) configuration, wait for HA sync to complete before continuing.*

6. Select **Commit** > **Push to Devices** and push the changes to the Collector Group the Log Collector belongs to.

STEP 7 | (New Panorama deployments in Panorama mode only) Configure Panorama to receive logs.

If you are adding logging disks to an existing Panorama virtual appliance, skip to step 8.

1. Configure a Collector Group.
2. Configure Log Forwarding to Panorama.

STEP 8 | Verify that the Panorama Log Storage capacity is increased.

1. Log in to the Panorama web interface.
2. Select the Collector Group to which the Panorama virtual appliance belongs (**Panorama** > **Collector Groups**).
3. Verify that the **Log Storage** capacity accurately displays the disk capacity.

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Add a Virtual Disk to Panorama on Hyper-V

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Panorama virtual appliance operating in Panorama mode or Log Collector mode on Hyper-V • Initial logging disk of exactly 2TB

After you Install Panorama on Hyper-V, add virtual logging disks to the Panorama™ virtual appliance instance to provide storage for logs generated by managed firewalls. You can add virtual disks to a local log Collector for a Panorama virtual appliance in Panorama mode or for a Dedicated Log Collector. The Panorama virtual appliance on Hyper-V supports only 2TB logging disks and, in total, supports up to 24TB of log storage. You cannot add a logging disk smaller than 2TB or a logging disk of a size that is not evenly divisible by 2TB because the Panorama virtual appliance partitions logging disks in to 2TB partitions. For example, if you attach a 4TB logging disk, Panorama will create two 2TB partitions. However, you cannot add a 5TB logging disk because the leftover 1TB is not supported as a partition.

STEP 1 | Power off the Panorama virtual appliance.

1. On the Hyper-V Manager, select the Panorama virtual appliance instance from the list of **Virtual Machines**.
2. Select **Action** > **Turn Off** to power off the Panorama virtual appliance.

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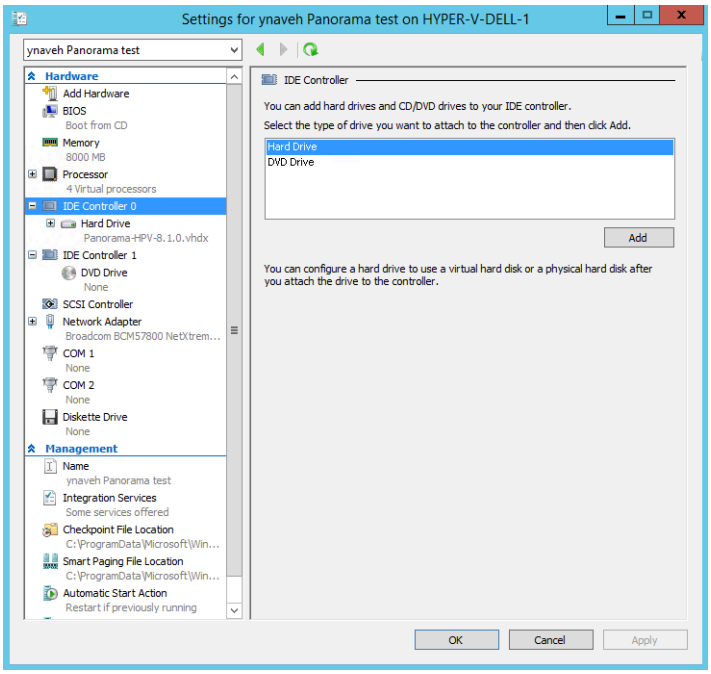
STEP 2 | Add the virtual logging disk. Repeat this step as many times as needed.

In all modes, the first logging disk on the Panorama VM must be 2TB to add additional disks. If the first logging disk is smaller than 2TB, you will be unable to add additional disk space.

The IDE Controller supports only one additional disk besides the system disk. To add more disks, choose a SCSI Controller.

For information on adding a disk larger than 2TB to Panorama on Hyper-V, see this document.

1. Select the Panorama virtual appliance from the list of **Virtual Machines**, and select **Action** > **Settings**.
2. In the **Hardware** list, select **IDE Controller 0** or **SCSI Controller**.
3. From the **IDE Controller** or **SCSI Controller** drives list, select **Hard Drive** and **Add** the new virtual logging disk.



4. Select the new **Hard Drive** created under **IDE Controller 0** or **SCSI Controller**.
5. Under **Media**, add a **New** hard disk.

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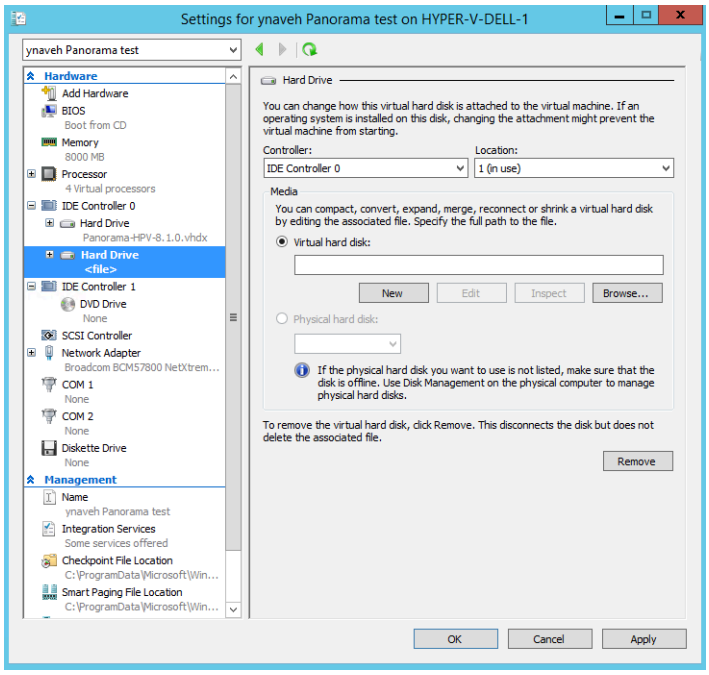

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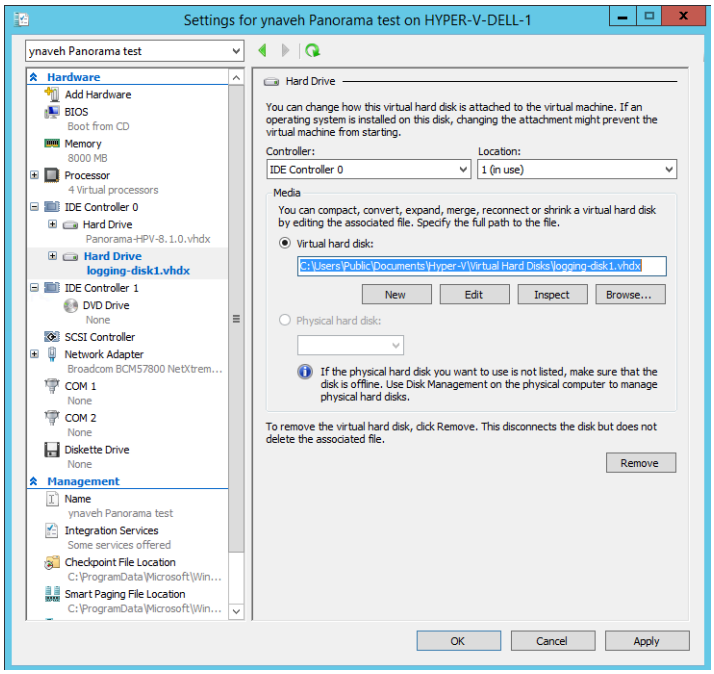
STEP 3 | Configure the new virtual logging disk.

1. If you see the Before You Begin prompt, click **Next** to begin adding the virtual logging disk
2. For the Disk Format, select **VHDX**. Click **Next** to continue
3. For the Disk Type, select **Dynamically Expanding** based on your needs. Click **Next** to continue.
4. Specify the **Name** and **Location** for the virtual logging disk file. Click **Next** to continue.
5. To configure the disk, select **Create a new virtual hard disk** and enter the disk size. Click **Next** to continue.
6. Review the Summary and **Finish** adding the virtual hard logging disk.
7. **Apply** the new hard disk addition.



STEP 4 | Power on the Panorama virtual appliance.

1. Select the Panorama virtual appliance instance from the list of **Virtual Machines**.
2. Select **Action** > **Start** to power on the Panorama virtual appliance.

STEP 5 | Configure each disk.

The following example uses the sdc virtual disk.

1. Log in to the Panorama CLI.
2. Enter the following command to view the disks on the Panorama virtual appliance:
   `show system disk details`

   The user will see the following response:

```
 Name
: sdb
 State : Present
```

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```
Size : 2048000 MB
Status : Available
Reason : Admin enabled
Name : sdc
State : Present
Size : 2048 MB
Status : Available
Reason : Admin disabled
```

3. Enter the following command and confirm the request when prompted for all disks with the `Reason : Admindisabled` response:

   `request system disk add sdc`

   *The **`request system disk add`** command is not available on a Panorama management server in Management Only mode because logging is not supported in this mode. If you do not see the command, Set up a Panorama Virtual Appliance in Panorama Mode to enable the logging disks. Once in Panorama mode, Log in to the Panorama CLI and continue to Step 4 to verify the disk addition.*

4. Enter the **`show system disk details`** command to verify the status of the disk addition. Continue to the next step when all newly added disk responses display `Reason : Admin enabled`.

STEP 6 | Make disks available for logging.

1. Log in to the Panorama web interface.
2. Edit a Log Collector (**Panorama** > **Managed Collectors**).
3. Select **Disks** and **Add** each newly added disk.
4. Click **OK**.
5. Select **Commit** > **Commit to Panorama**.

   For Panorama in an Active/Passive high availability (HA) configuration, wait for HA sync to complete before continuing.

6. Select **Commit** > **Push to Devices** and push the changes to the Collector Group the Log Collector belongs to.

STEP 7 | (New Panorama deployments in Panorama mode only) Configure Panorama to receive logs.

If you are adding logging disks to an existing Panorama virtual appliance, skip to Step 8.

1. Configure a Collector Group.
2. Configure Log Forwarding to Panorama.

STEP 8 | Verify that the Panorama Log Storage capacity is increased.

1. Log in to the Panorama web interface.
2. Select the Collector Group to which the Panorama virtual appliance belongs (**Panorama** > **Collector Groups**).
3. Verify that the **Log Storage** capacity accurately displays the disk capacity.

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Add a Virtual Disk to Panorama on Oracle Cloud Infrastructure (OCI)

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Panorama virtual appliance operating in Panorama mode or Log Collector mode on OCI • Initial logging disk of exactly 2TB

After you Install Panorama on Oracle Cloud Infrastructure (OCI), add additional virtual logging disks to expand log storage capacity on the Panorama™ virtual appliance for logs generated by managed firewalls. You can add virtual disks to a local Log Collector for a Panorama virtual appliance in Panorama mode or for a Dedicated Log Collector. To add virtual disks, you must have access to the OCI console, the Panorama command-line interface (CLI), and the Panorama web interface.

The Panorama virtual appliance on OCI supports only 2TB logging disks and, in total, supports up to 24TB of log storage. You cannot add a logging disk smaller than 2TB or a logging disk of a size that is not evenly divisible by 2TB because the Panorama virtual appliance partitions logging disks in to 2TB partitions. For example, if you attach a 4TB logging disk, Panorama will create two 2TB partitions. However, you cannot add a 5TB logging disk because the leftover 1TB is not supported as a partition.

STEP 1 | Log in to the Oracle Cloud Infrastructure console.

STEP 2 | Create a 2TB block volume.

1. Select **Block Storage** > **Block Volumes** and **Create Block Volume**.
2. Enter a descriptive **Name** for the volume.
3. Select the same **Availability Domain** as the Panorama virtual appliance instance.
4. Select the **Custom** volume size.
5. For the Volume Size, enter **`2000`**.
6. **Create Block Volume**.

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STEP 3 | Attach a virtual logging disk to the Panorama virtual appliance instance.

In all modes, the first logging disk on the Panorama VM must be at least 2TB in order to add additional disks. If the first logging disk is smaller than 2TB, you will be unable to add additional disk space.

1. Select **Compute** > **Instances** and click the name of the Panorama virtual appliance instance.
2. Under resources, select **Attached Block Volumes** and **Attach Block Volume**.
3. For the Volume, **Select volume** and select the virtual logging disk.
4. For the Attachment type, select **Paravirtualized**.
   This is required for the Panorama virtual appliance to recognize the virtual logging disk.
5. For the Access, select **Read/Write**.
6. **Attach** the virtual logging disk.

STEP 4 | Configure each disk.

The following example uses the sdc virtual disk.

1. Log in to the Panorama CLI.
2. Enter the following command to view the disks on the Panorama virtual appliance:
   `show system disk details`
   The user will see the following response:

```
 Name : sdb
State : Present
Size : 2048000 MB
Status : Unavailable
Reason : Admin disabled
```

3. Enter the following command and confirm the request when prompted for all disks with the `Reason : Admin disabled` response:
   `request system disk add sdc`

   *The **`request system disk add`** command is not available on a Panorama management server in Management Only mode because logging is not supported in this mode. If you do not see the command, Set up a Panorama Virtual Appliance in Panorama Mode to enable the logging disks. Once in Panorama mode, Log in to the Panorama CLI and continue to the next step to verify the disk addition.*
4. Enter the **`show system disk details`** command to verify the status of the disk addition. Continue to the next step when all newly added disk responses display `Reason : Admin enabled`.

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STEP 5 | Make disks available for logging.

1. Log in to the Panorama web interface.
2. Edit a Log Collector (**Panorama** > **Managed Collectors**).
3. Select **Disks** and **Add** each newly added disk.
4. Click **OK**.
5. Select **Commit** > **Commit to Panorama**.

   For Panorama in an Active/Passive high availability (HA) configuration, wait for HA sync to complete before continuing.
6. Select **Commit** > **Push to Devices** and push the changes to the Collector Group the Log Collector belongs to.

STEP 6 | (New Panorama deployments in Panorama mode only) Configure Panorama to receive logs.

If you are adding logging disks to an existing Panorama virtual appliance, skip to step 6.

1. Configure a Collector Group.
2. Configure Log Forwarding to Panorama.

STEP 7 | Verify that the Panorama Log Storage capacity is increased.

1. Log in to the Panorama web interface.
2. Select the Collector Group to which the Panorama virtual appliance belongs (**Panorama** > **Collector Groups**).
3. Verify that the **Log Storage** capacity accurately displays the disk capacity.

Increase CPUs and Memory on the Panorama Virtual Appliance

When you Perform Initial Configuration of the Panorama Virtual Appliance, you specify the memory and number of CPUs based on whether the appliance is in Panorama mode or Management Only mode and based on the log storage capacity or number of managed firewalls. If you later add storage capacity or managed firewalls, you must also increase the memory and CPUs. A Panorama virtual appliance in Log Collector mode must meet the system requirements, and does not need to have the CPU and memory increased beyond the minimum requirement. Review the Setup Prerequisites for the Panorama Virtual Appliance for the CPU and memory requirements for each Panorama mode.

- Increase CPUs and Memory for Panorama on an ESXi Server
- Increase CPUs and Memory for Panorama on vCloud Air
- Increase CPUs and Memory for Panorama on Alibaba Cloud
- Increase CPUs and Memory for Panorama on AWS
- Increase CPUs and Memory for Panorama on Azure
- Increase CPUs and Memory for Panorama on Google Cloud Platform
- Increase CPUs and Memory for Panorama on KVM
- Increase CPUs and Memory for Panorama on Hyper-V
- Increase the CPUs and Memory for Panorama on Oracle Cloud Infrastructure (OCI)

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Increase CPUs and Memory for Panorama on an ESXi Server

For the minimum CPUs and memory that Panorama requires, see Increase CPUs and Memory on the Panorama Virtual Appliance.

STEP 1 | Access the VMware vSphere Client and select **Virtual Machines**.

STEP 2 | Right-click the Panorama virtual appliance and select **Power** > **Power Off**.

STEP 3 | Right-click the Panorama virtual appliance and select **Edit Settings**.

STEP 4 | Select **Memory** and enter the new **Memory Size**.

STEP 5 | Select **CPUs** and specify the number of CPUs (the **Number of virtual sockets** multiplied by the **Number of cores per socket**).

STEP 6 | Click **OK** to save your changes.

STEP 7 | Right-click the Panorama virtual appliance and select **Power** > **Power On**.

Increase CPUs and Memory for Panorama on vCloud Air

For the minimum CPUs and memory that Panorama requires, see Increase CPUs and Memory on the Panorama Virtual Appliance.

STEP 1 | Access the vCloud Air web console and select your **Virtual Private Cloud OnDemand** region.

STEP 2 | In the **Virtual Machines** tab, select the Panorama virtual machine and **Power Off**.

STEP 3 | Select **Actions** > **Edit Resources**.

STEP 4 | Set the **CPU** and **Memory**.

STEP 5 | **Save** your changes.

STEP 6 | Select the Panorama virtual machine and **Power On**.

Increase CPUs and Memory for Panorama on Alibaba Cloud

You can change the instance type of the Panorama™ virtual appliance to increase the CPUs and memory allocated to the Panorama virtual appliance instance. Be sure to review the supported Alibaba Cloud instance types and the Setup Prerequisites for the Panorama Virtual Appliance before changing the instance type.

STEP 1 | Log in to the Alibaba Cloud Console.

STEP 2 | Select **Elastic Compute Service** > **Instances & Images** > **Instances** and navigate to the Panorama virtual appliance instance.

STEP 3 | In the Actions column, select **More** > **Instance Status** > **Stop.**

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STEP 4 | Change the Panorama virtual appliance instance type.

1. Select the Panorama virtual appliance if not already selected.
2. In the Actions column, select **Change Instance Type**.
3. Select the desired instance type and **Change** the instance type.
4. When prompted, select **Console** to view your Panorama virtual appliance instance.

STEP 5 | In the Actions column for the Panorama virtual appliance instance, select **More** > **Instance Status** > **Start.**

STEP 6 | Verify the increased CPU and memory.

1. Log in to the Panorama CLI.
2. View the Panorma virtual appliance system information.

```
admin> show system info
```

3. Verify that the `num-cpus` and `ram-in-gb` display the correct number of CPUs and amount of memory as per the instance type you selected.

Increase CPUs and Memory for Panorama on AWS

For the minimum CPUs and memory that Panorama™ requires, see Increase CPUs and Memory on the Panorama Virtual Appliance.

STEP 1 | Log in to AWS Web Service console and select the EC2 Dashboard.

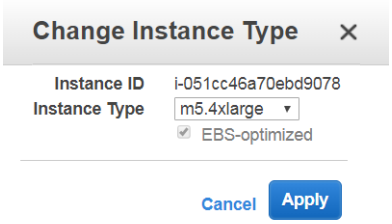
- Amazon Web Service Console
- AWS GovCloud Web Service Console

STEP 2 | On the EC2 Dashboard, select **Instances** and select the Panorama virtual appliance instance.

STEP 3 | Select **Actions** > **Instance State** > **Stop** to power off the Panorama virtual appliance instance.

STEP 4 | Select **Actions** > **Instance Settings** > **Change Instance Type** to change the Panorama virtual appliance instance type.

STEP 5 | Select the **Instance Type** to which you want to upgrade and **Apply** it.



STEP 6 | Select **Actions** > **Instance State** > **Start** to power on the Panorama virtual appliance instance.

Increase CPUs and Memory for Panorama on Azure

For the minimum CPUs and memory that Panorama™ requires, see Increase CPUs and Memory on the Panorama Virtual Appliance.

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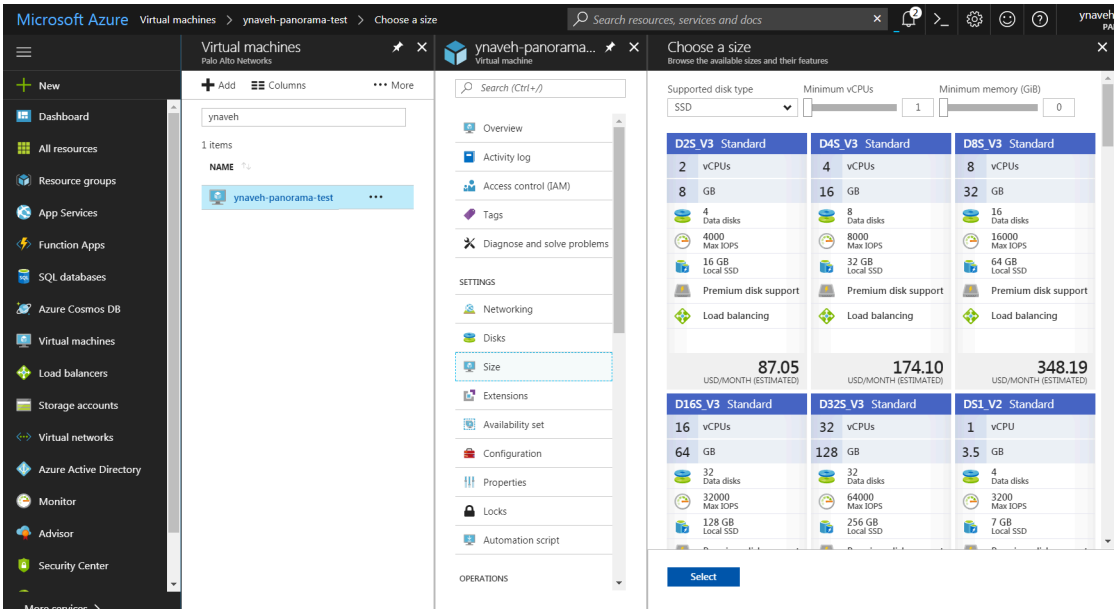
STEP 1 | Log in to the Microsoft Azure portal.

STEP 2 | On the Azure Dashboard, under **Virtual machines**, select the Panorama virtual appliance.

STEP 3 | Select **Overview** and **Stop** the Panorama virtual appliance.

STEP 4 | Choose the new virtual machine **Size** and then **Select** it.



STEP 5 | Select **Overview** and **Start** the Panorama virtual appliance.

Increase CPUs and Memory for Panorama on Google Cloud Platform

For the minimum CPUs and memory that Panorama™ requires, see Increase CPUs and Memory on the Panorama Virtual Appliance.

STEP 1 | Log in to the Google Cloud Console.

STEP 2 | Stop the Panorama virtual appliance instance.

1. Select the Panorama virtual appliance instance in the Products & Services menu (**Compute Engine** > **VM Instances**).
2. **Stop** the Panorama virtual appliance instance. It can take 2 to 3 minutes for the Panorama virtual appliance to completely shut down.

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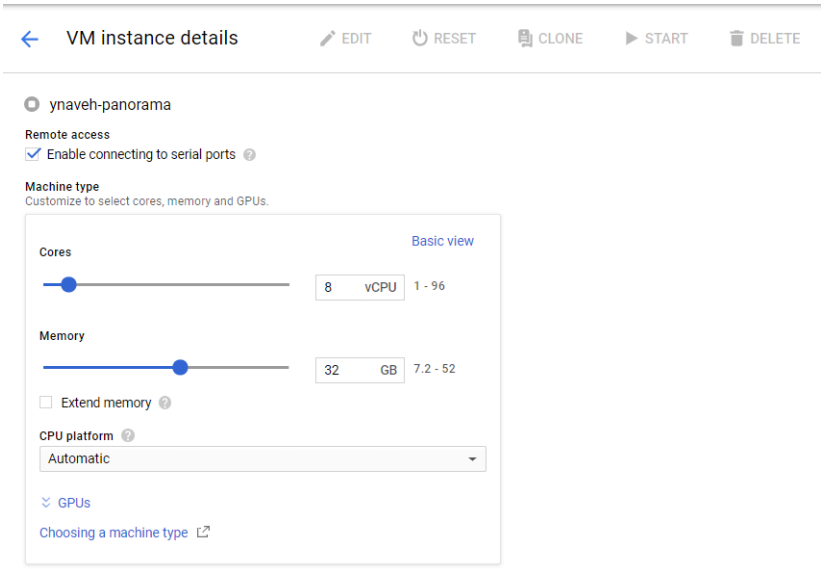
STEP 3 | Reconfigure the Panorama virtual appliance resources.

1. **Edit** the Panorama virtual appliance instance details.
2. Under Machine Type, **Customize** the Panorama virtual appliance CPU cores and memory.



STEP 4 | **Save** the changes to update the Panorama virtual appliance instance.

STEP 5 | **Start** the Panorama virtual appliance.

Increase CPUs and Memory for Panorama on KVM

For the minimum CPUs and memory that Panorama™ requires, see Increase CPUs and Memory on the Panorama Virtual Appliance.

STEP 1 | **Shutdown** the Panorama virtual appliance instance on the Virtual Machine Manager.

STEP 2 | Double-click the Panorama virtual appliance instance in the Virtual Machine Manager and **Show virtual hardware details**.

STEP 3 | Edit the allocated Panorama virtual appliance CPU cores.

1. Edit the currently allocated **CPUs**.
2. **Apply** the reconfigured CPU core allocation.

STEP 4 | Edit the allocated Panorama virtual appliance memory.

1. Edit the currently allocated **Memory**.
2. **Apply** the reconfigured memory allocation.

STEP 5 | **Power on** the Panorama virtual appliance instance.

Increase CPUs and Memory for Panorama on Hyper-V

For the minimum CPUs and memory that Panorama™ requires, see Increase CPUs and Memory on the Panorama Virtual Appliance.

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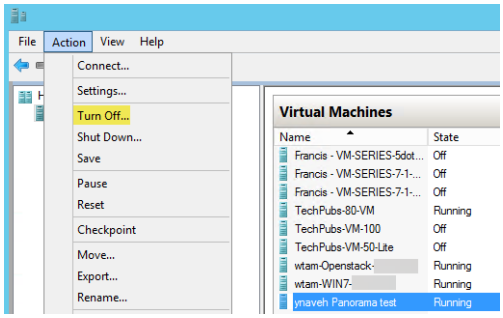
STEP 1 | Power off the Panorama virtual appliance.

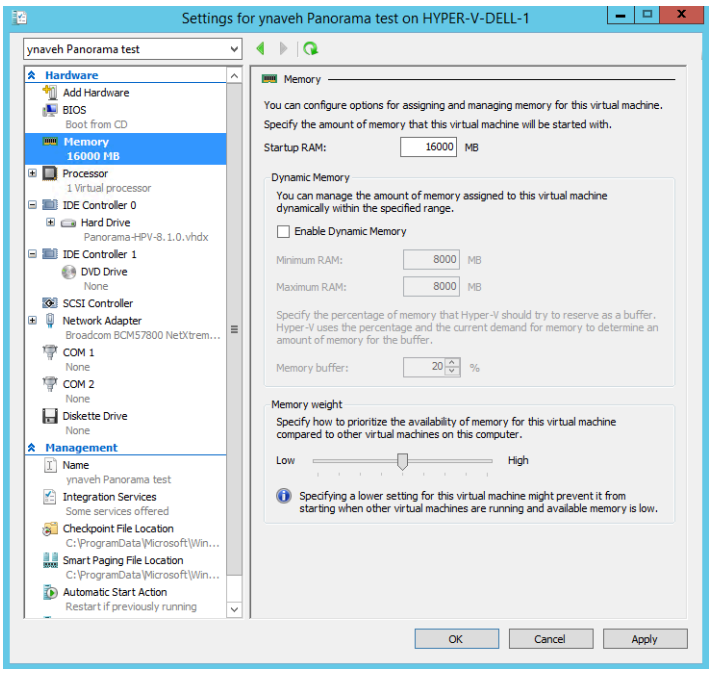
1. On the Hyper-V Manager, select the Panorama virtual appliance instance from the list of **Virtual Machines**.
2. Select **Action** > **Turn Off** to power off the Panorama virtual appliance.



STEP 2 | On the Hyper-V Manager, select the Panorama virtual appliance instance form the list of **Virtual Machines**, and select **Action** > **Settings** to edit the Panorama virtual appliance resources.

STEP 3 | Edit the allocated Panorama virtual appliance memory.

1. In the **Hardware** list, select **Memory**.
2. Edit the currently allocated **Startup RAM**.



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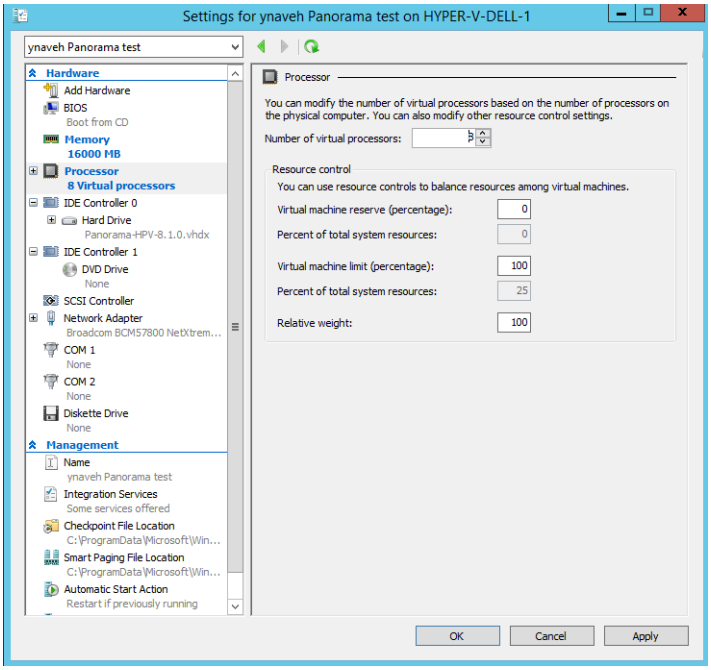
STEP 4 | Edit the allocated Panorama virtual appliance CPU cores.

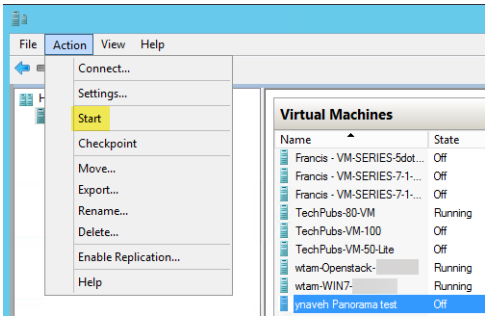
1. In the **Hardware** list, select **Processor**.
2. Edit the currently allocated **Number of virtual processors**.



STEP 5 | **Apply** the reallocated memory and CPU cores.

STEP 6 | Power on the Panorama virtual appliance.

1. Select the Panorama virtual appliance instance from the list of **Virtual Machines**.
2. Select **Action** > **Start** to power on the Panorama virtual appliance.



Increase the CPUs and Memory for Panorama on Oracle Cloud Infrastructure (OCI)

You can change the instance type of the Panorama™ virtual appliance to increase the CPUs and memory allocated to the Panorama virtual appliance instance. Be sure to review the Setup Prerequisites for the Panorama Virtual Appliance before modifying the Panorama virtual appliance instance CPUs and memory.

STEP 1 | Log in to the Oracle Cloud Infrastructure console.

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STEP 2 | Power off the Panorama virtual appliance instance.

1. Select **Compute** > **Instances** and click the name of the Panorama virtual appliance instance.
2. **Stop** the Panorama virtual appliance instance.

STEP 3 | Increase the CPUs and memory.

1. In the instance details, select **Edit** > **Edit Shape**.
2. Increase the number of CPUs and memory allocated to the instance.
3. **Save Changes**.

STEP 4 | In instance details, **Start** the Panorama virtual appliance.

STEP 5 | Verify the increased CPU and memory.

1. Log in to the Panorama CLI.
2. View the Panorma virtual appliance system information.

```
admin> show system info
```

3. Verify that the `num-cpus` and `ram-in-gb` display the correct number of CPUs and amount of memory as per the instance type you selected.

Increase the System Disk on the Panorama Virtual Appliance

Expand the system disk capacity to 224GB for the Panorama virtual appliance to support large datasets to allow for sufficient disk space for things such as dynamic updates when you Manage Large-Scale Firewall Deployments with Panorama. Additionally, a 224GB system disk expands storage for monitoring and reporting data for managed firewall health if you intended to use the Panorama virtual appliance in Panorama mode to manage your SD-WAN deployment

- Increase the System Disk for Panorama on an ESXi Server
- Increase the System Disk for Panorama on a KVM Server
- Increase the System Disk for Panorama on Google Cloud Platform
- Increase the System Disk for Panorama on AWS
- Increase the System Disk for Panorama on Azure
-

Increase the System Disk for Panorama on an ESXi Server

Add a 224GB system disk to replace the default 81GB system disk. For the minimum resource requirements for the Panorama virtual appliance, see Setup Prerequisites for the Panorama Virtual Appliance.

Decreasing the Panorama virtual appliance system disk back to 81GB is not supported.

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STEP 1 | (Best Practice) Save and Export Panorama and Firewall Configurations. Save and Export Panorama and Firewall Configurations

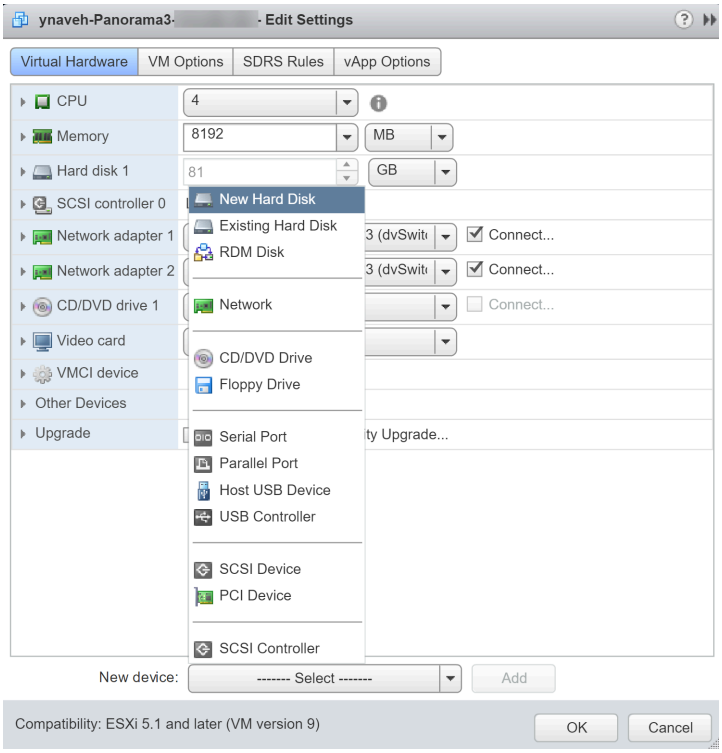
Save and export your Panorama and firewall configuration to ensure you can recover Panorama if you encounter any issues.

STEP 2 | Access the VMware vSphere Client and navigate to your Panorama virtual appliance.

STEP 3 | Right-click the Panorama virtual appliance and select **Power** > **Power Off**.

STEP 4 | Add the new 224GB system disk.

1. Right-click the Panorama virtual appliance and **Edit Settings**.
2. Select **New Hard Disk** as the **New Device** and **Add** the new device.
3. Configure the new hard disk with 224GB and click **OK**.



STEP 5 | Right-click the Panorama virtual appliance and select **Power** > **Power On**.

Panorama may take up to 30 minutes to initialize the new system disk. During this time the Panorama web interface and CLI are unavailable.

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STEP 6 | Migrate disk data from the old system disk to the new system disk.

In this example, we are migrating to the newly added system disk labeled sdb.

1. Log in to the Panorama CLI.
2. Enter the following command to view the available system disks for migration:

```
admin> request system clone-system-disk target ?
```

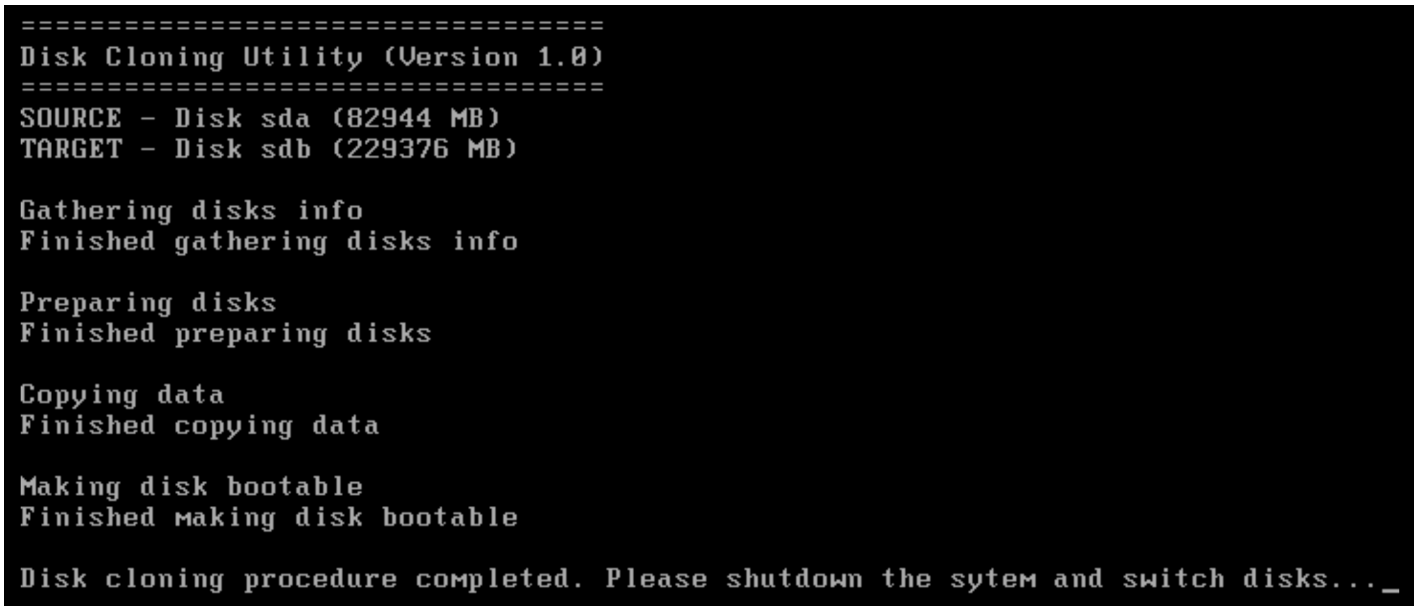
3. Migrate the disk data to the new system disk using the following command:

```
admin> request system clone-system-disk target sdb
```

Enter **Y** when prompted to begin the disk migration.

To begin the migration, Panorama reboots and takes at least 20 minutes to complete the disk migration. During this time the Panorama web interface and CLI are unavailable.

4. Monitor the disk migration from the web Console. Continue to the next step only after Panorama displays the following message to indicate the disk migration is complete.



STEP 7 | Delete the old system disk.

1. Access the VMware vSphere Client and navigate to your Panorama virtual appliance.
2. Right-click the Panorama virtual appliance and select **Power** > **Power Off**.
3. Right-click the Panorama virtual appliance and **Edit Settings**.
4. Delete the old 81GB system disk and click **OK**.

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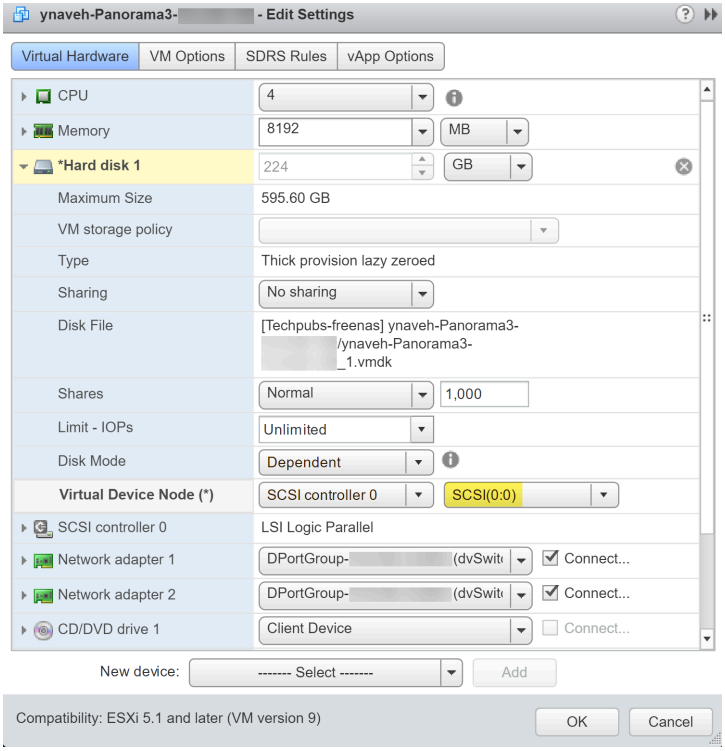
STEP 8 | Modify the Virtual Device Node for the new system disk.

1. Expand the settings options for the new system disk.
2. Select **SCSI(0:0)** as the **Virtual Device Node**.
3. Click **OK** to save your configuration changes.



STEP 9 | Right-click the Panorama virtual appliance and select **Power** > **Power On**.

STEP 10 | Verify that you successfully migrated to the new system disk.

1. Log in to the Panorama CLI.
2. Enter the following command to view the system disk partitions.

   You must examine the `/dev/root`, `/dev/sda5`, `/dev/sda6`, and `/dev/sda8` partitions to confirm the disk size is increased.

```
admin> show system disk-space
```

```
admin@Panorama-Ynaveh> show system disk-space

Filesystem      Size  Used Avail Use% Mounted on
/dev/root        16G  3.4G   12G  23% /
none            4.0G   60K  4.0G   1% /dev
/dev/sda5        76G  1.8G   71G   3% /opt/pancfg
/dev/sda6        23G  5.0G   17G  24% /opt/panrepo
tmpfs           4.0G  110M  3.8G   3% /dev/shm
cgroup_root     4.0G     0  4.0G   0% /cgroup
/dev/sda8        92G   52G   35G  60% /opt/panlogs
/dev/loop0       50G  7.4G   40G  16% /opt/mongobuffer
tmpfs            12M     0   12M   0% /opt/pancfg/mgmt/ssl/private
```

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Increase the System Disk for Panorama on a KVM Server

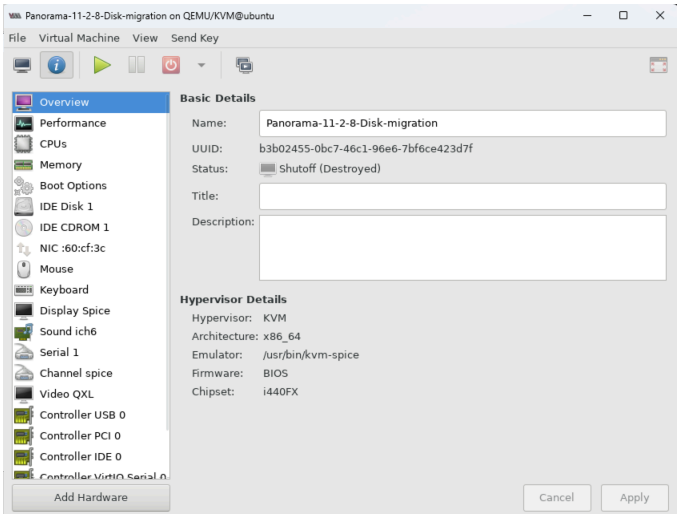
Add a 224GB system disk to replace the default 81GB system disk. For the minimum resource requirements for the Panorama virtual appliance, see Setup Prerequisites for the Panorama Virtual Appliance.

Decreasing the Panorama virtual appliance system disk back to 81GB is not supported.

STEP 1 | (Best Practice) Save and Export Panorama and Firewall Configurations.

Save and export your Panorama and firewall configuration to ensure you can recover Panorama if you encounter any issues.

STEP 2 | Access the KVM Client and navigate to your Panorama virtual appliance.



STEP 3 | Right-click the Panorama virtual appliance and select **Power** > **Power Off**.

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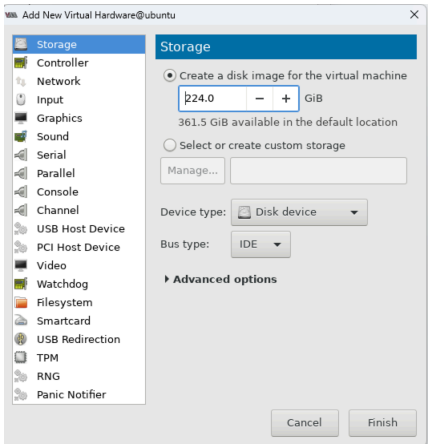
STEP 4 | Add the new 224GB system disk.

1. Open Virt-manager and click **Storage**.
2. Select **Create a disk image** for the virtual machine and configure the new hard disk with 224GB.
3. Configure the new hard disk with 224GB and click **OK**.



STEP 5 | **Power On** the Panorama.

Panorama may take up to 30 minutes to initialize the new system disk. During this time the Panorama web interface and CLI are unavailable.

STEP 6 | Execute the command `show system disk details` to verify that the new disk is added.

```
admin@Panorama> show system disk details


Name   : sdb
State  : Present
Size   : 229376 MB
Status : Unavailable
Reason : Admin disabled


admin@Panorama>
```

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STEP 7 | Log in to the Panorama CLI and migrate disk data from the existing system disk to the new system disk by executing the command `request system clone-system-disk target`.

In the example below, we are cloning the existing disk to the newly added system disk labeled sdb.

```
admin@Panorama>
admin@Panorama>
admin@Panorama>
admin@Panorama> request system clone-system-disk target
  sdb      New disk
  <value>  Select target disk

admin@Panorama> request system clone-system-disk target sdb
  <Enter>  Finish input

admin@Panorama> request system clone-system-disk target sdb
Executing this command will copy data from system disk and delete all data on the target disk. System wil

Broadcast message from root (Tue Oct 28 13:58:39 2025):

The system is going down for reboot NOW!

admin@Panorama>
admin@Panorama> Connection to 10.6.66.63 closed.
(mbpanenv) mb-parts@parts-vm:~/pa-vm/Eureka/tests$
```

Enter

Y

when prompted to begin the disk migration.

- *To begin the migration, Panorama reboots and takes at least 20 minutes to complete the disk migration. During this time the Panorama web interface and CLI are unavailable.*
- *Monitor the disk migration from the web Console. Continue to the next step only after Panorama displays a message to indicate the disk migration is complete.*

```
================================
Disk Cloning Utility (Version 1.0)
================================
SOURCE - Disk sda (82944 MB)
TARGET - Disk sdb (229376 MB)

Gathering disks info
Finished gathering disks info

Preparing disks
Finished preparing disks

Copying data
Finished copying data

Making disk bootable
Finished making disk bootable

Disk cloning procedure completed. Please shutdown the sytem and switch disks...
```

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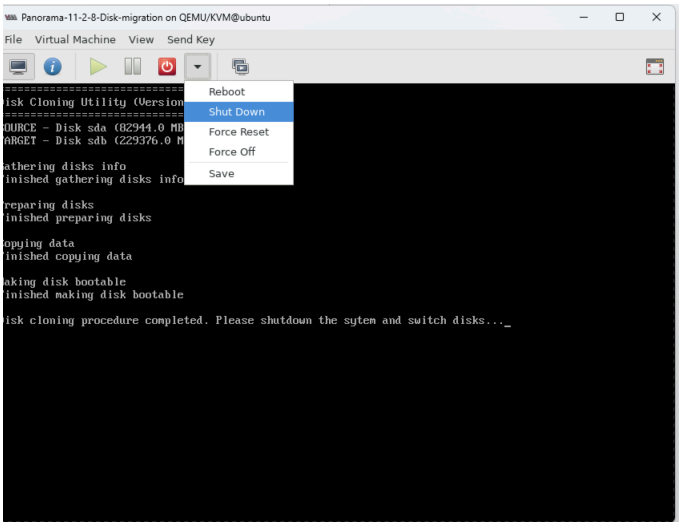
STEP 8 | Delete the old system disk.

1. Access the KVM Client and navigate to your Panorama virtual appliance.
2. Right-click the Panorama virtual appliance and select **Power** > **Power Off**.
3. Right-click the Panorama virtual appliance and **Edit Settings**.
4. Delete the old 81GB system disk and click **OK**.

STEP 9 | Shutdown the KVM instance.



STEP 10 | **Power On** the Panorama and verify that you successfully migrated to the new system disk.

1. Log in to the Panorama CLI.
2. Enter the following command to view the system disk partitions:

   ```
   show system disk-partition
   ```

3. Verify the new disk is of 224 GB disk and displays a root volume of 16GB instead of 8GB.

```
admin@Panorama> show system disk-partition

Disk /dev/sda: 224 GiB, 240518168576 bytes, 469762048 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x20f42fbf

Device     Boot     Start       End   Sectors   Size Id Type
/dev/sda1              63   4016249   4016187   1.9G 83 Linux
/dev/sda2         4016250  36033794  32017545  15.3G 83 Linux
/dev/sda3        36033795  68051339  32017545  15.3G 83 Linux
/dev/sda4        68051340 468857024 400805685 191.1G  f W95 Ext'd (LBA)
/dev/sda5        68051403 228074804 160023402  76.3G 83 Linux
/dev/sda6       228074868 276093089  48018222  22.9G 83 Linux
/dev/sda7       276093153 276109154     16002   7.8M 83 Linux
/dev/sda8       276109218 468857024 192747807  91.9G 83 Linux


Disk /dev/loop0: 36.3 GiB, 38948306944 bytes, 76070912 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

admin@Panorama>
```

Increase the System Disk on a KVM Server in Panorama-mode

To increase the system disk on a KVM Server in Panorama-mode:

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STEP 1 | Log in to the Panorama CLI.

STEP 2 | Execute the command `show system info | match system-mode` and verify the system-mode of the Panorama is listed as `panorama`.

```
admin@Panorama>
admin@Panorama> show system info | match system-mode
system-mode: panorama
admin@Panorama>
```

STEP 3 | Execute the command `show system disk details` and verify the following:

1. Status: Available
2. Reason: Admin enabled

```
admin@Panorama> show system disk details


Name   : sdb
State  : Present
Size   : 204800 MB
Status : Available
Reason : Admin enabled
```

STEP 4 | Shutdown the Panorama.

STEP 5 | Add a 224 GB disk using the virt-manager.

STEP 6 | Power ON the Panorama VM.

STEP 7 | Execute the command `show system disk details` and verify the new disk added is of 224 GB.

```
admin@Panorama> show system disk details


Name   : sdb
State  : Present
Size   : 204800 MB
Status : Available
Reason : Admin enabled

Name   : sdc
State  : Present
Size   : 229376 MB
Status : Unavailable
Reason : Admin disabled


admin@Panorama>
```

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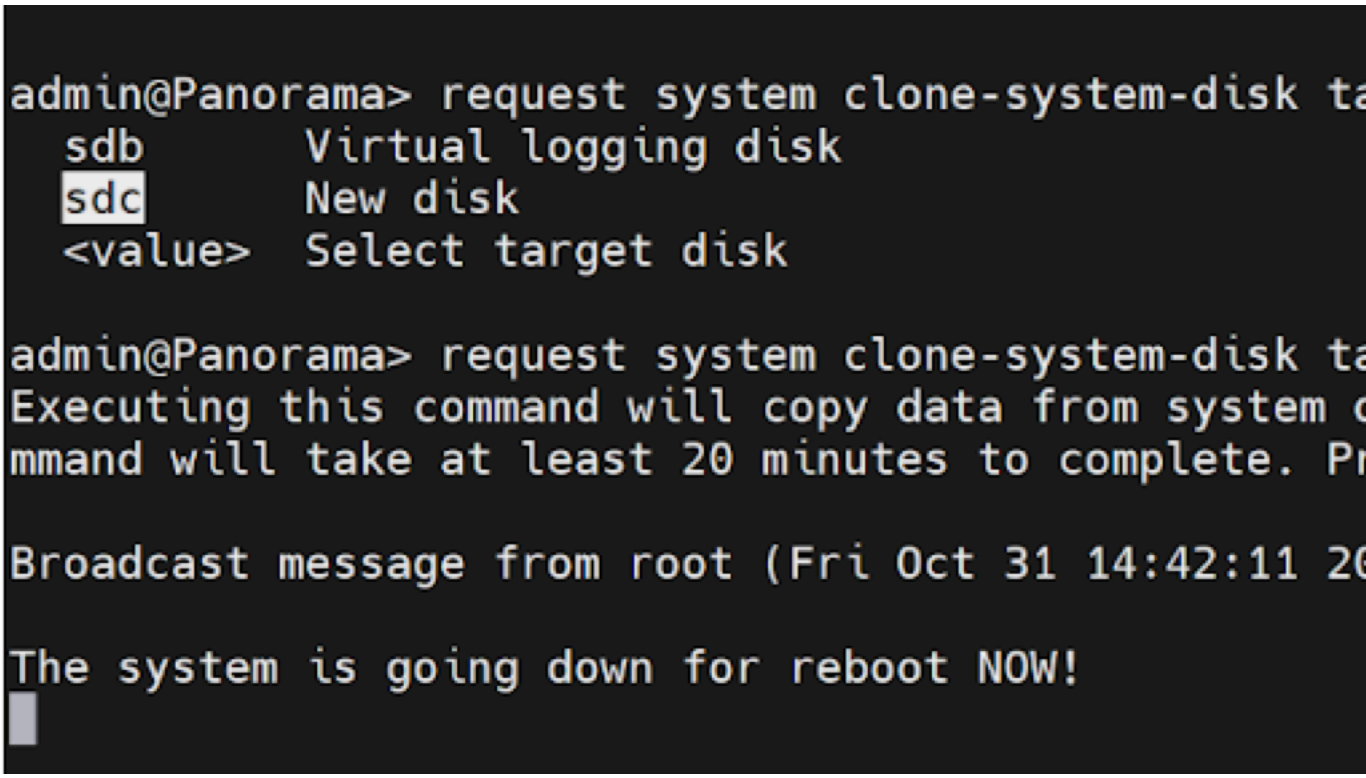
STEP 8 | Execute the command `request system clone-system-disk target` and select the required disk. This initiates a reboot on the Panorama following the disk cloning.



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STEP 9 | Verify the completion of disk cloning through virt-manager.

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STEP 10 | Initiate a shutdown on the VM and delete the 81GB disk.

STEP 11 | Power on the Panorama VM.

Increase the System Disk for Panorama on Google Cloud Platform

Add a 224GB system disk to replace the default 81GB system disk. For the minimum resource requirements for the Panorama virtual appliance, see Setup Prerequisites for the Panorama Virtual Appliance.

STEP 1 | (Best Practice) Save and Export Panorama and Firewall Configurations.

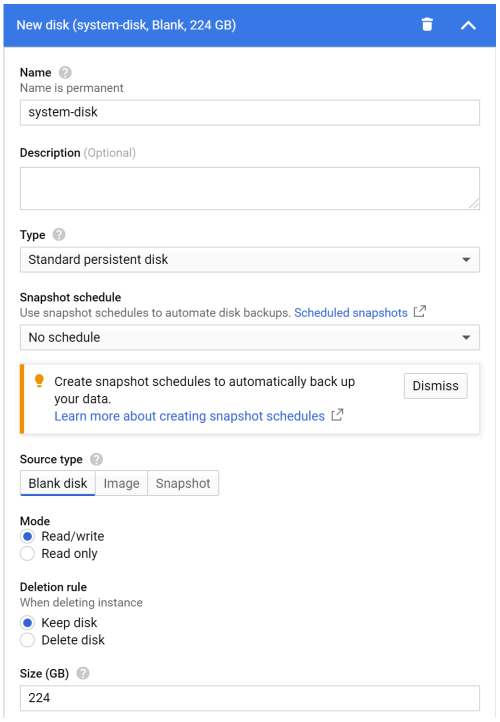
Save and export your Panorama and firewall configuration to ensure you can recover Panorama if you encounter any issues.

STEP 2 | Log in to the Google Cloud Console.

STEP 3 | In **VM Instances**, **Stop** the Panorama VM instance.

STEP 4 | Add the new 224GB system disk.

1. Select the Panorama VM instance and select **Edit**.
2. In the **Additional disks** section **Add new disk**.
3. Configure the new disk with 224GB and click **OK**.



STEP 5 | In **VM Instances**, **Start** the Panorama VM instance.

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STEP 6 | Migrate disk data from the old system disk to the new system disk.

In this example, we are migrating to the newly added system disk labeled sdb.

1. Log in to the Panorama CLI.
2. Enter the following command to view the available system disks for migration:

```
admin> request system clone-system-disk target ?
```

3. Migrate the disk data to the new system disk using the following command:

```
admin> request system clone-system-disk target sdb
```

Enter **Y** when prompted to begin the disk migration.

To begin the migration, Panorama reboots and takes at least 20 minutes to complete the disk migration. During this time the Panorama web interface and CLI are unavailable.

4. Monitor the disk migration by attempting to log in to the Panorama CLI. The Panorama management server is in maintenance mode after the system disk migration is completed and will allow you to log in to the Panorama CLI while in maintenance mode.

STEP 7 | Attach the new 224GB system disk.

1. In **VM Instances**, **Stop** the Panorama VM instance.
2. Select the Panorama VM instance and select **Edit**.
3. In the **Additional disks** section, detach the new 224GB system disk.
4. In the **Boot Disk** section, detach the old 81GB system disk.
5. In the **Boot Disk** section, **Add item** and select the new 224GB system disk.
6. **Save** your configuration changes.

STEP 8 | In **VM Instances**, **Start** the Panorama VM instance.

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STEP 9 | Verify that you successfully migrated to the new system disk.

1. Log in to the Panorama CLI.
2. Enter the following command to view the system disk partitions.

   You must examine the `/dev/root`, `/dev/sda5`, `/dev/sda6`, and `/dev/sda8` partitions to confirm the disk size is increased.

```
admin> show system disk-space
```

```
admin@Panorama-Ynaveh> show system disk-space

Filesystem      Size  Used Avail Use% Mounted on
/dev/root        16G  3.4G   12G  23% /
none            4.0G   60K  4.0G   1% /dev
/dev/sda5        76G  1.8G   71G   3% /opt/pancfg
/dev/sda6        23G  5.0G   17G  24% /opt/panrepo
tmpfs           4.0G  110M  3.8G   3% /dev/shm
cgroup_root     4.0G     0  4.0G   0% /cgroup
/dev/sda8        92G   52G   35G  60% /opt/panlogs
/dev/loop0       50G  7.4G   40G  16% /opt/mongobuffer
tmpfs            12M     0   12M   0% /opt/pancfg/mgmt/ssl/private
```

Increase the System Disk for Panorama on AWS

To increase the system disk on AWS, add a 224GB system disk to replace the default 81GB system disk. For the minimum resource requirements for the Panorama virtual appliance, see Setup Prerequisites for the Panorama Virtual Appliance.

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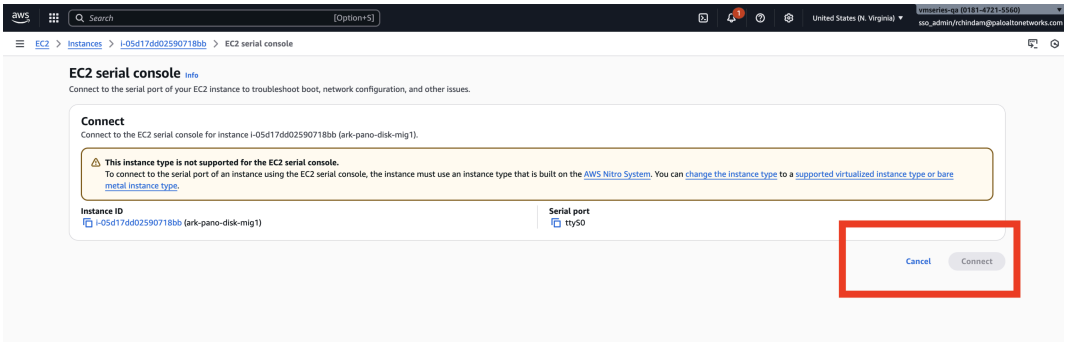
1. Save and Export Panorama and Firewall Configurations.

   Save and export your Panorama and firewall configuration to ensure you can recover Panorama if you encounter any issues.

2. *Before you start with the migration, ensure that the serial console is enabled for the instance type you are using.*



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STEP 1 | Log in to the AWS Console.

STEP 2 | From EC2 > Instances, select the Panorama VM, click **Instance state** and stop the instance.

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STEP 3 | Create a new volume with type = <based on current panorama volume type> and Size = 224GB in the Panorama's Availability Zone.

Ensure the new volume's type matches your existing EC2 instance. For example, a gp2 EC2 instance requires a gp2 volume type.

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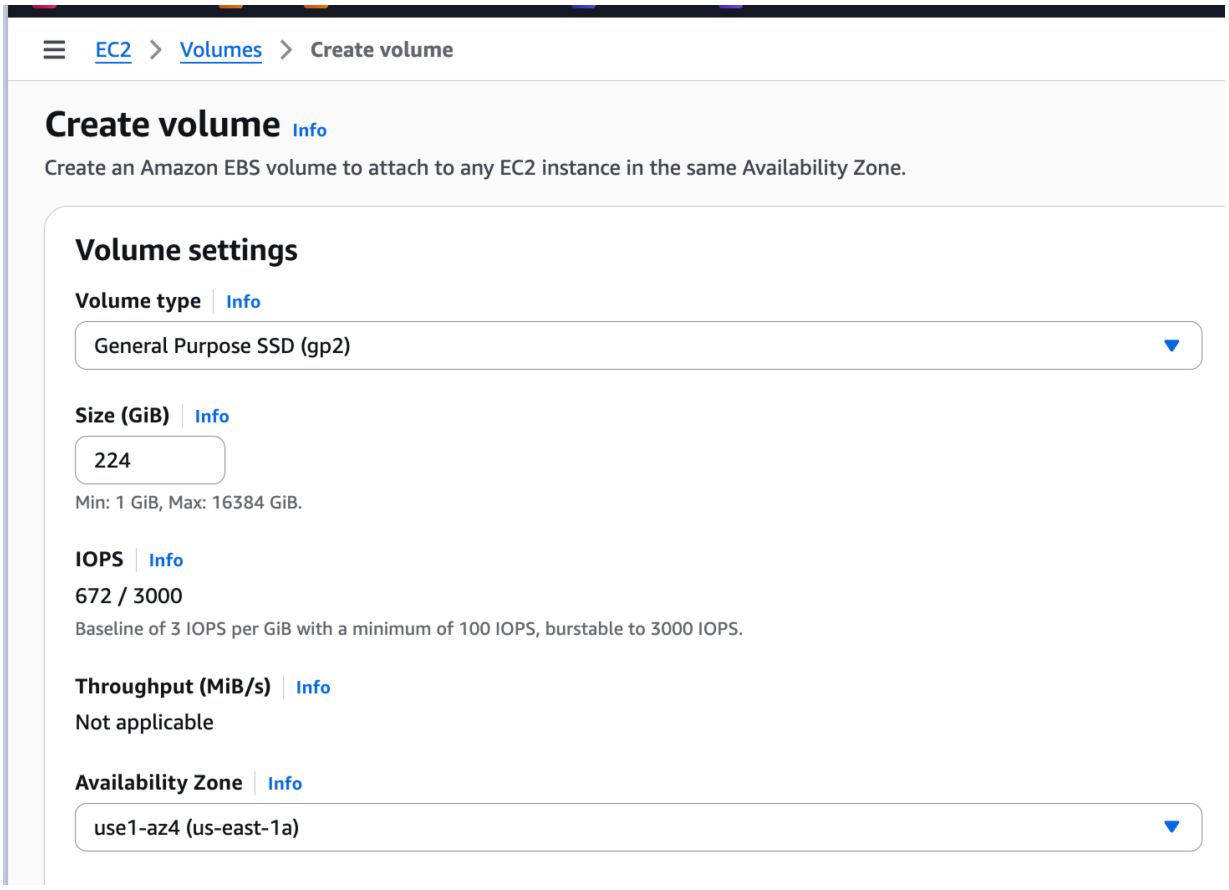

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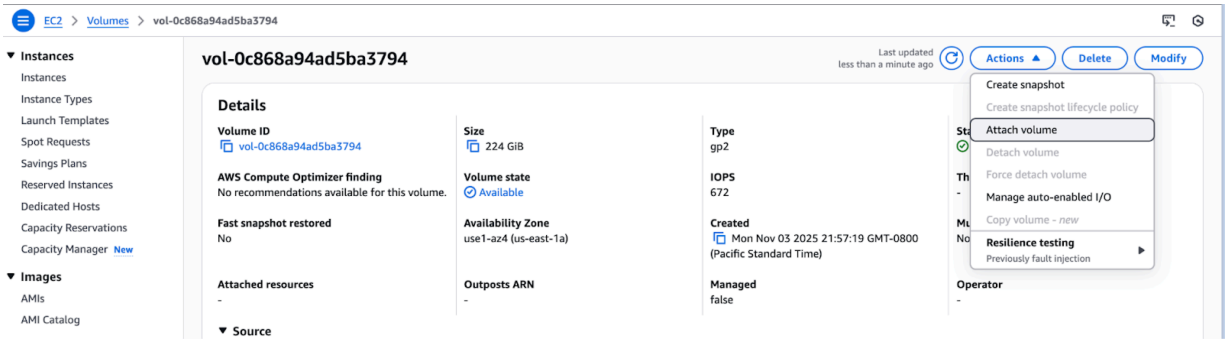
STEP 4 | Attach the created volume to the Panorama instance.



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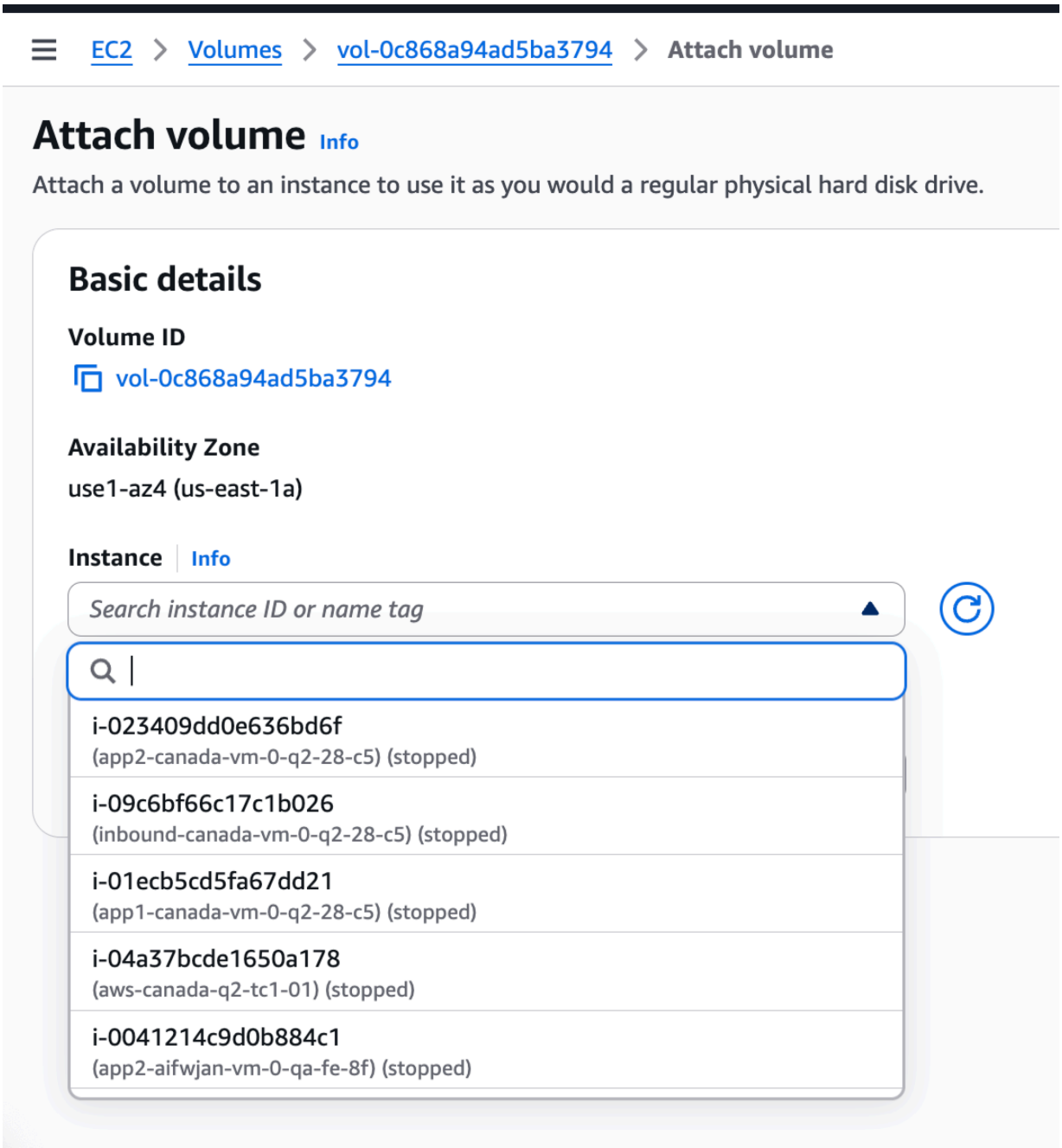

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STEP 5 | Confirm the instance has the new 224GiB volume attached.

1. From your EC2 instance, select **Instance** and then the Panorama instance that has the increased disk size.
2. Select the Storage tab and confirm that the disk with increased size is added.

STEP 6 | Start the Panorama Instance.

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STEP 7 | After the Panorama VM is up and running, confirm the disk is identified as a data disk using the command:

```
show system disk details
```

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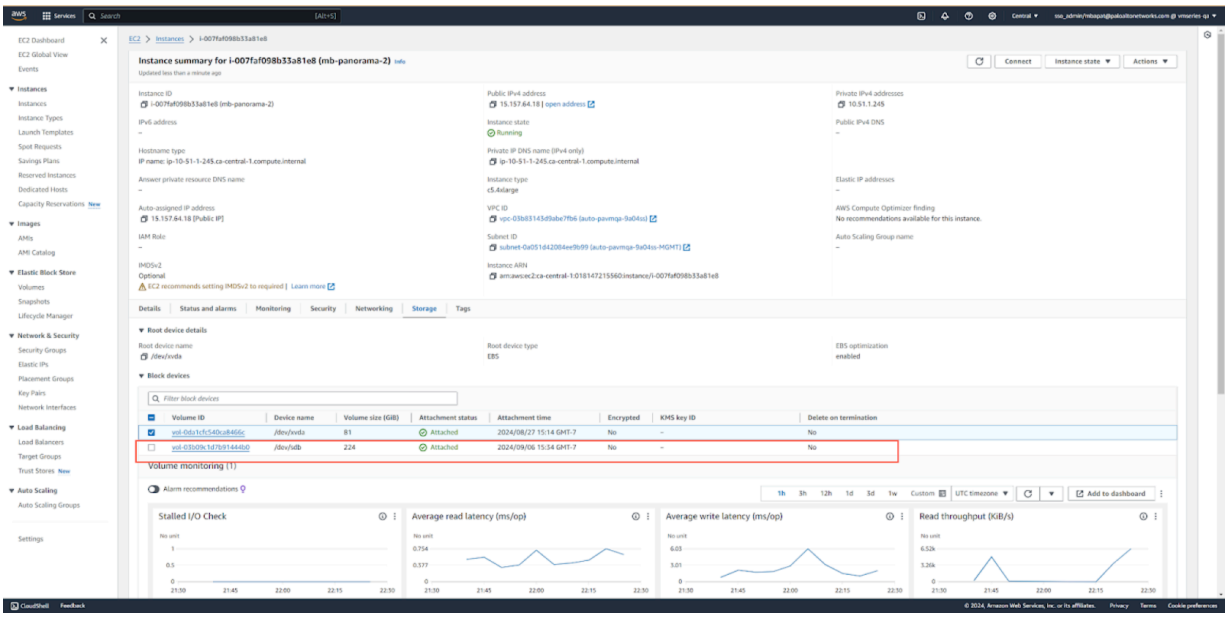

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```
Shows new 224 GB disk as below
Name   : nvme1n1
State  : Present
Size   : 229376 MB
Status : Unavailable
Reason : Admin disabled
```

STEP 8 | Execute the command `request system clone-system-disk target <new-disk>`

```
admin@Panorama> request system clone-system-disk target
  nvme1n1   Legacy logging disk
  <value>   Select target disk
admin@Panorama> request system clone-system-disk target nvme1n1
Executing this command will copy data from the system disk and
 delete all data on the target disk. The system will reboot and
 start the disk cloning procedure. The command will take at least
 20 minutes to complete. Progress information will be available
 only via the Console. Do you wish to continue? (y or n)

Broadcast message from root (Fri Sep  6 16:38:47 2024):

The system is going down for reboot NOW!
```

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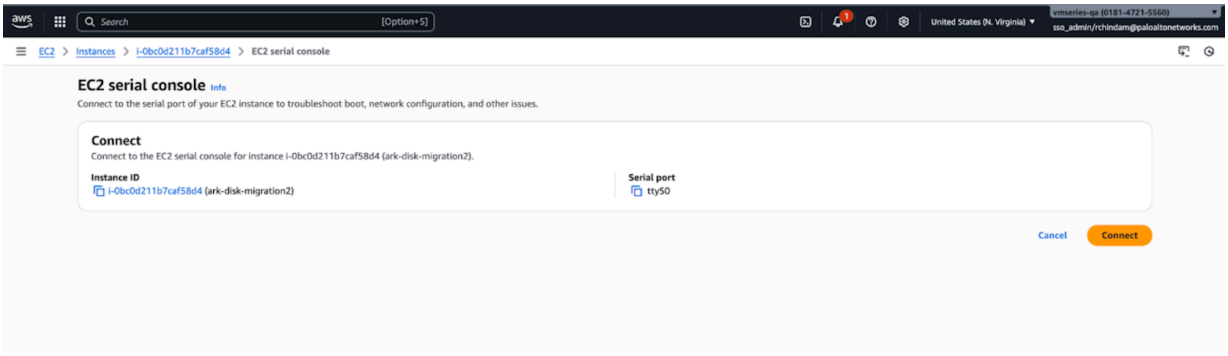
STEP 9 | Select **Instance state > Monitor and troubleshoot > EC2 serial console** and **Connect** to monitor the EC2 serial console and check the status of disk migration. Disk migration may take approximately 30 minutes. Confirm successful completion of the process.



The following image shows an example of successful completion of the disk migration.

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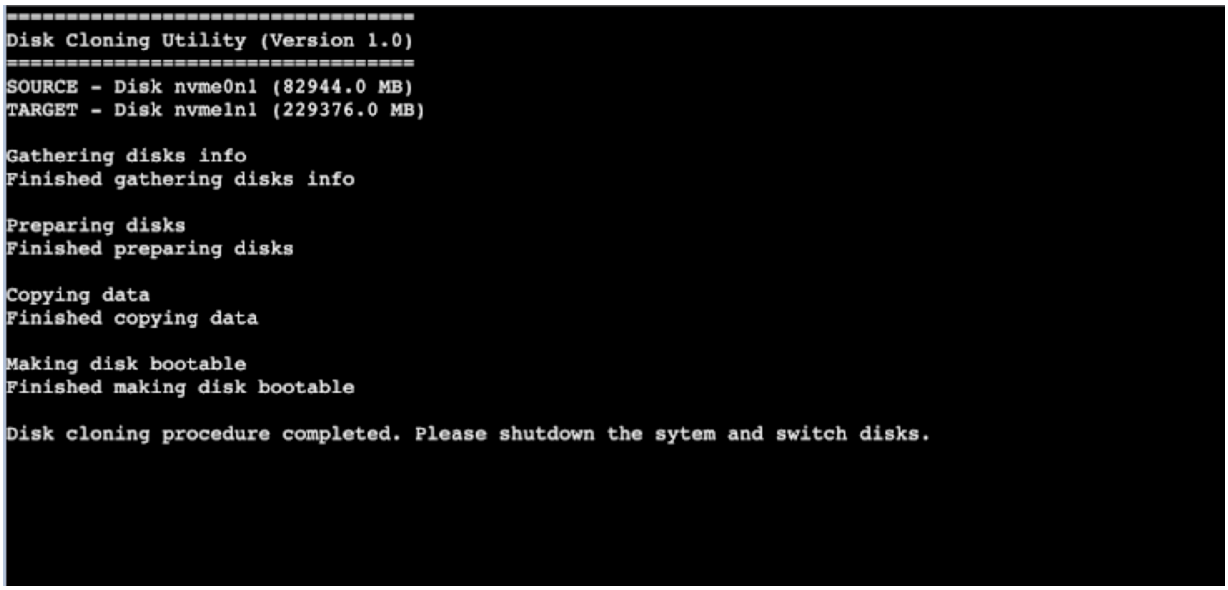

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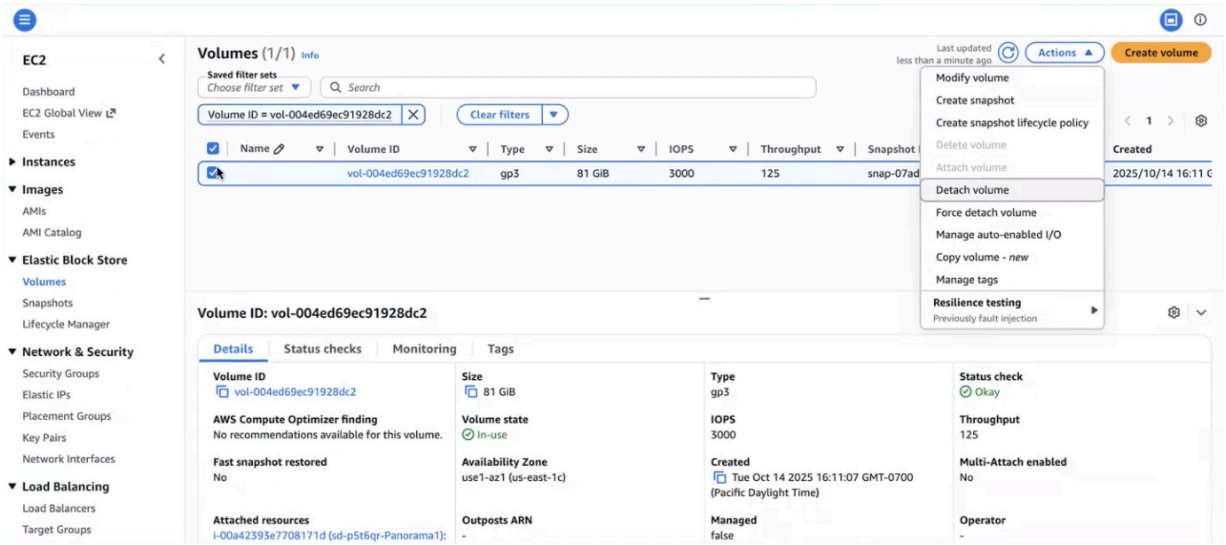
STEP 10 | Stop the Panorama instance after the disk migration is complete.

STEP 11 | Detach both disks (81GB and 224GB) individually. To detach the disks, select the **Volume ID** of the disk you need to detach, then **Actions** > **Detach volume**.



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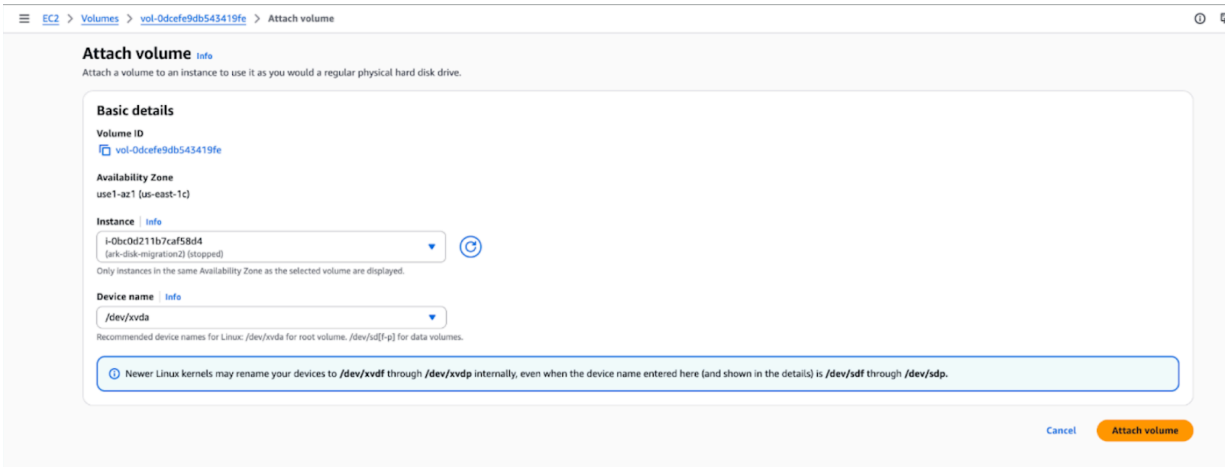
STEP 12 | Attach the new 224GB disk as the root volume (/dev/xvdp) to the Panorama instance. To attach the disk, select the **Volume ID** of the disk you need to attach, then **Actions** > **Attach volume**.



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STEP 13 | Start the instance.

STEP 14 | After the Panorama VM is running, execute the command `show system disk-partition` to confirm it displays root volumes of 224GB.

```
panadmin@Panorama> show system disk-partition

Disk /dev/nvme0n1: 224 GiB, 240518168576 bytes, 469762048 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 4096 bytes / 4096 bytes
Disklabel type: dos
Disk identifier: 0x05ef63be

Device         Boot     Start       End   Sectors   Size Id Type
/dev/nvme0n1p1             63   4016249   4016187   1.9G 83 Linux
/dev/nvme0n1p2        4016250  36033794  32017545  15.3G 83 Linux
/dev/nvme0n1p3       36033795  68051339  32017545  15.3G 83 Linux
/dev/nvme0n1p4       68051340 468857024 400805685 191.1G  f W95
 Ext'd (LBA)
/dev/nvme0n1p5       68051403 228074804 160023402  76.3G 83 Linux
/dev/nvme0n1p6      228074868 276093089  48018222  22.9G 83 Linux
/dev/nvme0n1p7      276093153 276109154     16002   7.8M 83 Linux
/dev/nvme0n1p8      276109218 468857024 192747807  91.9G 83 Linux

Disk /dev/loop0: 36.3 GiB, 38948306944 bytes, 76070912 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

Increase the System Disk for Panorama on Azure

For the minimum resource requirements for the Panorama virtual appliance, see Setup Prerequisites for the Panorama Virtual Appliance.

1. Save and Export Panorama and Firewall Configurations.

   Save and export your Panorama and firewall configuration to ensure you can recover Panorama if you encounter any issues.
2. *Power off the Panorama virtual appliance.*

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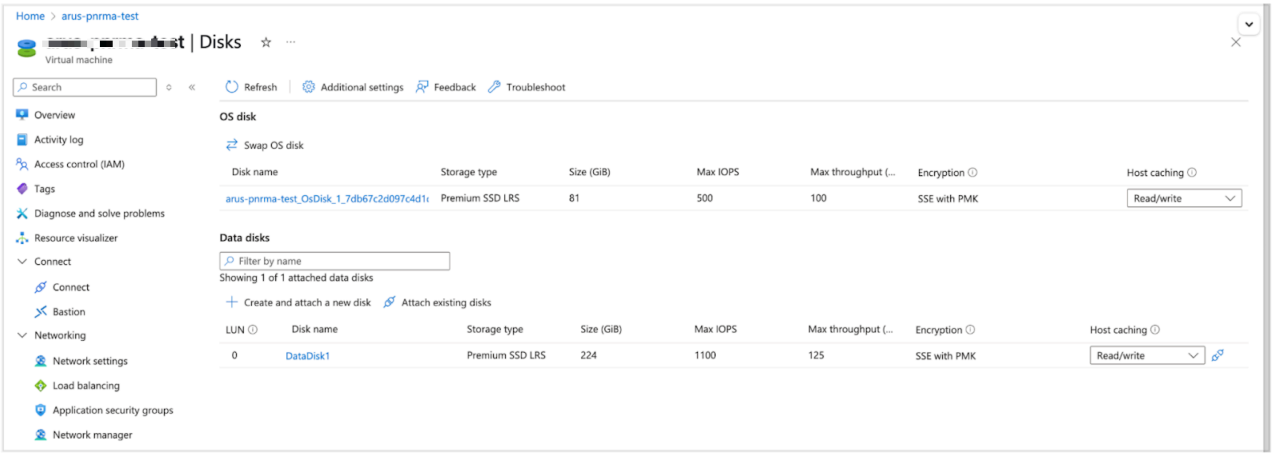
STEP 1 | Add a 224GB system disk to replace the default 81GB system disk.

1. Log in to the Azure Console.
2. Expand **Settings** and click **Disks** > **Create and attach a new disk**.



3. Enter a **Disk name** to the new disk you are creating.
4. Select the **Storage type** and **Encryption** similar to the ones you had selected for the 81 GB disk.
5. Enter **224** as the **Size (GB)**.
6. Select **Read/write** option for **Host caching**.
7. Click **Apply**.

STEP 2 | Check the disk size in the Panorama.

```
panorama@arus-pnrma-test>
panorama@arus-pnrma-test> show system disk
> disk             Disks information
> disk-partition   Show disk partion information
> disk-space       Report filesystem disk space usage

panorama@arus-pnrma-test> show system disk details


Name   : sdc
State  : Present
Size   : 229376 MB
Status : Unavailable
Reason : Admin disabled


panorama@arus-pnrma-test>
```

```
[root@arus-pnrma-test ~]# lsblk
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
loop0    7:0    0 12.7G  0 loop /opt/mongobuffer
sda      8:0    0   81G  0 disk
|-sda1   8:1    0 1003M  0 part
|-sda2   8:2    0    8G  0 part /
|-sda3   8:3    0    8G  0 part
|-sda4   8:4    0    1K  0 part
|-sda5   8:5    0   24G  0 part /opt/pancfg
|-sda6   8:6    0    6G  0 part /opt/panrepo
|-sda7   8:7    0    2G  0 part
`-sda8   8:8    0   32G  0 part /opt/panlogs
sdb      8:16   0  128G  0 disk
`-sdb1   8:17   0  128G  0 part
sdc      8:32   0  224G  0 disk
[root@arus-pnrma-test ~]#
```

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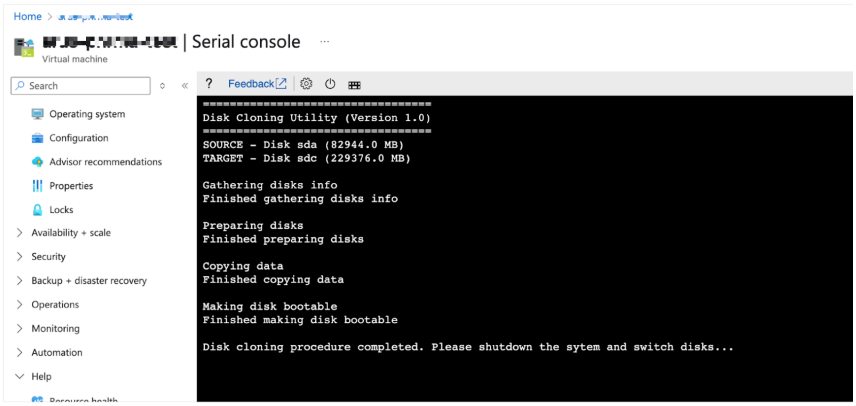
STEP 3 | Initiate disk cloning.

1. Initiate disk cloning by executing the following command:

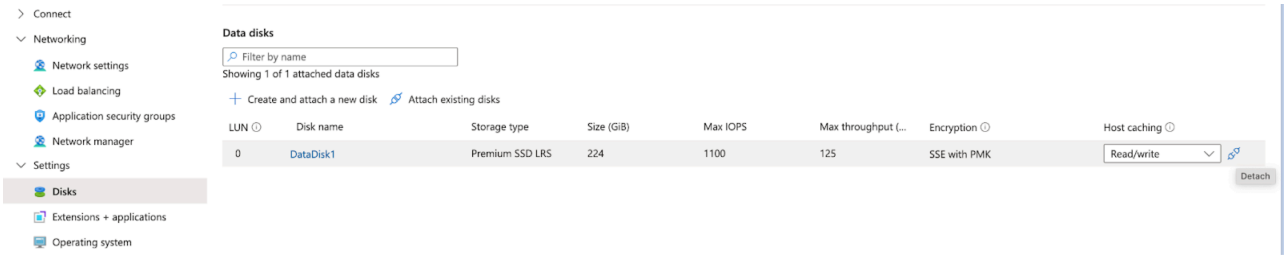
   ```
   request system clone-system-disk target <disk name>
   ```

2. Verify the ongoing cloning status from the console.



3. Shutdown the Panorama VM from the Azure console.

STEP 4 | Detach the data disk from the Panorama VM to which it was added initially, and convert the data disk to OS disk using powershell commands.




```
$managedDisk = Get-AzDisk -DiskName pnrma_new_data_disk -
ResourceGroupName arus-cpt-pnrma-rg
# Create disk configuration for the copy
$diskConfig = New-AzDiskConfig -SourceResourceId $managedDisk.Id -
Location $managedDisk.Location -CreateOption Copy -OsType Linux
New-AzDisk  -Disk $diskConfig -DiskName  pnrma_new_data_os_disk -
ResourceGroupName arus-cpt-pnrma-rg

pnrma_new_data_disk     -  name of the Data Disk that was initially
 created in the previous step
arus-cpt-pnrma-rg            -  name of the resource group where
 disk is present
```

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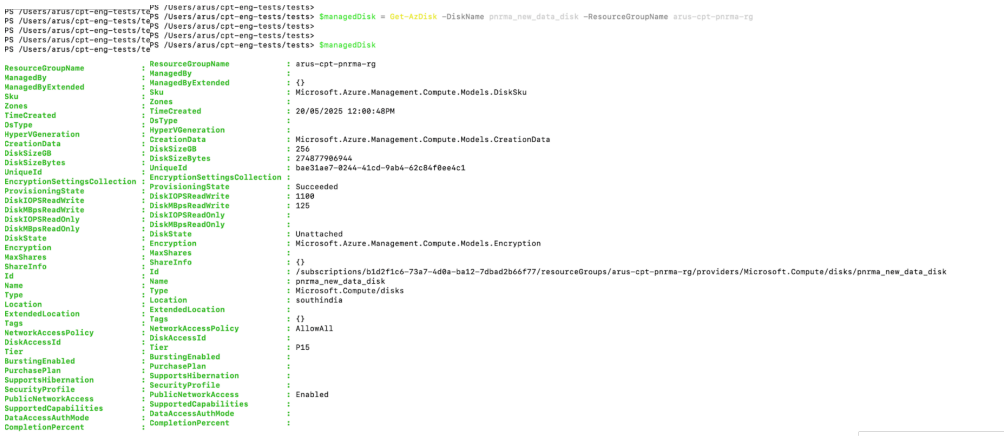
```
pnrma_new_data_os_disk -  name of the new os disk that will be
 created
```

```
$managedDisk = Get-AzDisk -DiskName pnrma_new_data_disk -
ResourceGroupName arus-cpt-pnrma-rg
```




```
$diskConfig = New-AzDiskConfig -SourceResourceId $managedDisk.Id -
Location $managedDisk.Location -CreateOption
```

```
New-AzDisk -Disk $diskConfig -DiskName pnrma_new_data_os_disk -
ResourceGroupName arus-cpt-pnrma-rg
```

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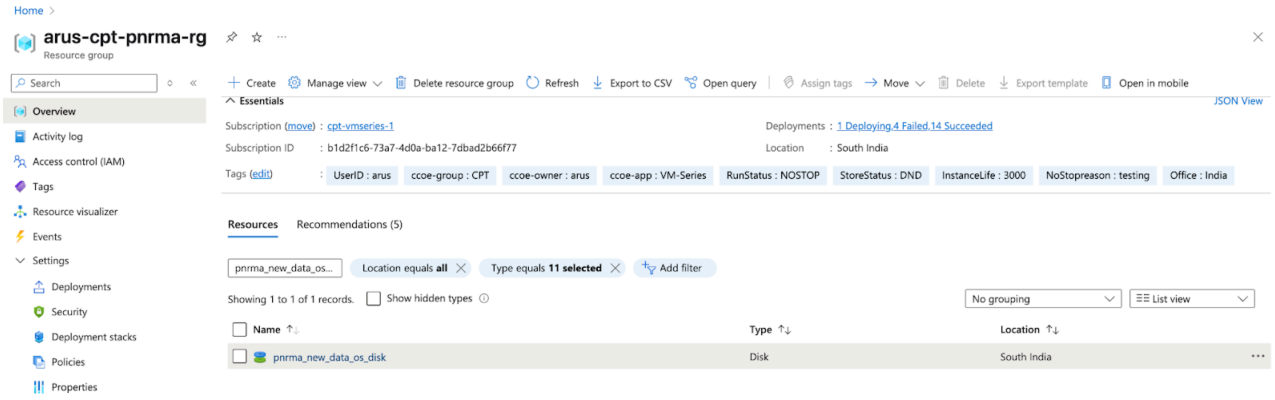
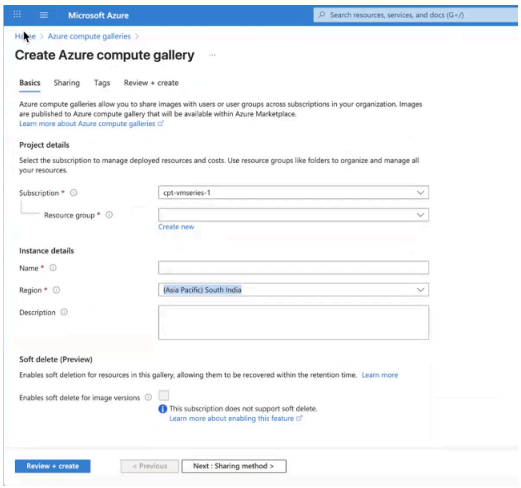
STEP 5 | Verify the OS disk you created.



STEP 6 | Create an Azure compute gallery.

1. From the Azure Console home page, click **Azure compute galleries**.
2. In the Azure compute gallery page, Click **Create**.
3. Select a **Resource group**.
4. You may retain the default settings for **Sharing** and **Tags** tabs.
5. Click **Review+create**.



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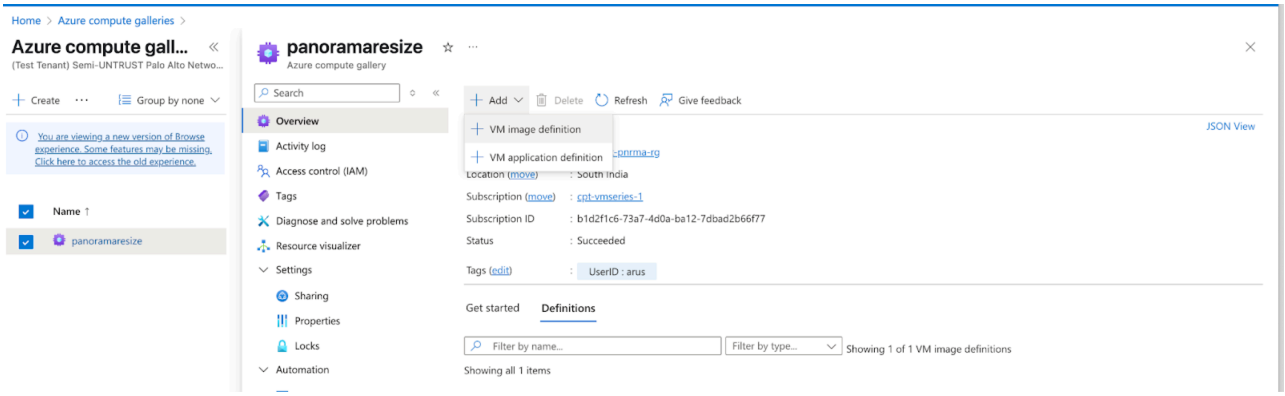
STEP 7 | Add a VM image definition from the Azure compute gallery.



1. Add the following details:
   - Subscription
   - Resource Group
   - Region
   - Target Azure Compute Gallery
   - VM image definition name
   - OS type
   - Security type
   - VM generation
   - Higher storage performance with NVMe
   - Accelerated networking
   - VM architecture
   - Hibernation supported
   - OS state
   - Publisher - Enter publisher as paloaltonetworks
   - Offer - Enter offer as panorama
   - SKU - Enter SKU as byol

   *Ensure that you select **Specialized** for the OS state to avoid VM deployment failures.*

   - *Ensure that the region, publisher, SKU, offer, and VM-Deployment sections are similar to that of the existing Panorama. You can view this information in the Source image details section of the VM Overview page in Azure Portal page.*
   - *The OS disk must be the one you created in step 5. Ensure that you select this OS disk under the version tab while creating a VM definition.*
   - *The CPU and memory values must be the same as that of the Panorama.*

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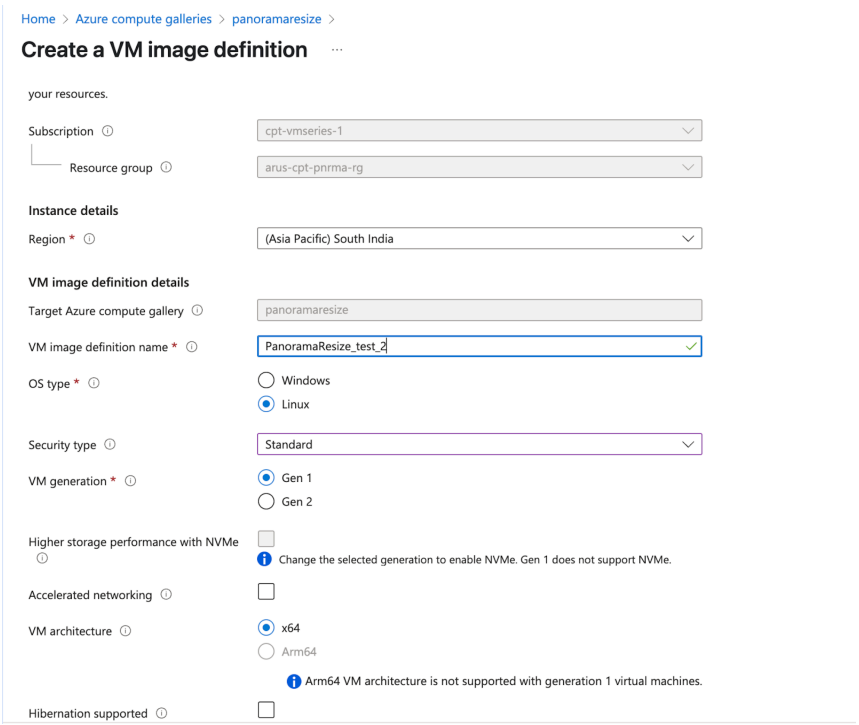

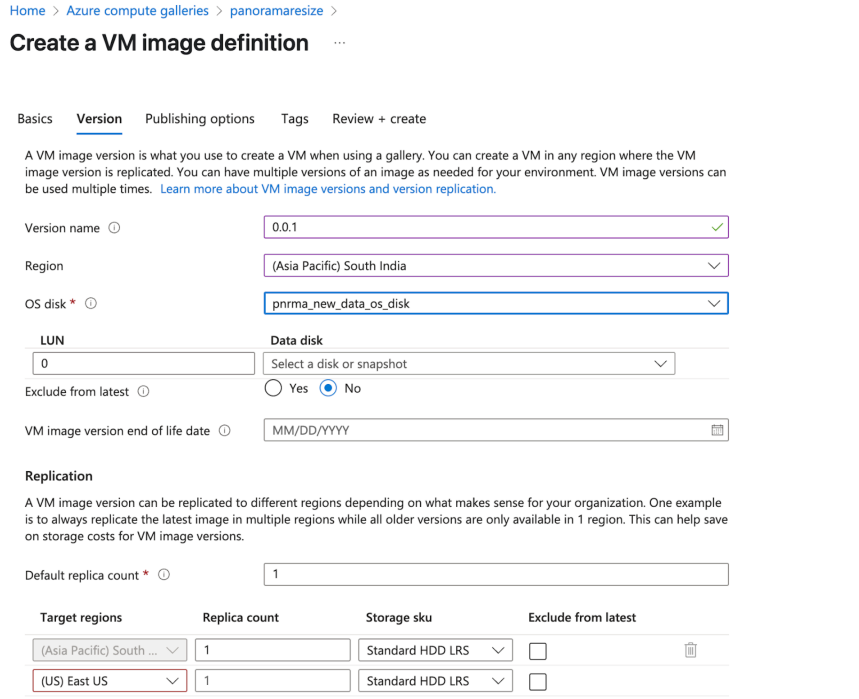

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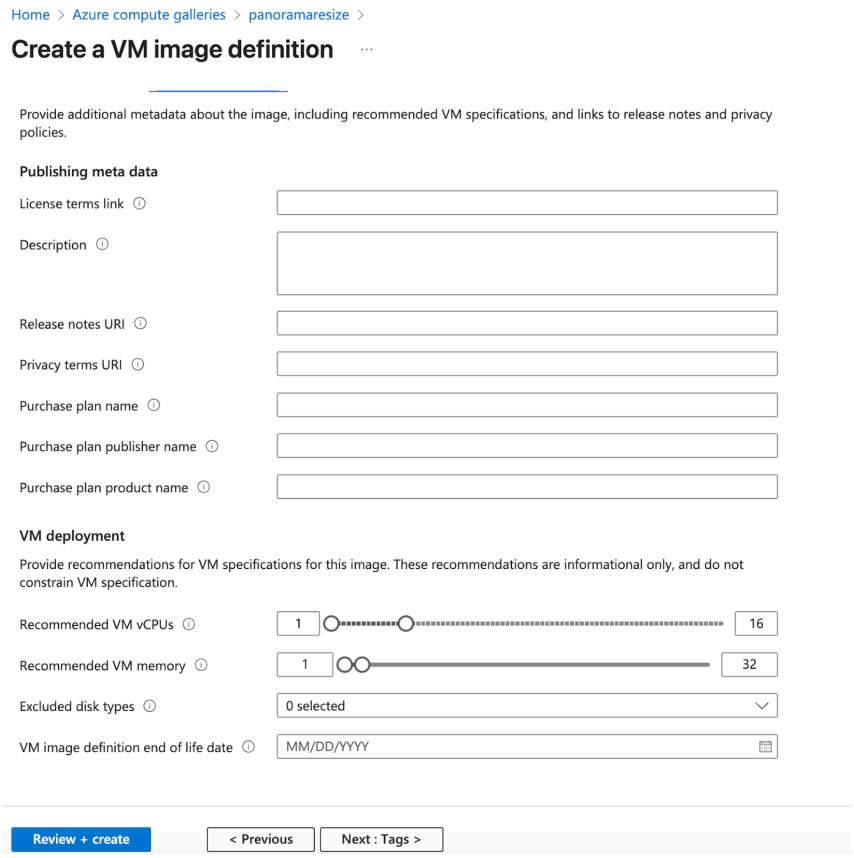

2. Click **Review+Create** to create the Azure compute gallery.

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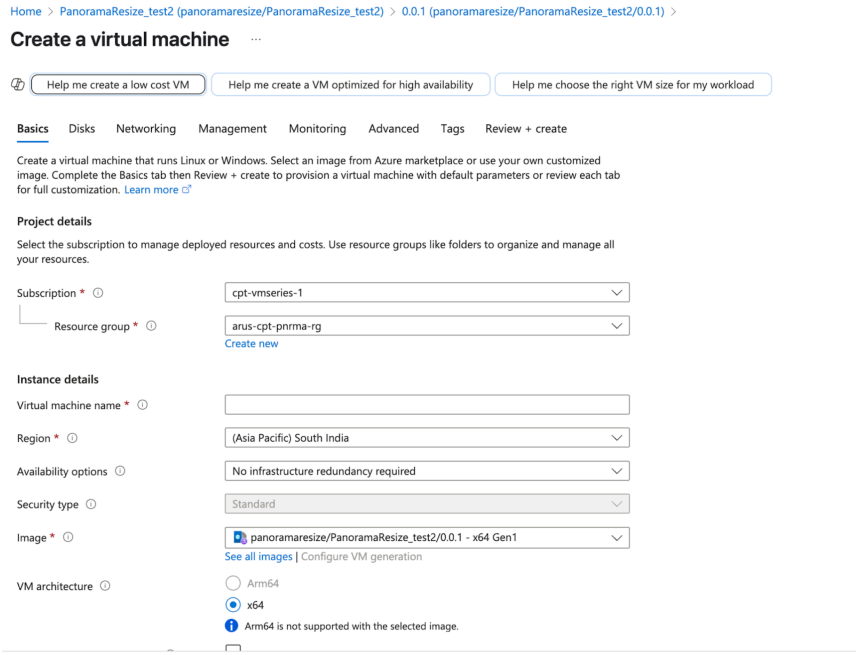
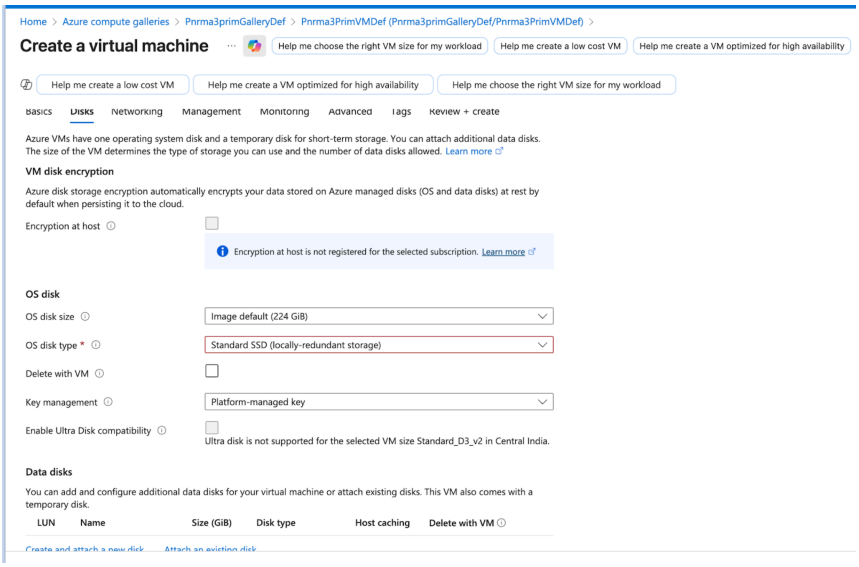
STEP 8 | Create a VM from the Azure compute gallery.

1. Click the VM definition and click **Create VM** to create a VM from the Azure compute gallery. This will create a new Panorama instance which will boot from the VM definition.
1. Configure the **Basics** settings.

   1. *When choosing the image, select the relevant Gallery Image definition that was created in previous steps.*
   2. *Ensure that the Azure infrastructure-related options align with the existing Panorama configuration.*



2. Add the Disks settings and ensure that the OS Disk selected is 224GB or more.



3. Add the Networking settings similar to that of the existing Panorama. This will bring up a new panorama with the resized OS disk details.

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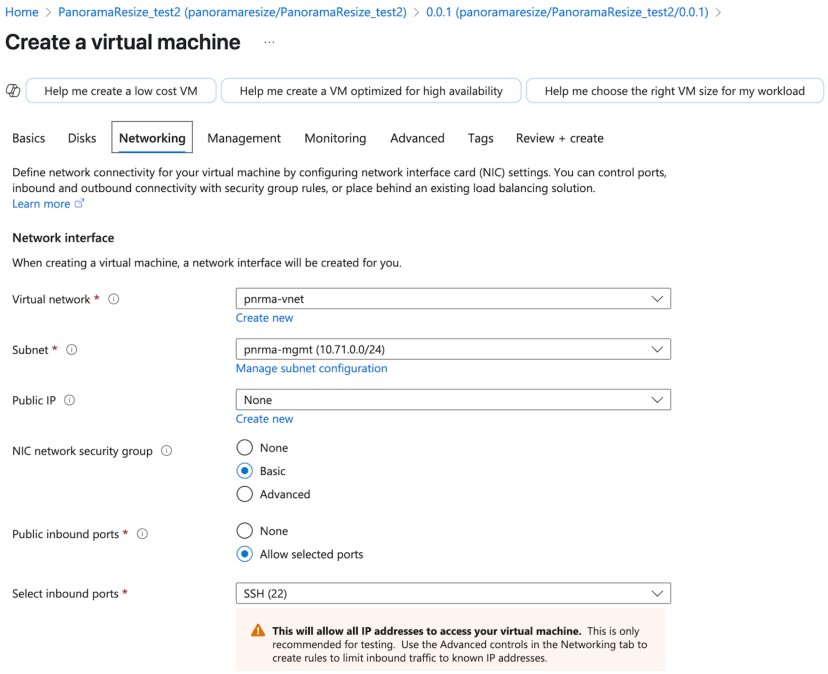

STEP 9 | Execute the following command to verify if the panrepo folder size has increased:

```
Show system disk space
```

```
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        16G  4.9G  9.6G  34% /
none             32G  116K   32G   1% /dev
/dev/sda5        76G   25G   48G  35% /opt/pancfg
/dev/sda6        23G   13G  8.6G  61% /opt/panrepo
tmpfs            32G  360M   32G   2% /dev/shm
tmpfs            12M   40K   12M   1% /opt/pancfg/mgmt/ssl/private
cgroup_root      32G     0   32G   0% /cgroup
/dev/sda8        91G  8.3G   78G  10% /opt/panlogs
/dev/loop0      9.8G   28K  9.3G   1% /opt/logbuffer
/dev/sdc1       1.8T  958G  774G  56% /opt/panlogs/ld1
```

STEP 10 | The new Panorama will have a different private IP address. You need to configure it with the same IP address as the existing Panorama. To make the existing Panorama's IP address available, assign a temporary, unused IP address to the existing Panorama. The existing Panorama's IP address can only be changed when the VM is shut down. From the Azure portal, assign the existing Panorama's IP address to the new Panorama's interface. Once

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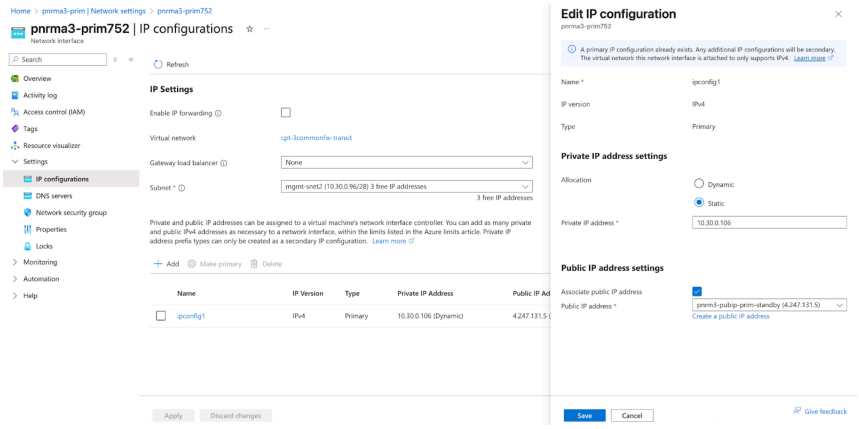
the IP addresses are swapped, the Network Interface Card (NIC) will reset, and the new Panorama will then have the IP address of the existing Panorama.

To change private/public IP of the Panorama, click **PanoramaVM** > **Network Settings** > **IP configurations**. Select the network for which you need to change the IP address and click **Edit**.



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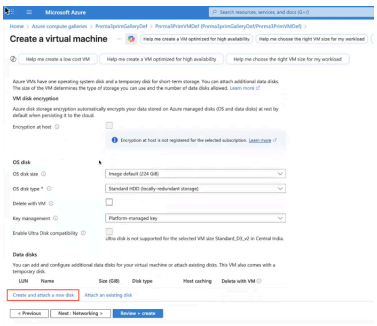
STEP 11 | If your existing Panorama had any data disks, detach them and then reattach them to the new Panorama. Finally, boot the Panorama. The new Panorama should now have all devices connected and function identically to the existing Panorama.



For an Active-Passive Panorama setup, apply these steps first to the Passive Panorama, then repeat them for the Active Panorama.

In case of a Cloud NGFW set-up, the newly deployed Panorama instance will be assigned a different Private IP address. It is necessary to configure this new instance to use the same IP address as the existing Panorama.

To make the existing Panorama IP available to the new Panorama instance, you must first assign a temporary, unused IP address to the current Panorama. The existing Panorama IP address can only be changed while the VM is shut down.

Assign the existing Panorama IP to the new Panorama interface from the Azure portal. Interchanging the IP addresses will cause the NIC to reset, resulting in the new Panorama instance adopting the IP address of the existing Panorama instance. After replacing the existing Panorama, complete the following steps:

1. *If the original Panorama had any data disks, detach them and reattach them to the new Panorama instance.*
2. *Start the new Panorama instance. The new Panorama should now be fully functional, with all devices connected, and operating the same way as the original Panorama.*
3. *After the Panorama is back online, you must re-fetch the device certificate and licenses.*

Complete the Panorama Virtual Appliance Setup

After you Perform Initial Configuration of the Panorama Virtual Appliance, continue with the following tasks for additional configuration:

- Activate a Panorama Support License
- Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is Internet-connected
- Install Content and Software Updates for Panorama
- Access and Navigate Panorama Management Interfaces
- Configure Administrative Access to Panorama
- Manage Firewalls

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Convert Your Panorama Virtual Appliance

You can convert your evaluation Panorama™ virtual appliance to a production Panorama virtual appliance to preserve its existing configuration and begin leveraging the management platform.

If you are utilizing Enterprise License Agreement (ELA) licensing, you can convert an existing production Panorama virtual appliance to leverage the benefits of ELA licensing.

- Convert Your Evaluation Panorama to a Production Panorama with Local Log Collector
- Convert Your Evaluation Panorama to a Production Panorama without Local Log Collector
- Convert Your Evaluation Panorama to VM-Flex Licensing with Local Log Collector
- Convert Your Evaluation Panorama to VM-Flex Licensing without Local Log Collector
- Convert Your Production Panorama to an ELA Panorama

Convert Your Evaluation Panorama to a Production Panorama with Local Log Collector

If you have an evaluation Panorama™ virtual appliance in Panorama mode configured with a local Log Collector, you can convert it to a production Panorama by migrating the configuration from the evaluation Panorama to the production Panorama and modifying as needed.

Logs ingested by the Log Collector on a Panorama virtual appliance cannot be migrated.

If you need to maintain access to the logs stored on your evaluation Panorama virtual appliance, after you migrate the evaluation Panorama configuration to the production Panorama, keep your evaluation Panorama powered on to access the logs locally for the remainder of the evaluation license lifetime. Adding the evaluation Panorama to the production Panorama as a managed collector is not supported.

STEP 1 | Plan the migration.

- ☐ Upgrade the software on the Panorama virtual appliance before you convert your evaluation Panorama virtual appliance to a production Panorama virtual appliance. Review the Compatibility Matrix for the minimum PAN-OS version required for your hypervisor. For important details about software versions, see Panorama, Log Collector, Firewall, and WildFire Version Compatibility.
- ☐ Schedule a maintenance window for the migration.

STEP 2 | Set up your production Panorama virtual appliance.

1. Set Up the Panorama Virtual Appliance.
2. Register the Panorama virtual appliance with the Palo Alto Networks Customer Support Portal (CSP).

   The Panorama serial number and authorization code are found in the Order Summary email from Palo Alto Networks.
3. Install Content and Software Updates for Panorama.

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STEP 3 | Activate the device management license on the Palo Alto Networks Custer Support Portal (CSP) for the production Panorama virtual appliance.

1. Log in to the Palo Alto Networks CSP.
2. Select **Assets** > **Devices** and locate your Panorama virtual appliance.
3. In the **Action** column, click the pencil icon to edit the device licenses.
4. Select **Activate Auth-Code** and enter the **Authorization Code**.
5. Select **Agree and Submit** to activate the device management license.

STEP 4 | Export the Panorama configuration from the evaluation Panorama virtual appliance.

1. Log in to the Panorama Web Interface.
2. Select **Panorama** > **Setup** > **Operations**.
3. Click **Export named Panorama configuration snapshot**, select `running-config.xml` and click **OK**. Panorama exports the configuration to your client system as an XML file.
4. Locate the `running-config.xml` file you exported and rename the XML file. This is required to import the configuration as Panorama does not support importing an XML file with the name `running-config.xml`.

STEP 5 | Load the Panorama configuration snapshot that you exported from the evaluation Panorama virtual appliance into the production Panorama virtual appliance.

1. Log in to the Panorama Web Interface of the production Panorama virtual appliance.
2. Select **Panorama** > **Setup** > **Operations**.
3. Click **Import named Panorama configuration snapshot**, **Browse** to the Panorama configuration file you exported from the Panorama virtual appliance, and click **OK**.
4. Click **Load named Panorama configuration snapshot**, select the **Name** of the configuration you just imported, leave the **Decryption Key** blank (empty), and click **OK**. Panorama overwrites its current candidate configuration with the loaded configuration. Panorama displays any errors that occur when loading the configuration file.
5. If errors occurred, save them to a local file. Resolve each error to ensure the migrated configuration is valid.

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STEP 6 | Modify the configuration on the production Panorama virtual appliance.

1. Select **Panorama** > **Setup** > **Management**.
2. Edit the General Settings, modify the **Hostname**, and click **OK**.
3. Edit the Management Interface Settings to configure the management IP address and click **OK**.

   The most efficient approach is to assign a new IP address to the evaluation Panorama virtual appliance and reuse its old IP address for the production Panorama virtual appliance. This ensures that the evaluation Panorama virtual appliance remains accessible and that firewalls can point to the production Panorama virtual appliance without you reconfiguring the Panorama IP address on each firewall.
4. Remove the Log Collector configuration imported from the evaluation Panorama.
   1. Select **Panorama** > **Collector Group** and **Delete** all configured collector groups.
   2. Select **Panorama** > **Managed Collectors** and **Delete** all configured Log Collectors.
5. Select **Commit** > **Commit to Panorama** and **Commit** your changes to the Panorama configuration.

STEP 7 | Configure your Log Collectors and collector groups.

You must add the managed collectors, collector group configuration, and log forwarding configurations you deleted in the previous step, as well as add the local Log Collector.

1. Configure a Managed Collector.
2. Configure a Collector Group.
3. Configure Log Forwarding to Panorama.

STEP 8 | Verify that the support and device management licenses are successfully activated.

1. Select **Panorama** > **Licenses** and **Retrieve license keys from license server**.
2. Verify the **Device Management License** displays the correct number of devices.
3. Select **Panorama** > **Support** and verify that the correct support **Level** and **Expiry Date** are displayed.

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STEP 9 | Synchronize the production Panorama virtual appliance with the firewalls to resume firewall management.

Complete this step during a maintenance window to minimize network disruption.

1. On the production Panorama virtual appliance, select **Panorama** > **Managed Devices** and verify that the Device State column displays **Connected** for the firewalls.

   At this point, the Shared Policy (device groups) and Template columns display **Out of sync** for the firewalls.
2. Push your changes to device groups and templates:
   1. Select **Commit** > **Push to Devices** and **Edit Selections**.
   2. Select **Device Groups**, select every device group, **Include Device and Network Templates**, and click **OK**.
   3. **Push** your changes.
3. In the **Panorama** > **Managed Devices** page, verify that the Shared Policy and Template columns display **In sync** for the firewalls.

Convert Your Evaluation Panorama to a Production Panorama without Local Log Collector

Change the serial number of your evaluation Panorama virtual appliance in Management Only mode or in Panorama mode with no local Log Collector configured to convert it to a production Panorama virtual appliance.

If a local Log Collector is configured, see Convert Your Evaluation Panorama to a Production Panorama with Local Log Collector.

STEP 1 | Log in to the Panorama web interface.

STEP 2 | Select **Panorama** > **Setup** > **Management** and edit the General Settings.

STEP 3 | Enter the **Serial Number** provided by Palo Alto Networks.

The Panorama serial number and authorization code are obtained from the deployment profile you created in the previous step.

STEP 4 | Click **OK**.

STEP 5 | Select **Commit** and **Commit to Panorama**.

STEP 6 | Restart management server on the Panorama virtual appliance.

1. Log in to the Panorama CLI.
2. Restart the management server.

```
admin> debug software restart process management-server
```

All administrators are logged out of the Panorama web interface and CLI when you restart the management server.

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STEP 7 | Verify that the support and device management licenses are successfully activated.

1. Log in to the Panorama web interface.
2. Select **Panorama** > **Licenses** and **Retrieve license keys from license server**.
3. Verify the **Device Management License** displays the correct number of devices.
4. Select **Panorama** > **Support** and verify that the correct support **Level** and **Expiry Date** are displayed.

STEP 8 | Synchronize the production Panorama virtual appliance with the firewalls to resume firewall management.

Complete this step during a maintenance window to minimize network disruption.

1. On the production Panorama virtual appliance, select **Panorama** > **Managed Devices** and verify that the Device State column displays **Connected** for the firewalls.

   At this point, the Shared Policy (device groups) and Template columns display **Out of sync** for the firewalls.
2. Push your changes to device groups and templates:
   1. Select **Commit** > **Push to Devices** and **Edit Selections**.
   2. Select **Device Groups**, select every device group, **Include Device and Network Templates**, and click **OK**.
   3. **Push** your changes.
3. In the **Panorama** > **Managed Devices** page, verify that the Shared Policy and Template columns display **In sync** for the firewalls.

Convert Your Evaluation Panorama to VM-Flex Licensing with Local Log Collector

If you have an evaluation Panorama™ virtual appliance in Panorama mode configured with a local Log Collector, you can convert it to a production Panorama with VM Flex licensing by migrating the configuration from the evaluation Panorama to the production Panorama and modifying as needed.

If a local Log Collector is not configured, see Convert Your Evaluation Panorama to VM-Flex Licensing without Local Log Collector.

Logs ingested by the Log Collector on a Panorama virtual appliance cannot be migrated.

If you need to maintain access to the logs stored on your evaluation Panorama virtual appliance, after you migrate the evaluation Panorama configuration to the production Panorama, keep your evaluation Panorama powered on to access the logs locally for the remainder of the evaluation license lifetime. Adding the evaluation Panorama to the production Panorama as a managed collector is not supported.

STEP 1 | Plan the migration.

- ☐ Upgrade the software on the Panorama virtual appliance before you convert your evaluation Panorama virtual appliance to a production Panorama virtual appliance. Review the Compatibility Matrix for the minimum PAN-OS version required for your hypervisor.

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For important details about software versions, see Panorama, Log Collector, Firewall, and WildFire Version Compatibility.

- ☐ Schedule a maintenance window for the migration.

STEP 2 | Obtain the Panorama serial number and auth code from your flexible VM-Series licensing deployment profile.

1. Log in to the Palo Alto Networks Customer Support Portal (CSP).
2. Create a deployment profile that enables a Panorama virtual appliance.
3. Provision Panorama to generate the a serial number for Panorama.
4. Copy the **Serial Number** and **Auth Code**.

STEP 3 | Set up your production Panorama virtual appliance.

1. Log in to the Palo Alto Networks CSP.
2. Set Up the Panorama Virtual Appliance.
3. Register the Panorama virtual appliance with the Palo Alto Networks Customer Support Portal (CSP).

   The Panorama serial number and authorization code you generated in the previous step.
4. Install Content and Software Updates for Panorama.

STEP 4 | Activate the device management license on the Palo Alto Networks CSP for the production Panorama virtual appliance.

1. Select **Assets** > **Devices** and locate your Panorama virtual appliance.
2. In the **Action** column, click the pencil icon to edit the device licenses.
3. Select **Activate Auth-Code** and enter the **Authorization Code**.
4. Select **Agree and Submit** to activate the device management license.

STEP 5 | Export the Panorama configuration from the evaluation Panorama virtual appliance.

1. Log in to the Panorama Web Interface.
2. Select **Panorama** > **Setup** > **Operations**.
3. Click **Export named Panorama configuration snapshot**, select `running-config.xml` and click **OK**. Panorama exports the configuration to your client system as an XML file.
4. Locate the `running-config.xml` file you exported and rename the XML file. This is required to import the configuration as Panorama does not support importing an XML file with the name `running-config.xml`.

STEP 6 | Load the Panorama configuration snapshot that you exported from the evaluation Panorama virtual appliance into the production Panorama virtual appliance.

1. Log in to the Panorama Web Interface of the production Panorama virtual appliance.
2. Select **Panorama** > **Setup** > **Operations**.
3. Click **Import named Panorama configuration snapshot**, **Browse** to the Panorama configuration file you exported from the Panorama virtual appliance, and click **OK**.
4. Click **Load named Panorama configuration snapshot**, select the **Name** of the configuration you just imported, leave the **Decryption Key** blank (empty), and click **OK**.

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Panorama overwrites its current candidate configuration with the loaded configuration. Panorama displays any errors that occur when loading the configuration file.

5. If errors occurred, save them to a local file. Resolve each error to ensure the migrated configuration is valid.

STEP 7 | Modify the configuration on the production Panorama virtual appliance.

1. Select **Panorama** > **Setup** > **Management**.
2. Edit the General Settings, modify the **Hostname**, and click **OK**.
3. Edit the Management Interface Settings to configure the management IP address and click **OK**.

   The most efficient approach is to assign a new IP address to the evaluation Panorama virtual appliance and reuse its old IP address for the production Panorama virtual appliance. This ensures that the evaluation Panorama virtual appliance remains accessible and that firewalls can point to the production Panorama virtual appliance without you reconfiguring the Panorama IP address on each firewall.
4. Remove the Log Collector configuration imported from the evaluation Panorama.
   1. Select **Panorama** > **Collector Group** and **Delete** all configured collector groups.
   2. Select **Panorama** > **Managed Collectors** and **Delete** all configured Log Collectors.
5. Select **Commit** > **Commit to Panorama** and **Commit** your changes to the Panorama configuration.

STEP 8 | Reconfigure your Log Collectors and collector groups.

You must add the managed collectors, collector group configuration, and log forwarding configurations you deleted in the previous step, as well as add the local Log Collector.

1. Configure a Managed Collector.
2. Configure a Collector Group.
3. Configure Log Forwarding to Panorama.

STEP 9 | Verify that the support and device management licenses are successfully activated.

1. Select **Panorama** > **Licenses** and **Retrieve license keys from license server**.
2. Verify the **Device Management License** displays the correct number of devices.
3. Select **Panorama** > **Support** and verify that the correct support **Level** and **Expiry Date** are displayed.

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STEP 10 | Synchronize the production Panorama virtual appliance with the firewalls to resume firewall management.

Complete this step during a maintenance window to minimize network disruption.

1. On the production Panorama virtual appliance, select **Panorama** > **Managed Devices** and verify that the Device State column displays **Connected** for the firewalls.

   At this point, the Shared Policy (device groups) and Template columns display **Out of sync** for the firewalls.
2. Push your changes to device groups and templates:
   1. Select **Commit** > **Push to Devices** and **Edit Selections**.
   2. Select **Device Groups**, select every device group, **Include Device and Network Templates**, and click **OK**.
   3. **Push** your changes.
3. In the **Panorama** > **Managed Devices** page, verify that the Shared Policy and Template columns display **In sync** for the firewalls.

Convert Your Evaluation Panorama to VM-Flex Licensing without Local Log Collector

Change the serial number of your evaluation Panorama virtual appliance in Management Only mode or in Panorama mode with no local Log Collector configured to convert it to a production Panorama virtual appliance.

If a local Log Collector is configured, see Convert Your Evaluation Panorama to VM-Flex Licensing with Local Log Collector.

STEP 1 | Obtain the Panorama serial number and auth code from your flexible VM-Series licensing deployment profile.

1. Log in to the Palo Alto Networks Customer Support Portal (CSP).
2. Create a deployment profile that enables a Panorama virtual appliance.
3. Provision Panorama to generate the a serial number for Panorama.
4. Copy the **Serial Number** and **Auth Code**.

STEP 2 | Log in to the Panorama web interface.

STEP 3 | Select **Panorama** > **Setup** > **Management** and edit the General Settings.

STEP 4 | Enter the **Serial Number** provided by Palo Alto Networks.

The Panorama serial number and authorization code you generated in the previous step.

STEP 5 | Click **OK**.

STEP 6 | Select **Commit** and **Commit to Panorama**.

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STEP 7 | Restart management server on the Panorama virtual appliance.

1. Log in to the Panorama CLI.
2. Restart the management server.

```
admin> debug software restart process management-server
```

All administrators are logged out of the Panorama web interface and CLI when you restart the management server.

STEP 8 | Verify that the support and device management licenses are successfully activated.

1. Log in to the Panorama web interface.
2. Select **Panorama** > **Licenses** and **Retrieve license keys from license server**.
3. Verify the **Device Management License** displays the correct number of devices.
4. Select **Panorama** > **Support** and verify that the correct support **Level** and **Expiry Date** are displayed.

STEP 9 | Synchronize the production Panorama virtual appliance with the firewalls to resume firewall management.

Complete this step during a maintenance window to minimize network disruption.

1. On the production Panorama virtual appliance, select **Panorama** > **Managed Devices** and verify that the Device State column displays **Connected** for the firewalls.

   At this point, the Shared Policy (device groups) and Template columns display **Out of sync** for the firewalls.
2. Push your changes to device groups and templates:
   1. Select **Commit** > **Push to Devices** and **Edit Selections**.
   2. Select **Device Groups**, select every device group, **Include Device and Network Templates**, and click **OK**.
   3. **Push** your changes.
3. In the **Panorama** > **Managed Devices** page, verify that the Shared Policy and Template columns display **In sync** for the firewalls.

Convert Your Production Panorama to an ELA Panorama

You can convert your production Panorama™ virtual appliance to continue leveraging your Panorama with the benefits of ELA licensing. To convert your production deployment, Panorama must have out-bound Internet access.

Converting your production Panorama to ELA licensing is supported in Management Only and Panorama mode with or without a local Log Collector configured. If your Panorama has a local Log Collector configured, you must submit a support ticket with Palo Alto Networks to convert your Panorama to ELA licensing.

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During conversion from a production Panorama to ELA licensing, do not change the Panorama serial number if a local Log Collector is configured.

The log on the local Log collector become inaccessible and other Log Collectors in the Collector Group may become inaccessible and no longer ingest logs if the serial number of a Log Collector is changed.

STEP 1 | Covert your Panorama to ELA licensing.

- **Panorama virtual appliance in Panorama mode with a local Log Collector.**

  Submit support ticket with Palo Alto Networks to convert your Panorama to ELA licensing. This is required in order to preserve all existing logs on the local Log Collector when converting a Panorama with a local Log Collector to ELA licensing. An example is provided

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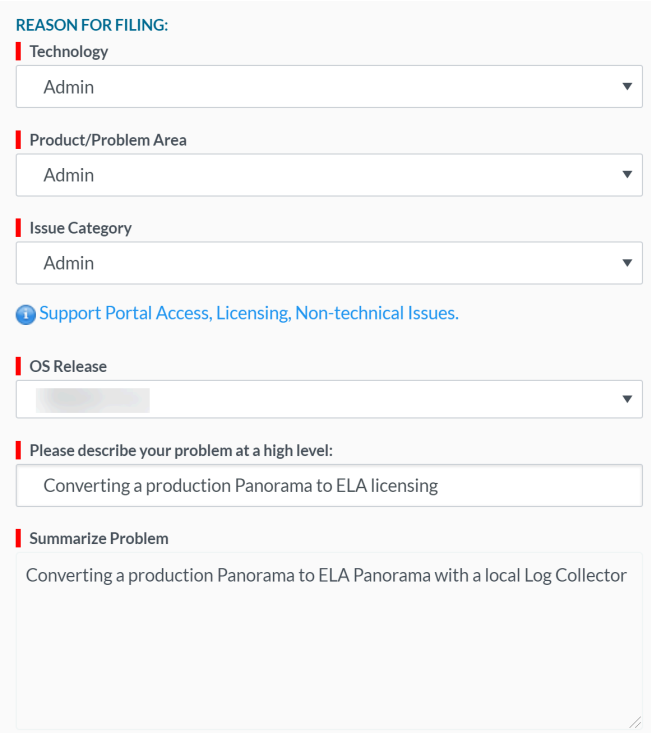
below to assist in filing the support ticket. Create the ticket exactly as displayed below, and select the **OS Release** your Panorama is running.

Continue to the next step only after Palo Alto Networks support successfully resolves your support ticket.



- **Panorama virtual appliance in Management Only mode or Panorama mode with no local Log Collector.**
  1. Generate a serial number from your ELA licensing pool.
     1. Log in to the Palo Alto Networks CSP.
     2. Select **Assets** > **VM-Series Auth-Codes** and locate your ELA licensing pool.
     3. In the Actions column, select **Panorama** and **Provision** a new serial number.

        Confirm the new serial number provision when prompted.
     4. Copy the newly provisioned serial number.
  2. Log in to the Panorama web interface.
  3. Select **Panorama** > **Setup** > **Management** and edit the General Settings.
  4. Enter the **Serial Number** you provisioned.
  5. Click **OK**.
  6. Select **Commit** and **Commit to Panorama**.

STEP 2 | Log in to the Panorama web interface if not already logged in.

STEP 3 | Select **Panorama** > **Licenses** and **Retrieve new licenses from the license server**.

STEP 4 | Verify that Panorama retrieved the new licenses as per your ELA agreement.

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STEP 5 | Verify that the support and device management licenses are successfully activated.

1. Select **Panorama** > **Licenses** and verify that the correct licenses are activated.
2. Select **Panorama** > **Support** and verify that the correct support **Level** and **Expiry Date** are displayed.

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Set Up the M-Series Appliance

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• M-series hardware appliance

The M-700, M-600, M-300, and M-200 appliances are high performance hardware appliances that you can deploy in Management Only mode (as Panorama management servers with no local log collection), Panorama mode (as Panorama management servers with local log collection) or in Log Collector mode (as Dedicated Log Collectors). The appliances provide multiple interfaces that you can assign to various Panorama services such as firewall management and log collection. Before setting up the appliance, consider how you can configure the interfaces to optimize security, enable network segmentation (in large-scale deployments), and load balance the traffic for Panorama services.

- M-Series Appliance Interfaces
- Perform Initial Configuration of the M-Series Appliance
- Perform Initial Configuration of an Air Gapped M-Series Appliance
- M-Series Setup Overview
- Set Up the M-Series Appliance as a Log Collector
- Increase Storage on the M-Series Appliance
- Configure Panorama to Use Multiple Interfaces

M-Series Appliance Interfaces

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• M-series hardware appliance

The Panorama M-700, M-600, M-300, and M-200 appliances have several interfaces for communicating with other systems such as managed firewalls and the client systems of Panorama administrators. Panorama communicates with these systems to perform various services, including managing devices (firewalls, Log Collectors, and WildFire appliances and appliance clusters), collecting logs, communicating with Collector Groups, deploying software and content updates to devices, and providing administrative access to Panorama. By default, Panorama uses its management (MGT) interface for all these services. However, you can improve security by reserving the MGT interface for administrative access and dedicating separate interfaces for the other services. In a large-scale network with multiple subnetworks and heavy log traffic, using multiple interfaces for device management and log collection also enables network segmentation and load balancing (see Configure Panorama to Use Multiple Interfaces).

When assigning Panorama services to various interfaces, keep in mind that only the MGT interface allows administrative access to Panorama for configuration and monitoring tasks. You can assign any interface to the other services when you Perform Initial Configuration of the M-

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Series Appliance. The M-Series Appliance Hardware Reference Guides explain where to attach cables for the interfaces.

The M-Series appliances, with the exception of the M-700, do not support Link Aggregation Control Protocol (LACP) for aggregating interfaces. The M-700 supports LACP for aggregate interface bond1.

Supported Interfaces

Interfaces can be used for device management, log collection, Collector Group communication, licensing and software updates. See Configure Panorama to Use Multiple Interfaces for more information on network segmentation.

Interface	Maximum Speed	M-700 Appliance	M-600 Appliance	M-300 Appliance	M-200 Appliance
Management (MGT)	1Gbps	Yes	Yes	No	Yes
	10Gbps	Yes	No	Yes	No
Ethernet 1 (Eth1)	1Gbps	Yes	Yes	No	Yes
	10Gbps	Yes	No	Yes	No
Ethernet 2 (Eth2)	1Gbps	No	Yes	No	Yes
	10Gbps	Yes	No	No	No
Ethernet 3 (Eth3)	1Gbps	No	Yes	No	Yes
	10Gbps	Yes	No	No	No
Ethernet 4 (Eth4)	1Gbps	N/A	Yes	N/A	N/A
Ethernet 5 (Eth5)	10Gbps	N/A	Yes	N/A	N/A
Ethernet 6 (Eth6)	10Gbps	N/A	Yes	N/A	N/A

The M-700 Appliance has two ports on its back panel labeled Ethernet 1/2 and Ethernet 1/3; however, the appliance uses a 20Gb aggregate software interface called ***bond1*** *instead of separate Eth2 and Eth3 subinterfaces.*

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Logging Rates

Review the logging rates for the all M-Series appliance models. To achieve the logging rates listed below, the M-Series appliance must be a single log collector in a collector group and you must install all the logging disks for your M-Series model.

Model Capacities and Features	M-700 Appliance	M-600 Appliance	M-300 Appliance	M-200 Appliance
Maximum Logging Rate for Panorama in Management Only mode	Local log storage is not supported			
Maximum Logging Rate for Panorama in Panorama Mode	36,500 logs/ second	25,000 logs/ second	16,500 logs/ second	10,000 logs/ second
Maximum Logging Rate for Panorama in Log Collector Mode	73,000 logs/ second	50,000 logs/ second	33,000 logs/ second	28,000 logs/ second
Maximum Log Storage on Appliance	48TB (12x8TB RAID disk)	48TB (12x8TB RAID disk)	16TB (4x8TB RAID disk)	16TB (4x8TB RAID disk)
Default Log Storage on Appliance	16TB (4x8TB RAID disks)	16TB (4x8TB RAID disks)	16TB (4x8TB RAID disks)	16TB (4x8TB RAID disks)
SSD Storage on Appliance (for logs that M-Series appliances generate)	240GB	240GB	240GB	240GB
NFS Attached Log Storage	Not available			

Perform Initial Configuration of the M-Series Appliance

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• M-series hardware appliance • Super user role

By default, Panorama has an IP address of 192.168.1.1 and a username/password of admin/ admin. For security reasons, you must change these settings before continuing with other

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configuration tasks. You must perform these initial configuration tasks either from the Management (MGT) interface or using a direct serial port connection to the console port on the M-700, M-600, M-500, M-300, or M-200 appliance.

If you are configuring an M-Series appliance in Log Collector mode with 10GB interfaces, you must complete this entire configuration procedure for the 10GB interfaces to display as Up.

STEP 1 | Gather the required interface and server information from your network administrator.

- Gather the IP address, netmask (for IPv4) or prefix length (for IPv6), and default gateway for each interface that you plan to configure (MGT, Eth1, Eth2, Eth3, Eth4, Eth5). Only the MGT interface is mandatory.

  Palo Alto Networks recommends that you specify all these settings for the MGT interface. If you omit values for some of these settings (such as the default gateway), you can access Panorama only through the console port for future configuration changes. You cannot commit the configurations for other interfaces unless you specify all these settings.

  If you plan to use the appliance as a Panorama management server, Palo Alto Networks recommends using the MGT interface only for managing Panorama and using other interfaces for managing devices, collecting logs, communicating with Collector Groups, and deploying updates to devices (see M-Series Appliance Interfaces).
- Gather the IP addresses of the DNS servers.

STEP 2 | Access the M-Series appliance from your computer.

1. Connect to the M-Series appliance in one of the following ways:
   - Attach a serial cable from a computer to the Console port on the M-Series appliance and connect using terminal emulation software (9600-8-N-1).
   - Attach an RJ-45 Ethernet cable from a computer to the MGT port on the M-Series appliance. From a browser, go to https://192.168.1.1. Enabling access to this URL might require changing the IP address on the computer to an address in the 192.168.1.0 network (for example, 192.168.1.2).
2. When prompted, log in to the appliance using the default username and password (admin/admin). The appliance starts initializing.

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STEP 3 | Change the default admin password.

Starting with PAN-OS 9.0.4, the predefined, default administrator password (admin/admin) must be changed on the first login on a device. The new password must be a minimum of eight characters and include a minimum of one lowercase and one uppercase character, as well as one number or special character.

Be sure to use the best practices for password strength to ensure a strict password and review the password complexity settings.

1. Click the **admin** link in the lower left of the web interface.
2. Enter the **Old Password**, **New Password**, and **Confirm New Password**, and then click **OK**. Store the new password in a safe location.

*To ensure that the MGT interface remains secure, configure Minimum Password Complexity settings (select **Panorama** > **Setup** > **Management**) and specify the interval at which administrators must change their passwords.*

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STEP 4 | Configure the network access settings for each interface that you will use to manage Panorama, manage devices, collect logs, communicate with Collector Groups, and deploy updates to devices.

To configure connectivity to Panorama using an IPv6 IP address, you must configure both an IPv4 and IPv6 to successfully configure Panorama using an IPv6 IP address. Panorama does not support configuring the management interface with only an IPv6 IP address.

1. Select **Panorama** > **Setup** > **Interfaces** and click the Interface Name.
2. (Non-MGT interfaces only) **Enable** the interface.
3. Edit the network access settings of each interface that Panorama will use. Only the MGT interface is required. The Eth1, Eth2, Eth3, Eth4, and Eth5 interfaces are optional and apply only if you plan to use the M-Series appliance as a Panorama management server.
   1. Complete one or both of the following field sets based on the IP protocols of your network:

      IPv4—**Public IP Address**, **IP Address**, **Netmask**, and **Default Gateway**

      *If your firewalls connect to the Panorama management server using a public IP address that is translated to a private IP address (NAT), enter the public IP in the **Public IP Address** field, and the private IP in the **IP Address** field to push both addresses to your firewalls.*

      IPv6—**IPv6 Address/Prefix Length** and **Default IPv6 Gateway**
   2. Select the Device Management Services that the interface supports:

      Device Management and Device Log Collection—You can assign one or more interfaces.

      Collector Group Communication—You can assign only one interface.

      Device Deployment (software and content updates)—You can assign only one interface.
   3. (Optional) Select the Network Connectivity Services that the interface supports.

      *(MGT interface only) Disable **Telnet** and **HTTP**; these services use plaintext and so are less secure than other services.*
   4. Click **OK** to save your changes.

STEP 5 | Configure the hostname, time zone, and general settings.

1. Select **Panorama** > **Setup** > **Management** and edit the General Settings.
2. Align the clock on Panorama and the managed firewalls to use the same **Time Zone**, for example GMT or UTC. If you plan to use the Strata Logging Service, you must configure NTP so that Panorama can stay in sync with the Strata Logging Service.

   The firewall records timestamps when it generate logs and Panorama records timestamps upon receiving the logs. Aligning the time zones ensures that the timestamps

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are synchronized and that the process of querying logs and generating reports on Panorama is harmonious.

3. Enter a **Hostname** for the server. Panorama uses this as the display name/label for the appliance. For example, this is the name that appears at the CLI prompt. It also appears in the Collector Name field if you add the appliance as a managed collector on the **Panorama** > **Managed Collectors** page.
4. (Optional) Enter the **Latitude** and **Longitude** to enable accurate placement of the M-Series appliance on the world map. The **App Scope** > **Traffic Maps** and **App Scope** > **Threat Maps** use these values.
5. Click **OK** to save your entries.

STEP 6 | Configure the DNS servers and Palo Alto Networks Update Server.

1. Select **Panorama** > **Setup** > **Services** and edit the settings.
2. Enter the IP address of the **Primary DNS Server** and (optionally) of the **Secondary DNS Server**.
3. Enter the URL or static address of the **Update Server** (default updates.paloaltonetworks.com).

   *Select **Verify Update Server Identity** if you want Panorama to verify that the Update Server from which it downloads software or content packages has an SSL certificate that a trusted authority signed. This option adds an additional level of security for communication between the Panorama management server and Update Server.*
4. Click **OK** to save your entries.

STEP 7 | Commit your configuration changes.

Select **Commit** > **Commit to Panorama** and **Commit** your changes.

*If you plan to use the M-Series appliance as a Panorama management server and you configured interfaces other than MGT, you must assign those interfaces to the **Device Log Collection** or **Collector Group Communication** functions when you Configure a Managed Collector.. To make the interfaces operational, you must then Configure a Collector Group for the managed collector and perform a Collector Group commit.*

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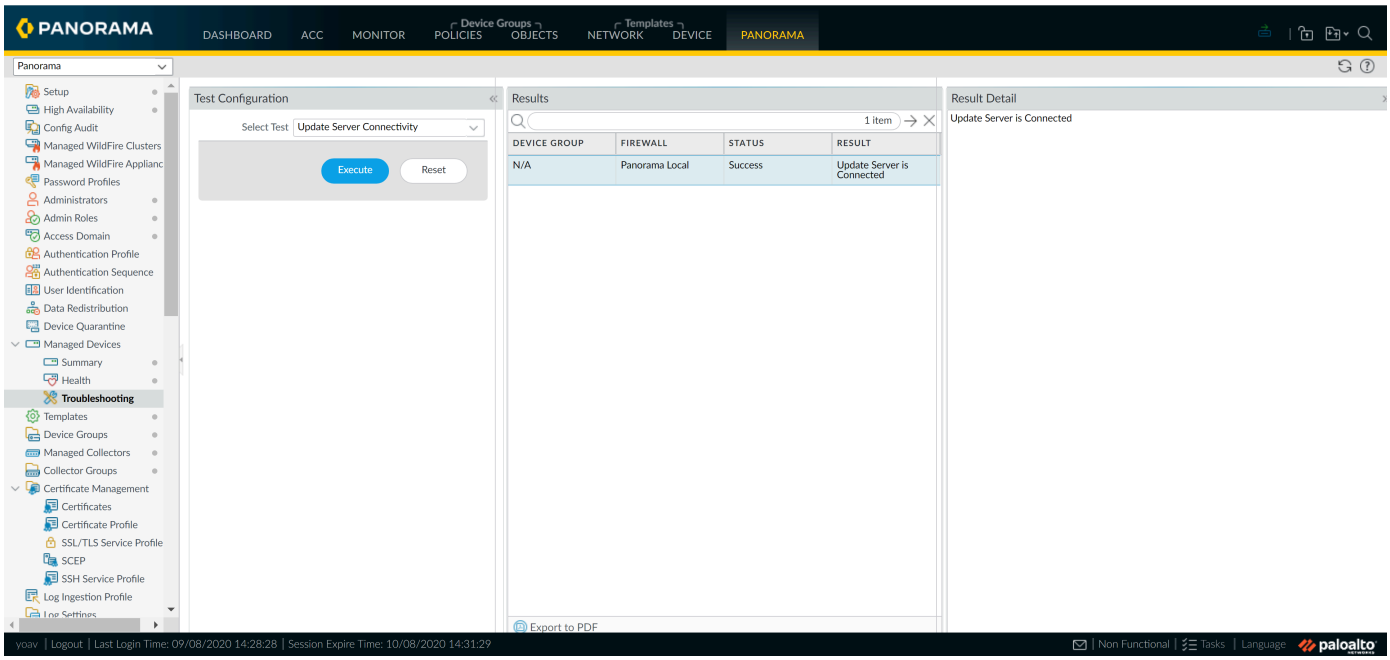
STEP 8 | Verify network access to external services required for Panorama management, such as the Palo Alto Networks Update Server.

1. Connect to the M-Series appliance in one of the following ways:
   - Attach a serial cable from your computer to the Console port on the M-Series appliance. Then use a terminal emulation software (9600-8-N-1) to connect.
   - Use terminal emulation software such as PuTTY to open an SSH session to the IP address that you specified for the MGT interface of the M-Series appliance during initial configuration.
2. Log in to the CLI when prompted. Use the default admin account and the password that you specified during initial configuration.
3. Use the Update Server Connectivity test to verify network connectivity to the Palo Alto Networks Update Server as shown in the following example.
   1. Select **Panorama** > **Managed Devices** > **Troubleshooting**, and select **Updates Server Connectivity** from the Select Test drop-down.
   2. **Execute** the update server connectivity test.



4. Use the following CLI command to retrieve information on the support entitlement for Panorama from the Update Server:

```
admin> request support check
```

If you have connectivity, the Update Server responds with the support status for Panorama. Because Panorama is not registered, the Update Server returns the following message:

```
Contact Us
https://www.paloaltonetworks.com/company/contact-us.html
Support Home
```

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```
https://www.paloaltonetworks.com/support/tabs/overview.html
Device not found on this update server
```

STEP 9 | Next steps...

1. Register Panorama and Install Licenses.
2. Install Content and Software Updates for Panorama.

As a best practice, replace the default certificate that Panorama uses to secure HTTPS traffic over the MGT interface.

Perform Initial Configuration of an Air Gapped M-Series Appliance

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• M-series hardware appliance • Super user role

Perform the initial configuration for an air gapped M-Series appliance. By default, Panorama has an IP address of 192.168.1.1 and a username/password of admin/admin. For security reasons, you must change these settings before continuing with other configuration tasks. You must perform these initial configuration tasks either from the Management (MGT) interface or using a direct serial port connection to the console port on the M-700, M-600, M-500, M-300, or M-200 appliance.

The air gapped Panorama cannot connect to the Palo Alto Networks update server because an outbound internet connection is required. To activate licenses, upgrade the PAN-OS software version, and install dynamic content updates you must upload the relevant files to the air gapped firewalls manually.

If you are configuring an M-Series appliance in Log Collector mode with 10GB interfaces, you must complete this entire configuration procedure for the 10GB interfaces to display as Up.

STEP 1 | Gather the required information from your network administrator.

- Private IP address for the management (MGT) port
- Netmask
- Default gateway
- DNS server address
- NTP server address

STEP 2 | Install and power on M-Series appliance.

Review your M-Series appliance hardware reference guide for details and best practices.

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STEP 3 | Connect to the M-Series appliance.

You must log in using the default **admin** username. You are immediately prompted to change the default `admin` password before you can continue. The new password must be a minimum of eight characters and include a minimum of one lowercase and one uppercase character, as well as one number or special character.

You can connect to the M-Series appliance in one of the following ways:

- Connect a serial cable from your computer to the Console port and connect to the M-Series appliance using terminal emulation software (9600-8-N-1). Wait a few minutes for the boot-up sequence to complete; when the M-Series appliance is ready, the prompt changes to the name of the M-Series, for example `M-500 login`.
- Log in to the Panorama CLI by connecting an RJ-45 Ethernet cable from your computer to the MGT interface on the M-Series appliance. From a browser, go to **https://192.168.1.1**.

  You may need to change the IP address on your computer to an address in the 192.168.1.0/24 network, such as 192.168.1.2, to access this URL.

STEP 4 | Configure the network settings for the air gapped M-Series appliance.

The following commands set the interface IP allocation to `static`, configures the IP address for the MGT interface, the Domain Name Server (DNS), and Network Time Protocol (NTP) server.

```
admin> configure
```

```
admin# set deviceconfig system type static
```

```
admin# set deviceconfig system ip-address <IP-Address> netmask
 <Netmask-IP> default-gateway <Gateway-IP>
```

```
admin# set deviceconfig system dns-settings servers primary <IP-
Address> secondary <IP-Address>
```

```
admin# set deviceconfig system ntp-servers primary-ntp-server ntp-
server-address <IP-Address>
```

```
admin# set deviceconfig system ntp-servers secondary-ntp-server
 ntp-server-address <IP-Address>
```

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STEP 5 | Register the M-Series appliance with the Palo Alto Networks Customer Support Portal (CSP).

1. Log in to the Palo Alto Networks CSP.
2. Click **Register a Device**.
3. Select **Register device using Serial Number** and click **Next**.
4. Enter the required `Device Information`.
   - Enter the M-Series appliance **Serial Number**.
   - Check (enable) **Device will be used offline**.
   - Select the PAN-OS **OS Release** running on the M-Series appliance.
5. Enter the required `Location Information`.
   - Enter the **City** the M-Series appliance is located in,
   - Enter the **Postal Code** the M-Series appliance is located in,
   - Enter the **Country** the M-Series appliance is located in.
6. **Agree and Submit**.
7. **Skip this step** when prompted to generate the optional `Day 1 Configuration` config file.

STEP 6 | Download the Panorama license keys.

The license key files are required to activate your Panorama licenses when air gapped.

1. Log in to the Palo Alto Networks CSP.
2. Select **Product** > **Devices** and locate the M-Series appliance you added.
3. Download all license keys files from the download links available `License` column.

   You must download a license key file for each license you want to active on Panorama.

STEP 7 | Active the Panorama licenses.

1. Log in to the Panorama web interface.
2. Select **Panorama** > **Licenses** and **Manually upload license key**.

   Click **Choose File** to select the license key file you downloaded in the previous step and click **OK**.
3. Repeat this step to uploaded and activate all licenses.

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STEP 8 | (Optional) Configure general Panorama settings as needed.

1. Select **Panorama** > **Setup** > **Management** and edit the General Settings.
2. Enter a **Hostname** for Panorama and enter your network **Domain** name. The domain name is just a label; it will not be used to join the domain.
3. Enter **Login Banner** text that informs users who are about to log in that they require authorization to access the Panorama management functions.

   As a best practice, avoid using welcoming verbiage. Additionally, you should ask your legal department to review the banner message to ensure it adequately warns that unauthorized access is prohibited.
4. Enter the **Latitude** and **Longitude** to enable accurate placement of the M-Series on the world map.
5. Click **OK**.
6. **Commit** and **Commit to Panorama**.

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STEP 9 | Upgrade the PAN-OS and dynamic content versions on Panorama.

Review the PAN-OS Upgrade Guide and PAN-OS Release Notes for detailed information about your target PAN-OS upgrade version.

1. Log in to the Palo Alto Networks CSP.
2. Download dynamic content updates.

   Alternatively, you can use a Secure Copy Protocol (SCP) server to automatically download dynamic content updates for Panorama, managed firewalls, Log Collectors, and WildFire appliances. An outbound internet connection is required for the SCP server to download dynamic content updates from the Palo Alto Networks Update Server.

   1. Select **Updates** > **Dynamic Updates**.
   2. Select the dynamic `Content type` you want to install.
   3. **Download** the dynamic content update to your local device.
   4. Repeat this step to download all required dynamic content updates.
3. Download a PAN-OS software update.
   1. Select **Updates** > **Software Updates**.
   2. For the `Content type`, select **Panorama M Base**. For the `Release type`, select **All**(default) or **Preferred**.
   3. In the `Download` column, click the PAN-OS version to download the software image to your local device.
4. Log in to the Panorama web interface.
5. Select **Panorama** > **Dynamic Updates** and **Upload** the dynamic content updates you downloaded.

   Repeat this step to **Browse** and select all the dynamic content release versions.
6. **Install** the dynamic content updates.
7. Select **Panorama** > **Software** and **Upload** the PAN-OS software image you download.
8. **Install** the PAN-OS software version.

   Panorama needs to restart to finish installing the PAN-OS software upgrade.

STEP 10 | Connect Panorama to your network.

1. Disconnect Panorama from your computer.
2. Connect the MGT port to a switch port on your management network using an RJ-45 Ethernet cable. Make sure that the switch port you cable on Panorama is configured for autonegotiation.

M-Series Setup Overview

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• M-series hardware appliance

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When setting up an M-Series appliance, you can deploy it using one of three operational modes:

- **Management Only Mode:** Dedicates the appliance entirely to managing firewalls and Dedicated Log Collectors, providing no local log collection capabilities aside from basic system and configuration logs.
- **Panorama Mode:** Deploys the appliance as a management server that also collects logs locally, which requires you to configure RAID disk arrays to make them available for logging.
- **Log Collector Mode:** Sets up the appliance as a Dedicated Log Collector, which also requires you to configure the RAID disk arrays for logging before finalizing the Log Collector setup,
- Set Up an M-Series Appliance in Management Only Mode
- Set Up an M-Series Appliance in Panorama Mode
- Set Up an M-Series Appliance in Log Collector Mode

Set Up an M-Series Appliance in Management Only Mode

Set up the Panorama management server in Management Only mode to dedicate Panorama to managing firewalls and Dedicated Log Collectors. Panorama in Management Only mode have no log collection capabilities, except for config and system logs, and requires a Dedicated Log Collector to store logs.

*If you configured a local Log Collector, the local Log Collector still exists on Panorama when you change to Management Only mode despite having no log collection capabilities. Deleting the local Log Collector (**Panorama** > **Managed Collectors**) deletes the Eth1/1 interface configuration the local Log Collector uses by default. If you decide to delete the local Log Collector, you must* reconfigure the Eth1/1 interface.

STEP 1 | Rack mount the M-Series appliance. Refer to the M-Series Appliance Hardware Reference Guide for instructions.

STEP 2 | Perform Initial Configuration of the M-Series Appliance.

STEP 3 | Register Panorama and Install Licenses.

STEP 4 | Install content and software updates on Panorama.

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STEP 5 | Change to Management Only mode.

1. Log in to the Panorama CLI.
2. Switch from Panorama mode to Management Only mode:

   `request system system-mode management-only`
3. Enter **`Y`** to confirm the mode change. The Panorama management server reboots. If the reboot process terminates your terminal emulation software session, reconnect to the Panorama management server to see the Panorama login prompt.

   If you see a `CMS Login` prompt, this means the Panorama management server has not finished rebooting. Press Enter at the prompt without typing a username or password.
4. Log back in to the CLI.
5. Verify that the switch to Management Only mode succeeded:

   `show system info | match system-mode`

   If the mode change succeeded, the output displays:

   `system mode:management-only`

STEP 6 | Configure Administrative Access to Panorama

STEP 7 | Manage Firewalls

STEP 8 | Manage Log Collection

Set Up an M-Series Appliance in Panorama Mode

STEP 1 | Rack mount the M-Series appliance. Refer to the M-Series Appliance Hardware Reference Guide for instructions.

STEP 2 | Perform Initial Configuration of the M-Series Appliance.

STEP 3 | Register Panorama and Install Licenses.

STEP 4 | Install Content and Software Updates for Panorama.

STEP 5 | Configure each array. This task is required to make the RAID disks available for logging. Optionally, you can add disks to Increase Storage on the M-Series Appliance.

STEP 6 | Configure Administrative Access to Panorama.

STEP 7 | Manage Firewalls.

STEP 8 | Manage Log Collection.

Set Up an M-Series Appliance in Log Collector Mode

STEP 1 | Rack mount the M-Series appliance. Refer to the M-Series Appliance Hardware Reference Guide for instructions.

STEP 2 | Perform Initial Configuration of the M-Series Appliance

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STEP 3 | Register Panorama and Install Licenses

STEP 4 | Install Content and Software Updates for Panorama

STEP 5 | Configure each array. This task is required to make the RAID disks available for logging. Optionally, you can add disks to Increase Storage on the M-Series Appliance.

STEP 6 | Set Up the M-Series Appliance as a Log Collector

STEP 7 | Manage Log Collection

Set Up the M-Series Appliance as a Log Collector

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• M-series hardware appliance • Log Collector License

If you want a dedicated appliance for log collection, configure an M-200, M-300, M-600, or M-700 appliance in Log Collector mode. To do this, you first perform the initial configuration of the appliance in Panorama mode, which includes licensing, installing software and content updates, and configuring the management (MGT) interface. You then switch the M-Series appliance to Log Collector mode and complete the Log Collector configuration. Additionally, if you want to use dedicated M-Series Appliance Interfaces (recommended) instead of the MGT interface for log collection and Collector Group communication, you must first configure the interfaces for the Panorama management server, then configure them for the Log Collector, and then perform a Panorama commit followed by a Collector Group commit.

Perform the following steps to set up a new M-Series appliance as a Log Collector or to convert an existing M-Series appliance that was previously deployed as a Panorama management server.

If you are configuring an M-Series appliance in Log Collector mode with 10GB interfaces, you must complete this entire configuration procedure for the 10GB interfaces to display as Up.

Switching the M-Series appliance from Panorama mode to Log Collector mode reboots the appliance, deletes the local Log Collector, deletes any existing log data, and deletes all configurations except the management access settings. Switching the mode does not delete licenses, software updates, or content updates.

STEP 1 | Set up the Panorama management server that will manage the Log Collector if you have not already done so.

Perform one of the following tasks:

- Set Up the Panorama Virtual Appliance
- Set Up the M-Series Appliance

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STEP 2 | Record the management IP addresses of the Panorama management server.

If you deployed Panorama in a high availability (HA) configuration, you need the IP address of each HA peer.

1. Log in to the web interface of the Panorama management server.
2. Record the **IP Address** of the solitary (non-HA) or active (HA) Panorama by selecting **Panorama** > **Setup** > **Management** and checking the Management Interface Settings.
3. For an HA deployment, record the **Peer HA IP Address** of the passive Panorama by selecting **Panorama** > **High Availability** and checking the Setup section.

STEP 3 | Set up the M-Series appliance that will serve as a Dedicated Log Collector.

If you previously deployed this appliance as a Panorama management server, you can skip this step because the MGT interface is already configured and the licenses and updates are already installed.

The M-Series appliance in Log Collector mode does not have a web interface for configuration tasks, only a CLI. Therefore, before changing the mode on the M-Series appliance, use the web interface in Panorama mode to:

1. Perform Initial Configuration of the M-Series Appliance.
2. Register Panorama and Install Licenses.
3. Install Content and Software Updates for Panorama.

STEP 4 | Access the CLI of the M-Series appliance.

1. Connect to the M-Series appliance in one of the following ways:
   - Attach a serial cable from your computer to the Console port on the M-Series appliance. Then use terminal emulation software (9600-8-N-1) to connect.
   - Use terminal emulation software such as PuTTY to open an SSH session to the IP address that you specified for the MGT interface of the M-Series appliance during initial configuration.
2. Log in to the CLI when prompted. Use the default admin account and the password that you specified during initial configuration.

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STEP 5 | Switch from Panorama mode to Log Collector mode.

1. Switch to Log Collector mode by entering the following command:

   ```
   > request system system-mode logger
   ```

2. Enter **Y** to confirm the mode change. The M-Series appliance reboots. If the reboot process terminates your terminal emulation software session, reconnect to the M-Series appliance to see the Panorama login prompt.

   *If you see a **CMS Login** prompt, this means the Log Collector has not finished rebooting. Press Enter at the prompt without typing a username or password.*

3. Log back in to the CLI.
4. Verify that the switch to Log Collector mode succeeded:

   ```
   > show system info | match system-mode
   ```

   If the mode change succeeded, the output displays:

   ```
   system-mode: logger
   ```

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STEP 6 | Configure the logging disks as RAID1 pairs.

If you previously deployed the appliance as a Panorama management server, you can skip this step because the disk pairs are already configured and available.

The time required to configure the drives varies from several minutes to a couple of hours, based on the amount of data on the drives.

1. Determine which disk pairs are present for configuring as RAID pairs on the M-Series appliance:

   ```
   > show system raid detail
   ```

   Perform the remaining steps to configure each disk pair that has `present` disks. This example uses disk pair A1/A2.
2. To add the first disk in the pair, enter the following command and enter **y** when prompted to confirm the request:

   ```
   > request system raid add A1
   ```

   Wait for the process to finish before adding the next disk in the pair. To monitor the progress of the RAID configuration, re-enter:

   ```
   > show system raid detail
   ```

   After the process finishes for the first disk, the output displays the disk pair status as `Available` but `degraded`.
3. Add the second disk in the pair:

   ```
   > request system raid add A2
   ```
4. Verify that the disk setup is complete:

   ```
   > show system raid detail
   ```

   After the process finishes for the second disk, the output displays the disk pair status as `Available` and `clean`:

   ```
   Disk Pair A      Available
   Status         clean
   ```

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STEP 7 | Enable connectivity between the Log Collector and Panorama management server.

Enter the following commands at the Log Collector CLI, where <*IPaddress1*> is for the MGT interface of the solitary (non-HA) or active (HA) Panorama and <*IPaddress2*> is for the MGT interface of the passive (HA) Panorama, if applicable.

```
> configure
# set deviceconfig system panorama-server <IPaddress1> panorama-
server-2 <IPaddress2>
# commit
# exit
```

STEP 8 | Record the serial number of the Log Collector.

You need the serial number to add the Log Collector as a managed collector on the Panorama management server.

1. At the Log Collector CLI, enter the following command to display its serial number.

   ```
   > show system info | match serial
   ```

2. Record the serial number.

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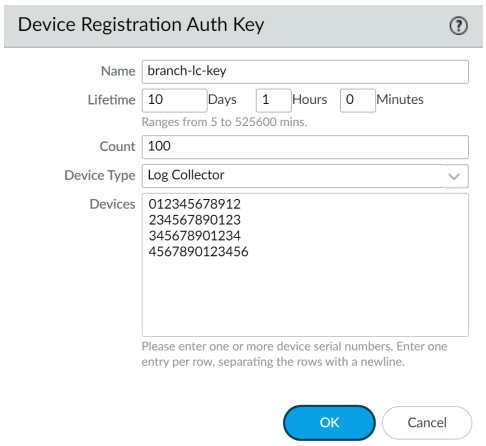
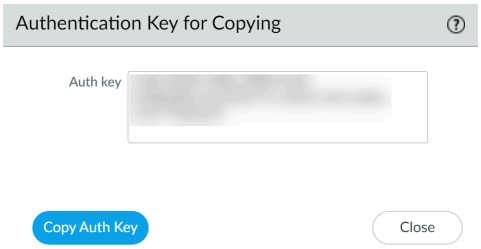
STEP 9 | Create a device registration authentication key.

1. Select **Panorama** > **Device Registration Auth Key** and **Add** a new authentication key.
2. Configure the authentication key.
   - **Name**—Add a descriptive name for the authentication key.
   - **Lifetime**—Specify the key lifetime for how long you can use the authentication key to onboard new Log Collectors.
   - **Count**—Specify how many times you can use the authentication key to onboard new Log Collectors.
   - **Device Type**—Specify that this authentication key is used to authenticate only a **Log Collector**.

     *You can select **Any** to use the device registration authentication key to onboard firewalls, Log Collectors, and WildFire appliances.*
   - (Optional) **Devices**—Enter one or more device serial numbers to specify for which Log Collectors the authentication key is valid.
3. Click **OK**.



4. **Copy Auth Key** and **Close**.



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STEP 10 | Add the Log Collector as a managed collector to the Panorama management server.

1. Select **Panorama** > **Managed Collectors** and **Add** a managed collector.
2. In the **General** settings, enter the serial number (**Collector S/N**) you recorded for the Log Collector.
3. In the **Panorama Server IP** field, enter the IP address or FQDN of the solitary (non-HA) or active (HA) Panorama. For HA deployments, enter the IP address or FQDN of the passive Panorama peer in the **Panorama Server IP 2** field.

   These IP addresses must specify a Panorama interface that has **Device Management and Device Log Collection** services enabled. By default, these services are enabled only on the MGT interface. However, you might have enabled the services on other interfaces when you Set Up the M-Series Appliance that is a Panorama management server.
4. Select **Interfaces**, click **Management**, and configure one or both of the following field sets for the MGT interface based on the IP protocols of your network.
   - IPv4—**IP Address**, **Netmask**, and **Default Gateway**
   - IPv6—**IPv6 Address/Prefix Length** and **Default IPv6 Gateway**
5. Click **OK** twice to save your changes to the Log Collector.
6. Select **Commit** > **Commit to Panorama** and **Commit** your changes to the Panorama configuration.

   This step is required before you can enable logging disks.
7. Verify that **Panorama** > **Managed Collectors** lists the Log Collector you added. The Connected column displays a check mark to indicate that the Log Collector is connected to Panorama. You might have to wait a few minutes before the page displays the updated connection status.

   At this point, the Configuration Status column displays Out of Sync and the Run Time Status column displays disconnected. The status will change to In Sync and connected after you configure a Collector Group (Step Assign the Log Collector to a Collector Group.).

STEP 11 | Add the device registration authentication key to the Log Collector.

Add the device registration authentication key only to a Dedicated Log Collector. A Panorama in Panorama mode does not need to authenticate its own local Log Collector.

1. Log in to the Log Collector CLI.
2. Add the device registration authentication key.

```
admin> request authkey set <auth-key>
```

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STEP 12 | Enable the logging disks.

1. Select **Panorama** > **Managed Collectors** and edit the Log Collector.
2. Select **Disks** and **Add** each RAID disk pair.
3. Click **OK** to save your changes.
4. Select **Commit** > **Commit to Panorama** and **Commit** your changes to the Panorama configuration.

STEP 13 | (Recommended) Configure the **Ethernet1**, **Ethernet2**, **Ethernet3**, **Ethernet4**, and **Ethernet5** interfaces if the Panorama management server and Log Collector will use them for **Device Log Collection** (receiving logs from firewalls) and **Collector Group Communication**.

If you previously deployed the Log Collector as a Panorama management server and configured these interfaces, you must reconfigure them because switching to Log Collector

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mode (Switch from Panorama mode to Log Collector mode.) would have deleted all configurations except the management access settings.

1. Configure each interface on the Panorama management server (other than the MGT interface) if you haven't already:
   1. Select **Panorama** > **Setup** > **Interfaces** and click the Interface Name.
   2. Select <*interface-name*> to enable the interface.
   3. Complete one or both of the following field sets based on the IP protocols of your network:

      IPv4—**IP Address**, **Netmask**, and **Default Gateway**

      IPv6—**IPv6 Address/Prefix Length** and **Default IPv6 Gateway**
   4. Select the Device Management Services that the interface supports:

      Device Management and Device Log Collection—You can assign one or more interfaces.

      Collector Group Communication—You can assign only one interface.

      Device Deployment (software and content updates)—You can assign only one interface.
   5. Click **OK** to save your changes.
2. Configure each interface on the Log Collector (other than the MGT interface):
   1. Select **Panorama** > **Managed Collectors** and edit the Log Collector.
   2. Select **Interfaces** and click the name of the interface.
   3. Select <*interface-name*> to enable the interface.
   4. Complete one or both of the following field sets based on the IP protocols of your network:

      IPv4—**IP Address**, **Netmask**, and **Default Gateway**

      IPv6—**IPv6 Address/Prefix Length** and **Default IPv6 Gateway**
   5. Select the Device Management Services that the interface supports:

      Device Log Collection—You can assign one or more interfaces.

      Collector Group Communication—You can assign only one interface.
   6. Click **OK** to save your changes to the interface.
3. Click **OK** to save your changes to the Log Collector.
4. Select **Commit** > **Commit to Panorama** and **Commit** your changes to the Panorama configuration.

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STEP 14 | (Optional) If your deployment is using custom certificates for authentication between Panorama and managed devices, deploy the custom client device certificate. For more information, see Configure Authentication Using Custom Certificates.

1. Select **Panorama** > **Certificate Management** > **Certificate Profile** and choose the certificate profile from the drop-down or click **New Certificate Profile** to create one.
2. Select **Panorama** > **Managed Collectors** > **Add** > **Communication** for a Log Collector.
3. Select the **Secure Client Communication** check box.
4. Select the type of device certificate the Type drop-down.
   - If you are using a local device certificate, select the **Certificate** and **Certificate Profile** from the respective drop-downs.
   - If you are using SCEP as the device certificate, select the **SCEP Profile** and **Certificate Profile** from the respective drop-downs.
5. Click **OK**.

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STEP 15 | (Optional) Configure Secure Server Communication on a Log Collector. For more information, see Configure Authentication Using Custom Certificates.

1. Select **Panorama** > **Managed Collectors** > **Add** > **Communication**.
2. Verify that the **Custom Certificate Only** check box is not selected. This allows you to continue managing all devices while migrating to custom certificates.

   When the Custom Certificate Only check box is selected, the Log Collector does not authenticate and cannot receive logs from devices using predefined certificates.
3. Select the SSL/TLS service profile from the **SSL/TLS Service Profile** drop-down. This SSL/TLS service profile applies to all SSL connections between the Log Collector and devices sending it logs.
4. Select the certificate profile from the **Certificate Profile** drop-down.
5. Select **Authorize Client Based on Serial Number** to have the server check clients against the serial numbers of managed devices. The client certificate must have the special keyword $UDID set as the CN to authorize based on serial numbers.
6. In **Disconnect Wait Time (min)**, enter the number of minutes Panorama should wait before breaking and reestablishing the connection with its managed devices. This field is blank by default and the range is 0 to 44,640 minutes.

   The disconnect wait time does not begin counting down until you commit the new configuration.
7. (Optional) Configure an authorization list.
   1. Click **Add** under Authorization List.
   2. Select the **Subject** or **Subject Alt Name** as the Identifier type.
   3. Enter an identifier of the selected type.
   4. Click **OK**.
   5. Select **Check Authorization List** to enforce the authorization list.
8. Click **OK**.
9. Select **Commit** > **Commit to Panorama**.

STEP 16 | Assign the Log Collector to a Collector Group.

1. Configure a Collector Group. You must perform a Panorama commit and then a Collector Group commit to synchronize the Log Collector configuration with Panorama and to put

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the Eth1, Eth2, Eth3, Eth4, and Eth5 interfaces (if you configured them) in an operational state on the Log Collector.

In any single Collector Group, all the Log Collectors must run on the same Panorama model: all M-700 appliances, all M-600 appliances, all M-500 appliances, all M-300 appliances, all M-200 appliances, or all Panorama virtual appliances.

*As a best practice, **Enable log redundancy across collectors** if you add multiple Log Collectors to a single Collector group. This option requires each Log Collector to have the same number of logging disks.*

2. Select **Panorama** > **Managed Collectors** to verify that the Log Collector configuration is synchronized with Panorama.

   The Configuration Status column should display In Sync and the Run Time Status column should display connected.

3. Access the Log Collector CLI and enter the following command to verify that its interfaces are operational:

   ```
   > show interface all
   ```

   The output displays the `state` as `up` for each interface that is operational.

4. If the Collector Group has multiple Log Collectors, troubleshoot connectivity to network resources to verify they can communicate with each other by performing a Ping connectivity test for each interface that the Log Collectors use. For the `source` IP address, specify the interface of one of the Log Collectors. For the `host` IP address, specify the matching interface of another Log Collector in the same Collector Group.

STEP 17 | Next steps...

To enable the Log Collector to receive firewall logs:

1. Configure Log Forwarding to Panorama.
2. Verify Log Forwarding to Panorama.

Increase Storage on the M-Series Appliance

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• M-series hardware appliance

After you Perform Initial Configuration of the M-Series Appliance, you can increase log storage capacity of the appliance by upgrading the existing drive pairs to larger capacity drives or by installing additional drive pairs in empty drive bays. For example, you can choose to upgrade the existing 1TB drives to 2TB on an M-500 appliance, or you can add 2TB drives to the empty drive bays (B1 through D2).

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The M-Series appliances leverage RAID 1 for data redundancy in the event of disk failure. Therefore, the pair of drives in a RAID 1 array need to be identical. However, you are free to mix drive capacities across different RAID 1 arrays. For example, the drives in the A1/A2 RAID 1 array can be 1TB drives, and the drives in the B1/B2 RAID 1 array can be 2TB drives.

The following table lists the maximum number of drive bays (disks) and the available drive capacities supported on M-Series appliances.

Because each drive pair (A1/A2 for example) is in a RAID 1 array, the total storage capacity is half of the total drives installed. For example, if an M-500 appliance has 2TB drives installed in drive bays A1/A2 and B1/B2, the A1/A2 array provides 2TB total storage and the B1/B2 array provides another 2TB for a total of 4TB.

Appliance	Number of Supported Drive Bays (Disks)	Supported Drive Capacity
M-200 Appliance	4	8TB
M-300 Appliance	4	8TB
M-500 Appliance	24	1TB or 2TB
M-600 Appliance	12	8TB
M-700 Appliance	12	8TB

Before expanding log storage capacity, Determine Panorama Log Storage Requirements. If you need more log storage than a single M-Series appliance supports, you can add Dedicated Log Collectors (see Configure a Managed Collector) or you can Configure Log Forwarding from Panorama to External Destinations.

*You don't need to take the M-Series appliance offline to expand the storage when adding drives to an M-Series appliance that is already deployed. When the additional drives are configurable and available, the M-Series appliance redistributes the logs among all available drives. This log redistribution process happens in the background and does not impact uptime or the availability of the M-Series appliance. However, the process does diminish the maximum logging rate. The Redistribution State column (**Panorama** > **Collector Groups**) indicates the completion status of the process as a percentage.*

- Add Additional Drives to an M-Series Appliance
- Upgrade Drives on an M-Series Appliance

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Add Additional Drives to an M-Series Appliance

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• M-series hardware appliance

STEP 1 | Install the new drives in the appropriate drive bays.

Make sure to add the drives sequentially in the next open drive bays. For example, add drives to B1 and B2 before adding drives to C1 and C2.

STEP 2 | Access the command line interface (CLI) on the M-Series appliance.

Connect to the M-Series appliance in one of two ways:

- Connect a serial cable from your computer to the Console port and connect to the M-Series appliance using terminal emulation software (9600-8-N-1).
- Use terminal emulation software (such as PuTTY) to open a Secure Shell (SSH) session to the IP address of the M-Series appliance.

STEP 3 | When prompted, log in to the appliance.

Use the default administrator account and the assigned password.

STEP 4 | Configure each array.

The time required to mirror the data on the drive can take minutes, a few hours, or more than a day depending on the amount of data on the drive.

The following example uses the drives in bays B1 and B2.

1. Enter the following commands and confirm the request when prompted. This command deletes any existing data on the disks you are adding to the M-Series appliance.

```
> request system raid add B1
> request system raid add B2
```

(RMA only) If you want to try and preserve the data on the disks, try the following commands instead:

```
> request system raid add B1 force no-format
> request system raid add B2 force no-format
```

2. To monitor the progress of the RAID configuration, enter the following command:

```
> show system raid detail
```

When the RAID set up is complete, the following response displays:

```
Disk Pair A       Available
```

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```
   Status          clean
   Disk id A1      Present
       model         : ST91000640NS
       size          : 953869 MB
       status        : active sync
   Disk id A2      Present
       model         : ST91000640NS
       size          : 953869 MB
       status        : active sync
Disk Pair B         Available
   Status           clean
   Disk id B1       Present
       model         : ST91000640NS
       size          : 953869 MB
       status        : active sync
   Disk id B2      Present
       model         : ST91000640NS
       size          : 953869 MB
       status        : active sync
```

STEP 5 | Make the array available for logging.

To enable the array for logging, you must first add the appliance as a managed collector on Panorama. If not already added, see Configure a Managed Collector.

1. Log in to the web interface of the Panorama management server that manages this Log Collector.
2. Select **Panorama** > **Managed Collectors** and edit the Log Collector.
3. Select **Disks** and **Add** each array.
4. Click **OK** to save your changes.
5. Select **Commit** > **Commit to Panorama** and **Commit** your changes.
6. Select **Commit** > **Push to Devices**, select the Collector Group, and **Push** your changes.

Upgrade Drives on an M-Series Appliance

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• M-200, M-300, M-500, M-600, or M-700 hardware appliance

STEP 1 | Access the command line interface (CLI) on the M-Series appliance.

Connect to the M-Series appliance in one of two ways:

- Connect a serial cable from your computer to the Console port and connect to the M-Series appliance using terminal emulation software (9600-8-N-1).
- Use terminal emulation software (such as PuTTY) to open a Secure Shell (SSH) session to the IP address of the M-Series appliance.

STEP 2 | When prompted, log in to the appliance.

Use the default administrator account and the assigned password.

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STEP 3 | Verify that the RAID 1 status for the installed drives shows there are at least two functioning RAID 1 arrays. During the upgrade, you will upgrade one RAID 1 array at a time and there must be at least one other RAID 1 array that is available to the appliance. The appliance will show an abort error if you try to remove the only functioning array from the configuration.

Enter the following command to view RAID status:

```
> show system raid detail
```

For example, the following shows an output from an M-500 appliance with two available arrays (Disk Pair A and Disk Pair B). If there is only one available array, you must add a second array as described in Add Additional Drives to an M-Series Appliance before you upgrade the drives.

```
Disk Pair A                              Available
   Status                                    clean
   Disk id A1                              Present
       model        : ST91000640NS
       size         : 953869 MB
       status       : active sync
   Disk id A2                              Present
       model        : ST91000640NS
       size         : 953869 MB
       status       : active sync
Disk Pair B                              Available
   Status                                    clean
   Disk id B1                              Present
       model        : ST91000640NS
       size         : 953869 MB
       status       : active sync
   Disk id B2                              Present
       model        : ST91000640NS
       size         : 953869 MB
       status       : active sync
```

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STEP 4 | Remove the first 1TB drive and replace it with a 2TB drive.

1. To remove the first drive from the RAID 1 array configuration (A1 in this example), enter the following command and enter **y** when prompted to confirm the request:

   ```
   > request system raid remove A1
   ```

2. Physically remove the first drive from the drive bay. Press the ejector button on the drive carrier in drive bay A1 to release the ejector handle. Then pull the handle toward you and slide the drive out of the appliance.
3. Remove a 2TB drive from its packaging and place the drive on a table next to the drive you just removed. Take note of how the drive is installed in the carrier because you will install the 2TB drive in this same carrier.
4. Remove the four screws holding the 1TB drive in the carrier and remove the drive from the carrier.
5. Attach the 2TB drive to the carrier using the same four screws you removed from the 1TB drive and then reinsert the carrier with the 2TB drive into drive bay A1.
6. Enter the following command to verify the 2TB drive is recognized:

   ```
   show system raid detail
   ```

   Verify that the A1 disk shows the correct model and size (about 2TB). If the model and size are not correct, run the above command again until the correct model and size are shown.

   If the wrong model and size are consistently shown, enter the following command:

   ```
   request system raid remove A1
   ```

   Wait for 30 seconds once you run the above command, then remove the disk and reinsert it and repeat the **show system raid detail** command to verify the size and model.

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STEP 5 | Copy the data from the remaining installed 1TB drive in the RAID 1 array to the newly installed 2TB drive in that array.

The time required to copy the data may vary from several minutes to a few hours, depending on the amount of data on the drive.

1. To copy the data from the 1TB drive in drive bay A2 to the newly installed 2TB drive in drive bay A1, enter the following command and enter **y** when prompted:

```
> request system raid copy from A2 to A1
```

2. To view the status of the copy process, run the following command:

```
> show system raid detail
```

Continue running this command to view the RAID detail output until you see that the array (A1/A2 in this example) shows `Available`.

At this point, drive A2 will show `not in use` because there is a drive size mismatch.

STEP 6 | Upgrade the second drive in the RAID 1 array to a 2TB drive.

1. Remove the second 1TB drive (from drive bay A2 in the current example) for the RAID 1 array configuration:

```
> request system raid remove A2
```

2. Insert the carrier with the newly installed 2TB drive into drive bay A2 and add it to the RAID 1 array configuration:

```
> request system raid add A2
```

The system will copy the data from A2 to A1 to mirror the drives.

3. To view the status of the copy process, run the following command:

```
> show system raid detail
```

Continue to view the RAID detail output until you see that the array (A1/A2 in this example) shows `Available` and both disks show `active sync`.

```
Disk Pair A       Available
   Status         clean
   Disk id A1     Present
       model        : ST2000NX0253
       size         : 1907138 MB
       status       : active sync
Disk id A2     Present
       model        : ST2000NX0253
```

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```
        size          : 1907138 MB
        status        : active sync
```

STEP 7 | Upgrade drives for additional RAID 1 arrays as needed.

To upgrade additional RAID 1 arrays to 2TB drives, repeat this procedure replacing the drive designators as applicable. For example, replace A1 with B1 and A2 with B2 to upgrade the drives in the B1/B2 RAID 1 array.

Configure Panorama to Use Multiple Interfaces

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• M-series hardware appliance

In a large-scale network, you can improve security and reduce congestion by implementing network segmentation, which involves segregating the subnetworks based on resource usage, user roles, and security requirements. Panorama supports network segmentation by enabling you to use multiple M-Series Appliance Interfaces for managing devices (firewalls, Log Collectors, and WildFire appliances and appliance clusters) and collecting logs; you can assign separate interfaces to the devices on separate subnetworks.

Using multiple interfaces to collect logs also provides the benefit of load balancing, which is particularly useful in environments where the firewalls forward logs at high rates to the Log Collectors. If you enable the **forward to all Log Collectors** setting in the Collector Group log forwarding preference list, logs are sent on all configured interfaces configured. Otherwise, logs are forwarded over a single interface, and if that interface goes down, log forwarding continues over the next configured interface. For example, you configure Eth1/1, Eth1/2, and Eth1/3 for log forwarding. In the event the Eth1/1 interface goes down, log forwarding continues over Eth1/2.

Because administrators access and manage Panorama over the MGT interface, securing that interface is especially important. One method for improving the security of the MGT interface is to offload Panorama services to other interfaces. In addition to device management and log collection, you can also offload Collector Group communication and deployment of software and content updates to firewalls, Log Collectors, and WildFire appliances and appliance clusters. By offloading these services, you can reserve the MGT interface for administrative traffic and assign it to a secure subnetwork that is segregated from the subnetworks where your firewalls, Log Collectors, and WildFire appliances and appliance clusters reside.

- Multiple Interfaces for Network Segmentation Example
- Configure Panorama for Network Segmentation

Multiple Interfaces for Network Segmentation Example

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• M-series hardware appliance

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Figure 1 illustrates a deployment that uses multiple interfaces on M-500 appliances in Panorama mode and Log Collector mode. In this example, the interfaces support network segmentation as follows:

- **Panorama management network**—To protect the Panorama web interface, CLI, and XML API from unauthorized access, the MGT interface on Panorama connects to a subnetwork that only administrators can access.
- **Internet**—Panorama uses the MGT interface to communicate with external services such as the Palo Alto Networks Update Server.
- **Perimeter Gateway** and **Data Center**—Panorama uses a separate pair of interfaces to manage the firewalls and Log Collectors in each of these subnetworks. Managing firewalls typically generates less traffic than querying Log Collectors for report information. Therefore, Panorama uses 1Gbps interfaces (Eth1 and Eth2) for managing the firewalls and uses 10Gbps interfaces (Eth4 and Eth5) for querying and managing the Log Collectors. Each Log Collector uses its MGT interface to respond to the queries but uses its Eth4 and Eth5 interfaces for the heavier traffic associated with collecting logs from the firewalls.
- **Software and content updates**—The firewalls and Log Collectors in both subnetworks retrieve software and content updates over the Eth3 interface on Panorama.

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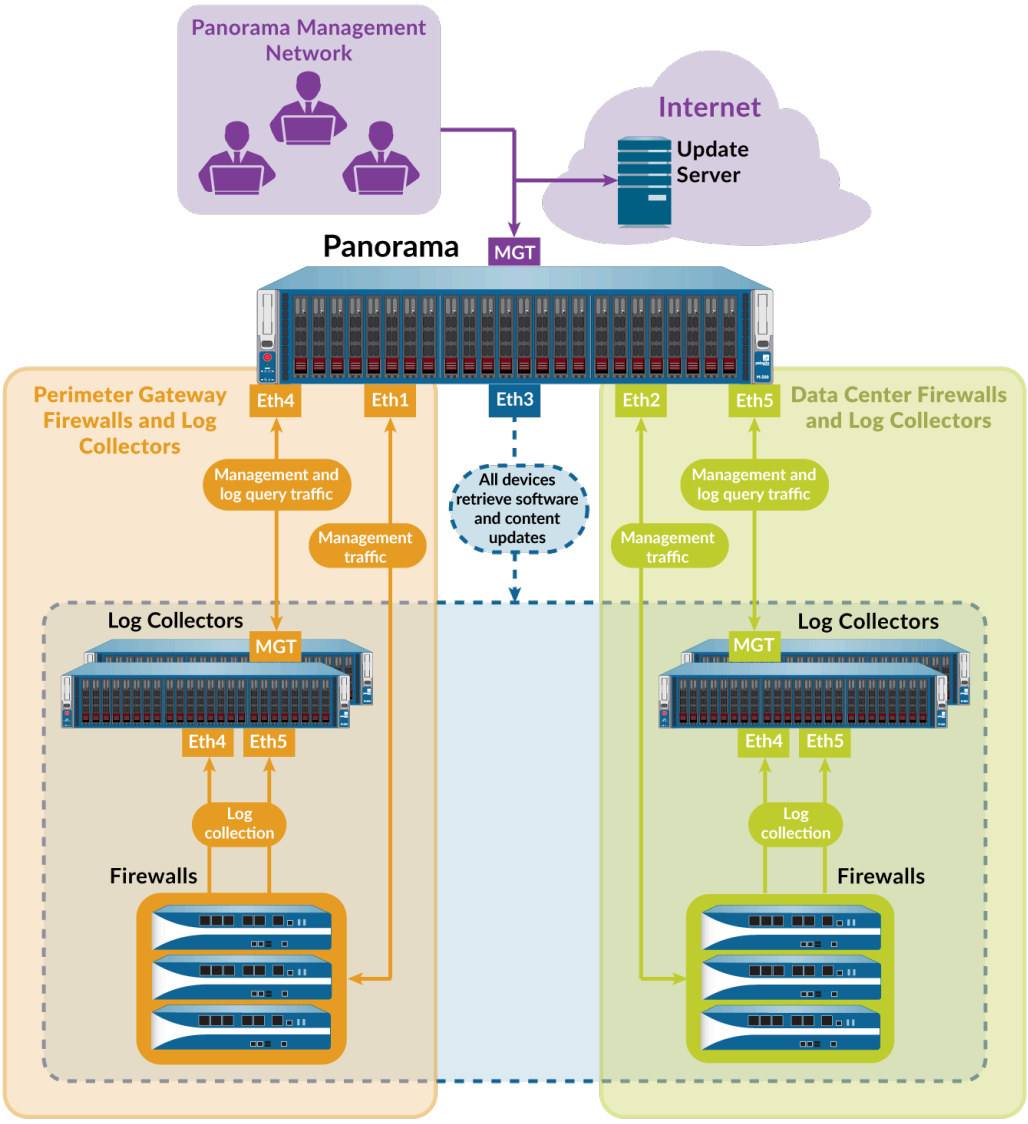

Figure 9: Multiple Panorama Interfaces

Configure Panorama for Network Segmentation

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• M-series hardware appliance

To offload Panorama services from the MGT interface to other interfaces, start by configuring the interfaces on the Panorama management server. If your network has heavy log traffic, remember that the Eth4 and Eth5 interfaces on the M-600, and M-700 appliances support higher throughput (10Gbps) than the other interfaces (1Gbps). Then, configure the Log Collectors in each subnetwork to connect with specific interfaces on Panorama. For each Log Collector, you also select an interface for Collector Group communication and one or more interfaces for collecting logs from firewalls. Finally, configure the firewalls in each subnetwork to connect with interfaces on Panorama.

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If you are configuring an M-Series appliance in Log Collector mode with 10GB interfaces, you must complete this entire configuration procedure for the 10GB interfaces to display as Up.

Palo Alto Networks recommends that you specify the IP address, netmask (for IPv4) or prefix length (for IPv6), and default gateway for the MGT interface. If you omit one of these settings (such as the default gateway), you can access the M-Series appliance only through the console port for future configuration changes.

Perform the following steps to configure Panorama and Dedicated Log Collectors to use multiple interfaces:

STEP 1 | Verify that the Panorama appliances and firewalls support multiple interfaces, and have the prerequisite software versions and configurations.

- The M-Series appliances must run Panorama 8.0 or later to use a separate interface for deploying updates and to use multiple interfaces for device management and log collection. The M-200 and M-600 appliances must run Panorama 8.1 or later while the M-300 and M-700 must run Panorama 11.1 or later. Panorama appliances deployed on ESXi, vCloud, Air, Hyper-V and KVM must run Panorama 8.1 or later.
- If you deployed a Panorama or Log Collector as a virtual appliance, verify the Supported Interfaces for the Panorama Virtual Appliance.
- The M-Series appliances must run Panorama 6.1 or later to use separate interfaces for log collection or Collector Group communication.
- The initial configuration of each Panorama management server is complete. This includes configuration of the MGT interface.

  To configure an IPv6 IP address for the Panorama MGT interface, you must configure both an IPv4 and IPv6 to successfully configure Panorama using an IPv6 IP address. Panorama does not support configuring the MGT interface with only an IPv6 IP address.
- Log Collectors and Collector Groups are configured. This includes configuration of the MGT interface on the Log Collectors.

  To configure an IPv6 IP address for the MGT interface of a Log Collector, you must configure both an IPv4 and IPv6 to successfully configure Panorama using an IPv6 IP address. Panorama does not support configuring the MGT interface with only an IPv6 IP address.
- The initial configuration of the firewalls is complete, you have added the firewalls to Panorama as managed devices, and the firewalls in each subnetwork are assigned to a separate template.
- The initial configuration of WildFire appliances is complete and you have added WildFire appliances to Panorama as managed devices.

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STEP 2 | Configure the interfaces on the solitary (non-HA) or active (HA) Panorama management server.

Because the MGT interface was configured during initial Panorama configuration, you don't have to configure it again.

Perform these steps for each interface:

1. Log in to the Panorama Web Interface of the solitary (non-HA) or active (HA) Panorama management server.
2. Select **Panorama** > **Setup** > **Interfaces**.
3. Click an Interface Name to edit the interface.
4. Select <*interface-name*> to enable the interface.
5. Configure one or both of these field sets based on the IP protocols of your network:
   - IPv4—**IP Address**, **Netmask**, and **Default Gateway**
   - IPv6—**IPv6 Address/Prefix Length** and **Default IPv6 Gateway**
6. Select the services that the interface supports:
   - **Device Management and Device Log Collection**—Manage firewalls, Log Collectors, and WildFire appliances and appliance clusters, collect logs that the Log Collectors generate, and query the Log Collectors for report information. To support a segmented network, you can enable these services on multiple interfaces.
   - **Collector Group Communication**—Communicate with the Collector Groups that Panorama manages across all subnetworks.
   - **Device Deployment**—Deploy software and content updates to managed firewalls, Log Collectors, and WildFire appliances and appliance clusters across all subnetworks.
7. Click **OK** to save your changes to the interface.
8. Click **Commit** > **Commit to Panorama** and **Commit** your changes.
9. Click **Commit** > **Push to Devices** and push the changes to the Collector Group that contain the Log Collectors you modified.

STEP 3 | (HA only) Configure the interfaces on the passive Panorama management server.

1. Log in to the Panorama Web Interface of the active Panorama management server.
2. Select **Panorama** > **Managed Collectors** and select the passive HA peer.
3. Select **Interfaces** and click an interface to edit.
4. Check the **Enable Interface** box to enable the interface.
5. Configure one or both of these field sets based on the IP protocols of your network:
   - IPv4—**IP Address**, **Netmask**, and **Default Gateway**
   - IPv6—**IPv6 Address/Prefix Length** and **Default IPv6 Gateway**
6. Select the services that the interface supports:
   - **Device Management and Device Log Collection**—Manage firewalls, Log Collectors, and WildFire appliances and appliance clusters, collect logs that the Log Collectors

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generate, and query the Log Collectors for report information. To support a segmented network, you can enable these services on multiple interfaces.

- **Collector Group Communication**—Communicate with the Collector Groups that Panorama manages across all subnetworks.
- **Device Deployment**—Deploy software and content updates to managed firewalls, Log Collectors, and WildFire appliances and appliance clusters across all subnetworks.

7. Click **OK** to save your changes to the interface.
8. Select **Commit** > **Commit and Push** to commit your changes to Panorama and to push the changes to Collector Groups that contain the passive HA peer you modified.

STEP 4 | Configure each Log Collector to connect with a Panorama interface.

To support a segmented network, you can connect the Log Collectors in each subnetwork to separate Panorama interfaces. The interfaces must have **Device Management and Device Log Collection** enabled, as described in the previous step.

1. Log in to the Panorama Web Interface of the solitary (non-HA) or active (HA) Panorama management server.
2. Select **Panorama** > **Managed Collectors** and edit the Log Collector.
3. In the **Panorama Server IP** field, enter the IP address of an interface on the solitary (non-HA) or active (HA) Panorama.
4. (HA only) In the **Panorama Server IP 2** field, enter the IP address of an interface on the passive Panorama that will support **Device Management and Device Log Collection** if failover occurs on the active Panorama.
5. Click **OK** to save your changes.
6. Select **Commit** > **Commit and Push** to commit your changes to Panorama and to push the changes to Collector Groups that contain the Log Collector you modified.
7. Perform the following steps on each Dedicated Log Collector:
   1. Access the Log Collector CLI by using emulation software such as PuTTY to open a SSH session to the Log Collector using its MGT interface IP address. When prompted, log in using Panorama administrator credentials.
   2. Run the following commands, where *<IPaddress1>* is for the solitary (non-HA) or active (HA) Panorama and *<IPaddress2>* is for the passive Panorama (if applicable).

```
> configure
# set deviceconfig system panorama-server <IPaddress1>
 panorama-server-2 <IPaddress2>
# commit
```

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STEP 5 | (HA only) Configure an interface on the passive Panorama management server to deploy updates in case the active Panorama fails over.

1. Log in to the Panorama Web Interface of the passive Panorama management server.
2. Select **Panorama** > **Setup** > **Interfaces**.
3. Click an Interface Name to edit the interface.
4. Select *<interface-name>* to enable the interface.
5. Configure one or both of these field sets based on the IP protocols of your network:
   - IPv4—**IP Address**, **Netmask**, and **Default Gateway**
   - IPv6—**IPv6 Address/Prefix Length** and **Default IPv6 Gateway**
6. Select **Device Deployment**.
7. Click **OK** to save your changes.
8. Click **Commit** > **Commit to Panorama** and **Commit** your changes.

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STEP 6 | Configure the interfaces that the Log Collectors will use to collect logs from firewalls and communicate with other Log Collectors.

Because the MGT interface was configured during initial configuration of the Log Collectors, you don't have to configure it again.

1. Log in to the Panorama Web Interface of the solitary (non-HA) or active (HA) Panorama management server.
2. Select **Panorama** > **Managed Collectors** and edit the Log Collector.
3. Select **Interfaces** and perform the following steps for each interface:
   1. Click an interface name to edit that interface.
   2. Select *<interface-name>* to enable the interface.
   3. Configure one or both of the following field sets based on the IP protocols of your network.

      IPv4—**IP Address**, **Netmask**, and **Default Gateway**

      IPv6—**IPv6 Address/Prefix Length** and **Default IPv6 Gateway**
   4. Select the functions that the interface supports:

      Device Log Collection—Collect logs from firewalls. You can load balance the logging traffic by enabling multiple interfaces to perform this function.

      Collector Group Communication—Communicate with other Log Collectors in the Collector Group.
   5. Click **OK** to save your changes to the interface.
4. Click **OK** to save your changes to the Log Collector.
5. Select **Commit** > **Commit and Push** to commit your changes to Panorama and to push the changes to Collector Groups that contain the Log Collectors you modified.
6. Select **Panorama** > **Managed Collectors** to verify that the Log Collectors are synchronized and connected with Panorama.

   The Configuration Status column should display `InSync` and the Run Time Status column should display `connected`.

STEP 7 | Configure the firewalls to connect with a Panorama interface.

To support a segmented network, you can connect the firewalls in each subnetwork to separate Panorama interfaces. The interfaces must have **Device Management and Device**

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Log Collection enabled. This step assumes that you use separate templates to configure the firewalls in separate subnetworks.

In this example deployment, Panorama uses these interfaces to manage the firewalls but not to collect firewall logs. You specify which Dedicated Log Collectors will collect firewall logs when you configure Collector Groups.

1. Log in to the Panorama Web Interface of the solitary (non-HA) or active (HA) Panorama management server.
2. On Panorama, select **Device** > **Setup** > **Management**, select a **Template** and edit the Panorama Settings.
3. In the first **Panorama Servers** field, enter the IP address of an interface on the solitary (non-HA) or active (HA) Panorama.
4. (HA only) In the second **Panorama Servers** field, enter the IP address of an interface on the passive Panorama that will support device management if failover occurs.
5. Click **OK** to save your changes.
6. Select **Commit** > **Commit and Push** to commit your changes to Panorama and push the template changes to firewalls.
7. Select **Panorama** > **Managed Devices** to verify that the firewalls are synchronized and connected with Panorama.

   The Device State column should display `Connected`. The Shared Policy and Template columns should display `InSync`.

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Register Panorama and Install Licenses

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Customer Support Portal • Panorama serial number

Before you can begin using Panorama for centralized management, logging, and reporting, you are required to register, activate, and retrieve the Panorama device management and support licenses. Every instance of Panorama requires valid licenses that entitle you to manage firewalls and obtain support. The firewall device management license enforces the maximum number of firewalls that Panorama can manage. This license is based on firewall serial numbers, not on the number of virtual systems on each firewall. The support license enables Panorama software updates and dynamic content updates (for the latest Applications and Threats signatures, as an example). Additionally, Panorama virtual appliances on AWS and Azure must be purchased from Palo Alto Networks, and cannot be purchased on the AWS or Azure marketplaces.

After upgrading your Panorama virtual appliance to PAN-OS 8.1, you are prompted if a capacity license has not been successfully installed or if the total number of firewalls being managed by Panorama exceeds the device management license. You have 180 days from the date of upgrade to install a valid device management license if no license has been installed. If the number of managed firewalls exceeds the device management license, you have 180 days to delete firewalls to meet the device management license requirements or upgrade your device management license. All commits fail if a valid device management license is not installed, or the existing device management license limit is not met, within 180 days of upgrade. To purchase a device management license, contact your Palo Alto Networks sales representative or authorized reseller.

If you want to use the cloud-based Strata Logging Service, you require a Strata Logging Service license, in addition to the firewall management license and premium support license. To purchase licenses, contact your Palo Alto Networks Systems Engineer or reseller.

If you are running an evaluation license for firewall management on your Panorama virtual appliance and want to apply a Panorama license that you purchased, perform the tasks Register Panorama *and* Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is Internet-connected.

- Register Panorama
- Activate a Panorama Support License
- Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is Internet-connected
- Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is not Internet-connected
- Activate/Retrieve a Firewall Management License on the M-Series Appliance

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Register Panorama

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Customer Support Portal • Panorama serial number

STEP 1 | Record the Panorama serial number or auth-code and record your Sales Order Number or Customer ID.

For the auth-code, Sales Order Number, or Customer ID, see the order fulfillment email that Palo Alto Networks Customer Service sent when you placed your order for Panorama.

For the serial number, the location depends on the model:

- M-Series appliance—Log in to the Panorama web interface and record the **Serial #** value in the **Dashboard** tab, General Information section.
- Panorama virtual appliance—See the order fulfillment email or refer to the serial number generated when provisioning Panorama using VM Flex licensing.

The Panorama virtual appliance is automatically registered when you allocate a serial number using VM Flex licensing.

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STEP 2 | Register Panorama in the Palo Alto Networks Customer Support Portal (CSP).

The steps depend on whether you already have a login for the Palo Alto Networks CSP.

- If this is the first Palo Alto Networks appliance you are registering and you do not yet have a CSP login:
  1. Go to the Palo Alto Networks CSP.
  2. Click **Create my account**.
  3. Enter **Your Email Address** and respond to the reCAPTCHA prompt.
  4. Click **Submit** after you successfully respond to the reCAPTCHA prompt.
  5. Select **Register device using Serial Number or Authorization Code** and click **Submit**
  6. Complete the fields in the **Create Contact Details** and **Create UserID and Password** sections.
  7. Enter the Panorama **Device Serial Number** or **Auth Code**.
  8. Enter your **Sales Order Number** or **Customer ID**.
  9. Respond to the reCAPTCHA prompt.
  10. Click **Submit** after you successfully respond to the reCAPTCHA prompt.
- If you already have a CSP login:
  1. Log in to the Palo Alto Networks CSP.
  2. Click **Products** > **Devices** > **Register New Device**.

     *You can also **Register a Device** in the CSP **Support Home**.*
  3. Select **Register device using Serial Number** and click **Next**.
  4. Enter the Panorama **Serial Number**.
  5. Enter the **Device Name** to apply a name to search for and identify your Panorama.
  6. (Optional) Select a **Device Tag** to group Panorama with any other devices for which you have selected a device tag.

     The device tag must first be created at the account level (**Products** > **Devices** > **Device Tag**) before it can be selected when you register Panorama.
  7. If the Panorama management server is not internet-connected, check **Device will be used offline** and select the **OS Release** version.
  8. Enter the required Location Information (as indicated by the asterisks) if you have purchased the 4 hour RMA.
  9. **Agree and Submit** the EULA.

     After you see the registration complete message, close the Device Registration dialog.

Activate a Panorama Support License

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Customer Support Portal

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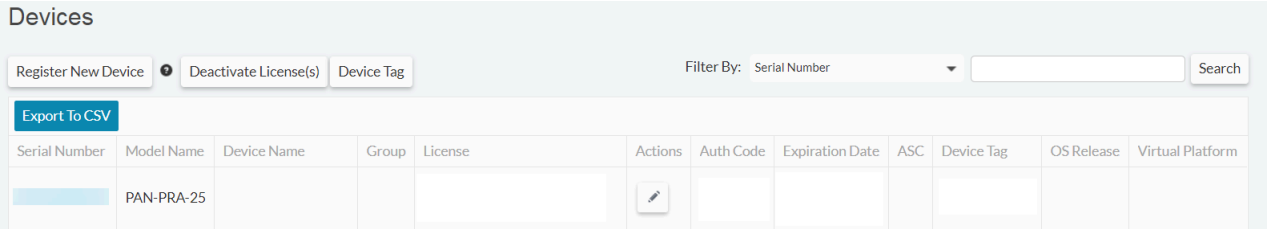
Where Can I Use This?	What Do I Need?
	• Registered Panorama management server

Before activating a Panorama support license on a Panorama M-Series appliance or Panorama virtual appliance, you must Register Panorama.

If the support license expires, Panorama can still manage firewalls and collect logs, but software and content updates will be unavailable. The software and content versions on Panorama must be the same as or later than the versions on the managed firewalls, or else errors will occur. For details, see Panorama, Log Collector, Firewall, and WildFire Version Compatibility.

STEP 1 | Log in to the Palo Alto Networks customer support portal to activate the auth-code.

1. Select **Products** > **Devices** and enter your Panorama serial number to Filter by the **Serial Number**.



2. Select the pencil icon in the Action column, select **Activate Auth-Code** and enter your support license **Authorization Code**, and click **Agree and Submit**.

STEP 2 | Log in to the Panorama web interface, and select **Panorama** > **Support** > **Activate feature using authorization code**.

STEP 3 | Enter the **Authorization Code** and click **OK**.

STEP 4 | Verify that the subscription is activated. Check the details (for example, the **Expiry Date**, support **Level**, and **Description**) in the Support section of the page.

Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is Internet-connected

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Panorama virtual appliance • Customer Support Portal • Outbound internet access • Registered Panorama management server

In order to manage devices on Panorama, you need to activate a firewall management license generated by PAN-OS. The device management license you activate determines the number of devices Panorama can manage. Log Collectors and WildFire appliances are not treated

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as managed devices and do not count toward the number of devices allotted by the device management license.

Before activating and retrieving a firewall management license on the Panorama virtual appliance, you must Register Panorama. If you are running an evaluation license and want to apply a license that you purchased, you must still register and activate/retrieve the purchased license. Additionally, you must then change the serial number of the Panorama from the evaluation serial number to the production serial number.

STEP 1 | Log in to the Panorama Web Interface.

STEP 2 | Select **Panorama** > **Setup** > **Management** and edit the General Settings.

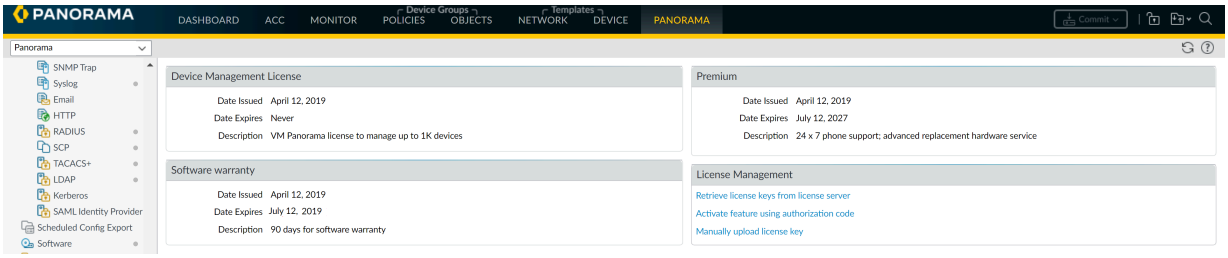
STEP 3 | Enter the Panorama **Serial Number** (included in the order fulfillment email) and click **OK**.

STEP 4 | Select **Panorama** > **Licenses** to activate or retrieve the firewall management license:

- **Retrieve license keys from license server**—Panorama automatically retrieves and activates the firewall management license from the Panorama Update Server.
- **Activate feature using authorization code**—Enter the firewall management license authorization code and click **OK** to activate the license. The authorization code can be obtained from the order fulfillment email or by logging in to the Palo Alto Networks Customer Support web site by finding the Panorama management server.
- **Manually upload license key**—Log in to the Palo Alto Networks Customer Support web site, find your Panorama management server, and download the firewall management license key to your local device. After you download the license key, click **Choose File** to select the license key and click **OK**.

STEP 5 | Verify the firewall management license is activated.

The Device Management License section now appears displaying the date the license was issued, when the license expires, and a description of the firewall management license.



Activate/Retrieve a Firewall Management License when the Panorama Virtual Appliance is not Internet-connected

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• Panorama virtual appliance • Customer Support Portal • Registered Panorama management server

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Before activating and retrieving a firewall management license on the Panorama virtual appliance, you must Register Panorama. In order to manage devices on Panorama, you will need to activate a device management license. The device management license you activate will determine the number of devices Panorama can manage. Log Collectors and WildFire appliances are not treated as managed devices and will not count toward the number of devices allotted by the device management license. If you are running an evaluation license and want to apply a license that you purchased, you must still register and activate/retrieve the purchased license.

After upgrading to PAN-OS 8.1, you will be prompted to retrieve a valid Panorama management license when you first log in to the Panorama web interface when Panorama has finished rebooting. To activate or retrieve the valid management license if the Panorama virtual appliance is offline or unable to reach the Palo Alto Networks update server, you must get the relevant appliance information for the Panorama virtual appliance and upload it to the Customer Support web site.

STEP 1 | Log in to the Panorama Web Interface.

STEP 2 | (Initial Deployment only) Enter the Panorama **Serial Number**.

1. Select **Panorama** > **Setup** > **Management** and edit the General Settings.
2. Enter the Panorama **Serial Number** (included in the order fulfillment email) and click **OK**.
3. Select **Commit** > **Commit to Panorama** and **Commit** your changes.

STEP 3 | Upload the Panorama virtual appliance information to the Customer Support website.

1. On the Retrieve Management License dialogue, click the **here** link to gather the UUID, CPUID, Panorama Version and Virtual Platform information. Click **Download Link** to

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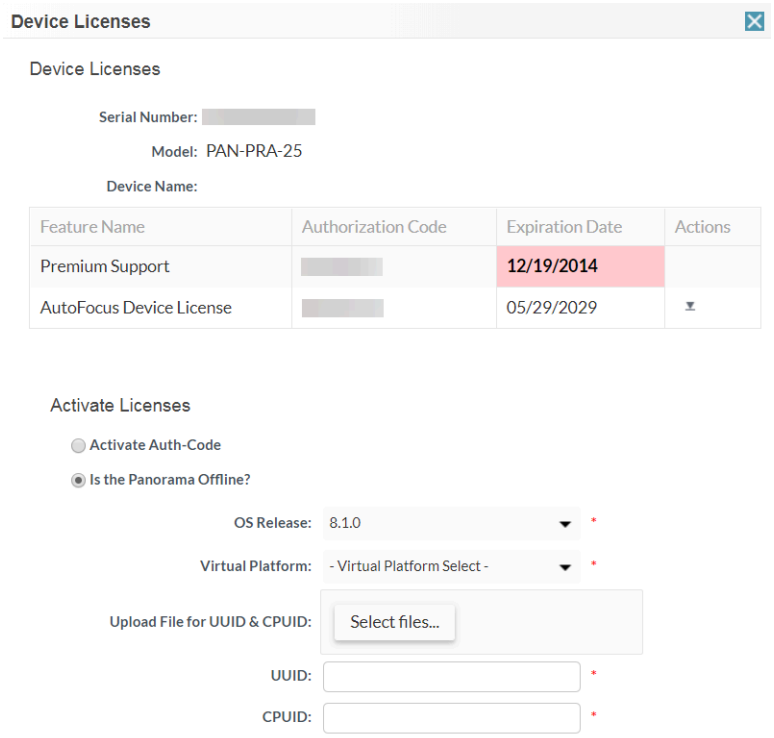
download a XML file of the required Panorama information that can be uploaded to the Customer Support Portal.

On initial deployment, may need to log out and back in to the web interface to see the dialogue.

2. Log in to the Palo Alto Networks Customer Support web site.
3. Click **Get Support** in the upper right-hand corner.
4. Select **Products** > **Devices**, find your Panorama virtual appliance and in the Action column, click the edit icon (✎).
5. Select **Is the Panorama Offline?** and enter the Panorama information gathered in Step 2, or click **Select files...** to upload the downloaded XML file.
6. **Agree and Submit** the EULA.



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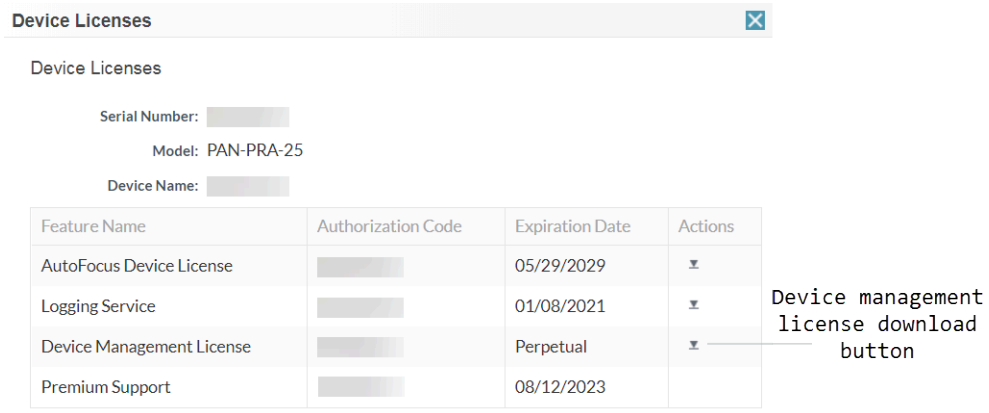
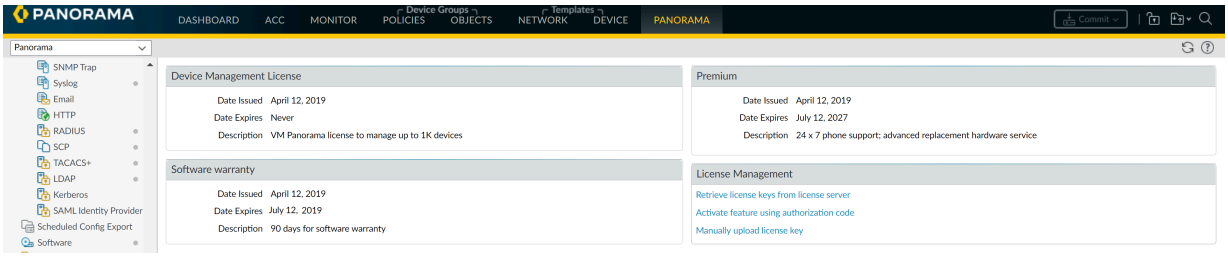
STEP 4 | Install the device management license.

1. In the Actions column, download the device management license.



2. In the Panorama web interface, click **Panorama** > **Licenses** and **Manually upload license key**.
3. Click **Choose file**, locate the downloaded device management license key and click **OK**.

STEP 5 | Confirm that the device management license was successfully uploaded by verify that the Device Management License displays with the license information.



Activate/Retrieve a Firewall Management License on the M-Series Appliance

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	• M- series hardware appliance • Customer Support Portal • Registered Panorama management server

In order to manage devices on Panorama, you need to activate a Capacity License. The Capacity License determines the number of devices Panorama can manage. Log Collectors and WildFire appliances are not treated as managed devices and do not count toward the number of devices allotted by the Capacity License.

Before activating and retrieving a Panorama firewall management license on the M-Series appliance:

- Register Panorama.

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- Locate the auth-codes for the product/subscription you purchased. When you placed your order, Palo Alto Networks Customer Service sent you an email that listed the auth-code associated with the purchase. If you cannot locate this email, contact Palo Alto Networks Customer Support to obtain your codes before proceeding.

After you activate and retrieve the license, the **Panorama** > **Licenses** page displays the associated issuance date, expiration date, and the number of firewalls that the license enables Panorama to manage.

To activate and retrieve the license, the options are:

- Use the web interface to activate and retrieve the license.

  Select this option if Panorama is ready to connect to the Palo Alto Networks update server (you completed the task Perform Initial Configuration of the M-Series Appliance) but you have not activated the license on the Palo Alto Networks Customer Support web site.

  1. Select **Panorama** > **Licenses** and click **Activate feature using authorization code**.
  2. Enter the **Authorization Code** and click **OK**. Panorama retrieves and activates the license.

- Retrieve the license key from the license server.

  If Panorama is not ready to connect to the update server (for example, you have not completed the initial M-Series appliance setup), you can activate the license on the Support website so that, when Panorama is ready to connect, you can then use the web interface to retrieve the activated license. The process of retrieving an activated license is faster than the process of both retrieving and activating.

  1. Activate the license on the Palo Alto Networks Customer Support web site.
     1. On a host with internet access, use a web browser to access the Palo Alto Networks Customer Support web site and log in.
     2. Select **Products** > **Devices**, find your M-Series appliance and, in the Action column, click the edit icon ( ).
     3. Select **Activate Auth-Code**, enter the **Authorization Code** and click **Agree and Submit** to activate the license.
  2. Configure Panorama to connect to the update server: see Perform Initial Configuration of the M-Series Appliance.
  3. Select **Panorama** > **Licenses** and click **Retrieve license keys from the license server**. Panorama retrieves the activated license.

- Manually upload the license from a host to Panorama. Panorama must have access to that host.

  If Panorama is set up (you completed the task Perform Initial Configuration of the M-Series Appliance) but does not have a connection to the update server, activate the license on the

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Support website, download it to a host that has a connection to the update server, then upload it to Panorama.

1. Activate and download the license from the Palo Alto Networks Customer Support web site.
    1. On a host with internet access, use a web browser to access the Palo Alto Networks Customer Support web site and log in.
    2. Select **Products** > **Devices**, find your M-Series appliance and, in the Action column, click the edit icon ( ).
    3. Select **Activate Auth-Code**, enter the **Authorization Code** and click **Agree and Submit** to activate the license.
    4. In the Action column, click the download icon and save the license key file to the host.
2. In the Panorama web interface, select **Panorama** > **Licenses**, click **Manually upload license key** and click **Browse**.
3. Select the key file you downloaded to the host and click **Open**.
4. Click **OK** to upload the activated license key.

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Install the Panorama Device Certificate

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	❑ Device management license ❑ Support license ❑ Outbound internet access ❑ Customer Support Portal (CSP) account with one of the following user roles: Super User, Standard User, Limited User, Threat Researcher, AutoFocus Trial Role, Group Super User, Group Standard User, Group Limited User, Group Threat Researcher, Authorized Support Center (ASC) User, and ASC Full Service User. ❑ Panorama superuser role

You must install the device certificate on the Panorama™ management server to use one or more cloud services. You only need to install a device certificate once. The device certificate has a 90-day lifetime. The firewall reinstalls the device certificate 15 days before the certificate expires. In the event Panorama is unable to reinstall the device certificate on its own, you may need to manually restore an expired device certificate.

To successfully install the device certificate, Panorama must have an outbound internet connection and the following Fully Qualified Domain Names (FQDN) and ports must be allowed on your network.

FQDN	Ports
• http://ocsp.paloaltonetworks.com • http://crl.paloaltonetworks.com • http://ocsp.godaddy.com	TCP 80
• https://api.paloaltonetworks.com • http://apitrusted.paloaltonetworks.com • https://certificatetrusted.paloaltonetworks.com • https://certificate.paloaltonetworks.com	TCP 443
• *.gpcloudservice.com	TCP 444 and TCP 443

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M-300 and M-700 appliances automatically install the device certificate when they first connect to the Palo Alto Networks CSP during the initial registration process. You do not need to manually install the device certificate for these M-Series appliances.

STEP 1 | Generate the One Time Password (OTP).

An OTP lifetime is 60 minutes and expires if not used within the 60 minute lifetime.

Panorama may only attempt to retrieve the OTP from the CSP one time. If Panorama fails for any reason to fetch the OTP, the OTP expires and you must generate a new OTP.

1. Log in to the Customer Support Portal with a user role that has permission to generate an OTP.
2. Select **Products** > **Device Certificates** and **Generate OTP**.
3. For the **Device Type**, select **Generate OTP for Panorama** and click **Next**.
4. Select the **Panorama Device** serial number and **Generate OTP**.
5. **Generate OTP** and copy the OTP.

STEP 2 | Log in to the Panorama Web Interface as a Superuser.

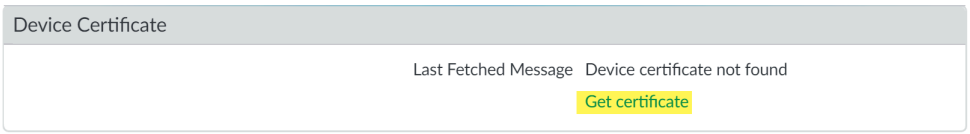
A Panorama admin with Superuser access privileges is required to required to apply the OTP used to install the device certificate on Panorama.

STEP 3 | Configure the Network Time Protocol (NTP) server.

An NTP server is required to validate the device certification expiration date, ensure the device certificate does not expire early or become invalid.

1. Select **Panorama** > **Setup** > **Services**.
2. Select **NTP** and enter the hostname or IP address of the **Primary NTP Server**.
3. (Optional) Enter a the hostname or IP address of the **Secondary NTP Server**.
4. (Optional) To authenticate time updates from the NTP server(s), for **Authentication Type**, select one of the following for each server.
   - **None** (default)—Disables NTP authentication.
   - **Symmetric Key**—Firewall uses symmetric key exchange (shared secrets) to authenticate time updates.
     - **Key ID**—Enter the Key ID (1-65534)
     - **Algorithm**—Select the algorithm to use in NTP authentication (**MDS** or **SHA1**)
5. Click **OK** to save your configuration changes.
6. Select **Commit** and **Commit to Panorama**.

STEP 4 | Select **Panorama** > **Setup** > **Management** > **Device Certificate Settings** and **Get certificate**.



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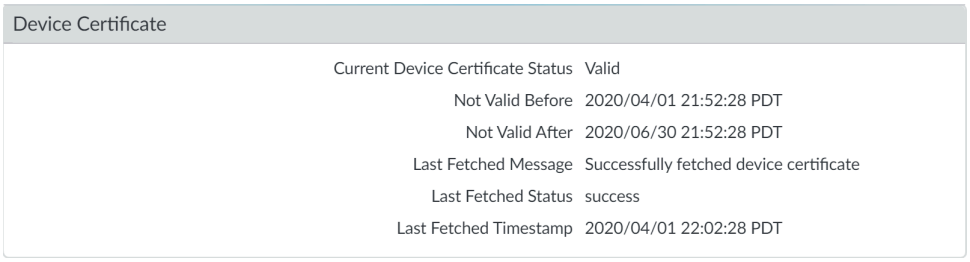
STEP 5 | Enter the **One-time Password** you generated and click **OK**.

STEP 6 | Panorama successfully retrieves and installs the certificate.



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Install the Device Certificate for a Dedicated Log Collector

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	❑ Device management license ❑ Support license ❑ Outbound internet access ❑ Customer Support Portal (CSP) account with one of the following user roles: Super User, Standard User, Limited User, Threat Researcher, AutoFocus Trial Role, Group Super User, Group Standard User, Group Limited User, Group Threat Researcher, Authorized Support Center (ASC) User, and ASC Full Service User. ❑ Panorama superuser role

You must install the device certificate on the Dedicated Log Collector to use Device Telemetry. You only need to install a device certificate once. The device certificate has a 90-day lifetime. The Dedicated Log Collector reinstalls the device certificate 15 days before the certificate expires. In the event the Dedicated Log Collector is unable to reinstall the device certificate on its own, you may need to manually restore an expired device certificate.

To successfully install the device certificate, the Dedicated Log Collector must have an outbound internet connection and the following Fully Qualified Domain Names (FQDN) and ports must be allowed on your network.

You must manually install the device certificate on each Dedicated Log Collector individually. Installing the device certificate from the Panorama™ management server is not supported.

FQDN	Ports
• http://ocsp.paloaltonetworks.com • http://crl.paloaltonetworks.com • http://ocsp.godaddy.com	TCP 80
• https://api.paloaltonetworks.com • http://apitrusted.paloaltonetworks.com • https://certificatetrusted.paloaltonetworks.com • https://certificate.paloaltonetworks.com	TCP 443

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FQDN	Ports
• *.gpcloudservice.com	TCP 444 and TCP 443

M-300 and M-700 appliances automatically install the device certificate when they first connect to the Palo Alto Networks CSP during the initial registration process. You do not need to manually install the device certificate for these M-Series appliances.

STEP 1 | Log in to the Dedicated Log Collector CLI as a Superuser.

An admin with Superuser access privileges is required to required to apply the OTP used to install the device certificate on Panorama.

STEP 2 | View the current device certificate status on the Dedicated Log Collector.

```
admin>show device-certificate status
```

The Dedicated Log Collector displays one of the following responses:

- **Device certificate was never installed**—`No device certificate found`
- **Device certificate expired**—`Current device certificate status: Expired`

  The response also displays the lifetime of the previous device certificate and the date and time the last device certificate fetch was attempted.
- **Device certificate fetch failed**—Response displays the last time the device certificate fetch was attempted.

STEP 3 | Generate the One Time Password (OTP).

An OTP lifetime is 60 minutes and expires if not used within the 60 minute lifetime.

The Dedicated Log Collector may only attempt to retrieve the OTP from the CSP one time. If the Dedicated Log Collector fails for any reason to fetch the OTP, the OTP expires and you must generate a new OTP.

1. Log in to the Customer Support Portal with a user role that has permission to generate an OTP.
2. Select **Products** > **Device Certificates** and **Generate OTP**.
3. For the **Device Type**, select **Generate OTP for Panorama** and click **Next**.
4. Select the **Panorama Device** serial number and **Generate OTP**.
5. **Generate OTP** and copy the OTP.

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STEP 4 | Configure the Network Time Protocol (NTP) server.

An NTP server is required to validate the device certification expiration date, ensure the device certificate does not expire early or become invalid.

1. Log in to the Dedicated Log Collector CLI as a Superuser.

   An admin with Superuser access privileges is required to required to apply the OTP used to install the device certificate on Panorama.
2. configure the NTP server.

```
admin>configure
```

```
admin#set deviceconfig system ntp-servers primary-ntp-server
 ntp-server-address <ip_address>
```

```
admin#set deviceconfig system ntp-servers secondary-ntp-server
 ntp-server-address <ip_address>
```

```
admin>commit
```

```
admin>exit
```

STEP 5 | Install the device certificate.

```
admin>request certificate fetch otp <otp_value>
```

STEP 6 | Verify the device certificate successfully installed.

```
admin> show device-certificate status
```

A successful device certificate installation displays the following response:

```
Device Certificate information:
Current device certificate status: Valid
Not valid before: 2022/11/30 15:17:47 PST
Not valid after: 2023/02/28 15:17:47 PST
Last fetched timestamp: 2022/11/30 15:29:42 PST
Last fetched status: success
Last fetched info: Successfully fetched Device Certificate
```

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Access and Navigate Panorama Management Interfaces

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	❑ This is a core network security feature and does not have prerequisites

Panorama provides three management interfaces:

- **Web interface**—The Panorama web interface has a look and feel similar to the firewall web interface. If you are familiar with the latter, you can easily navigate, complete administrative tasks, and generate reports from the Panorama web interface. This graphical interface enables you to access Panorama using HTTPS and it is the best way to perform administrative tasks. See Log in to the Panorama Web Interface and Navigate the Panorama Web Interface. If you need to enable HTTP access to Panorama, edit the Management Interface Settings on the **Panorama** > **Setup** > **Management** tab.
- **Command line interface (CLI)**—The CLI is a no-frills interface that allows you to type commands in rapid succession to complete a series of tasks. The CLI supports two command modes—operational and configuration—and each has its own hierarchy of commands and statements. When you become familiar with the nesting structure and the syntax for the commands, the CLI enables quick response times and administrative efficiency. See Log in to the Panorama CLI.
- **XML API**—The XML-based API is provided as a web service that is implemented using HTTP/HTTPS requests and responses. It enables you to streamline your operations and integrate with existing, internally developed applications and repositories. For details on using the Panorama API, refer to the PAN-OS and Panorama XML API Usage Guide.

Log in to the Panorama Web Interface

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	❑ This is a core network security feature and does not have prerequisites

The Panorama web interface for the M-600 appliance in Panorama or Management Only mode is inaccessible if you have 6 or more concurrent API calls. A `504 Gateway Timeout` error is displayed when you try to log in to the Panorama web interface of an M-600 appliance with 6 or more concurrent API calls.

STEP 1 | Launch an internet browser and enter the Panorama IP address using a secure connection (https://<IP address>).

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STEP 2 | Log in to Panorama according to the type of authentication used for your account. If logging in to Panorama for the first time, use the default value **admin** for your username and password.

- **SAML**—Click **Use Single Sign-On** (SSO). If Panorama performs authorization (role assignment) for administrators, enter your **Username** and **Continue**. If the SAML identity provider (IdP) performs authorization, **Continue** without entering a **Username**. In both cases, Panorama redirects you to the IdP, which prompts you to enter a username and password. After you authenticate to the IdP, the Panorama web interface displays.
- **Any other type of authentication**—Enter your user **Name** and **Password**. Read the login banner and select **I Accept and Acknowledge the Statement Below** if the login page has the banner and check box. Then click **Login**.

STEP 3 | Read and **Close** any messages of the day.

Navigate the Panorama Web Interface

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	❑ This is a core network security feature and does not have prerequisites

Use the Panorama web interface to configure Panorama, manage and monitor firewalls, Log Collectors, and WildFire appliances and appliance clusters, and access the web interface of each firewall through the **Context** drop-down. Refer to the Panorama online help for details on the options and fields in each web interface tab. The following is an overview of the tabs:

Tab	Description
Dashboard	View general information about the Panorama model and network access settings. This tab includes widgets that display information about applications, logs, system resources, and system settings.
ACC	View the overall risk and threat level on the network, based on information that Panorama gathered from the managed firewalls.
Monitor	View and manage logs and reports.
Device Groups > **Policies**	Create centralized policy rules and apply them to multiple firewalls/device groups. You must add a device group for this tab to display.
Device Groups > **Objects**	Define policy objects that policy rules can reference and that managed firewalls/device groups can share. You must add a device group for this tab to display.

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Tab	Description
Templates > **Network**	Configure network setting, such as network profiles, and apply them to multiple firewalls. You must add a template for this tab to display.
Templates > **Device**	Configure device settings, such as server profiles and admin roles, and apply them to multiple firewalls. You must add a template for this tab to display.
Panorama	Configure Panorama, manage licenses, set up high availability, access software updates and security alerts, manage administrative access, and manage the deployed firewalls, Log Collectors, and WildFire appliances and appliance clusters.

Log in to the Panorama CLI

Where Can I Use This?	What Do I Need?
• NGFW (Managed by Panorama)	☐ This is a core network security feature and does not have prerequisites

You can log in to the Panorama CLI using a serial port connection or remotely using a Secure Shell (SSH) client.

◉ Use SSH to log in to the Panorama CLI.

The same instructions apply to an M-Series appliance in Log Collector mode.

Optionally, you can Configure an Administrator with SSH Key-Based Authentication for the CLI.

1. Ensure the following prerequisites are met:
   - You have a computer with network access to Panorama.
   - You know the Panorama IP address.
   - The Management interface supports SSH, which is the default setting. If an administrator disabled SSH and you want to re-enable it: select **Panorama** > **Setup**

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> **Interfaces**, click **Management**, select **SSH**, click **OK**, select **Commit** > **Commit to Panorama**, and **Commit** your changes to the Panorama configuration.

2. To access the CLI using SSH:
   1. Enter the Panorama IP address in the SSH client and use port 22.
   2. Enter your administrative access credentials when prompted. After you log in, the message of the day displays, followed by the CLI prompt in Operational mode. For example:

      ```
      admin@ABC_Sydney>
      ```

- Use a serial port connection to log in to the Panorama CLI.
  1. Make sure that you have the following:
     - A null-modem serial cable that connects Panorama to a computer with a DB-9 serial port
     - A terminal emulation program running on the computer
  2. Use the following settings in the terminal emulation software to connect: 9600 baud; 8 data bits; 1 stop bit; No parity; No hardware flow control.
  3. Enter your administrative access credentials when prompted. After you log in, the message of the day displays, followed by the CLI prompt in Operational mode.

- Change to Configuration mode.

  To switch to Configuration mode, enter the following command at the prompt:

  ```
  admin@ABC_Sydney> configure
  ```

  The prompt changes to `admin@ABC_Sydney#`.

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